



Marketing Research

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Preface



Chapter 1

Prerequisites

No prerequisites required to read this book

Chapter 2

Introduction

This is not an actual book or guide on marketing research, but more like a placeholder for major constructs in marketing as well as interesting articles in marketing.

Feel free to comment or suggest changes to the content of this book.

Differences between goods and services:

- Intangibility
- Complexity
- Heterogeneity

Dimensions of Qualitative Research:

1. Ontology: “is a philosophical belief system about the nature of social reality- what can be known and how”.
2. Epistemology: “is a philosophical belief system about who can be a knower”.
3. Methodology (theoretical perspective): “an account of social reality or some component of it that extends further than what has been empirically investigated”:
 - Post-positivist: causal relationship can be tested (i.e., disprove)
 - Interpretive: assumes the world is constantly being constructed through group interactions; hence social reality can be understood via the perspectives of social actors enmeshed in meaning-making

activities”.

- Critical: view social reality as an ongoing construction, and suggest that discourses created in shifting fields of social power shape social reality and its study.

4. Methods: “technique for gathering evidence”.

Chapter 3

Advertising

People don't like intrusive ads, especially when they are vulnerable (Tybout and calkins, 2005, p.143)

Visual Persuasion

- (vis, 1997) and (vis, 2003) are two books on this subject

Prior distribution for TV advertising elasticity for consumer packaged goods can be found in (Shapiro et al., 2018). A substantial portion of products has statistically insignificant or negative estimates for advertising elasticity. TV ads might not be the best vehicle to reach the customer now.

(Gordon et al., 2019) reframe our idea of advertising effect, which should be thought in the sense of incremental effect above the baseline of an ad on consumer behavior.

(Gordon et al., 2017) and (Lewis and Rao, 2015) provided evidence that observational data can hardly detect advertising effect from noise (i.e., we could either over or under-estimate).

The goal standard to measure advertising is random experiment, but it not possible. An improvement is to have panel data from different regions of your market to estimate the baseline for each market. Then, you can uncover the advertising effect.

Spillover effect in advertising:

- (Rutz and Bucklin, 2011)

(McGuire, 1978):

- Information processing model of social influence (e.g., advertising effectiveness)

- Behavioral chain of information profession model. (p. 158) (Markov process) - persuasion
- $P(\text{presentation of message}) \times P(\text{attention to message}) \times P(\text{Comprehension of Conclusion}) \times P(\text{Yielding to the Conclusion}) \times P(\text{retention of the new belief}) \times P(\text{Behaving on the basis of belief})$
 - Presentation of message = # of reach (advertising investment)
 - attention to message = # of true reach or Recognition test
 - comprehension = recall test or semantic differential profiles or checklist tests
 - yielding = attitude change
 - retention = attitude change after time interval
 - behavior = actual observation
- Independent variables (components of persuasive communication): source, message, channel, receiver, and destination (Table 1, p. 167)
- Matrix of persuasion
- Advertising appeals:
 - Positive (gain): reduce anxiety
 - Negative (loss): increase anxiety
- Development in the field: this model assume 1 hierarchy, but later developments change the idea of hierarchy model.

Court et al. (2009)

Traditional funnel:

1. Awareness
2. Familiarity
3. Consideration
4. Purchase
5. Loyalty
 1. Active loyalist
 2. Passive loyalist

Consumer decision journey McKinsey (2009):

1. Initial consideration set
2. Active evaluation
3. Decision at moment of purchase
4. Postpurchase experience

- Customer Decision Journey is from the perspective of consumers
- Customer Funnel is still important but it's from the company's perspective.
- Analogy here is that you look from 3D (Customer Decision Journey) to 2D (Customer Funnel).

(MacInnis and Jaworski, 1989)

1. Hierarchy of effects model (1960s)
2. Multiattribute attitude model and cognitive response models (1970s)

Proposed model:

- Antecedents:
 - Types of needs
 - * Utilitarian needs
 - * Expressive needs
 - Socially expressive needs: to express, or reflect self-image
 - Experiential needs: to satisfy one's cognitive or sensory
 - Motivation (in CB **involvement**): “the desire to process brand information in the ad.” (p. 4)
 - * situational
 - * enduring
- Processing
 - Attention:
 - * higher processing motivation leads to higher attention
 - * higher utilitarian needs, more focused on brand attributes
 - * higher expressive needs, more focused on symbolic/ experiential value
 - Processing capacity:
 - * greater processing motivation leads to greater processing capacity for analyzing the ad
 - * greater the processing capacity results in less processing capacity for other task.
 - Level of brand processing matches processing operations
 1. feature analysis: salient properties/features
 2. basic categorization

3. meaning analysis: basic understanding
4. information integration: Is this brand association?
5. role-taking
6. constructive processes

– Ability and opportunity are moderators of processing

- Consequences
 - Cognitive responses (thoughts)
 - Emotional responses (feelings)

Another classifications of responses:

- message-related (brand-relevant)
- execution-related (brand-irreverent)
- viewing-context-related (environment-related)

Advantages of the proposed model: Incorporate

- Elaboration Likelihood model: (Petty and Cacioppo, 1986)
- Mitchell's brand processing model
- Greenwald and Leavitt's model
- Lutz's typology

Note:

- A_{Ad} mediates A_B based on levels of processing (specifically, meaning analysis, information integration, and role taking - high levels).
- Tradeoff between brand/message-related elements and ad-execution elements
- Cohen and chestnut 1990, behavioral loyalty and attitudinal loyalty stem did not stem from the gap between attitude and actual behavioral.

(Vakratsas and Ambler, 1999)

- Advertising input -> Filters -> Consumer (Cognition, Affect, Experience) -> Behavior
- Market Response Models:
 - Aggregate level (Bass' model): market data
 - * 3 exposures per purchase cycle is optimal (p. 29)
 - Individual level: individual brand choice
- Cognitive Information Models (C)

- There is a differential effect between price (increase) and non-price (decrease) advertising on price sensitivity
- Pure Affect Models (A)
 - Mere exposure theories
 - * Response competition and Optimal arousal theories: “wear-in” effect which means it takes time to getting familiar to advertising messages to reach optimal effectiveness.
 - * Two-factor theory: wear-out effect: after optimal exposures, the effect of advertising starts to decrease
 - * Hence, advertising response has an inverted-U shape
 - Affective responses to advertising:
 - * Attitude towards the brand
 - * Attitude towards the ad
- Persuasive Hierarchy Models (CA)
 - Cognition -> Affect -> behavior. (CA)
 - Elaboration Likelihood Model (ELM): (Petty and Cacioppo, 1986)
 - Another model is (MacInnis and Jaworski, 1989)
 - Fishbein-Ajzen (1975) involvement model
 - Batra and Ray (1985): Utilitarian and hedonic effect on attitudes towards the brand
 - Involvement moderates the effect of ad evaluation (persuasion is found in low-involvement consumers with attitude towards the ad).
 - Little support for CA model, but the authors still believe in it.
- Low-involvement (motivation) Hierarchy Models (CEA)
 - Cognition -> Experience -> Affect
- Integrative Models (CAE)
 - Information Integration Response Model (IIRM)
 - Deighton’s (1984, 1986) two-stage model:
 - * first experience = expectations
 - * second stage = product trial/experience
- Hierarchy-Free Models (NH)
- Generalizations:
 1. Experience, affect, cognition are mediators of advertising effects.

2. "Short-term ad elasticities are small and decrease during the product life cycle (p. 35).

(Wind and Sharp, 2009)

- 23 Empirical Generalizations
- Gaps:
 - Boundary conditions
 - Advertising properties
 - Measurement issues

(Batra and Keller, 2016)

- Media:
 - search
 - display
 - mobile
 - TV
 - social media
- Considerations for a well-integrated marketing communications program
 - Consistency
 - Complementarity
 - Cross-effects
- Dynamic Expanded Customer Decision Journey
 - traditional media
 - newer media
 - * Search ads
 - * display ads
 - * websites
 - * email
 - * social media
 - * Mobile
- Interaction and Cross-effects
 - Traditional media synergies
 - sales force and personal selling interactions

3.1. COGNITIVE AND AFFECTIVE MEDIATORS OF ADVERTISING EFFECTS 19

- Online and offline synergies
- Drawbacks
 - Limited outcome variables
 - Limited longitudinal studies
 - Did not account for consumer decision stage
- Media type
 - Paid (TV, print, direct)
 - owned (websites, blogs, apps, social media)
 - earned (WOM, press coverage)
- Factors that affect consumer communication processing (model) (p. 130)
- Communication Matching Model: “matches the expected main and interactive effects of different media options with the communications objectives for a brand”
- Communications Optimization Model
 - Coverage
 - Cost
 - Contribution
 - Commonality: different communications share the same meaning
 - Complementarity
 - Cross-Effects
 - Conformability
- IMC Conceptual Framework

3.1 Cognitive and Affective mediators of Advertising Effects

Evaluative responses = attitude

Cognitive approach -> Affective approach (not only comes from cognitive) -> Behavioral approach (Fishbein & Ajzen, 1975 - theory of planned behavior)

System 2: is kinda of independent of Cognitive, but Affective is system 1.

- Expanded model added subjective norm
- Zajonc (1980)

- Evaluation from cognitive approach is multi-attribute model
- Anthony Greenwald: Cognitive Response Theory: what important is what is in the consumer mind when they see the ad.

Conditioning:

- Classical (Pavlovian) conditioning: physiological automatic reaction occurred after being exposed to an unconditioned stimulus.
- Evaluative conditioning: direct transfer of affect from one stimulus to another via a conditioning paradigm.

Affective route:

Moods = diffuse, hard to pin down the source, more long-lasting

Emotion = specific, discrete

(Wright, 1973)

- Three modes of spontaneous cognitive responses to advertising stimulus:
 - Counterargument
 - Source Derogation
 - Support Argument

$$Acceptance = w_{SA} \sum_i SA_i - w_{CA} \sum_j CA_j - w_{SD} \sum_k SD_k$$

Situational Factors

- Content-processing involvement: “stemming from receiver’s perception of the relevancy”
- Message Modality: audio, print

(Batra and Ray, 1986b)

- Advertising repetition increases brand attitude and purchase intention when support and counter argument production are low; while under high level of such production, brand attitude and purchase intention level off.
- What happen to make the downturn of advertising repetition earlier or later?
- Appropriate interval: purchase cycle (number of exposure per purchase cycle). 3 exposures per purchase cycle is the optimal number
- Wear-in: how many times it takes for the ad to take effect?
- Wear-out: how many times it takes for the ad to bore you?
 - If you change the ad execution, the wear-out is pushed back.

3.1. COGNITIVE AND AFFECTIVE MEDIATORS OF ADVERTISING EFFECTS²¹

- Traditional thoughts advertising repetition would always wear out (inverted-U curve between repetition and impact on customer's attitude) because of wearout and mere exposure
- Ability, motivation and opportunity are antecedents of cognitive processing

(Kupor and Tormala, 2015)

- Momentary interruptions can promote persuasion
 - higher for low need for cognitive individuals (motivation to engage in thoughtful processing) than high ones
- In other words, interruptions can increase consumers' processing of a message.
- Interruption amplified arousal (need for completion/ goal pursuit and curiosity)

(Gibson, 2008): Affective Responses Mediating Acceptance of Advertising

- Using Implicit Association Test (???)
- Evaluative conditioning only influences explicit attitudes when there is no previous strong preference or priori

Batra and Ray (1986a)

- Antecedents of attitude towards the ad:
- Attitude toward the ad leads changes in brand attitudes (MacKenzie et al., 1986; Mitchell and Olson, 1981)
- In low involvement context, execution cues and source likeability (message-oriented and communicator-oriented) have greater impact on persuasion
- Affect typologies (p. 237)

Pham et al. (2013)

- ad-evoked feelings positively influence brand attitudes both directly and indirectly (via changes in attitude toward the ad), regardless of involvement with the product category, products types (e.g., durables, nondurables, services, search or experience goods).
- This effect is greater among hedonic products than utilitarian ones.

Holbrook and Batra (1987)

- emotional reactions mediate the effect of advertising on attitudes toward ad or brand.
- Why divided two articles? the second study claimed that the last paper's list of positive affective mediators was limited, the second one expands to range of emotions.

- Is there a difference between affect and emotions?

(Batra and Ahtola, 1991; Voss et al., 2003) offer scale to measure the hedonic and utilitarian dimensions of consumer attitude

Chapter 4

Brand Equity

- Quantitative measure of brand equity: Kamakura and Russell (1993)

brand and psychological distance: Congruence effect: “information is easier to process, which leads consumers to respond more favorably.” (Connors et al., 2020) “consumer spending higher when distance between consumer and brand matched with construal level of marketing communications.”

Value-added is the difference between a product’s price to consumers and the cost of producing it.

Whereas, brand equity is the difference between a branded product’s price to a generic product’s price.

This generic product can also have value-added in it in other forms (e.g., convenience)

Brand identity offers a value proposition to customers, where “a brand’s value proposition is a statement of the functional, emotional, and self-expressive benefits delivered by the brand that provide value to the customers” (Aaker, 1996, pp.95).

Functional benefits are product attributes that deliver functional utility to customers, while emotional benefits are positive feelings customers feel when using the brand. Ads that provide emotional benefits are higher in effectiveness score compared to functional benefits ads (Aaker, 1996, pp.98).

For example, Evian – “Another day, another chance to feel healthy” ad - which provides emotional benefits of feeling satisfied after a workout. Moreover, self-expressive benefits are when brands provide a means for consumers to express their self-image. A person/ consumer can have multiple roles which associate with multiple self-concepts. Such as a woman, a mother, an author. The difference between emotional benefits and self-expressive benefits can be subtle. Emotional benefits focus on feelings, private setting (e.g., feeling of watching

a TV show), past-oriented (memories), transitory, after-use emotional; In contrast, self-expressive benefits focus on self, public settings (e.g., using cars or clothes to signal), aspiration or future-oriented, permanent (i.e., the self links to person's personality), and during-use (wearing fancy clothes to signify a successful person). (cf. (Aaker, 1996, pp.99))

With the advancement in mass production of high -quality products, quality is no longer a dimension of status signal (Holt, 1998)

In the virality framework, the first and foremost driver is social currency. Social currency refers to things worth sharing. People share so that they can let others know that they are in the know. Sharing a new product to assimilate his or her identity with the product and its brand identity, customers are acquiring social currency. Moreover, brand identity is a driver of brand equity, which refers to the value portion that customers are willing to pay above the product's utility value (practical value). Brand identity offers a value proposition to customers, where "a brand's value proposition is a statement of the functional, emotional, and self-expressive benefits delivered by the brand that provide value to the customers" (Aaker, 1996, p. 95). Functional benefits are product attributes that deliver functional utility to customers, while emotional benefits are positive feelings customers feel when using the brand. Ads that provide emotional benefits are higher in effectiveness score compared to functional benefits ads (Aaker, 1996, p. 99). For example, Evian – "Another day, another chance to feel healthy" ad - which provides emotional benefits of feeling satisfied after a workout. Moreover, self-expressive benefits are when brands provide a means for consumers to express their self-image. A person/ consumer can have multiple roles which associate with multiple self-concepts. Such as a woman, a mother, an author. The difference between emotional benefits and self-expressive benefits can be subtle. Emotional benefits focus on feelings, private setting (e.g., feeling of watching a TV show), past-oriented (memories), transitory, after-use emotional; In contrast, self-expressive benefits focus on self, public settings (e.g., using cars or clothes to signal), aspiration or future-oriented, permanent (i.e., the self links to person's personality), and during-use (wearing fancy clothes to signify a successful person). (cf. (Aaker, 1996, p. 99)) Hence, one can see that the brand identity portion that provides self-expressive benefits is social currency while exhaustively different from the brand identity portion that provides functional benefits (i.e., the practical value/ usefulness/ utility) .

We acknowledge that some might view usefulness/ utility as a dimension of brand equity (perceived quality): the perceived quality dimension can signal the true practical value. However, the practical value that drives virality is usually understood, experimented, experienced; while perceived quality is subjective comprehension that is hard to demonstrate. Moreover, quality used to be the status signal. However, with the advancement in mass production of high-quality products, quality is no longer a status signal dimension (Holt, 1998). Thus, we believe that the distinction between the two understandings should be appreciated.

4.1 Brand Association

“different groups of people can have different or even contradictory associations with the same stimulus” (Wheeler & Berger, 2007)

4.2 Brand Loyalty

- (Berkowitz et al., 1978; Berkowitz, 1978) reviews on Brand Loyalty: Measurement and Management by Jacoby and Chestnut 1978, which pioneered the idea of **attitudinal** and **behavioral loyalty**.

Chapter 5

Celebrity Endorsement

Based on Power Distance Theory, power distance beliefs moderates the positive effect of celebrity endorsers on evaluations of advertising (Winterich et al., 2018)

Chapter 6

Innovation

(Chandrasekaran et al., 2020) introduce a model for successive technologies adoption that accounts for:

- Different rate of technology disengagement
- Different rate of technology adoption

Diffusion

Bass Model (Bass, 1969)

$$h(t) = p + qF(t)$$

where

- $h(t)$ = rate of adoption, given not adopted so far
- $F(t)$ = cumulative fraction of adopters
- p = coefficient of innovation, tendency to adopt independent social contagion
- q = coefficient of imitation tendency to adopt due to social contagion
- m = market size

After going public, firms increase their innovation levels (i.e., have more innovation), but those innovation are less risky (i.e., fewer breakthrough). (Wies and Moorman, 2015)

Chapter 7

WOM / Virality

To understand virality, we first need to comprehend its root, which is WOM. Acknowledging that there are subtle differences among WOM, virality, and sharing, we use the three terms interchangeably in this paper. Previous authors provide strong supports for the similarity between virality and WOM (Camarero & San José, 2011; Phelps, Lewis, Mobilio, Perry, & Raman, 2004).

Formally, (Westbrook, 1987, p. 261) defined WOM as “informational communication directed at other consumers about the ownership, usage, or characteristics of particular goods and services or their sellers.” In the age of the Internet, WOM evolves to “word of mouse” (or e-WOM). Both virality and WOM take advantages of the network effect to reach wide audiences (Vilpponen, Winter, & Sundqvist, 2006), but virality does it better due to its competitive cost, rapid diffusion, and better targeting (Bampo, Ewing, Mather, Stewart, & Wallace, 2008; Khan & Vong, 2014). Hence, virality is more synonymous with e-WOM.

Virality comes up as a by-product of the Internet, social media platforms, and network theory developments. In network theory, researchers from the domain of sociology or computer science study the structural characteristics/properties of a node (an individual) or a network, such as a node’s position in a network or network’s cluster. While epidemiologists, management, and marketing researcher study the characteristics of the entity being transmitted, such as virus or message itself. Thus, virality can be driven by content or context of the message (Berger & Milkman, 2012), nature of the spreader, and audience (Hennig-Thurau, Gwinner, Walsh, & Gremler, 2004; Iyengar, van den Bulte, & Valente, 2011), seeding strategies (Kalish, Mahajan, & Muller, 1995; Libai, Muller, & Peres, 2005), and social network structure (Bampo et al., 2008). In this conceptual development, we focus on the content and context drivers of virality.

Virality is defined as the probability of an entity being passed along (Hansen, Arvidsson, Nielsen, Colleoni, & Etter, 2011). The entity can be a message,

content, video, knowledge, or disease. Alternatively, (Tucker, 2015) defines virality as “a process whereby an ad is successively shared by viewers.” These two definitions focus on virality as a process; instead, we focus on viral as a construct. Hence, we are in line with (Chandler & Munday, 2016) defining virality (social media virality, social shareability, social media shareability) as “the potential for spreadability of any given content; or particular qualities which are considered to have led to content ‘going viral’”

For WOM to work, people must share brand-related content, and attention needs to be translated into sales. With these positive associations towards a brand, marketers can increase a customer’s positive brand attitude and later purchase behavior (Faircloth, Capella, & Alford, 2001).

81 percent of a telecom firm indicated that they are likely to refer other to the company; however, only 33 percent actually did. And of those who are referred, only 8 percent became profitable customer. (Kumar et al., 2007)

Even though suffering from the same fate as brand equity, WOM has made it come back under a different alias - virality. The attention that virality has garnered in recent years can be attributed to several reasons:

- (1) the dawn of the Internet helps propel marketing into a new paradigm - digital marketing - where earned media is the new king
- (2) the introduction of social network platforms such as Facebook, Twitter that facilitate the transmission of information and messages seamlessly and provide an unlimited source of data
- (3) The advancement in network theory both theoretically, and empirically.

One platform that can be representative of this digital revolution is Twitter. According to a recent report, Twitter has 145 million monetizable daily active users. Twelve percent of Americans get their news from Twitter. And Twitter has 2 billion video impressions per day (25 Twitter Stats All Marketers Need to Know in 2020. There are more than 500 million Tweets are sent per day. (The 2014 #YearOnTwitter, 2014). One cannot deny the power of online WOM as well as its potential reach; hence, viral marketing has been born with these platforms.

The age group from 18-24 can be more challenging to target since they reported fewer positive responses compared to other age groups (Libert, 2014). Since their threshold is higher due to the overwhelming digital content they encounter, their emotions are harder to activate.

Wide gaps between two opposing opinions can increase future discussion (reviews); however, fluctuation around one opinion does not. Higher informational content moderate positively the relationship between opposing opinions and future reviews. (Nagle and Riedl, 2014)

WOM should be taken under the context of the evolution of conversation. (Berger, 2014)

Contagion/ Virality used to be thought of in term of S-shaped curve, but there are problems with this notion:

1. S-shaped could result from different processes that are not contagion
 1. Population heterogeneity (Chatterjee and Eliashberg, 1990)
 2. Marketing efforts (den Bulte and Lilien, 2001)
2. Selection bias (or survival bias)

Questions:

- When diffusion is “viral”?
- What does viral diffusion look like?
 - (Goel et al., 2012):
 - * 90% events have no adoption
 - * less than 99% events has less than 2 viral adoptions
- Can we predict it?

Viral is different from Popular (Goel et al., 2015)

- Popular = Broadcast
- Viral = Cascades
 - needs diversity (i.e., differences/ distinction)
- Total number of retweet over time is not a good approximate of virality

“Big data” help understand rare events.

Valuable virality (Akpinar and Berger, 2017)

Establishes causality of appeals and brand integralness to virality.

- Compared to informative appeal ads (focus on product features), emotional appeal ads (drama, mood, music, emotion-eliciting strategies) are more likely to be shared.
- Informative appeals boost brand evaluations and purchase because the brand is an integral part of the ad content.
- Hence, to combine the best of both worlds, emotional brand-integral ads boost sharing while also bolstering brand-related outcomes (embedded the brand into the stories).
- The mechanism that ads affect brand evaluation through two simultaneous mechanisms: persuasive attempt and brand knowledge.

“keeping viewers involved depends in large part on two emotions: joy and surprise”(Teixeira, 2012) - “Research: The Emotions that Make Marketing Campaigns Go Viral”

“ads that produce stable emotional states generally aren’t effective at engaging viewers for very long.” And “shock may get people to watch an ad privately, it often works against their desire to share the spot. Like Clothing Drive that try to mimic”Swear Jar” by Bud Light” (Teixeira, 2012)

(Westbrook, 1987, p. 261) defined WOM as “informational communication directed at other consumers about the ownership, usage, or characteristics of particular goods and services or their sellers.” WOM evolves to “word of mouse” (or e-WOM).

(Berger, 2014):

WOM is goal-driven and serves five key functions:

- Impression Management
- Emotion regulation
- Information acquisition
- Social bonding
- Persuasion

The motivation behind this process is self -serving and even subconscious level. Hence, we can predict the types of news and information people are most likely to discuss.

- Over 70% of everyday speech is about the self, such as personal experiences or relationships (Dunbar et al., 1997). Consistent with previous research, online behaviors exhibit the same pattern: over 70% of social media posts are about self (Naaman et al., 2010).
- At any given point in time, multiple drivers can be present.

7.1 Structural Virality

(Goel et al., 2015)

$$S = f(Q) + E$$

where

- Q = skill
- E = luck
- S = successful viral process

Measure fraction of variance remaining after conditioning on Q

$$F = \frac{E(\text{var}(S|Q))}{\text{var}(S)} = 1 - R^2$$

Hence, if

- $R^2 \rightarrow 1$, you can succeed in the world we created based on “pure skill”
- $R^2 \rightarrow 0$ you can’t succeed in the world we created (because it’s based on luck)

The more variations in the initial seed, the more likely there is a larger variation in the prediction, which means lower R-squared

They show there is a theoretical limit to the predictive power of predicting cascade event due to its intrinsic dynamic properties.

7.2 Mechanisms/ Processes

7.2.1 Impression Management

Consumers buy to signal their derived identities to achieve desired impressions (J. Berger & Heath, 2007)

Interpersonal communication (WOM) influences impression management in 3 ways: (1) self-enhancement, (2) identity -signaling, and (3) filling conversational space.

- **Self-enhancement:** human has a tendency to self-enhance (Markus, 1987). Hence, people like to share things to help them look good (Chung & Darke, 2006; Hennig-Thurau et al., 2004). Since bragging too much about oneself might have an opposite effect. People sometimes engage in “humblebragging”: brag but self-deprecating at the same time. (Sezer et al., 2018)
- **Identity-signalling:** they talk about themselves to signal they have certain traits or expertise (G. Packard & Wooten, 2013), such as opinion leaders talk to signal their identity (share to show their knowledge).
- **Filling conversational space:** (small talk) people engage in small talk to avoid pauses and silence in between conversations so that the other party would not make lousy inferences about the person. (J. Berger, 2014)

Hence impression management induces people to share things that are

- **Entertaining** (i.e., interesting, surprising, funny or extreme) so that they shared look more interesting, and in-the-know. Interesting products get more word of mouth (J. Berger & Iyengar, 2013; J. Berger & Milkman, 2012; J. Berger & Schwartz, 2011). Moderate controversy is a catalyst for word of mouth to spread (Chen & Berger, 2013). And sometimes, people even exaggerate stories to make things more interesting (C. Heath, 1996); around 60% of the stories are distorted (Marsh & Tversky, 2004). Things

that can be considered interesting are those that are either novel, exciting, or violate previous expectations (i.e., surprise) (Berlyne, 2006; Silvia, 2008). Nevertheless, comprehensibility can be a moderate of whether a novel thing is interesting: novel and comprehensible equals interesting, while novel and incomprehensible equal confusing (Silvia, 2008). Novelty refers to things that are new, surprising, exciting, or complex (Berlyne, 1960). Comprehensibility refers to the fact that the novelty must be understandable.

- **Useful:** because it makes the sharer smart and helpful. Empirical evidence show that useful stories (J. Berger & Milkman, 2012) and higher quality brands (Lovett et al., 2013) are more likely to be shared.
- **Self-concept relevant:** different people see different domain as the optimal signal amplifier such as politics, economics, etc. hence symbolic products are more likely to be shared than utilitarian ones (Chung & Darke, 2006) (even though useful information is helpful, but the signal to be smart and helpful are not appreciated as being interesting and entertaining).
- **Status related:** premium brands are talked about more (Lovett et al., 2013). Knowledge is can also be considered as a status symbol, and people share to signal that they re in the know (Ritson & Elliott, 1999)
- **Unique:** unique products are more likely to be shared, but people with a high need for uniqueness will be less likely to share positive WOM because they do not want people to adopt the product and reduces their uniqueness (Cheema & Kaikati, 2010). Sometimes they even scare others by citing product complexity (Moldovan et al., 2015)
- **Common ground:** sharing common things induces listener to convey that there is interpersonal similarity and facilitates conversation.
- **Emotional valence:** people prefer to share positive things than negative news (J. Berger & Milkman, 2012). However, in some special cases, negative WOM can be perceived as a desired component such as reviewers are seen as more intelligent, (Amabile, 1983). Which is moderated by whether people are refereeing to themselves or others will affect WOM valence (positive vs. negative) (Kamins et al., 1997). Positive WOM when talking oneself conveys a positive image, while negative WOM about others convey that they are relatively better than the other party.
- **Incidental arousal:** unrelated arousal (e.g., running in place) can increase sharing in general (J. Berger, 2011) because of the increase in arousal levels.
- **Accessible things:** products with more cued or triggered will be discussed more (J. Berger & Schwartz, 2011). Hence, more advertised products generate more WOM (Onishi & Manchanda, 2012). “top of mind” and “tip of the tongue.” Availability bias means that the easier we can

recall something, the more likely we are going to talk about it. Publicly visible products also have more WOM (J. Berger & Schwartz, 2011)

7.2.2 Emotion regulation

WOM help consumers regulate their emotions. Emotion regulation refers to “a person’s ability to effectively manage and respond to an emotional experience” (Rolston & Lloyd-Richardson, 2017). and sharing with other (WOM) can facilitate emotion regulation by:

- **Generating social support:** sharing can give you comfort and consolation (Rimé, 2009). For example, after a negative emotional experience, sharing can help increase well-being through perceived social support (J. A. Berger & Buechel, 2012)
- **Venting:** WOM (sharing) helps people achieve catharsis when experiencing negative experiences (Alicke et al., 1992): consumers express when they are angry (Wetzer et al., 2007) or dissatisfied (Anderson, 1998).
- **Sensemaking:** “talking can help people understand what they feel and why” (J. Berger, 2014)
- **Reducing dissonance:** even after making a decision, we want to make sure we make the right choice by talking to others to reduce feelings of doubt (Rosnow, 1980)
- **Taking vengeance:** to punish the company (different from venting, which makes you feel better), consumers share negative consumption experience (Ward & Ostrom, 2006)
- **Rehearsal:** people talk to relive positive experiences (Rimé, 2009)

Emotion regulation drives people to share (1) emotional content (2) influence the valence of the content shared, (3) lead people to share more emotionally arousing content:

- **Emotionality:** more emotional intensity, more sharing (J. Berger & Milkman, 2012), people share more emotional social anecdotes (Peters et al., 2009). But not all emotions increase sharing: for example, shame and guilt decreases sharing (Finkenauer & Rimé, 1998) since sharing those things makes people look bad.
- **Valence:** “Emotion regulation tends to focus on the management of negative emotions” (J. Berger, 2014), but a counter vailing effect is that under impression management, people avoid sharing negative stories since it might reflect on the shares. Sharing negative things can decrease willingness to share (Chen & Berger, 2013)
- **Emotional arousal:**
 - **Negative** side: low arousal emotion (sadness), high arousal (anger or anxiety) should increase the need to vent.
 - **Positive** side: low arousal (contentment), high arousal (awe, excitement or amusement) increase desires for rehearsal.

7.2.3 Information acquisition

WOM helps acquire information.

- **Seeking advice:** (Hennig-Thurau et al., 2004; Rimé, 2009). Rather than trial and error, or direct observation, gossip serves as another form of learning (Baumeister et al., 2004)
- **Resolving problems:** online forums or other online opinion platforms help customers find a solution quickly from others (Hennig-Thurau et al., 2004)

Hence the process of information acquisition would induce to talk more about

- **Risky, important, complex, or uncertainty-ridden decision:** talking to others can reduce risk, simplify complexity, and increase consumer's confidence (Hennig-Thurau et al., 2004)
- **Lack of (trustworthy) information:**

7.2.4 Social bonding

“people have a fundamental desire for social relationships (Baumeister & Leary, 1995), and interpersonal communication fills that need (Hennig-Thurau et al., 2004)”. Empirically, people in brand communities because want to connect with others like them (Muniz & O’Guinn, 2001). Sharing facilitates social bonding through

- **Reinforce shared views:** group memberships, and one’s place in the social hierarchy. WOM (sharing) allows people to fit in as a group member (J. Berger & Heath, 2007; Ritson & Elliott, 1999)
- **Reducing loneliness and social exclusion:** sharing decrease interpersonal distance and decreases loneliness and exclusion (Maner et al., 2007)

Social bonding drives what people share: people share things are :

- **Common ground:** social bonding drives people to talk about things they have in common because it makes them feel socially connected (Clark & Kashima, 2007)
- **Emotionality:** “sharing an emotional story or narrative increases the chance that others will feel similarly” (J. Berger, 2014). Emotional similarity increase group cohesiveness (Barsade & Gibson, 2007). And emotion sharing and social bonding can be a simultaneous process since previous works found that emotion sharing bonds people (Peters & Kashima, 2007) and high arousal emotion increase social bonding needs (Chen & Berger, 2013)

7.2.5 Persuading others

people use WOM to persuade others to affect their satisfaction or choices (J. Berger, 2014). Persuasive drive people to share things that are

- **Emotionally polarized** (polarized valence): since the goal is to convince something is good or bad (i.e., people share extremely rather than moderately positive(negative) information
- **Arousing content**: “people who want to persuade others may find shared arousing content to incite others to take desired actions” (J. Berger, 2014)

7.3 Moderators

- Two key moderators when different WOM functions have a greater impact:
 - **Communication audience**:
 - * Tie strength
 - * Audience size
 - * Tie status
 - **Communication Channel**
 - * Written vs. oral
 - * Identifiability: whether communicators are identifiable. (anonymously)
 - * Audience salience: whether the audience is salient during communication (online communication, the audience is less salient).

Social motivation: why we share :(why_some_videos_go_viral_2015)

- Impression management:
 - authority (e.g., “I want to demonstrate my knowledge”)
 - Social utility (e.g., this could be useful to my friends)
 - coolhunter (e.g., I want to be the first o tell my friends)
 - Zeitgeist (e.g., It’s about a current trend or event)
 - Conversation starting (e.g., I want to start an online conversation)
 - Self-expression (e.g., it says something about me)
 - Social good (e.g., It’s for a good cause, and I want to help”).
- Information acquisition:
 - Opinion seeking (e.g., ”I want to see what my friends think)
- Social bonding:
 - Shared passion (e.g, ”it lets me connect with my friends bout a shared interest)
 - Social in real life (e.g., it will help me socialize with my friends of-fline).

More frequent exposure to perceptually or conceptually related cues increases product accessibility and makes the product easier to process. In turn, this increased accessibility influences product evaluation and choice, which are found to vary directly with the frequency of exposure to conceptually related cues. These results support the hypothesis that conceptual priming effects can have a strong impact on real-world consumer judgment” (Berger and Fitzsimons, 2008)

7.4 Drivers of Virality

7.4.1 Social Currency

Different people see different domain as the optimal signal amplifier, such as politics, economics

Even though useful information is helpful, the signal of being smart and helpful is not appreciated as being interesting and entertaining (Berger, 2014).

People talk more about status-related products because they enhance their impression.

In extreme cases, they even scare others by citing product complexity (Moldovan, Steinhart, & Ofen, 2015).

Interesting things are discussed more in online settings (J. Berger & Milkman, 2012), but this effect fails to replicate in the case of face-to-face settings (Berger & Schwartz, 2011). Hence, research has found that written communication (e.g., texting, messaging, posting online) leads people to talk more about interesting products or brands than oral communication because they have more time to think and polish what to say, leading to more prevalent self-enhancement motives (Berger & Iyengar, 2013). In other words, not because people do not have self-enhancement motives, but because synchronous conversations do not allow for much time to think and polish ideas, people might talk about mundane things to fill the gaps between conversational turns (Berger, 2014).

7.4.2 Accessibility

Accessibility drives ongoing WOM (i.e., publicly visible products are cued more in the environment – top of mind), regardless of how mundane it might be.

From the theory of semantic process (Collins & Loftus, 1975), priming can help activate related perceptual or conceptual constructs in memory spontaneously (Berger & Fitzsimons, 2008). For example, prior exposure affects the brand choice of both target brand and competing brands (Nedungadi, 1990). In other words, the activation of one brand can lead to others more accessible, which increases the chance of remembered brands to be included in the consideration set. (Lee & Labroo, 2004) argues that ease of processing drives our positive evaluation, which is also consistent with the discrepancy -attribution hypothesis. The Discrepancy-attribution hypothesis states that if people cannot easily attribute

the ease of processing to a source, they will likely attribute it to the positive quality of the target. Additionally, (Nedungadi, 1990) found that recently exposed brands are more likely to be included and chosen from consumers' consideration set. Repeated exposure to an object increases favorable ratings (e.g., an object, or person) (Zajonc, 1968), which also apply to attractive people (Moreland & Beach, 1992), and brand (Baker, 1999). Behavioral can be influenced by incidental exposure to stimuli because the primes can activate relevant constructs associated in memory (Christian Wheeler & Petty, 2001; Dijksterhuis & Bargh, 2001; Wheeler & Berger, 2007)

7.4.3 Emotions

(Milkman & Berger, 2014) found that contents that are (1) emotional aroused, (2) useful, or (3) interesting have a higher chance of being shared. Consistently, (Milkman & Berger, 2014) also found that women produce articles that are more likely to be shared because female authors write more comprehensible, useful, and interesting articles.

People share their extreme emotional experience (i.e., either highly satisfied or highly dissatisfied) (Anderson, 1998)

Lastly, people use WOM to persuade others to affect their satisfaction or choice (Berger, 2014). Thus, emotionally polarized (polarized valence) are more likely to be shared due to persuasion reasons. Since the goal is to convince something is good or bad, people share extremely rather moderately positive (negative) information.

Virality is partially driven by physiological arousal (Berger, 2011; Berger & Milkman, 2012). These results hold even when the authors control how surprising, interesting, or practically useful content is (all of which are positively linked to virality) and external drivers of attention (e.g., how prominently content was featured) (Berger & Milkman, 2012). Different emotion evokes different levels of physiological arousal (i.e., activation) (Smith & Ellsworth, 1985). Arousal is an excitatory state that increases action-related behavior, such as helping others (Gaertner & Dovidio, 1977). Hence, we see contents that are physiological arousal get shared more. Specifically, Negative high arousal emotions (e.g., anger or anxiety) are shared more than negative low arousal (e.g., sadness). Moreover, sharing negative experiences can help people vent to regulate their emotions (Berger, 2014). On the other hand, positive emotion high arousal emotions (e.g., awe, excitement, or amusement) are shared more than positive low arousal (e.g., contentment). Furthermore, sharing for the purpose of rehearsal (i.e., people talk to relive positive experiences) helps people regulate their emotions (Rimé, 2009).

7.5 Other variables

7.5.1 Contractuality

(Lisjak et al., 2021) defined contractuality as ” the extend to which a perk is perceived to be conditional on specific behaviors and contingencies dictated by a company.”

- Consumers like to have perks with less salient contractuality.
- The existing relationship with the company (e.g., dislike, distrusted) may giving consumers the wrong intention, or let them interpret the park as a manipulative act with ulterior motives.
- Low contractuality perks can foster “real” authentic WOM, but you have to trade off with effectiveness and efficiency (e.g., as in the case of filling out satisfaction survey).

7.5.2 Locus of Control

(Lam and Mizerski, 2005)

- An individual personality construct is locus of control
- Locus of control is defined as ”individuals’ general and daily expectancies about the causes of their reward and punishment (Lam and Mizerski, 2005, pp. 216), consisting of internal (i.e., one can control his or her lives) and external (i.e., external factors can control their lives such as luck, or fate).
 - Internals tended to be more educated (Lachman and Leff, 1989), higher household incomes (Hoffman et al., 2003), men and those in senior positions (Smith et al., 1997). Internals are more action-oriented, risk-taking.
 - Externals perform avoidance behavior, greater needs for affiliation
- Individuals with high internal locus of control are more likely to engage in WOM communication with their out-groups.
- Individuals with high external locus of control are more likely to engage in WOM with their in-group.

7.5.3 Horizontal/Vertical Individualism

(Choi and Kim, 2019)

- Antecedents of WOM: homophily, tie strength, trust.
- Culture affects individuals’ communication attitudes and styles.
- Vertical (i.e., emphasizing hierarchy) and horizontal (i.e., emphasizing equality) dimensions of individualism and collectivism (HVIC) (Singelis et al., 1995)

- opinion leadership: “the tendency that an individual attempts to influence the decisions of others by giving his or her opinion about products, services, or firms” (Choi and Kim, 2019, pp. 294)
 - * socially appropriate (horizontal orientation)
 - * show off knowledge (vertical orientation)
- opinion seeking “the tendency that an individual seeks information or opinions from more knowledgeable others to find out about and/or evaluate products or services” (Choi and Kim, 2019, pp. 294)
 - * well-informed decisions (individualism)
 - * find values and beliefs of the reference group (collectivism)
- Horizontal individualism is “a cultural pattern where an autonomous self is postulated but the individual is more or less equal in status with others” (Singelis et al., 1995, pp. 245)
- Vertical individualism is “a cultural pattern in which an autonomous self is postulated, but individuals see each other as different, and inequality is expected” (Singelis et al., 1995, pp. 245).
- Horizontal collectivism is “a cultural pattern in which the individual sees the self as an aspect of an in-group” (Singelis et al., 1995, pp. 244).
- Vertical collectivism is “a cultural pattern in which the individual sees the self as an aspect of an in-group, but the members of the in-group are different from each other, some having more status than others” (Singelis et al., 1995, pp. 244).
- Collectivist culture will have greater levels of opinion seeking, whereas individualist culture has greater information-giving. (Fong and Burton, 2008)
- Results:
 - “horizontal individualism to opinion leadership, vertical individualism to opinion leadership and opinion seeking”
 - “horizontal collectivism to opinion leadership, and vertical collectivism to opinion leadership and opinion seeking”

7.6 Articles

- Empirical evidence
 - contagion in consumer packaged goods (Du and Kamakura, 2011)
- Interesting stat:

- “Articles by business academics, psychologists, and economists, for example, are more likely to be shared than articles by physicists, geneticists, and biochemists” (Milkman & Berger, 2014)
“Specifically, men see the same scientific summaries as more comprehensible ($P < 0.001$), interesting ($P < 0.001$), and useful” hence they share more than women (Milkman & Berger, 2014)
- “though intentionally outrageous videos command attention (Tellis 2004), an ad design of this type ultimately detracts from the ad’s persuasiveness” (Tellis, 2004)
- (Godes & Mayzlin, 2004) the impact of dispersion declines over time; hence should measure WOM early in a product’s life. Main finding: higher WOM dispersion leads to higher future ratings. WOM is both precursors and consequences of consumer behavior.
- Promotional giveaways increase WOM (Berger & Schwartz, 2011)

Chapter 8

Customer Lifetime Value (CLV)

also known as Lifetime Value (LTV)

“A year’s projected business gives a number that is normally about half of a customer’s full lifetime value, although that can vary by industry.” (Kumar et al., 2007)

(Kumar et al., 2008) CLV model

$$CLV_i = \sum_{j=T+1}^{T+N} \frac{p(Buy_{ij} = 1) \times \hat{C}M_{ij}}{(1+r)^{j-T}} - \frac{\hat{M}C_{ij}}{(1-r)^{j-T}}$$

where

- CLV_i = lifetime value for customer i,
- $p(Buy_{ij})$ = predicted probability that customer i will purchase in period j,
- $\hat{C}M_{ij}$ = predicted contribution margin provided by customer i in period j,
- $\hat{M}C_{ij}$ = predicted marketing costs directed toward customer i in period j,
- j = index for time periods (semiannual or annual)
- T = the end of the calibration or observation time frame (usually 1 year projection),

- N = total number of prediction periods,
- r = semiannual discount factor (e.g., 0.07238 (Kumar et al., 2008) = 15% annual rate)

8.1 Example

This example is subscription customers by Sergey Bryl' in which he adapted model Fader and Hardie (2007)

```
library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.0.5
## -- Attaching packages ----- tidyverse 1.3.1 --
## v ggplot2 3.3.5      v purrr   0.3.4
## v tibble  3.1.2      v dplyr  1.0.6
## v tidyr   1.1.3      v stringr 1.4.0
## v readr   2.0.1      v forcats 0.5.1
## Warning: package 'ggplot2' was built under R version 4.0.5
## Warning: package 'readr' was built under R version 4.0.5
## Warning: package 'dplyr' was built under R version 4.0.5
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
library(reshape2)

##
## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##      smiths
library(MLmetrics)

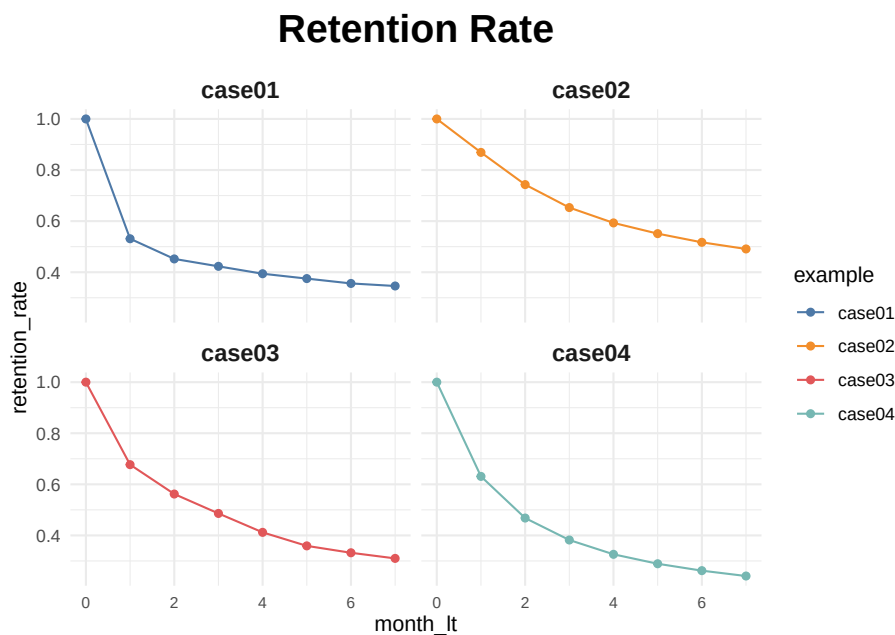
##
## Attaching package: 'MLmetrics'
## The following object is masked from 'package:base':
##
##      Recall
# retention rate data
df_ret <- data.frame(month_lt = c(0:7),
                     case01 = c(1, .531, .452, .423, .394, .375, .356, .346),
```

```

case02 = c(1, .869, .743, .653, .593, .551, .517, .491),
case03 = c(1, .677, .562, .486, .412, .359, .332, .310),
case04 = c(1, .631, .468, .382, .326, .289, .262, .241)
) %>%
melt(., id.vars = c('month_lt'), variable.name = 'example', value.name = 'retention_rate')

ggplot(df_ret, aes(x = month_lt, y = retention_rate, group = example, color = example)) +
  theme_minimal() +
  facet_wrap(~ example) +
  scale_color_manual(values = c('#4e79a7', '#f28e2b', '#e15759', '#76b7b2')) +
  geom_line() +
  geom_point() +
  theme(plot.title = element_text(size = 20, face = 'bold', vjust = 2, hjust = 0.5),
        axis.text.x = element_text(size = 8, hjust = 0.5, vjust = .5, face = 'plain'),
        strip.text = element_text(face = 'bold', size = 12)) +
  ggtitle('Retention Rate')

```



Predicition when we only have values on certain months

```

# functions for sBG distribution
churnBG <- Vectorize(function(alpha, beta, period) {
  t1 = alpha / (alpha + beta)
  result = t1
  if (period > 1) {
    result = churnBG(alpha, beta, period - 1) * (beta + period - 2) / (alpha + beta + period - 2)
  }
})

```

```

    }
    return(result)
}, vectorize.args = c("period"))

survivalBG <- Vectorize(function(alpha, beta, period) {
  t1 = 1 - churnBG(alpha, beta, 1)
  result = t1
  if(period > 1){
    result = survivalBG(alpha, beta, period - 1) - churnBG(alpha, beta, period)
  }
  return(result)
}, vectorize.args = c("period"))

MLL <- function(alphabeta) {
  if(length(activeCust) != length(lostCust)) {
    stop("Variables activeCust and lostCust have different lengths: ",
         length(activeCust), " and ", length(lostCust), ".")
  }
  t = length(activeCust) # number of periods
  alpha = alphabeta[1]
  beta = alphabeta[2]
  return(-as.numeric(
    sum(lostCust * log(churnBG(alpha, beta, 1:t))) +
    activeCust[t]*log(survivalBG(alpha, beta, t))
  ))
}

df_ret <- df_ret %>%
  group_by(example) %>%
  mutate(activeCust = 1000 * retention_rate,
         lostCust = lag(activeCust) - activeCust,
         lostCust = ifelse(is.na(lostCust), 0, lostCust)) %>%
  ungroup()

ret_preds01 <- vector('list', 7)
for (i in c(1:7)) {

  df_ret_filt <- df_ret %>%
    filter(between(month_lt, 1, i) == TRUE & example == 'case01')

  activeCust <- c(df_ret_filt$activeCust)
  lostCust <- c(df_ret_filt$lostCust)

```



```

    opt <- optim(c(1, 1), MLL)
    retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2], c(1:7))),

    df_pred <- data.frame(month_lt = c(0:7),
                          example = 'case01',
                          fact_months = i,
                          retention_pred = retention_pred)
    ret_preds01[[i]] <- df_pred
  }

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
## Warning in log(survivalBG(alpha, beta, t)): NaNs produced
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
## Warning in log(survivalBG(alpha, beta, t)): NaNs produced
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
## Warning in log(survivalBG(alpha, beta, t)): NaNs produced
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
## Warning in log(survivalBG(alpha, beta, t)): NaNs produced
ret_preds01 <- as.data.frame(do.call('rbind', ret_preds01))

ret_preds02 <- vector('list', 7)
for (i in c(1:7)) {

  df_ret_filt <- df_ret %>%
    filter(between(month_lt, 1, i) == TRUE & example == 'case02')

  activeCust <- c(df_ret_filt$activeCust)
  lostCust <- c(df_ret_filt$lostCust)

  opt <- optim(c(1, 1), MLL)
  retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2], c(1:7))),

  df_pred <- data.frame(month_lt = c(0:7),
                        example = 'case02',
                        fact_months = i,
                        retention_pred = retention_pred)
  ret_preds02[[i]] <- df_pred
}

```

```

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

ret_preds02 <- as.data.frame(do.call('rbind', ret_preds02))

ret_preds03 <- vector('list', 7)
for (i in c(1:7)) {

  df_ret_filt <- df_ret %>%
    filter(between(month_lt, 1, i) == TRUE & example == 'case03')

  activeCust <- c(df_ret_filt$activeCust)
  lostCust <- c(df_ret_filt$lostCust)

  opt <- optim(c(1, 1), MLL)
  retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2],

  df_pred <- data.frame(month_lt = c(0:7),
                        example = 'case03',
                        fact_months = i,
                        retention_pred = retention_pred)
  ret_preds03[[i]] <- df_pred
}

ret_preds03 <- as.data.frame(do.call('rbind', ret_preds03))

ret_preds04 <- vector('list', 7)
for (i in c(1:7)) {

  df_ret_filt <- df_ret %>%
    filter(between(month_lt, 1, i) == TRUE & example == 'case04')

  activeCust <- c(df_ret_filt$activeCust)

```

```

lostCust <- c(df_ret_filt$lostCust)

opt <- optim(c(1, 1), MLL)
retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2], c(1:7))),

df_pred <- data.frame(month_lt = c(0:7),
                      example = 'case04',
                      fact_months = i,
                      retention_pred = retention_pred)
ret_preds04[[i]] <- df_pred
}

ret_preds04 <- as.data.frame(do.call('rbind', ret_preds04))

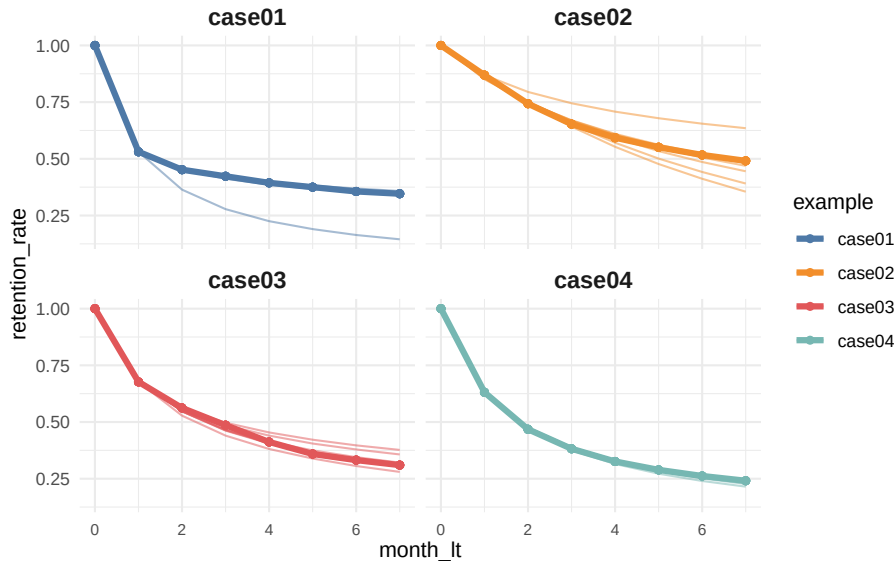
ret_preds <- bind_rows(ret_preds01, ret_preds02, ret_preds03, ret_preds04)

df_ret_all <- df_ret %>%
  select(month_lt, example, retention_rate) %>%
  left_join(., ret_preds, by = c('month_lt', 'example'))

ggplot(df_ret_all, aes(x = month_lt, y = retention_rate, group = example, color = example)) +
  theme_minimal() +
  facet_wrap(~ example) +
  scale_color_manual(values = c('#4e79a7', '#f28e2b', '#e15759', '#76b7b2')) +
  geom_line(size = 1.5) +
  geom_point(size = 1.5) +
  geom_line(aes(y = retention_pred, group = fact_months), alpha = 0.5) +
  theme(plot.title = element_text(size = 20, face = 'bold', vjust = 2, hjust = 0.5),
        axis.text.x = element_text(size = 8, hjust = 0.5, vjust = .5, face = 'plain'),
        strip.text = element_text(face = 'bold', size = 12)) +
  ggtitle('Retention Rate Projections')

```

Retention Rate Projections



calculate the average LTV for case03 based on two historical months with a forecast horizon of 24 months and a subscription price of \$1

```
### LTV prediction ###
df_ltv_03 <- df_ret %>%
  filter(between(month_lt, 1, 2) == TRUE & example == 'case03')

activeCust <- c(df_ltv_03$activeCust)
lostCust <- c(df_ltv_03$lostCust)

opt <- optim(c(1, 1), MLL)
retention_pred <- round(c(survivalBG(alpha = opt$par[1], beta = opt$par[2], c(3:24))),

df_pred <- data.frame(month_lt = c(3:24),
  retention_pred = retention_pred)

df_ltv_03 <- df_ret %>%
  filter(between(month_lt, 0, 2) == TRUE & example == 'case03') %>%
  select(month_lt, retention_rate) %>%
  bind_rows(., df_pred) %>%
  mutate(retention_rate_calc = ifelse(is.na(retention_rate), retention_pred, retention_rate),
    ltv_monthly = retention_rate_calc * 1,
    ltv_cum = round(cumsum(ltv_monthly), 2))
# average LTV of $9.33. actual data for the observed periods (from 0 to 2nd months) and
```

8.2 Referral value (CRV)

“Referrals made by customers after a referral-incentive marketing campaign can be attributed to that campaign for about a year.” (Kumar et al., 2007). Hence, we usually count those referrals made after one year of a campaign.

2 types of referral:

- Type-one referral: Would not join without referral.
- Type-two referral: still join without referral.

(Kumar et al., 2007) estimated that each customer made about half type-one and half type-two referrals.

Referral value = PV(Type-one referrals) + PV(Type-two referrals)

where

- PV(type-two referrals) = the present value of the savings in acquisition costs.

$$CRV_i = \sum_{t=1}^T \sum_{y=1}^{n1} \frac{A_{ty} - a_{ty} - M_{ty} + ACQ1_{ty}}{(1+r)^t} + \sum_{t=1}^T \sum_{y=n1+1}^{n2} \frac{ACQ2_{ty}}{(1+r)^t}$$

where

- T = the number of periods that will be predicted into the future (e.g., typically 1 year)
- A_{ty} = the gross margin contributed by customer y who otherwise would not have bought the product,
- a_{ty} = the cost of the referral for the customer y (this is what you set up)
- $n1$ = the number of customers who would not join without the referral,
- $n2 - n1$ = the number of customers who would have joined anyway,
- M_{ty} = the marketing costs needed to retain the referred customers,
- r = semiannual discount factor ((Kumar et al., 2008) used .07238 = 15% annual rate)
- $ACQ1_{ty}$ = the savings in acquisition costs from customers who would not join without the referral,
- $ACQ2_{ty}$ = the savings in acquisition cost from customers who would have joined anyway.

Customers with high CLV does not guarantee high CRV (Kumar et al., 2007). Hence, (Kumar et al., 2007) proposed the customer value matrix

```
## Warning: package 'jpeg' was built under R version 4.0.5
```

The Customer Value Matrix

When we segmented the customers in our telecom sample according to their lifetime values and referral values, it was easy to see which customers needed to be encouraged to buy more and which should be nudged to make more recommendations.

Average CRV after one year

Average CLV after one year		Average CRV after one year	
		LOW	HIGH
HIGH	AFFLUENTS 29% of customers CLV = \$1,219 CRV = \$49	CHAMPIONS 21% of customers CLV = \$370 CRV = \$590	
LOW	MISERS 21% of customers CLV = \$130 CRV = \$64	ADVOCATES 29% of customers CLV = \$180 CRV = \$670	

Note: Cutoff points for the low and high CLVs and CRVs were set at the median value for both measures.

Chapter 9

Finance-Marketing Interface

Market-based assets are: relational and intellectual.

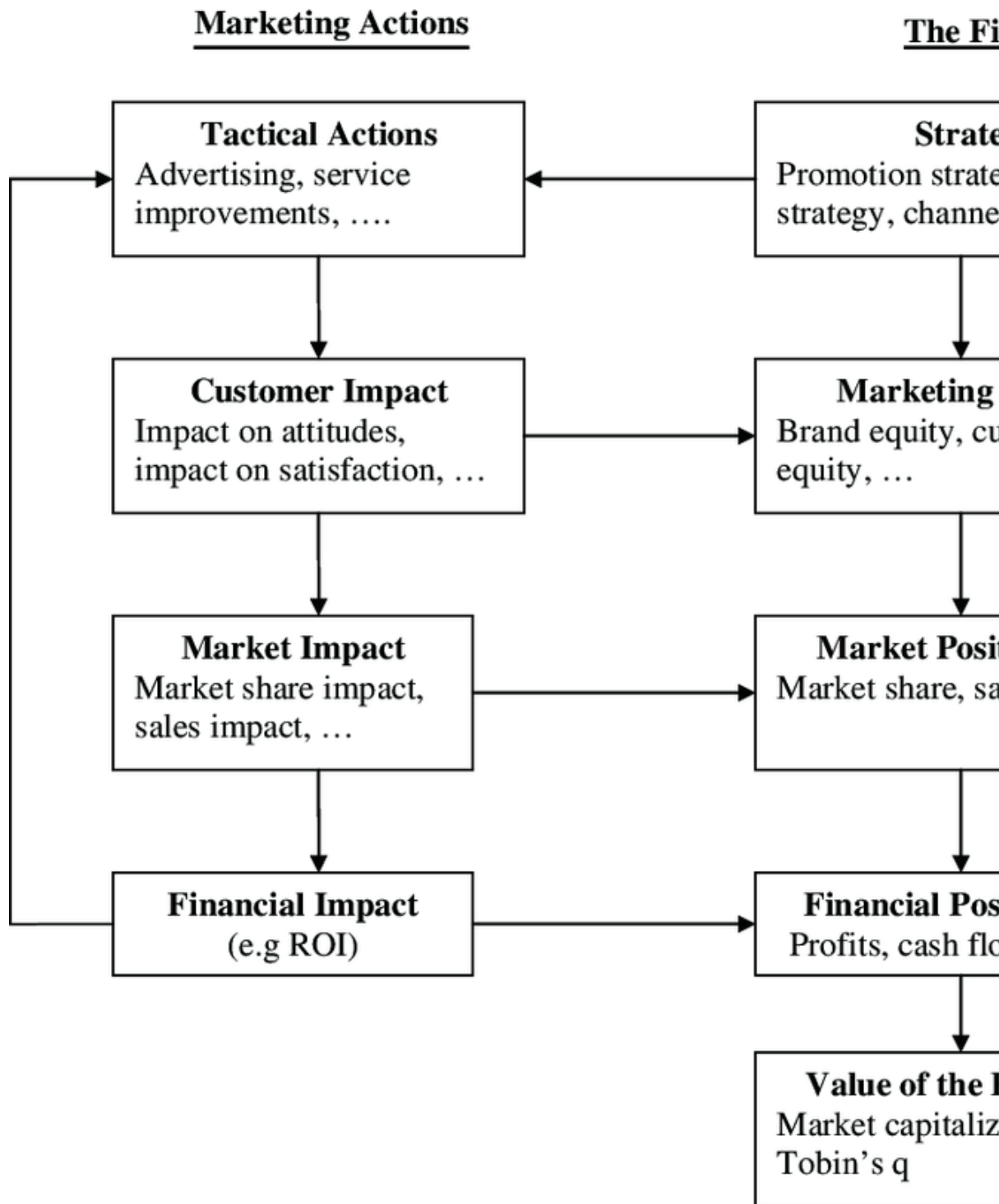
Relational market-based assets are “outcomes of the relationship between a firm and key external stakeholders, including distributors, retailers, end customers, other strategic partners”. (Srivastava et al., 1998)

For example, brand equity = relational assets between firms and their customers, while channel equity - relational assets between firms and their channel partners.

Intellectual market-based assets are “the types of knowledge a firm possesses about the environment, such as the emerging and the potential state of market conditions and the entities in it, including competitors, customers, channels, suppliers, and social and political interest group.” (Srivastava et al., 1998)

Marketing actions (e.g, advertizing, product innovation) can build long-term assets (e.g., brand equity, customer equity), in turn increase short-term profitability (e.g., increase marketing effectiveness for dollars spent on advertising) (Rust et al., 2004)

Warning: package 'jpeg' was built under R version 4.0.5



The Chain of Marketing Productivity (source (Rust et al., 2004))

Relationship between marketing and finance (Srivastava et al., 1998)

Stage 1: Market-based Assets

- Customer relationships: Brands, installed base.
- Partner Relationship: channels, co-branding network.

Stage 2:

- Faster Market penetration
- Price premium
- share premium
- extensions
- reduce sales/service costs
- Loyalty/ Retention

Stage 3: Shareholder value:

- Accelerate Cash Flows
- Enhance Cash Flows
- Reduce Volatility and Vulnerability of Cash flows
- Enhance Residual Value of Cash Flows.

(Grewal et al., 2010; McAlister et al., 2016; Morgan and Rego, 2009) used Tobin's Q in marketing research. since it is "forward-looking, risk-adjusted, and less easily manipulated by managers" (Wies et al., 2019). When probing factors affecting Tobin's Q, it was suggested to control for Financial Leverage, Cash flows

(Anderson et al., 2018) found that there is a significant improvements from increasing both business skills (marketing and finance). However, the pathway to improve profits from marketing skills is via growth focus (e.g., higher sales, more investments in products, and employees). The pathway to improve profits from the finance skills is via efficiency or cost focus. Hence, the recommendation is that marketing/sales skills have better fit for startups, while finance/accounting skills are more needed in mature companies.

Chapter 10

Communication

10.1 Information Theory

10.1.1 Entropy

Most of the examples are taken from `philentropy` package in R, which can be accessed via [this link](#)

10.1.1.1 Discrete Entropy

If $\mathbf{p}_X = (p_{x1}, p_{x2}, \dots, p_{xn})$ is the probability distribution of the discrete random variable X with $P(X = x) = p_{xi}$. Then, the **entropy** of the distribution is (Shannon, 1948)

$$H(\mathbf{p}_X) = - \sum_{x_i} p_{xi} \log_2 p_{xi}$$

where

- entropy unit is **bits**, and $H(p) \in [0, 1]$ and max at $p = 0.5$
- or the unit is **nats** ($\log_e$)
- or the unit is **dits** ($\log_{10}$)

In general, if you have different bases, you can convert between units. Given bases a and b

$$\log_a k = \log_a b \times \log_b k$$

where $\log_a b$ is a constant.

Generalize

$$H_b(X) = \log_b a \times H_a(X)$$

Equivalently,

$$H(X) = E\left(\frac{1}{\log p(x)}\right)$$

Entropy is the measure of how much information gained from learning the value of X (Zwillinger, 1995, pp.262). Entropy depends on the distribution of X .

Example:

If X has 2 values, then $\mathbf{p} = (p, 1 - p)$ and

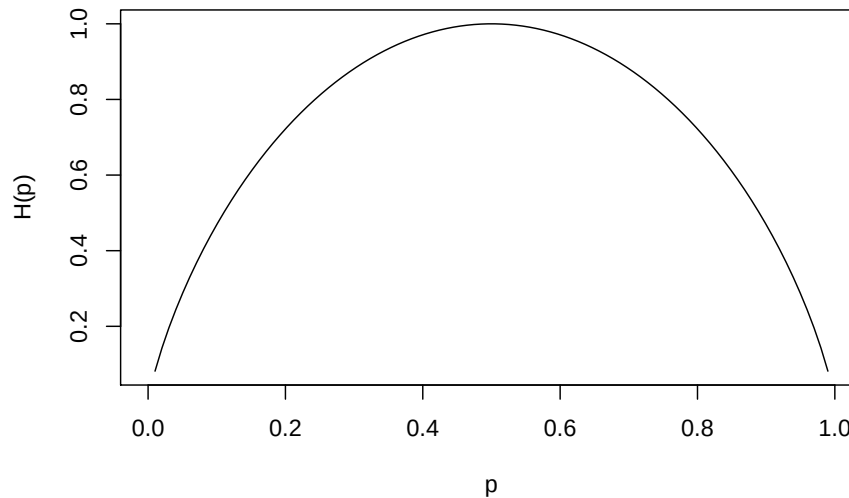
$$H(\mathbf{p}_X) = H(p, 1 - p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

which is known as binary entropy function

Plot of p vs. $H(p)$ where $\max H(\mathbf{p}_X) = \log_2 n$ when $X \sim U$

```
hb <- function(gamma) {
  -gamma * log2(gamma) - (1 - gamma) * log2(1 - gamma) # output in bits
}

xs <- seq(0, 1, len = 100)
plot(xs,
      hb(xs),
      type = 'l',
      xlab = "p",
      ylab = "H(p)")
```



```
library("phileentropy")

## Warning: package 'phileentropy' was built under R version 4.0.5
Prob <- 1:10/sum(1:10) # probabilities P(X)
H(Prob) # Shannon's Entropy

## [1] 3.103643
```

10.1.1.2 Mutual Information

If X and Y are two discrete random variables and $\mathbf{p}_{X \times Y}$ is their joint distribution, the **mutual information** (i.e., learning of a value of X gives information about the value of Y) of X and Y is

$$I(X, Y) = I(Y, X) = H(\mathbf{p}_X) + H(\mathbf{p}_Y) - H(\mathbf{p}_{X \times Y})$$

where $I(X, Y) \geq 0$ and $I(X, Y) = 0$ iff X and Y are independent (information about X gives no information about Y).

Equivalently,

$$MI(X, Y) = \sum_{i=1}^n \sum_{j=1}^m P(x_i, y_j) \times \log_b \left(\frac{P(x_i, y_j)}{P(x_i) \times P(y_j)} \right)$$

Properties:

- $I(X; X) = H(X)$ known as self-information
- $I(X; Y) \geq 0$ with equality iff $X \perp Y$

```
P_x <- 1:10/sum(1:10) # distribution P(X)
P_y <- 20:29/sum(20:29) # distribution P(Y)
P_xy <- 1:10/sum(1:10) # joint distribution P(X,Y)
MI(P_x, P_y, P_xy) # Mutual Information
```

```
## [1] 3.311973
```

10.1.1.3 Relative Entropy

The measure of the distance between distribution p and q is called **relative entropy**, denoted by $D(p||q)$

Equivalently, it measures the inefficiency of assuming distribution q when the true distribution is p .

The relative entropy between $p(x)$ and $q(x)$ is

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

where

- $D(p||q) \geq 0$ and equality iff $p = q$

10.1.1.4 Joint-Entropy

$$H(X, Y) = - \sum_{i=1}^n \sum_{j=1}^m P(x_i, y_j) \times \log_b(P(x_i, y_j))$$

```
P_xy <- 1:10/sum(1:10) # joint distribution P(X,Y)
JE(P_xy) # Joint-Entropy
```

```
## [1] 3.103643
```

10.1.1.5 Conditional Entropy

$$H(Y|X) = - \sum_{i=1}^n \sum_{j=1}^m P(x_i, y_j) \times \log_b\left(\frac{P(x_i)}{P(x_i, y_j)}\right)$$

```
P_x <- 1:10/sum(1:10) # distribution P(X)
P_y <- 1:10/sum(1:10) # distribution P(Y)
CE(P_x, P_y) # Joint-Entropy
```

```
## [1] 0
```

Relation among entropy measures

$$H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$$

Properties:

- Chain rule: $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i|X_{i-1}, \dots, X_1)$
- $0 \leq H(Y|X) \leq H(Y)$ Information on X does not increase the entropy measure of Y
- If $X \perp Y$, then $H(Y|X) = H(Y)$; $H(X, Y) = H(X) + H(Y)$
- $H(X) \leq \log|n_X|$ where n_X = number of elements in X
- $H(X) = \log(n_X)$ iff $X \sim U$
- $H(X_1, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$
- $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i)$ iff X_i 's are mutually independent.

10.1.1.6 Continuous Entropy

If X is a continuous variable, its entropy is

$$h(\mathbf{X}) = - \int_{R^d} p(\mathbf{x}) \log p(\mathbf{x}) d\mathbf{x}$$

where d is the dimension of X.

If X and Y are continuous random variables with density functions $p(\mathbf{x})$ and $q(\mathbf{y})$, the relative entropy is

$$H(X, Y) = \int_{R^d} p(x) \log \frac{p(x)}{q(x)} dx$$

More advance analysis can be access [here](#)

10.1.2 Divergence

Three metrics for divergence can be found [here](#):

- Kullback-Leibler
- Jensen-Shannon
- Generalized Jensen-Shannon

10.1.3 Channel Capacity

The information channel capacity of a discrete memoryless channel is

$$C = \max_{p(x)} I(X; Y)$$

which is the highest rate use at which information can be sent without much error.

More information can be found [here](#)

Chapter 11

Metrics

11.1 Return on Investment (ROI)

$$ROI = \frac{\text{Net Profit}}{\text{Investment}}$$

Similarly, return on marketing investment

$$ROIM = \frac{IRAM - CM}{MS}$$

where

- IRAM = Incremental Revenue Attributable to Marketing
- CM = Cost of the Marketing Investment
- MS = Marketing Spending

If ROMI is positive, then firm is on doing well.

11.2 Economic Value Added

- also known as economic profit.
- is a measure based on the residual wealth calculated by deducting its cost of capital from its operating profit, adjusted for taxes on a cash basis.
- measures the value a company generates from its invested funds.
- EVA relies heavily on invested capital. Hence, more suitable for asset-rich companies, whereas companies with intangible assets, such as technology businesses, may not be good candidates.

$$EVA = NOPAT - (InvestedCapital * WACC)$$

where

- NOPAT = Net Operating profit after taxes = Operating Profit x (1-Tax Rate)
- Invested capital = Debt + capital leases + shareholders' equity = Equity + long-term debt at the beginning of the period
- WACC = Weighted average cost of capital (average rate of return a company expects to pay its investors).

$$- WACC = \frac{K_e \times E}{E+D} + \frac{K_d \times (1-t) \times D}{E+D}$$

* K_e = required return on equity

* $K_d(1-t)$ = after tax return on debt.

- (WACC* capital invested) is also known as a **finance charge**

Invested capital can also be calculated as (Total Assets - Current Liabilities). Hence, the modified version of EVA is:

$$EVA = NOPAT - (\text{total assets} - \text{current liabilities}) * WACC$$

11.3 Market Value Added

- is the amount of wealth that a company is able to create for its stakeholders since its foundation.
- (the current market value of the company's stock - the initial invested capital)

MVA = market value of shares (or enterprise value)—book value of shareholders' equity

11.4 Unexpected size-adjusted advertising investments

(Chakravarty and Grewal, 2016; Kim and McAlister, 2011; Liu et al., 2017) used

11.5 Shareholder Complaints

(Wies et al., 2019) used data from RiskMetrics

11.6 Profitability

(Grewal et al., 2008; McAlister et al., 2016)

$$\text{profitability} = \frac{\text{operating income before depreciation}}{\text{total assets}}$$

11.7 Firm Size

(Grewal et al., 2010; McAlister et al., 2016; Nezami et al., 2018) used log of total asset in million as firm size

$$\text{firm size} = \log(\text{total assets})$$

Sales Growth (Grewal et al., 2010; Nezami et al., 2018; Rao et al., 2004)

Percentage change in gross sales = sales growth

11.8 Financial Flexibility

11.8.1 Cash flows

(Chakravarty and Grewal, 2011; Malshe and Agarwal, 2015) Cash flows = $\log(\text{operating cash flow in mil})$

11.8.2 Financial Leverage

(Kashmiri and Mahajan, 2017; Chakravarty and Grewal, 2011)

$$\text{financial leverage} = \frac{\text{long-term debt}}{\text{book value of assets}}$$

11.9 Stock return

(Markovitch et al., 2005; Chakravarty and Grewal, 2011, 2016) used dummy if stock returns exceed industry averaged stock returns

11.10 Financial Flexibility

11.11 Net Contribution

Net contribution = gross margin x sales - cost of marketing

$$NC = m \times S(a) - ka$$

Chapter 12

Data

12.1 MongoDB

in Mongo shell to check the location of your MongoDB files

```
db.adminCommand("getCmdLineOpts")
```

12.2 WRDS

Retrieve data from WRDS

```
## Warning: package 'RPostgres' was built under R version 4.0.5
```

```
## Warning: package 'dplyr' was built under R version 4.0.5
```

```
# to set up connection from R to WRDS (https://wrds-www.wharton.upenn.edu/pages/support/programming)
```

```
library(RPostgres)
```

```
library(dplyr)
```

```
# I've set up wrds connection before hand. Please use your username and password here.
```

```
# wrds <- dbConnect(Postgres(),  
#  
#  
#  
#  
#  
#  
#  
#  
#  
host='wrds-pgdata.wharton.upenn.edu',  
port=9737,  
dbname='wrds',  
sslmode='require',  
user='')
```

```
# Check variables (column headers) in COMP ANNUAL FUNDAMENTAL
```

```
#uses the already-established wrds connection to prepare the SQL query string and save the query
```

```
# check available databases: https://wrds-web.wharton.upenn.edu/wrds/tools/variable.cfm?vendor\_id
```

```

res <- dbSendQuery(wrds, "select column_name
                        from information_schema.columns
                        where table_schema='compa'
                        and table_name='funda'
                        order by column_name")
data <- dbFetch(res, n=-1) # fetches the data that results from running the query res
dbClearResult(res) # closes the connection
head(data)

##   column_name
## 1         acchg
## 2         acco
## 3         accrt
## 4        acctchg
## 5        acctstd
## 6         acdo

# select everything
res <- dbSendQuery(wrds, "select * from compa.funda")

# from compa.funda

# only select the following variables
res <- dbSendQuery(wrds, "select gvkey, datadate, fyear, indfmt, consol, popsrc, datafi")

## Warning in result_create(conn@ptr, statement): Closing open result set,
## cancelling previous query
# res <- dbSendQuery(wrds, "select gvkey, gp from compa.funda") #check variables from
data1 <- dbFetch(res, n=-1)
dbClearResult(res)

data = data1 %>%
  distinct(gvkey, datadate, fyear, tic, comm, .keep_all = T)

```

12.3 YouTube

- YouTube Ads
- YouTube Analytics and Reporting APIs: All YouTube Analytics and YouTube Reporting API requests must be authorized by the channel or content owner that owns the requested data

12.3.1 OAuth

Go to [link](#) to access your API

```
library(tuber)
app_id = "YOUR APP ID"
app_secret = "YOUR APP SECRET"
yt_oauth(app_id, app_secret)

get_stats(video_id = "N708P-A45D0")
```

Note for Ubuntu user

```
httr::set_config(httr::config( ssl_verifypeer = 0L ) )
```

Get info about a video

```
get_video_details(video_id="N708P-A45D0")
```

Get caption of a video

```
get_captions(video_id="yJTXN4xrI8")
```

Search Videos

```
res <- yt_search(term = "test")
head(res[, 1:3])
```

Get Comments of a video

```
res <- get_comment_threads(c(video_id="N708P-A45D0"))
head(res)
```

Get All the Comments Including Replies

```
get_all_comments(video_id = "a-UQz7fqR3w")
```

Get statistics of all the videos in a channel

```
a <- list_channel_resources(filter = c(channel_id = "UCT5Cx114IS3wHkJXNyuj4TA"), part="contentDetails")

# Uploaded playlists:
playlist_id <- a$items[[1]]$contentDetails$relatedPlaylists$uploads

# Get videos on the playlist
vids <- get_playlist_items(filter= c(playlist_id=playlist_id))

# Video ids
vid_ids <- as.vector(vids$contentDetails.videoId)

# Function to scrape stats for all vids
get_all_stats <- function(id) {
  get_stats(id)
```

```

}

# Get stats and convert results to data frame
res <- lapply(vid_ids, get_all_stats)
res_df <- do.call(rbind, lapply(res, data.frame))

head(res_df)

```

12.3.2 API

```

require(curl)
require(jsonlite)
library(kableExtra)
library(dplyr)
library(ggplot2)
library(plotly)
library(reshape2)

API_key = "YOUR API KEY"

getstats_video<-function(video_id,API_key){

  url=paste0("https://www.googleapis.com/youtube/v3/videos?part=snippet,statistics&id=")
  result <- fromJSON(txt=url)
  salida=list()
  return(data.frame(name=result$items$snippet$channelTitle, result$items$statistics,title=result$items$snippet$title))
}

get_playlist_canal<-function(id,API_key,topn=15){

  url=paste0('https://www.googleapis.com/youtube/v3/playlistItems?part=contentDetails&id=')
  result=fromJSON(txt=url)
  return(data.frame(result$items$contentDetails))
}

getstats_canal<-function(id,API_key){

  url=paste0('https://www.googleapis.com/youtube/v3/channels?part=snippet%2CcontentDetails&id=')
  result <- fromJSON(txt=url)
  return(data.frame(name=result$items$snippet$title,result$items$statistics,pl_list_id=result$items$snippet$playlistId))
}

```



```
getall_channels<-function(ids,API_key,topn=5){
  videos=lapply(ids,FUN=get_playlist_canal,API_key=API_key,topn=topn) %>% bind_rows()
  stats=lapply(videos[,1],FUN=getstats_video,API_key=API_key)
  stats=bind_rows(stats)
  stats$vid_id=videos[,1]
  return(stats)
}
```

Statistics per Channel

```
can_st=lapply(comp_data$cha_id,FUN = getstats_canal,API_key=API_key)
can_st=bind_rows(can_st)
can_st$viewCount=as.numeric(can_st$viewCount)
can_st[,1:6] %>% kable() %>%kable_styling()
```

```
can_st$viewCount=round(as.numeric(can_st$viewCount)/1000000,2)
p1=can_st %>% ggplot(aes(x=reorder(name,viewCount),y=viewCount,fill=name))+
  geom_bar(stat="sum")+guides(size=F)+coord_flip()+scale_fill_manual(values = c("red", "darkblue",
  geom_text(inherit.aes = T,aes(label=paste(viewCount,"M")),nudge_y =0,angle = 90)+
  labs(x="Total Visualizations(Millions)",y="Visualizations",fill="")+
  theme(legend.position = "top")+mytheme3

  ggplotly(p1,tooltip=c("name","viewCount")) %>%
    layout(legend = list(orientation = "h",x = 0.01, y = -0.1,autosize=F))
```

Information on Individual Videos per Channel

```
var_to_see="dislikeCount" #favoriteCount or commentCount
```

```
info=getall_channels(ids = can_st$pl_list_id.uploads,API_key = API_key,topn =20)
```

```
datacond=melt(info[,c(1:6,8)],id.vars = c("name","date"))
datacond$date=as.Date(datacond$date)
datacond$value=as.numeric(datacond$value)
```

```
ggplot(filter(datacond,variable==var_to_see),aes(x=as.Date(date),y=value,fill=name))+
  geom_bar(stat="sum")+labs(x=var_to_see,y="",fill="")+guides(size=FALSE)+scale_fill_manual(values
  scale_x_date(limits =as.Date(c(as.Date(min(datacond$date)),as.Date(Sys.time()))),date_breaks =
```

12.3.3 Python

Pytube

```
# from pytube import YouTube
# YouTube('https://youtube.com/watch?v=2lAe1cqCOXo').streams.first().download()

import requests
import json
r = requests.get("http://gdata.youtube.com/feeds/api/standardfeeds/top Rated?v=2&alt=json")
#r.text
data = json.loads(r.text)
data
# for item in data['data']['items']:
#     print "Video Title: %s" % (item['title'])
#     print "Video Category: %s" % (item['category'])
#     print "Video ID: %s" % (item['id'])
#     print "Video Rating: %f" % (item['rating'])
#     print "Embed URL: %s" % (item['player']['default'])
```

Using BeautifulSoup

```
from requests_html import HTMLSession
from bs4 import BeautifulSoup as bs # importing BeautifulSoup

# sample youtube video url
video_url = "https://www.youtube.com/watch?v=jNQXAC9IVRw"
# init an HTML Session
session = HTMLSession()
# get the html content
response = session.get(video_url)
# execute Java-script
response.html.render(sleep=1)
# create bs object to parse HTML
soup = bs(response.html.html, "html.parser")
# write all HTML code into a file
open("video.html", "w", encoding='utf8').write(response.html.html)
```

Generate random video

```
import json
import urllib.request
import string
import random

count = 50
API_KEY = 'your_key'
random = ''.join(random.choice(string.ascii_uppercase + string.digits) for _ in range(10))

urlData = "https://www.googleapis.com/youtube/v3/search?key={}&maxResults={}&part=snippet"
webURL = urllib.request.urlopen(urlData)
```

```

data = webURL.read()
encoding = webURL.info().get_content_charset('utf-8')
results = json.loads(data.decode(encoding))

for data in results['items']:
    videoId = (data['id']['videoId'])
    print(videoId)
    #store your ids

```

Random used by YouTube API, test

```

# -*- coding: utf-8 -*-

# Sample Python code for youtube.search.list
# See instructions for running these code samples locally:
# https://developers.google.com/explorer-help/guides/code_samples#python

import os

import google_auth_oauthlib.flow
import googleapiclient.discovery
import googleapiclient.errors

scopes = ["https://www.googleapis.com/auth/youtube.force-ssl"]

def main():
    # Disable OAuthlib's HTTPS verification when running locally.
    # *DO NOT* leave this option enabled in production.
    os.environ["OAUTHLIB_INSECURE_TRANSPORT"] = "1"

    api_service_name = "youtube"
    api_version = "v3"
    client_secrets_file = "YOUR_CLIENT_SECRET_FILE.json"

    # Get credentials and create an API client
    flow = google_auth_oauthlib.flow.InstalledAppFlow.from_client_secrets_file(
        client_secrets_file, scopes)
    credentials = flow.run_console()
    youtube = googleapiclient.discovery.build(
        api_service_name, api_version, credentials=credentials)

    request = youtube.search().list(
        part="snippet",
        maxResults=25,
        q="surfing"
    )

```

```

response = request.execute()

print(response)

if __name__ == "__main__":
    main()

```

Another way, but this is a dirty crawler, not for production

```

import re, urllib
from random import randint

def random_str(str_size):
    res = ""
    for i in xrange(str_size):
        x = randint(0,25)
        c = chr(ord('a')+x)
        res += c
    return res

def find_watch(text,pos):
    start = text.find("watch?v=",pos)
    if (start<0):
        return None,None
    end = text.find(" ",start)
    if (end<0):
        return None,None

    if (end-start > 200): #silly heuristics, probably not a must
        return None,None

    return text[start:end-1], start

def find_instance_links():
    base_url = 'https://www.youtube.com/results?search_query='
    url = base_url+random_str(3)
    #print url

    r = urllib.urlopen(url).read()

    links = {}

    pos = 0
    while True:
        link,pos = find_watch(r,pos)

```

```

        if link == None or pos == None:
            break
        pos += 1
        #print link
        if (";" in link):
            continue
        links[link] = 1

    items_list = links.items()

    list_size = len(items_list)
    selected = randint(list_size/2, list_size-1)
    return items_list[selected][0]

for i in xrange(1000):
    sleep(randint(7,20)) # pause randomly between 7 and 20 seconds
    link = find_instance_links()
    print link

```

12.4 Consumer Expenditure

- A consumer unit: “a single person, or group of person who live together and who share responsibilities for most major expenditures.”
- The “Reference Person” is **the first person named** when the respondent is asked, “Who is responsible for owning or renting this home?”
- 2 components to the Consumer Expenditure:
 - Interview survey
 - * for “big ticket” and recurring expenditures, and global estimates on others
 - * every three months for four quarter
 - * rotating-panel survey
 - * 6,000 consumer units each quarter
 - Diary survey
 - * small-ticket, frequently purchased items
 - * independent from the interview survey
 - * about 14,000 diaries within families each year
- Table format
 - annual

- * Jan to Dec: since 1984
 - * July to June: since 2013
 - two-year (Jan year 1 - Dec year 2)
 - * since 1986
- Experimental research
 - all consumer units (available on request)
 - E.g., income tables, generational groups (e.g., millennial, gen X)
-

Chapter 13

Modeling Application

13.1 Transformation

13.1.1 Log-transformation

(Manchanda et al., 2004; Wies et al., 2019) used 1 in place of 0 for log-transformation. And also use 0.5 and 0.0001 for sensitivity analysis.

To control for firm size effects, (Wies et al., 2019) scale advertising investments by the firm's total assets in the given year

13.2 Endogeneity

13.2.1 Control Function

In the context of consumer choice model (Petrin and Train, 2010)

Chapter 14

Analytical Models

This theoretical/analytical model part of this section comes mostly from professor Murali Mantrala's Marketing Model Seminar.

Marketing models consists of

1. Analytical Model: pure mathematical-based research
2. Empirical Model: data analysis.

"A model is a representation of the most important elements of a perceived real-world system".

Marketing model improves decision-making

- Econometric models
 - Description
 - Prediction
 - Simulation
- Optimization models
 - maximize profit using market response model, cost functions, or any constraints.
- Quasi- and Field experimental analyses
- Conjoint Choice Experiments.

"A decision calculus will be defined as a model-based set of procedures for processing data and judgments to assist a manager in his decision making"(Little, 1976):

- simple
- robust
- easy to control

- adaptive
- as complete as possible
- easy to communicate with

		Type of game	
Info Content	Complete	Static Nash	Dynamic Subgame perfect
	Incomplete	Bayesian Nash (Auctions)	Perfect Bayesian (signaling)

(Moorthy, 1993)

- Mathematical Theoretical Models
- Logical Experimentation
- An environment as a model, specified by assumptions
 - Math assumptions for tractability
 - Substantive assumptions for empirical testing
- Decision support modeling describe how things work, and theoretical modeling present how things should work.
- Compensation package including salaries and commission is a tradeoff between reduced income risk and motivation to work hard.
- Internal and External Validity are questions related to the boundaries conditions of your experiments.
- “Theories are tested by their predictions, not by the realism of their super model assumptions.” (Friedman, 1953)

(McAfee and McMillan, 1996)

- Competition is performed under uncertainty
- Competition reveals hidden information
 - Independent-private-values case: selling price = second highest valuation
 - It’s always better for sellers to reveal information since it reduces chances of cautious bidding that is resulted from the winner’s curse
- Competition is better than bargaining
 - Competition requires less computation and commitment abilities
- Competition creates effort incentives

(Leeflang et al., 2000)

- Types of model:
 - Predictive model
 - * Sales model: using time series data
 - * Trial rate: using exponential growth.
 - * Product growth model: Bass (1969)
 - Descriptive model
 - * Purchase incidence and purchase timing : use Poisson process
 - * Brand choice: Markov models or learning models.
 - * Pricing decisions in an oligopolistic market Howard and Morgenroth (1968)
 - Normative model
 - * Profit maximization based on price, adverting and quality (Dorfman and Steiner, 1976), extended by (Roberts et al., 1964; Lambin, 1970)

Later, Little (1970) introduced decision calculus and then multinomial logit model (Guadagni and Little, 1983)

Potential marketing decision automation:

- Promotion or pricing programs
- Media allocation
- Distribution
- Product assortment
- Direct mail solicitation

(Moorthy, 1985)

- Definitions:
 - Rationality = maximizing subjective expected utility
 - Intelligence = recognizing other firms are rational.
 - Rules of the game include
 - * # of firms
 - * feasible set of actions
 - * utilities for each combination of moves
 - * sequence of moves
 - * the structure of info (who knows what and when?)

- Incomplete info stems from
 - * unknown motivations
 - * unknown ability (capabilities)
 - * different knowledge of the world.
- Pure strategy = plan of action
- A mixed strategy = probability dist of pure strategies.
- Strategic form representation = sets of possible strategies for every firm and its payoffs.
- Equilibrium = a list of strategies in which “no firm would like unilaterally to change its strategy.”
- Equilibrium is not outcome of a dynamic process.
- Equilibrium Application
 - Oligopolistic Competition
 - * Cournot (1838): quantities supplied: Cournot equilibrium. Changing quantities is more costly than changing prices
 - * Bertrand (1883): Bertrand equilibrium: pricing.
 - Perfect competition
 - Product Competition: Hotelling (1929): Principle of Minimum Differentiation is invalid.
 - Entry:
 - * first mover advantage
 - * deterrent strategy
 - * optimal for entrants or incumbents
 - Channels
- Perfectness of equilibria
 - Subgame perfectness
 - Sequential rationality
 - Trembling-hand perfectness
 - Application
 - * Product and price competition in Oligopolies
 - * Strategic Entry Deterrence
- Dynamic games

- Long-term competition in oligopolies
- Implicit Collusion in practice : price match from leader firms
- Incomplete Information
 - Durable goods pricing by a monopolist
 - predatory pricing and limit pricing
 - reputation, product quality, and prices
 - Competitive bidding and auctions

(KIM and SERFES, 2006): A location model with preference variety

(Hotelling, 1929)

- Stability in competition
- Duopoly is inherently unstable
- Bertrand disagrees with Cournot, and Edgeworth elaborates on it.
 - because Cournot's assumption of absolutely identical products between firms.

seller try to $p_2 < p_1 c(l - a - b)$

the point of indifference

$$p_1 + cx = p_2 + cy$$

- c = cost per unit of time in each unit of line length
- p = price
- q = quantity
- x, y = length from A and B respectively

$$a + x + y + b = l$$

is the length of the street

Hence, we have

$$x = 0.5(l - a - b + \frac{p_2 - p_1}{c}) \quad y = 0.5(l - a - b + \frac{p_1 - p_2}{c})$$

Profits will be

$$\pi_1 = p_1 q_1 = p_1(a+x) = 0.5(l+a-b)p_1 - \frac{p_1^2}{2c} + \frac{p_1 p_2}{2c} \quad \pi_2 = p_2 q_2 = p_2(b+y) = 0.5(l+a-b)p_2 - \frac{p_2^2}{2c} + \frac{p_1 p_2}{2c}$$

To set the price to maximize profit, we have

$$\frac{\partial \pi_1}{\partial p_1} = 0.5(l + a - b) - \frac{p_1}{c} + \frac{p_2}{2c} = 0 \quad \frac{\partial \pi_2}{\partial p_2} = 0.5(l - a + b) - \frac{p_2}{c} + \frac{p_1}{2c} = 0$$

which equals

$$p_1 = c(l + \frac{a-b}{3})p_2 = c(l - \frac{a-b}{3})$$

and

$$q_1 = a + x = 0.5(l + \frac{a-b}{3})q_2 = b + y = 0.5(l - \frac{a-b}{3})$$

with the SOC satisfied

In case of deciding locations, socialism works better than capitalism

(d'Aspremont et al., 1979)

- Principle of Minimum Differentiation is invalid

$$\pi_1(p_1, p_2) = \begin{cases} ap_1 + 0.5(l - a - b)p_1 + \frac{1}{2c}p_1p_2 - \frac{1}{2c}p_1^2 & \text{if } |p_1 - p_2| \leq c(l - a - b) \\ lp_1 & \text{if } p_1 < p_2 - c(l - a - b) \\ 0 & \text{if } p_1 > p_2 + c(l - a - b) \end{cases}$$

and

$$\pi_2(p_1, p_2) = \begin{cases} bp_2 + 0.5(l - a - b)p_2 + \frac{1}{2c}p_1p_2 - \frac{1}{2c}p_2^2 & \text{if } |p_1 - p_2| \leq c(l - a - b) \\ lp_2 & \text{if } p_2 < p_1 - c(l - a - b) \\ 0 & \text{if } p_2 > p_1 + c(l - a - b) \end{cases}$$

14.1 Positioning Models

Tabuchi and Thisse (1995)

- Relax Hotelling's model's assumption of uniform distribution of consumers to non-uniform distribution.
- Assumptions:
 - Equal cost

- Consumers distributed over $[0,1]$
- $F(x)$ = cumulative distribution of consumers where $F(1) = 1$ = total population
- 2 distributions:
 - * Traditional uniform density: $f(x) = 1$
 - * New: triangular density: $f(x) = 2 - 2|2x - 1|$ which represents consumer concentration
- Transportation cost = quadratic function of distance.

Hence, marginal consumer is

$$\bar{x} = (p_2 - p_1 + x_2^2 - x_1^2)/2(x_2 - x_1)$$

then when $x_1 < x_2$ the profit function is

$$\Pi_1 = p_1 F(\bar{x})$$

and

$$\Pi_2 = p_2[1 - F(\bar{x})]$$

and vice versa for $x_1 > x_2$, and Bertrand game when $x_1 = x_2$

- If firms pick **simultaneously** their locations, and then **simultaneously** their prices, and consumer density function is log-concave, then there is a unique Nash price equilibrium
 - Under **uniform** distribution, firms choose to locate as far apart as possible (could be true when observing shopping centers are far away from cities), but then consumers have to buy products that are far away from their ideal.
 - Under **triangular** density, no symmetric location can be found, but two asymmetric Nash location equilibrium can still be possible (decrease in equilibrium profits of both firms)
- If firms pick **sequentially** their locations, and pick their prices **simultaneously**,
 - Under both uniform and triangular, first entrant will locate at the market center

Sajeesh and Raju (2010)

- Model satiation (variety-seeking) as a relative reduction in the willingness to pay of the previously purchased brand. also known as negative state dependence
- Previous studies argue that in the presence of variety seeking consumers, firms should enjoy higher prices and profits, but this paper argues that average prices and profits are lower.
 - Firms should charge lower prices in the second period to prevent consumers from switching.

Assumptions:

- Period 0, choose location simultaneously
- Period 1, choose prices simultaneously
- Period 2, firms choose prices simultaneously

14.2 Market Response Model

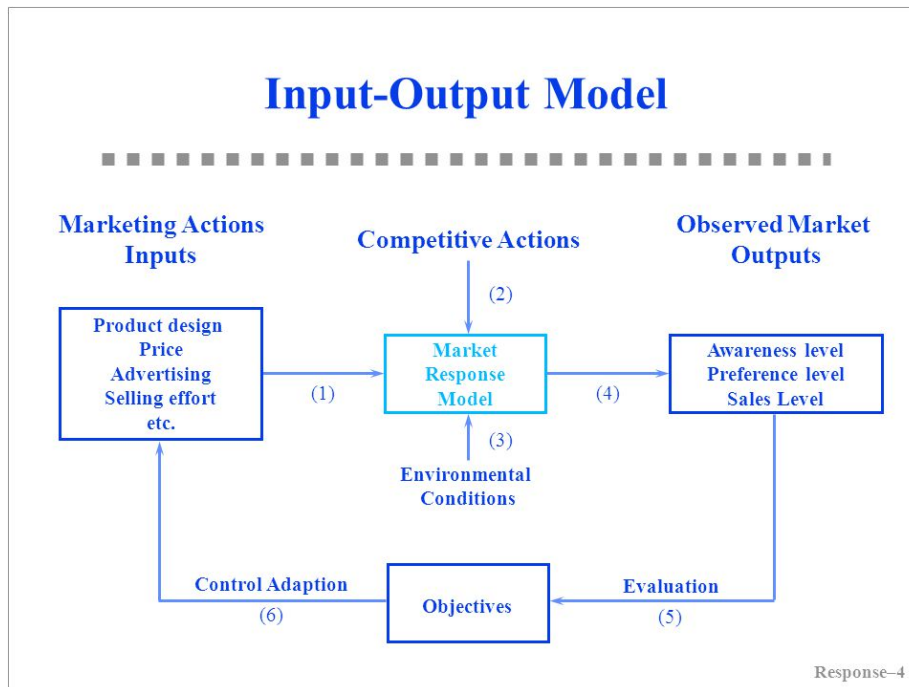
Marketing Inputs:

- Selling effort
- advertising spending
- promotional spending

↓

Marketing Outputs:

- sales
- share
- profit
- awareness



Give phenomena for a good model:

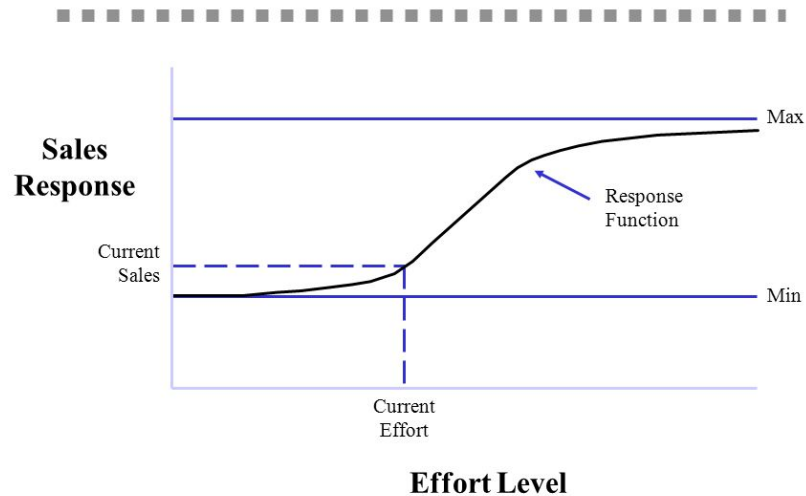
- P1: Dynamic sales response involves a sales **growth rate** and a sales **decay rate** that are different
- P2: Steady-state response can be **concave or S-shaped**. Positive sales at 0 advertising.
- P3: **Competitive effects**
- P4: Advertising effectiveness dynamics due to changes in media, copy, and other factors.
- P5: Sales still increase or fall off even as advertising is held constant.

Saunders (1987) phenomena

- P1: Output = 0 when Input = 0
- P2: The relationship between input and output is linear
- P3: Returns decrease as the scale of input increases (i.e., additional unit of input gives less output)
- P4: Output cannot exceed some level (i.e., saturation)
- P5: Returns increase as scale of input increases (i.e., additional unit of input gives more output)
- P6: Returns first increase and then decrease as input increases (i.e., S-shaped return)
- P7: Input must exceed some level before it produces any output (i.e., threshold)
- P8: Beyond some level of input, output declines (i.e., supersaturation)

point)

Response Function



Aggregate Response Models

- Linear model: $Y = a + bX$
 - Through origin
 - can only handle constant returns to scale (i.e., can't handle concave, convex, and S-shape)
- The Power Series/Polynomial model: $Y = a + bX + cX^2 + dX^3 + \dots$
 - can't handle saturation and threshold
- Fraction root model/ Power model: $Y = a + bX^c$ where c is prespecified
 - $c = 1/2$, called **square root model**
 - $c = -1$, called **reciprocal model**
 - c can be interpreted as elasticity if $a = 0$.
 - $c = 1$, linear
 - $c < 1$, decreasing return
 - $c > 1$, increasing returns
- Semilog model: $Y = a + b \ln X$

- Good when constant percentage increase in marketing effort (X) result in constant absolute increase in sales (Y)
- Exponential model: $Y = ae^{bX}$ where $X > 0$
 - $b > 0$, increasing returns and convex
 - $b < 0$, decreasing returns and saturation
- Modified exponential model: $Y = a(1 - e^{-bX}) + c$
 - Decreasing returns and saturation
 - upper bound = $a + c$
 - lower bound = c
 - typically used in selling effort
- Logistic model: $Y = \frac{a}{a + e^{-(b+cX)}} + d$
 - increasing return followed by decreasing return to scale, S-shape
 - saturation = $a + d$
 - good with saturation and s-shape
- Gompertz model
- ADBUDG model (Little, 1970) : $Y = b + (a - b) \frac{X^c}{d + X^c}$
 - $c > 1$, S-shaped
 - $0 < c < 1$
 - * Concave
 - * saturation effect
 - * upper bound at a
 - * lower bound at b
 - typically used in advertising and selling effort.
 - can handle, through origin, concave, saturation, S-shape
- Additive model for handling multiple Instruments: $Y = af(X_1) + bg(X_2)$
- Multiplicative model for handling multiple instruments: $Y = aX_1^bX_2^c$ where b and c are elasticities. More generally, $Y = af(X_1) \times bg(X_2)$
- Multiplicative and additive model: $Y = af(X_1) + bg(X_2) + cf(X_1)g(X_2)$
- Dynamic response model: $Y_t = a_0 + a_1X_t + \lambda Y_{t-1}$ where a_1 = current effect, λ = carry-over effect

Dynamic Effects

- Carry-over effect: current marketing expenditure influences future sales

- Advertising adstock/ advertising carry-over is the same thing: lagged effect of advertising on sales
- Delayed-response effect: delays between when marketing investments and their impact
- Customer holdout effects
- Hysteresis effect
- New trier and wear-out effect
- Stocking effect

Simple Decay-effect model:

$$A_t = T_t + \lambda T_{t-1}, t = 1, \dots,$$

where

- A_t = Adstock at time t
- T_t = value of advertising spending at time t
- λ = decay/ lag weight parameter

Response Models can be characterized by:

1. The number of marketing variables
2. whether they include competition or not
3. the nature of the relationship between the input variables
 1. Linear vs. S-shape
4. whether the situation is static vs. dynamic
5. whether the models reflect individual or aggregate response
6. the level of demand analyzed
 1. sales vs. market share

Market Share Model and Competitive Effects: $Y = M \times V$ where

- Y = Brand sales models
- V = product class sales models
- M = market-share models

Market share (attraction) models

$$M_i = \frac{A_i}{A_1 + \dots + A_n}$$

where A_i attractiveness of brand i

Individual Response Model:

Multinomial logit model representing the probability of individual i choosing brand l is

$$P_{il} = \frac{e^{A_{il}}}{\sum_j e^{A_{ij}}}$$

where

- A_{ij} = attractiveness of product j for individual i $A_{ij} = \sum_k w_k b_{ijk}$
- b_{ijk} = individual i 's evaluation of product j on product attribute k , where the summation is over all the products that individual i is considering to purchase
- w_k = importance weight associated with attribute k in forming product preferences.

14.3 Marketing Resource Allocation Models

This section is based on (Mantrala et al., 1992)

14.3.1 Case study 1

Concave sales response function

- Optimal vs. proportional at different investment levels
- Profit maximization perspective of aggregate function

$$s_i = k_i(1 - e^{-b_i x_i})$$

where

- s_i = current-period sales response (dollars / period)
- x_i = amount of resource allocated to submarket i
- b_i = rate at which sales approach saturation
- k_i = sales potential

Allocation functions

- Fixed proportion
 - R_i = Investment level (dollars/period)
 - w_i = fixed proportion or weights

$$\hat{x}_i = w_i R; \sum_{t=1}^2 w_t = 1; 0 < w_t < 1$$

- Informed allocator
 - optimal allocations via marginal analysis (maximize profits)

$$\max C = m \sum_{i=1}^2 k_i (1 - e^{-b_i x_i}) x_1 + x_2 \leq R; x_i \geq 0 \text{ for } i = 1, 2, x_1 = \frac{1}{(b_1 + b_2)(b_2 R + \ln(\frac{k_1 b_1}{k_2 b_2}))} x_2 = \frac{1}{(b_1 + b_2)(b_2 R + \ln(\frac{k_1 b_1}{k_2 b_2}))}$$

14.3.2 Case study 2

S-shaped sales response function:

- Optimal vs. proportional at different investment levels
- Profit maximization perspective of aggregate function

14.3.3 Case study 3

Quadratic-form stochastic response function

- Optimal allocation only with risk averse and risk neutral investors.

14.4 Meta-analyses of Econometric Marketing Models

14.5 Dynamic Advertising Effects and Spending Models

14.6 Marketing Mix Optimization Models

Check this post for implementation in Python

14.7 New Product Diffusion Models

14.8 Two-sided Platform Marketing Models

Example of Marketing Mix Model in practice: [link](#)

Chapter 15

Empirical Models

15.1 Attribution Models

15.1.1 Ordered Shapley

Based on (Zhao et al., 2018) (to access paper) Cooperative game theory: look at the marginal contribution of each player in the game, where **Shapley value** (i.e., the credit assigned to each individual) is the expected value value of the marginal contribution over all possible permutations (e.g., all possible sequences) of the players.

Shapely value considered:

- marginal contribution of each player (i.e., channel)
- sequence of joining the coalition (i.e., customer journey).

Typically, we can't apply the Shapley Value method due to computational burden (you need all possible permutations). And a drawback is that all the credit must be divided among your channels, if you have missing channels, then it will distort the estimates of other channels' estimates.

It's hard to use Shapley value model for model comparison since we have no "ground truth"

Marketing application:

- Pardot Einstein features and more

```
library("GameTheory")
```

```
## Loading required package: lpSolveAPI
```

```
## Loading required package: combinat
```

```
##
```

```
## Attaching package: 'combinat'

## The following object is masked from 'package:utils':
##
##      combn

## Loading required package: gtools
## Warning: package 'gtools' was built under R version 4.0.5

## Loading required package: ineq
## Loading required package: kappalab
## Loading required package: lpSolve
## Loading required package: quadprog
## Loading required package: kernlab

##
## Attaching package: 'kappalab'

## The following object is masked from 'package:ineq':
##
##      entropy

packages reference
```

15.1.2 Markov Model

Markov chains maps the movement and gives a probability distribution, for moving from one state to another state. A Markov Chain has three properties:

- **State space** – set of all the states in which process could potentially exist
- **Transition operator** – the probability of moving from one state to other state
- **Current state probability distribution** – probability distribution of being in any one of the states at the start of the process

In mathematically sense

$$w_{ij} = P(X_t = s_j | X_{t-1} = s_i), 0 \leq w_{ij} \leq 1, \sum_{j=1}^N w_{ij} = 1 \forall i$$

where

- The Transition Probability (w_{ij}) = The Probability of the Previous State (X_{t-1}) Given the Current State (X_t)
- The Transition Probability (w_{ij}) is No Less Than 0 and No Greater Than 1

- The Sum of the Transition Probabilities Equals 1 (i.e., Everyone Must Go Somewhere)

To examine a particular node in the Markov graph, we use **removal effect** (s_i) to see its contribution to a conversion. In another word, the Removal Effect is the probability of converting when a step is completely removed; all sequences that had to go through that step are now sent directly to the null node

Each node is called **transition states**

The probability of moving from one channel to another channel is called **transition probability**.

first-order or “memory-free” Markov graph is called “memory-free” because the probability of reaching one state depends only on the previous state visited.

- Order 0: Do not care about where the you came from or what step the you are on, only the probability of going to any state.
- Order 1: Looks back zero steps. You are currently at a state. The probability of going anywhere is based on being at that step.
- Order 2: Looks back one step. You came from state A and are currently at state B. The probability of going anywhere is based on where you were and where you are.
- Order 3: Looks back two steps. You came from state A after state B and are currently at state C. The probability of going anywhere is based on where you were and where you are.
- Order 4: Looks back three steps. You came from state A after B after C and are currently at state D. The probability of going anywhere is based on where you were and where you are.

15.1.2.1 Example 1

This section is by Analytics Vidhya

data link

```
# #Install the libraries
# install.packages("ChannelAttribution")
# install.packages("ggplot2")
# install.packages("reshape")
# install.packages("dplyr")
# install.packages("plyr")
# install.packages("reshape2")
# install.packages("markovchain")
# install.packages("plotly")

#Load the libraries
library("ChannelAttribution")
```

```
## Warning: package 'ChannelAttribution' was built under R version 4.0.5
```

```

## ChannelAttribution 2.0.4

## Looking for attribution at path level? Try ChannelAttributionPro! Visit www.channelattribution.com

library("ggplot2")

## Warning: package 'ggplot2' was built under R version 4.0.5

##
## Attaching package: 'ggplot2'

## The following object is masked from 'package:kernlab':
##
##      alpha

library("reshape")
library("dplyr")

## Warning: package 'dplyr' was built under R version 4.0.5

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:reshape':
##
##      rename

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union

library("plyr")

## -----

## You have loaded plyr after dplyr - this is likely to cause problems.
## If you need functions from both plyr and dplyr, please load plyr first, then dplyr:
## library(plyr); library(dplyr)

## -----

##
## Attaching package: 'plyr'

## The following objects are masked from 'package:dplyr':
##
##      arrange, count, desc, failwith, id, mutate, rename, summarise,
##      summarize

```

```
## The following objects are masked from 'package:reshape':
##
##      rename, round_any
library("reshape2")

##
## Attaching package: 'reshape2'

## The following objects are masked from 'package:reshape':
##
##      colsplit, melt, recast
library("markovchain")

## Package:  markovchain
## Version:  0.8.6
## Date:     2021-05-17
## BugReport: https://github.com/spedygiorgio/markovchain/issues
library("plotly")

## Warning: package 'plotly' was built under R version 4.0.5

##
## Attaching package: 'plotly'

## The following objects are masked from 'package:plyr':
##
##      arrange, mutate, rename, summarise

## The following object is masked from 'package:reshape':
##
##      rename

## The following object is masked from 'package:ggplot2':
##
##      last_plot

## The following object is masked from 'package:stats':
##
##      filter

## The following object is masked from 'package:graphics':
##
##      layout

#Read the data into R
channel = read.csv("images/Channel_attribution.csv", header = T) %>% select(-c(Output))
head(channel, n = 2)

##      R05A.01 R05A.02 R05A.03 R05A.04 R05A.05 R05A.06 R05A.07 R05A.08 R05A.09
```



```
Data = channel_fin
head(Data)
```

```
##           path conversion
## 1      1 > 1 > 1 > 20      1
## 2      1 > 1 > 12 > 12      1
## 3 1 > 1 > 14 > 13 > 12 > 20      1
## 4      1 > 1 > 3 > 13 > 3 > 20      1
## 5      1 > 1 > 3 > 17 > 17      1
## 6 1 > 1 > 6 > 1 > 12 > 20 > 12      1
```

```
heuristic model
```

```
H <- heuristic_models(Data, 'path', 'conversion', var_value='conversion')
H
```

```
##      channel_name first_touch_conversions first_touch_value
## 1             1             130             130
## 2            20              0              0
## 3            12             75             75
## 4            14             34             34
## 5            13            320            320
## 6             3            168            168
## 7            17             31             31
## 8             6             50             50
## 9             8             56             56
## 10           10            547            547
## 11           11             66             66
## 12           16            111            111
## 13            2            199            199
## 14            4            231            231
## 15            7             26             26
## 16            5             62             62
## 17            9            250            250
## 18           15             22             22
## 19           18              4              4
## 20           19             10             10
##      last_touch_conversions last_touch_value linear_touch_conversions
## 1             18             18          73.773661
## 2            1701            1701         473.998171
## 3             23             23          76.127863
## 4             25             25          56.335744
## 5             76             76         204.039552
## 6             21             21         117.609677
## 7             47             47          76.583847
## 8             20             20          54.707124
## 9             17             17          53.677862
```

## 10	42	42	211.822393
## 11	33	33	107.109048
## 12	95	95	156.049086
## 13	18	18	94.111668
## 14	88	88	250.784033
## 15	15	15	33.435991
## 16	23	23	74.900402
## 17	71	71	194.071690
## 18	47	47	65.159225
## 19	2	2	5.026587
## 20	10	10	12.676375
##	linear_touch_value		
## 1	73.773661		
## 2	473.998171		
## 3	76.127863		
## 4	56.335744		
## 5	204.039552		
## 6	117.609677		
## 7	76.583847		
## 8	54.707124		
## 9	53.677862		
## 10	211.822393		
## 11	107.109048		
## 12	156.049086		
## 13	94.111668		
## 14	250.784033		
## 15	33.435991		
## 16	74.900402		
## 17	194.071690		
## 18	65.159225		
## 19	5.026587		
## 20	12.676375		

- First Touch Conversion: credit is given to the first touch point.
- Last Touch Conversion: credit is given to the last touch point.
- Linear Touch Conversion: All channels/touch points are given equal credit in the conversion.

Markov model

```
M <- markov_model(Data, 'path', 'conversion', var_value='conversion', order = 1)

##
## Number of simulations: 100000 - Convergence reached: 2.05% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
```

M

##	channel_name	total_conversion	total_conversion_value
## 1	1	82.805970	82.805970
## 2	20	439.582090	439.582090
## 3	12	81.253731	81.253731
## 4	14	64.238806	64.238806
## 5	13	197.791045	197.791045
## 6	3	122.328358	122.328358
## 7	17	86.985075	86.985075
## 8	6	58.985075	58.985075
## 9	8	60.656716	60.656716
## 10	10	209.850746	209.850746
## 11	11	115.402985	115.402985
## 12	16	159.820896	159.820896
## 13	2	97.074627	97.074627
## 14	4	222.149254	222.149254
## 15	7	40.597015	40.597015
## 16	5	80.537313	80.537313
## 17	9	178.865672	178.865672
## 18	15	72.358209	72.358209
## 19	18	6.567164	6.567164
## 20	19	14.149254	14.149254

combine the two models

```
# Merges the two data frames on the "channel_name" column.
R <- merge(H, M, by='channel_name')
```

```
# Select only relevant columns
```

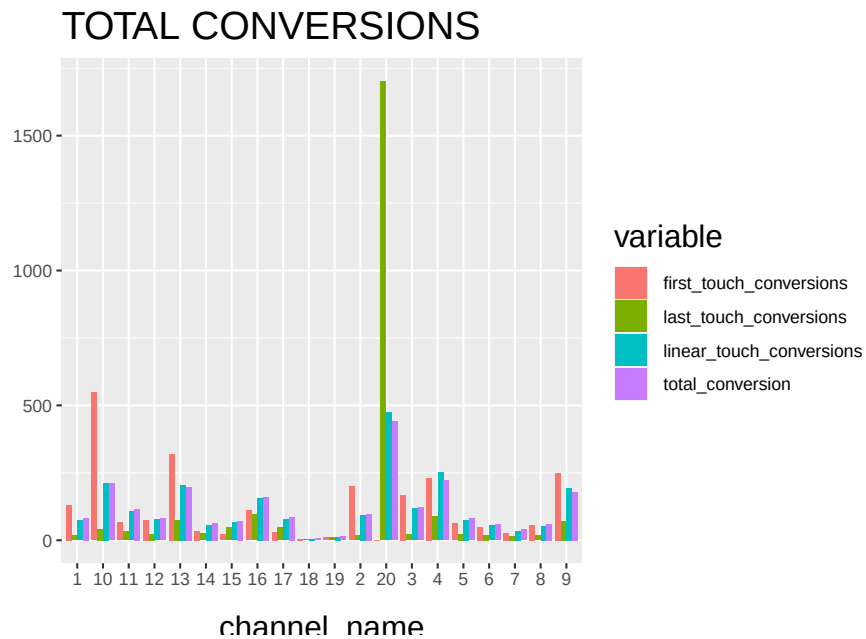
```
R1 <- R[, (colnames(R) %in% c('channel_name', 'first_touch_conversions', 'last_touch_conversions'))]
```

```
# Transforms the dataset into a data frame that ggplot2 can use to plot the outcomes
```

```
R1 <- melt(R1, id='channel_name')
```

```
# Plot the total conversions
```

```
ggplot(R1, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL CONVERSIONS') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
  theme(title = element_text(size = 16)) +
  theme(plot.title=element_text(size = 20)) +
  ylab("")
```



and then check the final results.

15.1.2.2 Example 2

Example code by Sergey Bryl'

```
library(dplyr)
library(reshape2)
library(ggplot2)
library(ggthemes)
library(ggrepel)
library(RColorBrewer)
library(ChannelAttribution)
library(markovchain)

##### simple example #####
# creating a data sample
df1 <- data.frame(path = c('c1 > c2 > c3', 'c1', 'c2 > c3'), conv = c(1, 0, 0), conv_n

# calculating the model
mod1 <- markov_model(df1,
                      var_path = 'path',
                      var_conv = 'conv',
                      var_null = 'conv_null',
                      out_more = TRUE)
```



```

# extracting the results of attribution
df_res1 <- mod1$result

# extracting a transition matrix
df_trans1 <- mod1$transition_matrix
df_trans1 <- dcast(df_trans1, channel_from ~ channel_to, value.var = 'transition_probability')

### plotting the Markov graph ###
df_trans <- mod1$transition_matrix

# adding dummies in order to plot the graph
df_dummy <- data.frame(channel_from = c('(start)', '(conversion)', '(null)'),
                      channel_to = c('(start)', '(conversion)', '(null)'),
                      transition_probability = c(0, 1, 1))
df_trans <- rbind(df_trans, df_dummy)

# ordering channels
df_trans$channel_from <- factor(df_trans$channel_from, levels = c('(start)', '(conversion)', '(null)'))
df_trans$channel_to <- factor(df_trans$channel_to, levels = c('(start)', '(conversion)', '(null)'))
df_trans <- dcast(df_trans, channel_from ~ channel_to, value.var = 'transition_probability')

# creating the markovchain object
trans_matrix <- matrix(data = as.matrix(df_trans[, -1]), nrow = nrow(df_trans[, -1]), ncol = ncol(df_trans[, -1]))
trans_matrix[is.na(trans_matrix)] <- 0
# trans_matrix1 <- new("markovchain", transitionMatrix = trans_matrix)
#
# # plotting the graph
# plot(trans_matrix1, edge.arrow.size = 0.35)

# simulating the "real" data
set.seed(354)
df2 <- data.frame(client_id = sample(c(1:1000), 5000, replace = TRUE),
                 date = sample(c(1:32), 5000, replace = TRUE),
                 channel = sample(c(0:9), 5000, replace = TRUE),
                 prob = c(0.1, 0.15, 0.05, 0.07, 0.11, 0.07, 0.13, 0.1, 0.06, 0.04))
df2$date <- as.Date(df2$date, origin = "2015-01-01")
df2$channel <- paste0('channel_', df2$channel)

# aggregating channels to the paths for each customer
df2 <- df2 %>%
  arrange(client_id, date) %>%
  group_by(client_id) %>%
  summarise(path = paste(channel, collapse = ' > '),
            # assume that all paths were finished with conversion
            conv = 1,

```

```

        conv_null = 0) %>%
    ungroup()

# calculating the models (Markov and heuristics)
mod2 <- markov_model(df2,
    var_path = 'path',
    var_conv = 'conv',
    var_null = 'conv_null',
    out_more = TRUE)

##
## Number of simulations: 100000 - Convergence reached: 1.40% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
# heuristic_models() function doesn't work for me, therefore I used the manual calculation
# instead of:
#h_mod2 <- heuristic_models(df2, var_path = 'path', var_conv = 'conv')

df_hm <- df2 %>%
    mutate(channel_name_ft = sub('>.*', '', path),
           channel_name_ft = sub(' ', '', channel_name_ft),
           channel_name_lt = sub('.*>', '', path),
           channel_name_lt = sub(' ', '', channel_name_lt))
# first-touch conversions
df_ft <- df_hm %>%
    group_by(channel_name_ft) %>%
    summarise(first_touch_conversions = sum(conv)) %>%
    ungroup()
# last-touch conversions
df_lt <- df_hm %>%
    group_by(channel_name_lt) %>%
    summarise(last_touch_conversions = sum(conv)) %>%
    ungroup()

h_mod2 <- merge(df_ft, df_lt, by.x = 'channel_name_ft', by.y = 'channel_name_lt')

# merging all models
all_models <- merge(h_mod2, mod2$result, by.x = 'channel_name_ft', by.y = 'channel_name',
    colnames(all_models)[c(1, 4)] <- c('channel_name', 'attrib_model_conversions'))

library("RColorBrewer")
library("ggthemes")
library("ggrepel")
##### visualizations #####
# transition matrix heatmap for "real" data

```

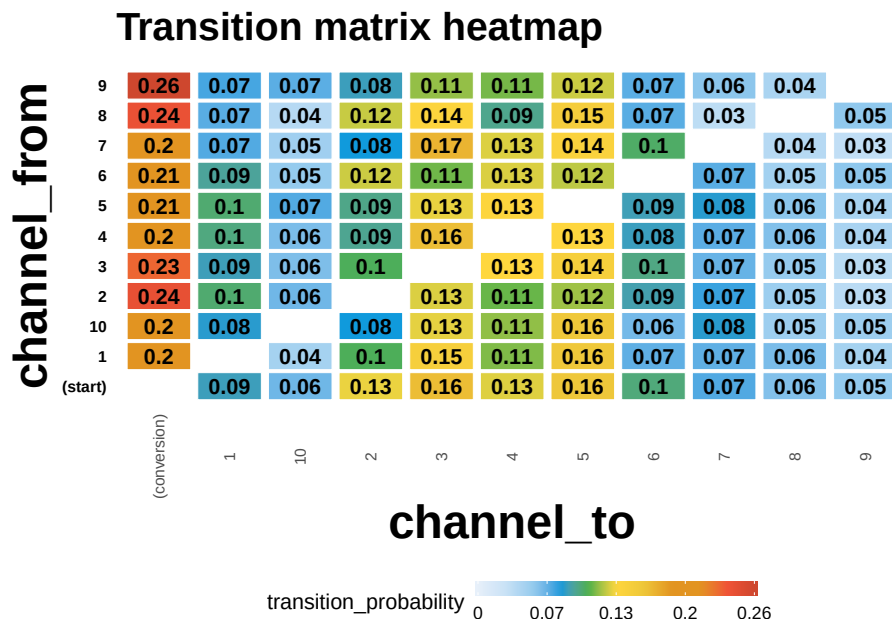
```

df_plot_trans <- mod2$transition_matrix

cols <- c("#e7f0fa", "#c9e2f6", "#95cbee", "#0099dc", "#4ab04a", "#ffd73e", "#eec73a",
          "#e29421", "#e29421", "#f05336", "#ce472e")
t <- max(df_plot_trans$transition_probability)

ggplot(df_plot_trans, aes(y = channel_from, x = channel_to, fill = transition_probability)) +
  theme_minimal() +
  geom_tile(colour = "white", width = .9, height = .9) +
  scale_fill_gradientn(colours = cols, limits = c(0, t),
                      breaks = seq(0, t, by = t/4),
                      labels = c("0", round(t/4*1, 2), round(t/4*2, 2), round(t/4*3, 2), round(t/4*4, 2)),
                      guide = guide_colourbar(ticks = T, nbin = 50, barheight = .5, label = "transition_probability")) +
  geom_text(aes(label = round(transition_probability, 2)), fontface = "bold", size = 4) +
  theme(legend.position = 'bottom',
        legend.direction = "horizontal",
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        plot.title = element_text(size = 20, face = "bold", vjust = 2, color = 'black', lineheight = 1),
        axis.title.x = element_text(size = 24, face = "bold"),
        axis.title.y = element_text(size = 24, face = "bold"),
        axis.text.y = element_text(size = 8, face = "bold", color = 'black'),
        axis.text.x = element_text(size = 8, angle = 90, hjust = 0.5, vjust = 0.5, face = "bold"),
        legend.title = element_text(size = 10, face = "bold", color = "black", lineheight = 1))
ggtitle("Transition matrix heatmap")

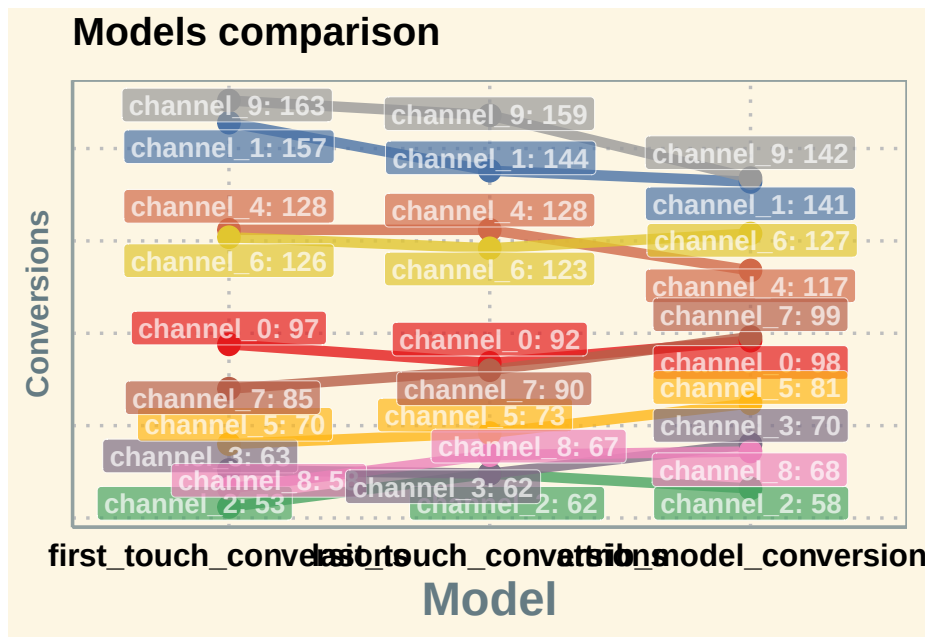
```



```
# models comparison
all_mod_plot <- reshape2::melt(all_models, id.vars = 'channel_name', variable.name = 'conv_type')
all_mod_plot$value <- round(all_mod_plot$value)
# slope chart
pal <- colorRampPalette(brewer.pal(10, "Set1"))
```

```
## Warning in brewer.pal(10, "Set1"): n too large, allowed maximum for palette Set1 is 9
## Returning the palette you asked for with that many colors
```

```
ggplot(all_mod_plot, aes(x = conv_type, y = value, group = channel_name)) +
  theme_solarized(base_size = 18, base_family = "", light = TRUE) +
  scale_color_manual(values = pal(10)) +
  scale_fill_manual(values = pal(10)) +
  geom_line(aes(color = channel_name), size = 2.5, alpha = 0.8) +
  geom_point(aes(color = channel_name), size = 5) +
  geom_label_repel(aes(label = paste0(channel_name, ': ', value), fill = factor(channel_name),
    alpha = 0.7,
    fontface = 'bold', color = 'white', size = 5,
    box.padding = unit(0.25, 'lines'), point.padding = unit(0.5, 'lines'),
    max.iter = 100) +
  theme(legend.position = 'none',
    legend.title = element_text(size = 16, color = 'black'),
    legend.text = element_text(size = 16, vjust = 2, color = 'black'),
    plot.title = element_text(size = 20, face = "bold", vjust = 2, color = 'black'),
    axis.title.x = element_text(size = 24, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x = element_text(size = 16, face = "bold", color = 'black'),
    axis.text.y = element_blank(),
    axis.ticks.x = element_blank(),
    axis.ticks.y = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_line(colour = "grey", linetype = "dotted"),
    panel.grid.minor = element_blank(),
    strip.text = element_text(size = 16, hjust = 0.5, vjust = 0.5, face = "bold", color = "white"),
    strip.background = element_rect(fill = "#f0b35f")) +
  labs(x = 'Model', y = 'Conversions') +
  ggtitle('Models comparison') +
  guides(colour = guide_legend(override.aes = list(size = 4)))
```



Additional concerns:

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.0.5
```

```
## -- Attaching packages ----- tidyverse 1.3.1 --
```

```
## v tibble 3.1.2      v purrr 0.3.4
```

```
## v tidyr 1.1.3      v stringr 1.4.0
```

```
## v readr 2.0.1      v forcats 0.5.1
```

```
## Warning: package 'readr' was built under R version 4.0.5
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x ggplot2::alpha() masks kernlab::alpha()
```

```
## x plotly::arrange() masks plyr::arrange(), dplyr::arrange()
```

```
## x purrr::compact() masks plyr::compact()
```

```
## x plyr::count() masks dplyr::count()
```

```
## x purrr::cross() masks kernlab::cross()
```

```
## x tidyr::expand() masks reshape::expand()
```

```
## x plyr::failwith() masks dplyr::failwith()
```

```
## x plotly::filter() masks dplyr::filter(), stats::filter()
```

```
## x plyr::id() masks dplyr::id()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## x plotly::mutate() masks plyr::mutate(), dplyr::mutate()
```

```
## x plotly::rename() masks plyr::rename(), dplyr::rename(), reshape::rename()
```

```
## x plotly::summarise() masks plyr::summarise(), dplyr::summarise()
```

```
## x plyr::summarize()   masks dplyr::summarize()

library(reshape2)
library(ggthemes)
library(ggrepel)
library(RColorBrewer)
library(ChannelAttribution)
library(markovchain)
library(visNetwork)
library(expm)

## Loading required package: Matrix
## Warning: package 'Matrix' was built under R version 4.0.5

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## The following object is masked from 'package:reshape':
##
##     expand

##
## Attaching package: 'expm'

## The following object is masked from 'package:Matrix':
##
##     expm

library(stringr)
library(purrr)
library(purrrlyr)

##### simulating the "real" data #####
set.seed(454)
df_raw <- data.frame(customer_id = paste0('id', sample(c(1:20000), replace = TRUE)), date = sample(
  group_by(customer_id) %>%
  mutate(conversion = sample(c(0, 1), n(), prob = c(0.975, 0.025), replace = TRUE))
  ungroup() %>%
  dmap_at(c(1, 3), as.character) %>%
  arrange(customer_id, date)

df_raw <- df_raw %>%
  mutate(channel = ifelse(channel == 'channel_2', NA, channel))
head(df_raw, n = 2)
```

```
## # A tibble: 2 x 4
##   customer_id date      channel conversion
##   <chr>      <date>    <chr>      <dbl>
## 1 id1       2016-01-02 channel_7      0
## 2 id1       2016-01-09 channel_4      0
```

15.1.2.2.1 1. Customers will be at different stage of purchase journey after each conversion. First-time buyer's journey will look different from n-times buyer's (e.g., he will not start at website)

You can create your own code to split data into customers in different stages.

```
##### splitting paths #####
df_paths <- df_raw %>%
  group_by(customer_id) %>%
  mutate(path_no = ifelse(is.na(lag(cumsum(conversion))), 0, lag(cumsum(conversion)) + 1)
  ungroup()
head(df_paths)
```

```
## # A tibble: 6 x 5
##   customer_id date      channel conversion path_no
##   <chr>      <date>    <chr>      <dbl>    <dbl>
## 1 id1       2016-01-02 channel_7      0        1
## 2 id1       2016-01-09 channel_4      0        1
## 3 id1       2016-01-18 channel_5      1        1
## 4 id1       2016-01-20 channel_4      1        2
## 5 id100     2016-01-01 channel_0      0        1
## 6 id100     2016-01-01 channel_0      0        1
```

attribution path for first-time buyers:

```
df_paths_1 <- df_paths %>%
  filter(path_no == 1) %>%
  select(-path_no)
```

15.1.2.2.2 2. Handle missing data We might have missing data on the channel or do not want to attribute a path (e.g., Direct Channel). We can either

- Remove NA/Channel
- Use the previous channel in its place.

In the first-order Markov chains, the results are unchanged because duplicated channels don't affect the calculation.

```
##### replace some channels #####
df_path_1_clean <- df_paths_1 %>%
  # removing NAs
  filter(!is.na(channel)) %>%
```

```

# adding order of channels in the path
group_by(customer_id) %>%
mutate(ord = c(1:n()),
       is_non_direct = ifelse(channel == 'channel_6', 0, 1),
       is_non_direct_cum = cumsum(is_non_direct)) %>%

# removing Direct (channel_6) when it is the first in the path
filter(is_non_direct_cum != 0) %>%

# replacing Direct (channel_6) with the previous touch point
mutate(channel = ifelse(channel == 'channel_6', channel[which(channel != 'chan

ungroup() %>%
select(-ord, -is_non_direct, -is_non_direct_cum)

```

15.1.2.2.3 3. one vs. multi-channel paths We need to calculate the weighted importance for each channel because the sum of the Removal Effects doesn't equal to 1. In case we have a path with a unique channel, the Removal Effect and importance of this channel for that exact path is 1. However, weighting with other multi-channel paths will decrease the importance of one-channel occurrences. That means that, in case we have a channel that occurs in one-channel paths, usually it will be underestimated if attributed with multi-channel paths.

There is also a pretty straight logic behind splitting – for one-channel paths, we definitely know the channel that brought a conversion and we don't need to distribute that value into other channels.

To account for one-channel path:

1. Split data for paths with one or more unique channels
2. Calculate total conversions for one-channel paths and compute the Markov model for multi-channel paths
3. Summarize results for each channel.

```

##### one- and multi-channel paths #####
df_path_1_clean <- df_path_1_clean %>%
  group_by(customer_id) %>%
  mutate(uniq_channel_tag = ifelse(length(unique(channel)) == 1, TRUE, FALSE)) %>%
  ungroup()

df_path_1_clean_uniq <- df_path_1_clean %>%
  filter(uniq_channel_tag == TRUE) %>%
  select(-uniq_channel_tag)

df_path_1_clean_multi <- df_path_1_clean %>%
  filter(uniq_channel_tag == FALSE) %>%

```



```

      select(-uniq_channel_tag)

### experiment ###
# attribution model for all paths
df_all_paths <- df_path_1_clean %>%
  group_by(customer_id) %>%
  summarise(path = paste(channel, collapse = ' > '),
            conversion = sum(conversion)) %>%
  ungroup() %>%
  filter(conversion == 1)

mod_attrib <- markov_model(df_all_paths,
                          var_path = 'path',
                          var_conv = 'conversion',
                          out_more = TRUE)

##
## Number of simulations: 100000 - Convergence reached: 1.28% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps (19) is reached
mod_attrib$removal_effects

##   channel_name removal_effects
## 1   channel_7      0.2812250
## 2   channel_4      0.4284428
## 3   channel_5      0.6056845
## 4   channel_0      0.5367294
## 5   channel_1      0.3820056
## 6   channel_3      0.2535028

mod_attrib$result

##   channel_name total_conversions
## 1   channel_7      192.8653
## 2   channel_4      293.8279
## 3   channel_5      415.3811
## 4   channel_0      368.0913
## 5   channel_1      261.9811
## 6   channel_3      173.8533

d_all <- data.frame(mod_attrib$result)

# attribution model for splitted multi and unique channel paths
df_multi_paths <- df_path_1_clean_multi %>%
  group_by(customer_id) %>%
  summarise(path = paste(channel, collapse = ' > '),

```

```

        conversion = sum(conversion)) %>%
    ungroup() %>%
    filter(conversion == 1)

mod_attrib_alt <- markov_model(df_multi_paths,
                              var_path = 'path',
                              var_conv = 'conversion',
                              out_more = TRUE)

##
## Number of simulations: 100000 - Convergence reached: 1.21% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
mod_attrib_alt$removal_effects

##   channel_name removal_effects
## 1   channel_7         0.3265696
## 2   channel_4         0.4844802
## 3   channel_5         0.6526369
## 4   channel_0         0.5814164
## 5   channel_1         0.4343546
## 6   channel_3         0.2898041

mod_attrib_alt$result

##   channel_name total_conversions
## 1   channel_7         150.9460
## 2   channel_4         223.9350
## 3   channel_5         301.6599
## 4   channel_0         268.7406
## 5   channel_1         200.7661
## 6   channel_3         133.9524

# adding unique paths
df_uniq_paths <- df_path_1_clean_uniq %>%
  filter(conversion == 1) %>%
  group_by(channel) %>%
  summarise(conversions = sum(conversion)) %>%
  ungroup()

d_multi <- data.frame(mod_attrib_alt$result)

d_split <- full_join(d_multi, df_uniq_paths, by = c('channel_name' = 'channel')) %>%
  mutate(result = total_conversions + conversions)

sum(d_all$total_conversions)

```

```
## [1] 1706
```

```
sum(d_split$result)
```

```
## [1] 1706
```

15.1.2.2.4 4. Higher Order Markov Chains Since the transition matrix stays the same in the first order Markov, having duplicates will not affect the result. But starting from the second order order Markov, you will have different results when skipping duplicates. In order to check the effect of skipping duplicates in the first-order Markov chain, we will use my script for “manual” calculation because the package skips duplicates automatically.

Higher order of Markov chains and consequent duplicated channels in the path

```
# computing transition matrix - 'manual' way
df_multi_paths_m <- df_multi_paths %>%
  mutate(path = paste0('(start) > ', path, ' > (conversion)'))
m <- max(str_count(df_multi_paths_m$path, '>')) + 1 # maximum path length

df_multi_paths_cols <- reshape2::colsplit(string = df_multi_paths_m$path, pattern = ' > ', names
colnames(df_multi_paths_cols) <- paste0('ord_', c(1:m))
df_multi_paths_cols[df_multi_paths_cols == ''] <- NA

df_res <- vector('list', ncol(df_multi_paths_cols) - 1)

for (i in c(1:(ncol(df_multi_paths_cols) - 1))) {

  df_cache <- df_multi_paths_cols %>%
    select(num_range("ord_", c(i, i+1))) %>%
    na.omit() %>%
    group_by(.dots = c(paste0("ord_", c(i, i+1)))) %>%
    summarise(n = n()) %>%
    ungroup()

  colnames(df_cache)[c(1, 2)] <- c('channel_from', 'channel_to')
  df_res[[i]] <- df_cache
}
```

```
## Warning: `group_by()` was deprecated in dplyr 0.7.0.
```

```
## Please use `group_by()` instead.
```

```
## See vignette('programming') for more help
```

```
## `summarise()` has grouped output by 'ord_1'. You can override using the `.groups` argument.
```

```
## `summarise()` has grouped output by 'ord_2'. You can override using the `.groups` argument.
```

```
## `summarise()` has grouped output by 'ord_3'. You can override using the `.groups` argument.
```

```
## `summarise()` has grouped output by 'ord_4'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_5'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_6'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_7'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_8'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_9'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_10'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_11'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_12'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_13'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_14'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_15'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_16'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_17'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_18'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_19'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_20'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_21'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'ord_22'. You can override using the `.groups` argument
```

```
df_res <- do.call('rbind', df_res)

df_res_tot <- df_res %>%
  group_by(channel_from, channel_to) %>%
  summarise(n = sum(n)) %>%
  ungroup() %>%
  group_by(channel_from) %>%
  mutate(tot_n = sum(n),
         perc = n / tot_n) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'channel_from'. You can override using the `.groups` argument
```

```
df_dummy <- data.frame(channel_from = c('(start)', '(conversion)', '(null)'),
                      channel_to = c('(start)', '(conversion)', '(null)'),
                      n = c(0, 0, 0),
                      tot_n = c(0, 0, 0),
                      perc = c(0, 1, 1))
```

```

df_res_tot <- rbind(df_res_tot, df_dummy)

# comparing transition matrices
trans_matrix_prob_m <- dcast(df_res_tot, channel_from ~ channel_to, value.var = 'perc', fun.aggregate = 'sum')
trans_matrix_prob <- data.frame(mod_attrib_alt$transition_matrix)
trans_matrix_prob <- dcast(trans_matrix_prob, channel_from ~ channel_to, value.var = 'transition_matrix')

# computing attribution - 'manual' way
channels_list <- df_path_1_clean_multi %>%
  filter(conversion == 1) %>%
  distinct(channel)
channels_list <- c(channels_list$channel)

df_res_ini <- df_res_tot %>% select(channel_from, channel_to)
df_attrib <- vector('list', length(channels_list))

for (i in c(1:length(channels_list))) {

  channel <- channels_list[i]

  df_res1 <- df_res %>%
    mutate(channel_from = ifelse(channel_from == channel, NA, channel_from),
           channel_to = ifelse(channel_to == channel, '(null)', channel_to)) %>%
    na.omit()

  df_res_tot1 <- df_res1 %>%
    group_by(channel_from, channel_to) %>%
    summarise(n = sum(n)) %>%
    ungroup() %>%

    group_by(channel_from) %>%
    mutate(tot_n = sum(n),
           perc = n / tot_n) %>%
    ungroup()

  df_res_tot1 <- rbind(df_res_tot1, df_dummy) # adding (start), (conversion) and (null) states

  df_res_tot1 <- left_join(df_res_ini, df_res_tot1, by = c('channel_from', 'channel_to'))
  df_res_tot1[is.na(df_res_tot1)] <- 0

  df_trans1 <- dcast(df_res_tot1, channel_from ~ channel_to, value.var = 'perc', fun.aggregate = 'sum')

  trans_matrix_1 <- df_trans1
  rownames(trans_matrix_1) <- trans_matrix_1$channel_from
  trans_matrix_1 <- as.matrix(trans_matrix_1[, -1])
}

```

```

inist_n1 <- dcast(df_res_tot1, channel_from ~ channel_to, value.var = 'n', fun
rownames(inist_n1) <- inist_n1$channel_from
inist_n1 <- as.matrix(inist_n1[, -1])
inist_n1[is.na(inist_n1)] <- 0
inist_n1 <- inist_n1['(start)', ]

res_num1 <- inist_n1 %*% (trans_matrix_1 %*% 100000)

df_cache <- data.frame(channel_name = channel,
                      conversions = as.numeric(res_num1[1, 1]))

df_attrib[[i]] <- df_cache
}

```

```

## `summarise()` has grouped output by 'channel_from'. You can override using the `.gr
## `summarise()` has grouped output by 'channel_from'. You can override using the `.gr
## `summarise()` has grouped output by 'channel_from'. You can override using the `.gr
## `summarise()` has grouped output by 'channel_from'. You can override using the `.gr
## `summarise()` has grouped output by 'channel_from'. You can override using the `.gr

```

```

df_attrib <- do.call('rbind', df_attrib)

# computing removal effect and results
tot_conv <- sum(df_multi_paths_m$conversion)

df_attrib <- df_attrib %>%
  mutate(tot_conversions = sum(df_multi_paths_m$conversion),
         impact = (tot_conversions - conversions) / tot_conversions,
         tot_impact = sum(impact),
         weighted_impact = impact / tot_impact,
         attrib_model_conversions = round(tot_conversions * weighted_impact)
  ) %>%
  select(channel_name, attrib_model_conversions)

```

Since with different transition matrices, the removal effects and attribution results stay the same, in practice we skip duplicates.

15.1.2.2.5 5. Non-conversion paths We incorporate null paths in this analysis.

```

##### Generic Probabilistic Model #####
df_all_paths_compl <- df_path_1_clean %>%
  group_by(customer_id) %>%
  summarise(path = paste(channel, collapse = ' > '),
            conversion = sum(conversion)) %>%

```

```

    ungroup() %>%
    mutate(null_conversion = ifelse(conversion == 1, 0, 1))

mod_attrib_complete <- markov_model(
  df_all_paths_compl,
  var_path = 'path',
  var_conv = 'conversion',
  var_null = 'null_conversion',
  out_more = TRUE
)

##
## Number of simulations: 100000 - Convergence reached: 4.05% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps (27) is reached

trans_matrix_prob <- mod_attrib_complete$transition_matrix %>%
  dmap_at(c(1, 2), as.character)

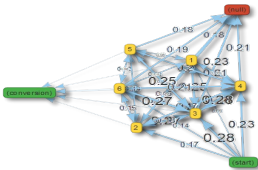
##### viz #####
edges <-
  data.frame(
    from = trans_matrix_prob$channel_from,
    to = trans_matrix_prob$channel_to,
    label = round(trans_matrix_prob$transition_probability, 2),
    font.size = trans_matrix_prob$transition_probability * 100,
    width = trans_matrix_prob$transition_probability * 15,
    shadow = TRUE,
    arrows = "to",
    color = list(color = "#95cbee", highlight = "red")
  )

nodes <- data_frame(id = c( c(trans_matrix_prob$channel_from), c(trans_matrix_prob$channel_to) ))
  distinct(id) %>%
  arrange(id) %>%
  mutate(
    label = id,
    color = ifelse(
      label %in% c('(start)', '(conversion)'),
      '#4ab04a',
      ifelse(label == '(null)', '#ce472e', '#ffd73e')
    ),
    shadow = TRUE,
    shape = "box"
  )

```

```
## Warning: `data_frame()` was deprecated in tibble 1.1.0.  
## Please use `tibble()` instead.  
visNetwork(nodes,  
            edges,  
            height = "2000px",  
            width = "100%",  
            main = "Generic Probabilistic model's Transition Matrix") %>%  
visIgraphLayout(randomSeed = 123) %>%  
visNodes(size = 5) %>%  
visOptions(highlightNearest = TRUE)
```


Generic Probabilistic model's Transition Matrix



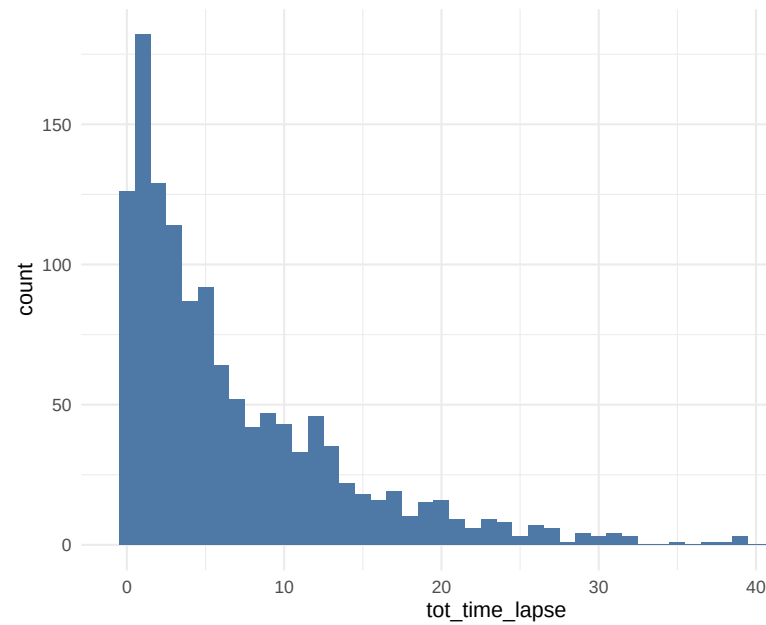
```
##### modeling states and conversions #####
# transition matrix preprocessing
trans_matrix_complete <- mod_attrib_complete$transition_matrix
trans_matrix_complete <- rbind(trans_matrix_complete, df_dummy %>%
                                mutate(transition_probability = perc) %>%
                                select(channel_from, channel_to, transition_prob))
trans_matrix_complete$channel_to <- factor(trans_matrix_complete$channel_to, levels = c(
trans_matrix_complete <- dcast(trans_matrix_complete, channel_from ~ channel_to, value = sum)
trans_matrix_complete[is.na(trans_matrix_complete)] <- 0
rownames(trans_matrix_complete) <- trans_matrix_complete$channel_from
trans_matrix_complete <- as.matrix(trans_matrix_complete[, -1])

# creating empty matrix for modeling
model_mtrx <- matrix(data = 0,
                     nrow = nrow(trans_matrix_complete), ncol = 1,
                     dimnames = list(c(rownames(trans_matrix_complete)), '(start)'))
# adding modeling number of visits
model_mtrx['channel_5', ] <- 1000

c(model_mtrx) %*% (trans_matrix_complete %*% 5) # after 5 steps
c(model_mtrx) %*% (trans_matrix_complete %*% 100000) # after 100000 steps
```

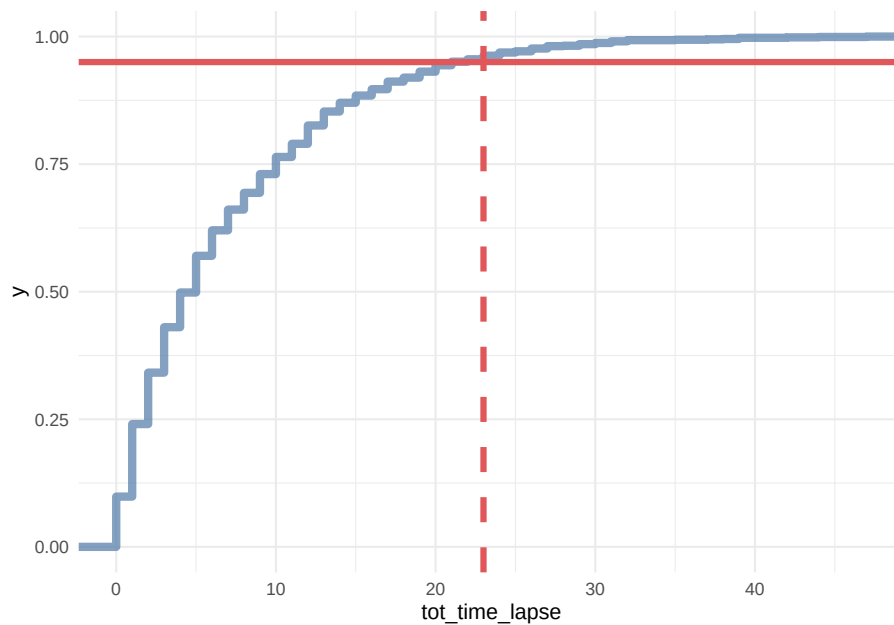
```
##### Customer journey duration #####
# computing time lapses from the first contact to conversion/last contact
df_multi_paths_tl <- df_path_1_clean_multi %>%
  group_by(customer_id) %>%
  summarise(path = paste(channel, collapse = ' > '),
            first_touch_date = min(date),
            last_touch_date = max(date),
            tot_time_lapse = round(as.numeric(last_touch_date - first_touch_date)),
            conversion = sum(conversion)) %>%
  ungroup()

# distribution plot
ggplot(df_multi_paths_tl %>% filter(conversion == 1), aes(x = tot_time_lapse)) +
  theme_minimal() +
  geom_histogram(fill = '#4e79a7', binwidth = 1)
```



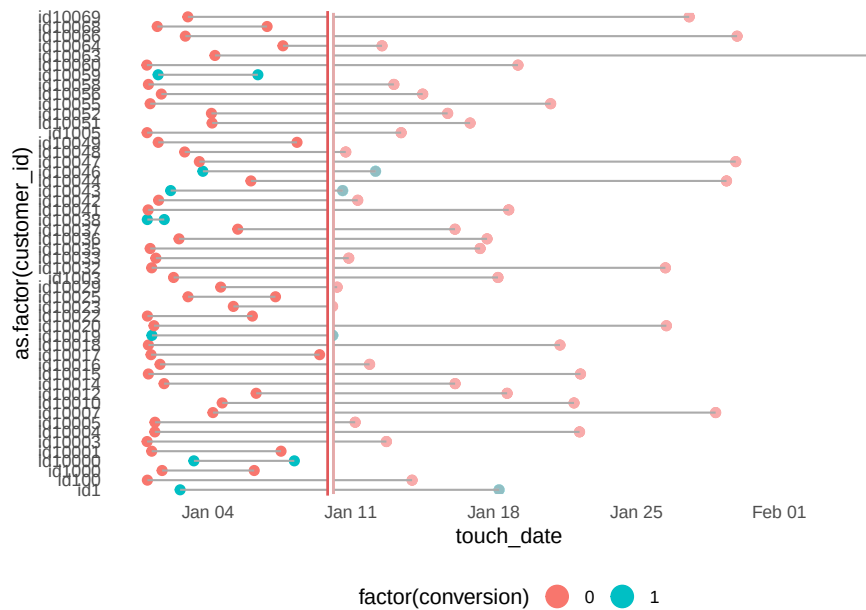
15.1.2.2.6 6. Customer Journey Duration

```
# cumulative distribution plot
ggplot(df_multi_paths_tl %>% filter(conversion == 1), aes(x = tot_time_lapse)) +
  theme_minimal() +
  stat_ecdf(geom = 'step', color = '#4e79a7', size = 2, alpha = 0.7) +
  geom_hline(yintercept = 0.95, color = '#e15759', size = 1.5) +
  geom_vline(xintercept = 23, color = '#e15759', size = 1.5, linetype = 2)
```



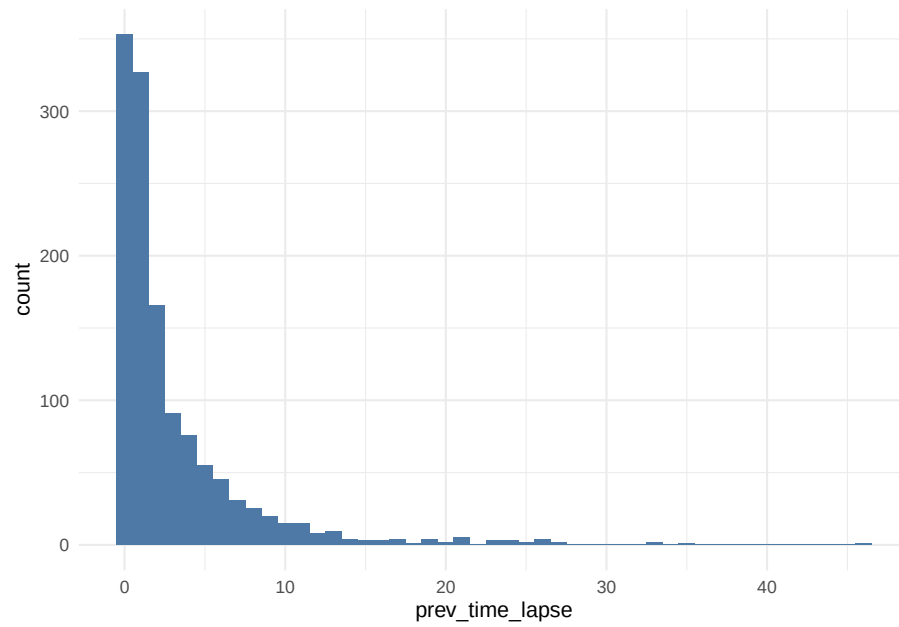
```
### for generic probabilistic model ###
df_multi_paths_tl_1 <- reshape2::melt(df_multi_paths_tl[c(1:50), ], %>% select(customer_id,
  id.vars = c('customer_id', 'conversion'),
  value.name = 'touch_date') %>%
  arrange(customer_id)
rep_date <- as.Date('2016-01-10', format = '%Y-%m-%d')

ggplot(df_multi_paths_tl_1, aes(x = as.factor(customer_id), y = touch_date, color = factor(customer_id))) +
  theme_minimal() +
  coord_flip() +
  geom_point(size = 2) +
  geom_line(size = 0.5, color = 'darkgrey') +
  geom_hline(yintercept = as.numeric(rep_date), color = '#e15759', size = 2) +
  geom_rect(xmin = -Inf, xmax = Inf, ymin = as.numeric(rep_date), ymax = Inf, alpha = 0.1) +
  theme(legend.position = 'bottom',
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    axis.ticks.x = element_blank(),
    axis.ticks.y = element_blank()) +
  guides(colour = guide_legend(override.aes = list(size = 5)))
```

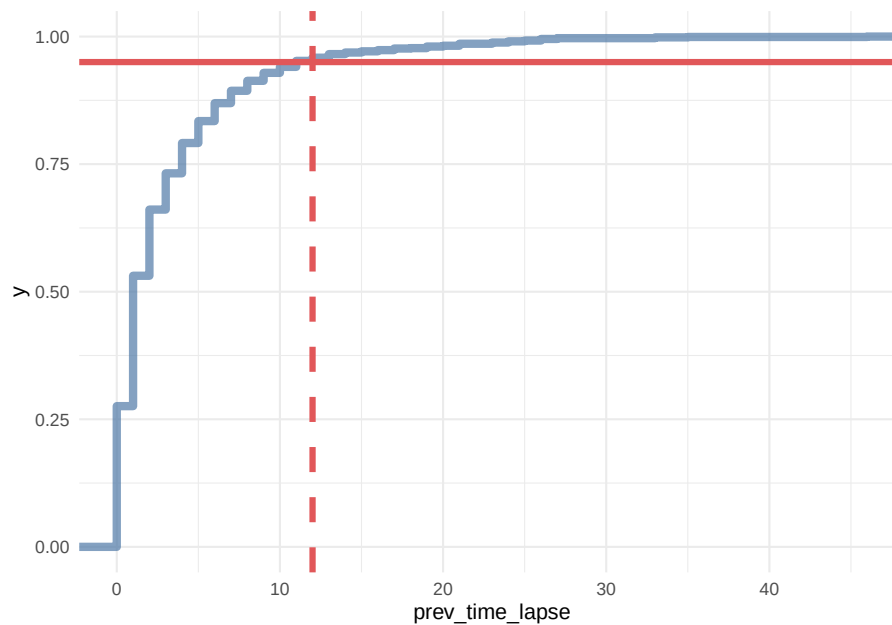


```
df_multi_paths_tl_2 <- df_path_1_clean_multi %>%
  group_by(customer_id) %>%
  mutate(prev_touch_date = lag(date)) %>%
  ungroup() %>%
  filter(conversion == 1) %>%
  mutate(prev_time_lapse = round(as.numeric(date - prev_touch_date)))

# distribution
ggplot(df_multi_paths_tl_2, aes(x = prev_time_lapse)) +
  theme_minimal() +
  geom_histogram(fill = '#4e79a7', binwidth = 1)
```



```
# cumulative distribution
ggplot(df_multi_paths_tl_2, aes(x = prev_time_lapse)) +
  theme_minimal() +
  stat_ecdf(geom = 'step', color = '#4e79a7', size = 2, alpha = 0.7) +
  geom_hline(yintercept = 0.95, color = '#e15759', size = 1.5) +
  geom_vline(xintercept = 12, color = '#e15759', size = 1.5, linetype = 2)
```



In conclusion, we say that if a customer made contact with a marketing channel the first time for more than 23 days and/or hasn't made contact with a marketing channel for the last 12 days, then it is a fruitless path.

```
# extracting data for generic model
df_multi_paths_tl_3 <- df_path_1_clean_multi %>%
  group_by(customer_id) %>%
  mutate(prev_time_lapse = round(as.numeric(date - lag(date)))) %>%
  summarise(path = paste(channel, collapse = ' > '),
            tot_time_lapse = round(as.numeric(max(date) - min(date))),
            prev_touch_tl = prev_time_lapse[which(max(date) == date)],
            conversion = sum(conversion)) %>%
  ungroup() %>%
  mutate(is_fruitless = ifelse(conversion == 0 & tot_time_lapse > 20 & prev_touch_tl > 10,
                               TRUE, FALSE))
  filter(conversion == 1 | is_fruitless == FALSE)
```

15.1.2.2.7 7. Channel Comparisons We can use `markov_model` with `var_value` to compare the gross margin among channels.

15.1.2.3 Example 3

This example is from Bounteous

```
# Install these libraries (only do this once)
# install.packages("ChannelAttribution")
# install.packages("reshape")
```

```
# install.packages("ggplot2")

# Load these libraries (every time you start RStudio)
library(ChannelAttribution)
library(reshape)
library(ggplot2)

# This loads the demo data. You can load your own data by importing a dataset or reading
data(PathData)
```

- Path Variable – The steps a user takes across sessions to comprise the sequences.
- Conversion Variable – How many times a user converted.
- Value Variable – The monetary value of each marketing channel.
- Null Variable – How many times a user exited.

Build the simple heuristic models (First Click / first_touch, Last Click / last_touch, and Linear Attribution / linear_touch):

```
H <- heuristic_models(Data, 'path', 'total_conversions', var_value='total_conversion_value')
```

Markov model

```
M <- markov_model(Data, 'path', 'total_conversions', var_value='total_conversion_value')
```

```
##
## Number of simulations: 100000 - Convergence reached: 1.46% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
# Merges the two data frames on the "channel_name" column.
R <- merge(H, M, by='channel_name')

# Selects only relevant columns
R1 <- R[, (colnames(R)%in%c('channel_name', 'first_touch_conversions', 'last_touch_conversions'))]

# Renames the columns
colnames(R1) <- c('channel_name', 'first_touch', 'last_touch', 'linear_touch', 'markov')

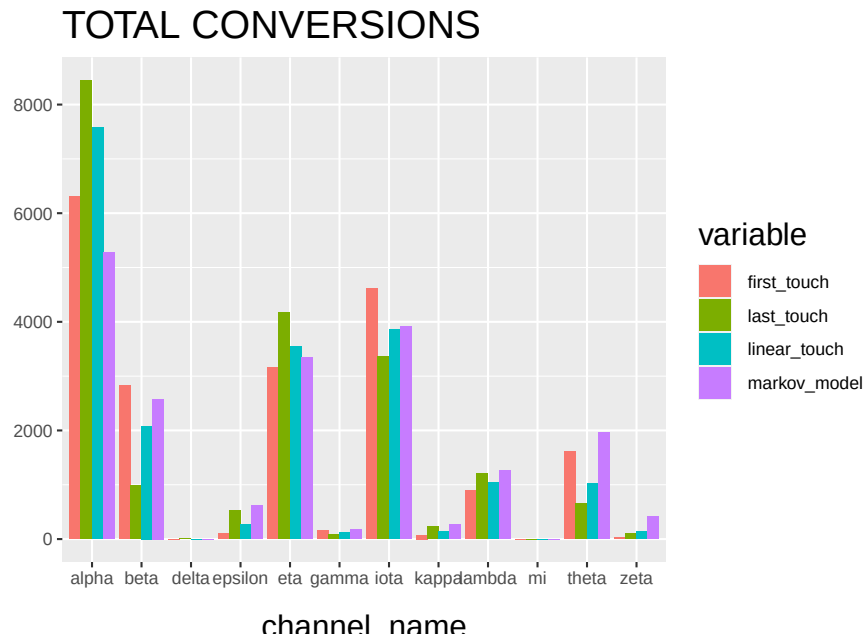
# Transforms the dataset into a data frame that ggplot2 can use to graph the outcomes
R1 <- melt(R1, id='channel_name')
```

Plot the total conversions

```
ggplot(R1, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL CONVERSIONS') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
```



```
theme(title = element_text(size = 16)) +
theme(plot.title=element_text(size = 20)) +
ylab("")
```



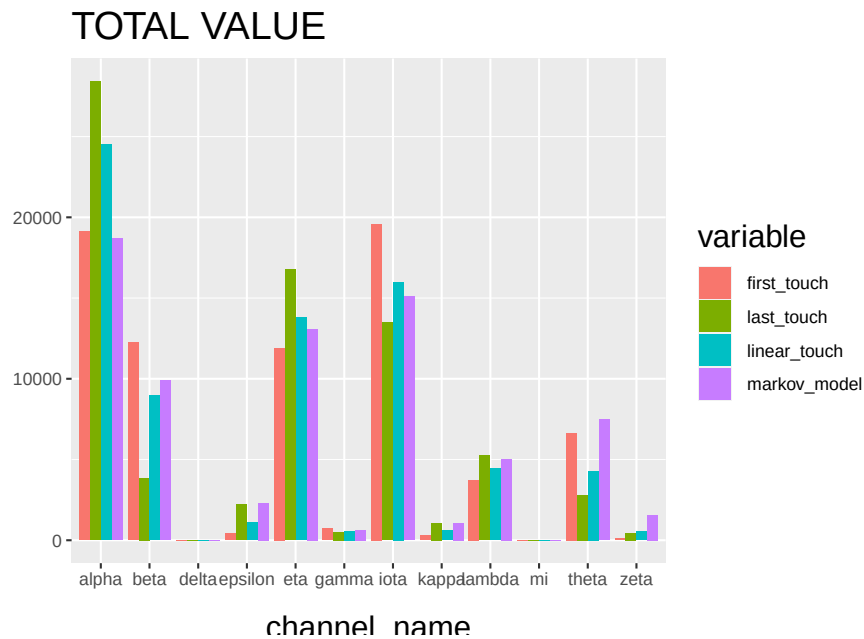
The “Total Conversions” bar chart shows you how many conversions were attributed to each channel (i.e. alpha, beta, etc.) for each method (i.e. first_touch, last_touch, etc.).

```
R2 <- R[, (colnames(R)%in%c('channel_name', 'first_touch_value', 'last_touch_value', 'linear_touch_value'))]

colnames(R2) <- c('channel_name', 'first_touch', 'last_touch', 'linear_touch', 'markov_model')

R2 <- melt(R2, id='channel_name')

ggplot(R2, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL VALUE') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
  theme(title = element_text(size = 16)) +
  theme(plot.title=element_text(size = 20)) +
  ylab("")
```



The “Total Conversion Value” bar chart shows you monetary value that can be attributed to each channel from a conversion.

15.2 Sales Funnel

15.2.1 Example 1

This example is based on Sergey Bryl

$$Awareness \rightarrow Interest \rightarrow Desire \rightarrow Action$$

Step in the funnel:

- 0 step (necessary condition) – customer visits a site for the first time
- 1st step (awareness) – visits two site’s pages
- 2nd step (interest) – reviews a product page
- 3rd step (desire) – adds a product to the shopping cart
- 4th step (action) – completes purchase

Simulate data

```
library(tidyverse)
library(purrrlyr)
library(reshape2)

##### simulating the "real" data #####
```

```

set.seed(454)
df_raw <-
  data.frame(
    customer_id = paste0('id', sample(c(1:5000), replace = TRUE)),
    date = as.POSIXct(
      rbeta(10000, 0.7, 10) * 10000000,
      origin = '2017-01-01',
      tz = "UTC"
    ),
    channel = paste0('channel_', sample(
      c(0:7),
      10000,
      replace = TRUE,
      prob = c(0.2, 0.12, 0.03, 0.07, 0.15, 0.25, 0.1, 0.08)
    )),
    site_visit = 1
  ) %>%

  mutate(
    two_pages_visit = sample(c(0, 1), 10000, replace = TRUE, prob = c(0.8, 0.2)),
    product_page_visit = ifelse(
      two_pages_visit == 1,
      sample(
        c(0, 1),
        length(two_pages_visit[which(two_pages_visit == 1)]),
        replace = TRUE,
        prob = c(0.75, 0.25)
      ),
      0
    ),
    add_to_cart = ifelse(
      product_page_visit == 1,
      sample(
        c(0, 1),
        length(product_page_visit[which(product_page_visit == 1)]),
        replace = TRUE,
        prob = c(0.1, 0.9)
      ),
      0
    ),
    purchase = ifelse(add_to_cart == 1,
                      sample(
                        c(0, 1),
                        length(add_to_cart[which(add_to_cart == 1)]),
                        replace = TRUE,

```

```

        prob = c(0.02, 0.98)
      ),
    0)
  ) %>%
  dmap_at(c('customer_id', 'channel'), as.character) %>%
  arrange(date) %>%
  mutate(session_id = row_number()) %>%
  arrange(customer_id, session_id)
df_raw <-
  reshape2::melt(
    df_raw,
    id.vars = c('customer_id', 'date', 'channel', 'session_id'),
    value.name = "trigger",
    variable.name = 'event'
  ) %>%
  filter(trigger == 1) %>%
  select(-trigger) %>%
  arrange(customer_id, date)

```

Preprocessing

```

### removing not first events ###
df_customers <- df_raw %>%
  group_by(customer_id, event) %>%
  filter(date == min(date)) %>%
  ungroup()

```

Assumption: all customers are first-time buyers. Hence, every next purchase as an event will be removed with the above code.

Calculate channel probability

```

### Sales Funnel probabilities ###
sf_probs <- df_customers %>%

  group_by(event) %>%
  summarise(customers_on_step = n()) %>%
  ungroup() %>%

  mutate(
    sf_probs = round(customers_on_step / customers_on_step[event == 'site_visit'],
    sf_probs_step = round(customers_on_step / lag(customers_on_step), 3),
    sf_probs_step = ifelse(is.na(sf_probs_step) == TRUE, 1, sf_probs_step),
    sf_importance = 1 - sf_probs_step,
    sf_importance_weighted = sf_importance / sum(sf_importance)
  )

```

Visualization

```
### Sales Funnel visualization ###
df_customers_plot <- df_customers %>%

  group_by(event) %>%
  arrange(channel) %>%
  mutate(pl = row_number()) %>%
  ungroup() %>%

  mutate(
    pl_new = case_when(
      event == 'two_pages_visit' ~ round((max(pl[event == 'site_visit']) - max(pl[event ==
      event == 'product_page_visit' ~ round((max(pl[event == 'site_visit']) - max(pl[event
      event == 'add_to_cart' ~ round((max(pl[event == 'site_visit']) - max(pl[event == 'add
      event == 'purchase' ~ round((max(pl[event == 'site_visit']) - max(pl[event == 'purcha
      TRUE ~ 0
    ),
    pl = pl + pl_new
  )

df_customers_plot$event <-
  factor(
    df_customers_plot$event,
    levels = c(
      'purchase',
      'add_to_cart',
      'product_page_visit',
      'two_pages_visit',
      'site_visit'
    )
  )

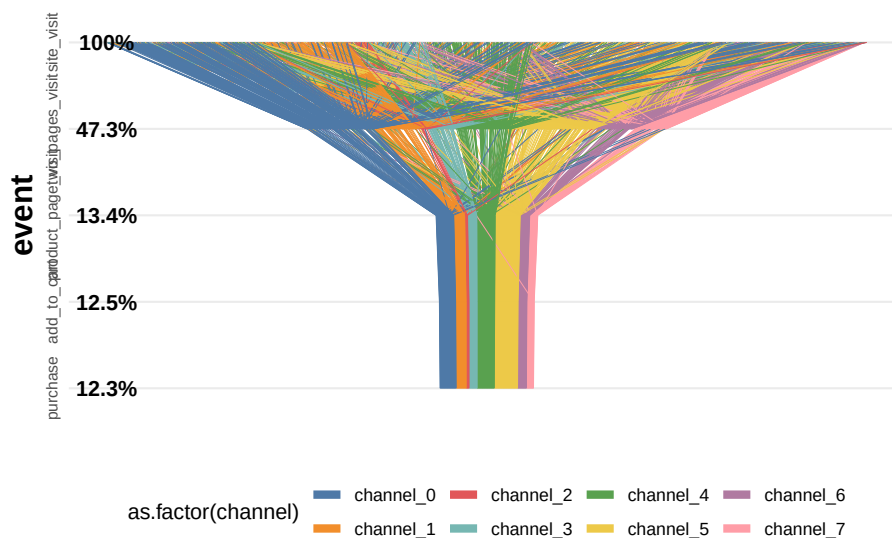
# color palette
cols <- c(
  '#4e79a7',
  '#f28e2b',
  '#e15759',
  '#76b7b2',
  '#59a14f',
  '#edc948',
  '#b07aa1',
  '#ff9da7',
  '#9c755f',
  '#bab0ac'
)
```

```

ggplot(df_customers_plot, aes(x = event, y = pl)) +
  theme_minimal() +
  scale_colour_manual(values = cols) +
  coord_flip() +
  geom_line(aes(group = customer_id, color = as.factor(channel)), size = 0.05) +
  geom_text(
    data = sf_probs,
    aes(
      x = event,
      y = 1,
      label = paste0(sf_probs * 100, '%')
    ),
    size = 4,
    fontface = 'bold'
  ) +
  guides(color = guide_legend(override.aes = list(size = 2))) +
  theme(
    legend.position = 'bottom',
    legend.direction = "horizontal",
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    plot.title = element_text(
      size = 20,
      face = "bold",
      vjust = 2,
      color = 'black',
      lineheight = 0.8
    ),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.text.y = element_text(
      size = 8,
      angle = 90,
      hjust = 0.5,
      vjust = 0.5,
      face = "plain"
    )
  ) +
  ggtitle("Sales Funnel visualization - all customers journeys")

```

Sales Funnel visualization – all customers jou



Calculate attribution

```
### computing attribution ###
df_attrib <- df_customers %>%
  # removing customers without purchase
  group_by(customer_id) %>%
  filter(any(as.character(event) == 'purchase')) %>%
  ungroup() %>%

  # joining step's importances
  left_join(., sf_probs %>% select(event, sf_importance_weighted), by = 'event') %>%

  group_by(channel) %>%
  summarise(tot_attribution = sum(sf_importance_weighted)) %>%
  ungroup()
```

15.2.2 Example 2

Code from Sergey Bryl

```
library(dplyr)
library(ggplot2)
library(reshape2)

# creating a data samples
# content
```

```

df.content <- data.frame(
  content = c(
    'main',
    'ad landing',
    'product 1',
    'product 2',
    'product 3',
    'product 4',
    'shopping cart',
    'thank you page'
  ),
  step = c(
    'awareness',
    'awareness',
    'interest',
    'interest',
    'interest',
    'interest',
    'desire',
    'action'
  ),
  number = c(150000, 80000,
             80000, 40000, 35000, 25000,
             130000,
             120000)
)
# customers
df.customers <- data.frame(
  content = c('new', 'engaged', 'loyal'),
  step = c('new', 'engaged', 'loyal'),
  number = c(25000, 40000, 55000)
)
# combining two data sets
df.all <- rbind(df.content, df.customers)

# calculating dummies, max and min values of X for plotting
df.all <- df.all %>%
  group_by(step) %>%
  mutate(totnum = sum(number)) %>%
  ungroup() %>%
  mutate(dum = (max(totnum) - totnum) / 2,
         maxx = totnum + dum,
         minx = dum)

# data frame for plotting funnel lines

```



```

df.lines <- df.all %>%
  distinct(step, maxx, minx)

# data frame with dummies
df.dum <- df.all %>%
  distinct(step, dum) %>%
  mutate(content = 'dummy',
         number = dum) %>%
  select(content, step, number)

# data frame with rates
conv <- df.all$totnum[df.all$step == 'action']

df.rates <- df.all %>%
  distinct(step, totnum) %>%
  mutate(
    prevnum = lag(totnum),
    rate = ifelse(
      step == 'new' | step == 'engaged' | step == 'loyal',
      round(totnum / conv, 3),
      round(totnum / prevnum, 3)
    )
  ) %>%
  select(step, rate)
df.rates <- na.omit(df.rates)

# creting final data frame
df.all <- df.all %>%
  select(content, step, number)

df.all <- rbind(df.all, df.dum)

# defining order of steps
df.all$step <-
  factor(
    df.all$step,
    levels = c(
      'loyal',
      'engaged',
      'new',
      'action',
      'desire',
      'interest',
      'awareness'
    )
  )

```

```

    )
df.all <- df.all %>%
  arrange(desc(step))
list1 <- df.all %>% distinct(content) %>%
  filter(content != 'dummy')
df.all$content <-
  factor(df.all$content, levels = c(as.character(list1$content), 'dummy'))

# calculating position of labels
df.all <- df.all %>%
  arrange(step, desc(content)) %>%
  group_by(step) %>%
  mutate(pos = cumsum(number) - 0.5 * number) %>%
  ungroup()

# creating custom palette with 'white' color for dummies
cols <- c(
  "#fec44f",
  "#fc9272",
  "#a1d99b",
  "#fee0d2",
  "#2ca25f",
  "#8856a7",
  "#43a2ca",
  "#fdbb84",
  "#e34a33",
  "#a6bddb",
  "#dd1c77",
  "#ffffff"
)

# plotting chart
ggplot() +
  theme_minimal() +
  coord_flip() +
  scale_fill_manual(values = cols) +
  geom_bar(
    data = df.all,
    aes(x = step, y = number, fill = content),
    stat = "identity",
    width = 1
  ) +
  geom_text(
    data = df.all[df.all$content != 'dummy',],
    aes(

```

```

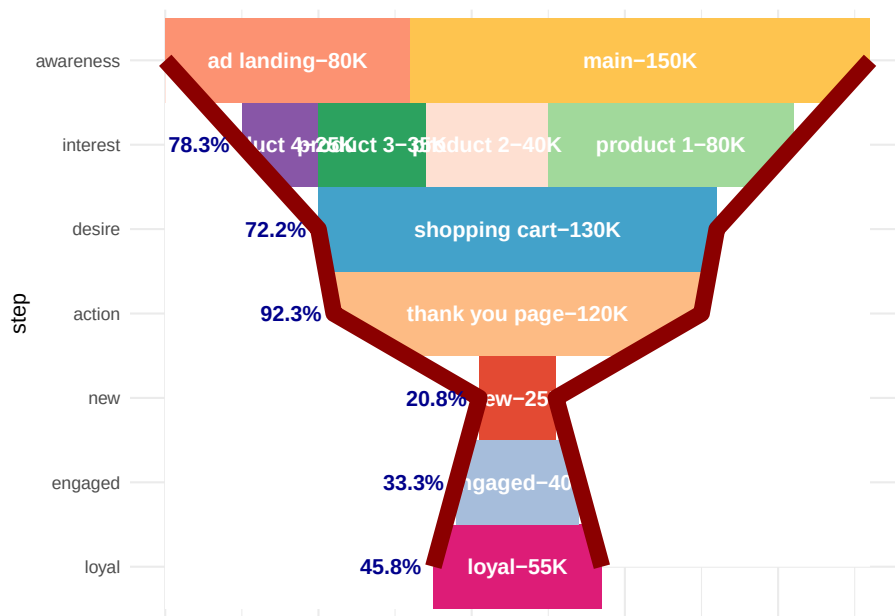
        x = step,
        y = pos,
        label = paste0(content, '-', number / 1000, 'K')
    ),
    size = 4,
    color = 'white',
    fontface = "bold"
) +
geom_ribbon(data = df.lines,
          aes(
            x = step,
            ymax = max(maxx),
            ymin = maxx,
            group = 1
          ),
          fill = 'white') +
geom_line(
  data = df.lines,
  aes(x = step, y = maxx, group = 1),
  color = 'darkred',
  size = 4
) +
geom_ribbon(data = df.lines,
          aes(
            x = step,
            ymax = minx,
            ymin = min(minx),
            group = 1
          ),
          fill = 'white') +
geom_line(
  data = df.lines,
  aes(x = step, y = minx, group = 1),
  color = 'darkred',
  size = 4
) +
geom_text(
  data = df.rates,
  aes(
    x = step,
    y = (df.lines$minx[-1]),
    label = paste0(rate * 100, '%')
  ),
  hjust = 1.2,
  color = 'darkblue',

```

```

    fontface = "bold"
  ) +
  theme(
    legend.position = 'none',
    axis.ticks = element_blank(),
    axis.text.x = element_blank(),
    axis.title.x = element_blank()
  )

```



15.3 RFM

RFM is calculated as:

- A recency score is assigned to each customer based on date of most recent purchase.
- A frequency ranking is assigned based on frequency of purchases
- Monetary score is assigned based on the total revenue generated by the customer in the period under consideration for the analysis

```

library("rfm")
rfm_data_customer

```

```

## # A tibble: 39,999 x 5
##   customer_id revenue most_recent_visit number_of_orders recency_days
##         <dbl>   <dbl> <date>                <dbl>         <dbl>

```

```
## 1      22086      777 2006-05-14      9      232
## 2      2290      1555 2006-09-08     16      115
## 3      26377      336 2006-11-19      5       43
## 4      24650      1189 2006-10-29     12       64
## 5      12883      1229 2006-12-09     12       23
## 6      2119       929 2006-10-21     11       72
## 7      31283      1569 2006-09-11     17      112
## 8      33815       778 2006-08-12     11      142
## 9      15972       641 2006-11-19      9       43
## 10     27650       970 2006-08-23     10      131
## # ... with 39,989 more rows
```

```
# a unique customer id
# number of transaction/order
# total revenue from the customer
# number of days since the last visit
```

```
rfm_data_orders # to generate data_orders, use rfm_table_order()
```

```
## # A tibble: 4,906 x 3
##   customer_id      order_date revenue
##   <chr>          <date>     <dbl>
## 1 Mr. Brion Stark Sr. 2004-12-20      32
## 2 Ethyl Botsford    2005-05-02      36
## 3 Hosteen Jacobi     2004-03-06     116
## 4 Mr. Edw Frami      2006-03-15      99
## 5 Josef Lemke        2006-08-14      76
## 6 Julisa Halvorson   2005-05-28      56
## 7 Judyth Lueilwitz   2005-03-09     108
## 8 Mr. Mekhi Goyette   2005-09-23     183
## 9 Hansford Moen PhD   2005-09-07      30
## 10 Fount Flatley      2006-04-12      13
## # ... with 4,896 more rows
```

```
# unique customer id
# date of transaction
# and amount
# customer_id: name of the customer id column
# order_date: name of the transaction date column
# revenue: name of the transaction amount column
# analysis_date: date of analysis
# recency_bins: number of rankings for recency score (default is 5)
# frequency_bins: number of rankings for frequency score (default is 5)
# monetary_bins: number of rankings for monetary score (default is 5)
```

```

analysis_date <- lubridate::as_date('2007-01-01')
rfm_result <-
  rfm_table_customer(
    rfm_data_customer,
    customer_id,
    number_of_orders,
    recency_days,
    revenue,
    analysis_date
  )
rfm_result

## # A tibble: 39,999 x 8
##   customer_id recency_days transaction_count amount recency_score
##   <dbl>         <dbl>          <dbl>    <dbl>         <int>
## 1      22086         232             9      777             2
## 2       2290         115            16     1555             4
## 3     26377          43             5      336             5
## 4     24650          64            12     1189             5
## 5     12883          23            12     1229             5
## 6       2119          72            11      929             5
## 7     31283         112            17     1569             4
## 8     33815         142            11      778             3
## 9     15972          43             9      641             5
## 10    27650         131            10      970             3
## # ... with 39,989 more rows, and 3 more variables: frequency_score <int>,
## #   monetary_score <int>, rfm_score <dbl>

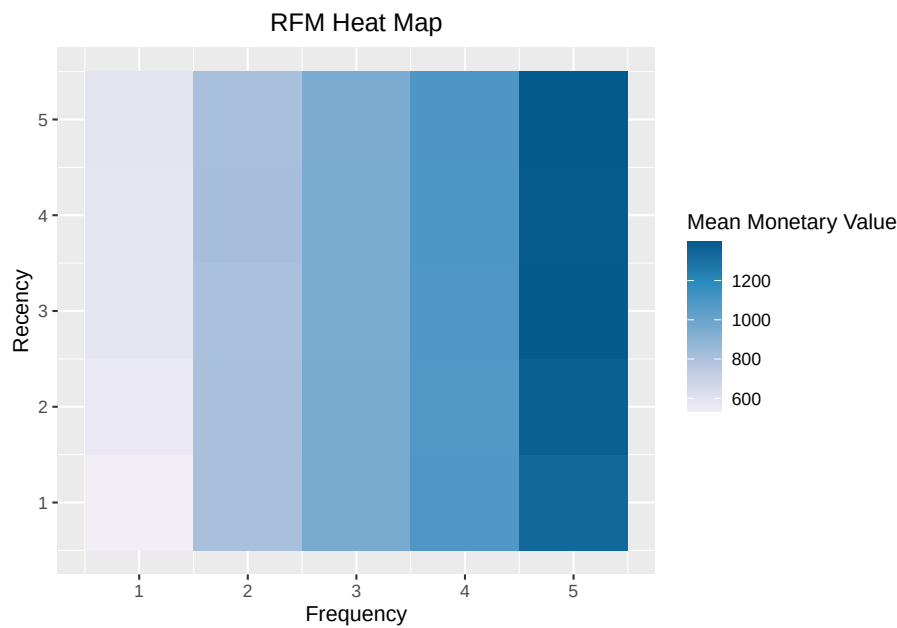
# customer_id: unique customer id
# date_most_recent: date of most recent visit
# recency_days: days since the most recent visit
# transaction_count: number of transactions of the customer
# amount: total revenue generated by the customer
# recency_score: recency score of the customer
# frequency_score: frequency score of the customer
# monetary_score: monetary score of the customer
# rfm_score: RFM score of the customer

```

15.3.1 Visualization

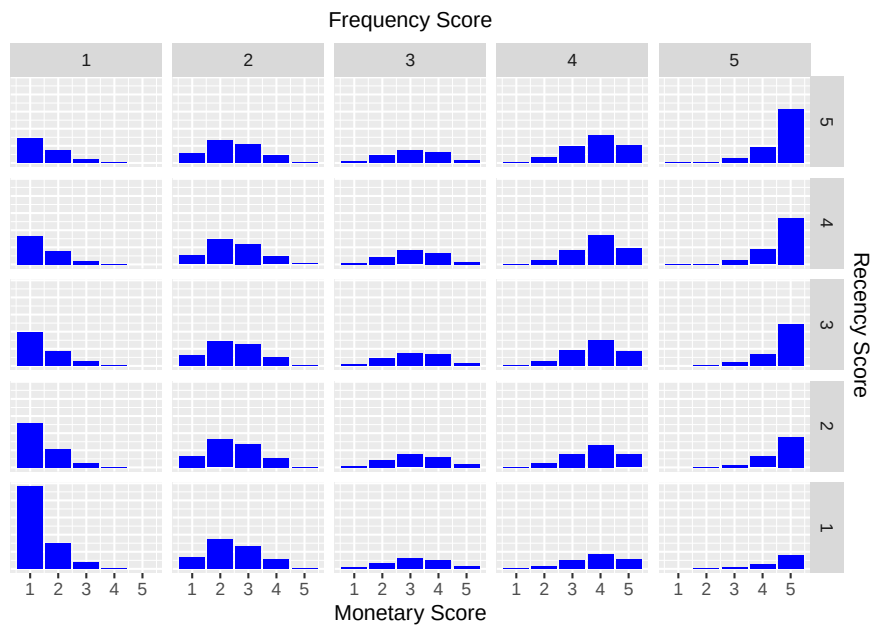
heat map shows the average monetary value for different categories of recency and frequency scores

```
rfm_heatmap(rfm_result)
```



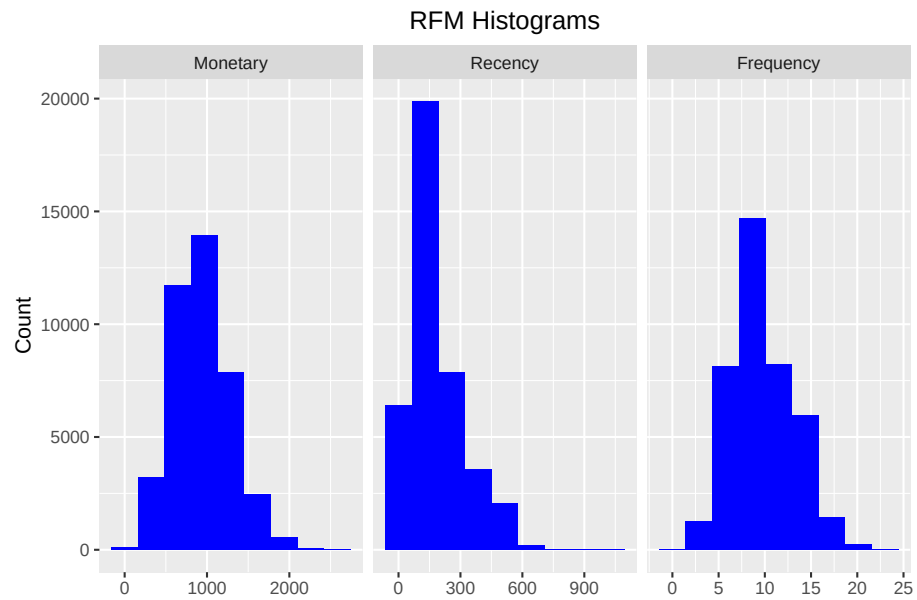
bar chart

```
rfm_bar_chart(rfm_result)
```



histogram

```
rfm_histograms(rfm_result)
```



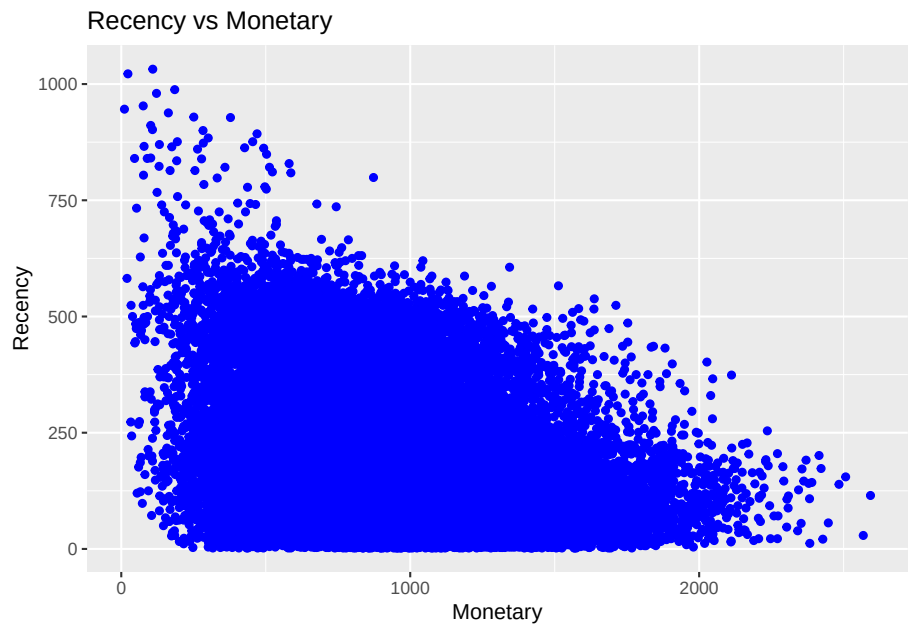
Customers by Orders

```
rfm_order_dist(rfm_result)
```

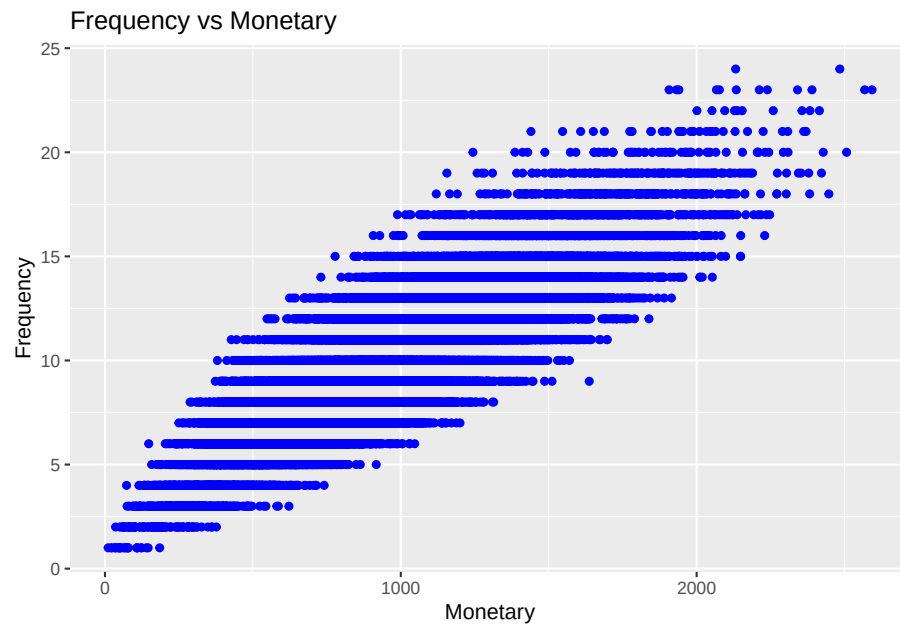



Scatter Plots

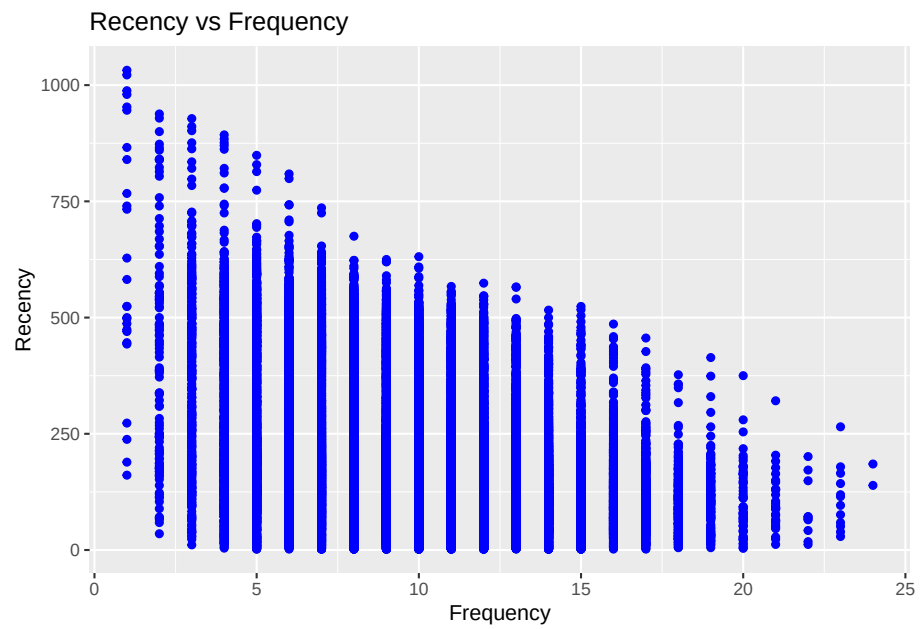
```
rfm_rm_plot(rfm_result)
```



```
rfm_fm_plot(rfm_result)
```



```
rfm_rf_plot(rfm_result)
```



15.3.2 RFMC

(C) clumpiness is defined as the degree of nonconformity to equal spacing (Zhang et al., 2015)

In finance, clumpiness can indicate high growth potential but large risk, Hence, it can be incorporated into firm acquisition decision. Originated from sports phenomenon - hot hand effect - where success leads to more success.

In statistics, clumpiness is the serial dependence or “non-constant propensity, specifically temporary elevations of propensity— i.e. periods during which one event is more likely to occur than the average level.” (Zhang et al., 2013)

Properties of clumpiness:

- Min (max) if events are equally spaced (close to one another)
- Continuity
- Convergence

15.4 Customer Segmentation

15.4.1 Example 1

Continue from the RFM

```
segment_names <-
  c(
    "Premium",
    "Loyal Customers",
    "Potential Loyalist",
    "New Customers",
    "Promising",
    "Need Attention",
    "About To Churn",
    "At Risk",
    "High Value Churners/Resurrection",
    "Low Value Churners"
  )

recency_lower <- c(4, 2, 3, 4, 3, 2, 2, 1, 1, 1)
recency_upper <- c(5, 5, 5, 5, 4, 3, 3, 2, 1, 2)
frequency_lower <- c(4, 3, 1, 1, 1, 2, 1, 2, 4, 1)
frequency_upper <- c(5, 5, 3, 1, 1, 3, 2, 5, 5, 2)
monetary_lower <- c(4, 3, 1, 1, 1, 2, 1, 2, 4, 1)
monetary_upper <- c(5, 5, 3, 1, 1, 3, 2, 5, 5, 2)

rfm_segments <-
  rfm_segment(
```

```

      rfm_result,
      segment_names,
      recency_lower,
      recency_upper,
      frequency_lower,
      frequency_upper,
      monetary_lower,
      monetary_upper
    )

head(rfm_segments, n = 5)

rfm_segments %>%
  count(rfm_segments$segment) %>%
  arrange(desc(n)) %>%
  rename(Count = n)

# median recency
rfm_plot_median_recency(rfm_segments)

# median frequency
rfm_plot_median_frequency(rfm_segments)

# Median Monetary Value
rfm_plot_median_monetary(rfm_segments)

```

15.4.2 Example 2

Example by Sergey

15.4.2.1 LifeCycle Grids

```

# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)

# creating data sample
set.seed(10)
data <- data.frame(
  orderId = sample(c(1:1000), 5000, replace = TRUE),
  product = sample(
    c('NULL', 'a', 'b', 'c'),
    5000,

```

```

        replace = TRUE,
        prob = c(0.15, 0.65, 0.3, 0.15)
      )
    )
order <- data.frame(orderId = c(1:1000),
                    clientId = sample(c(1:300), 1000, replace = TRUE))
gender <- data.frame(clientId = c(1:300),
                    gender = sample(
                      c('male', 'female'),
                      300,
                      replace = TRUE,
                      prob = c(0.40, 0.60)
                    ))
date <- data.frame(orderId = c(1:1000),
                  orderdate = sample((1:100), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, gender, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order, gender)

# reporting date
today <- as.Date('2012-04-11', format = '%Y-%m-%d')

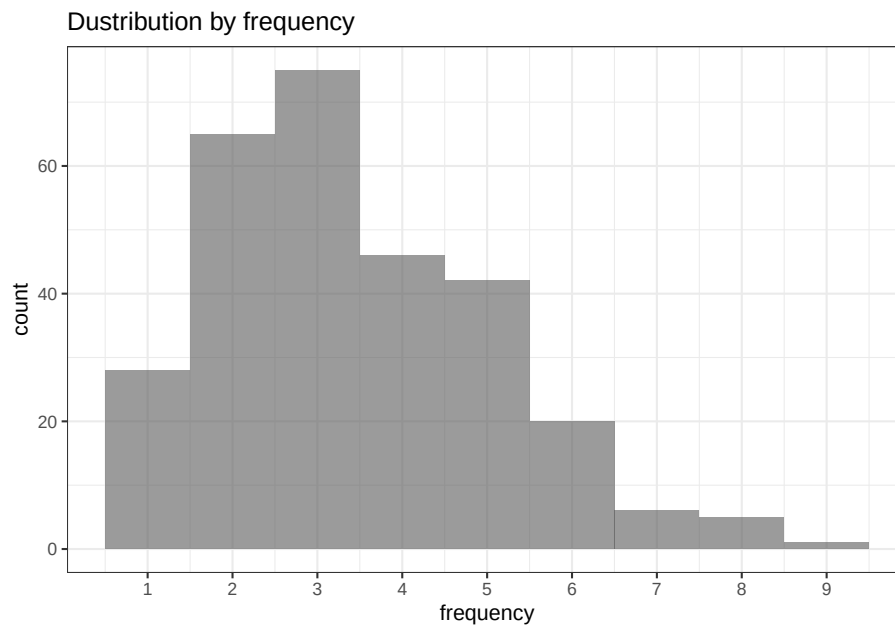
# processing data
orders <-
  dcast(
    orders,
    orderId + clientId + gender + orderdate ~ product,
    value.var = 'product',
    fun.aggregate = length
  )

orders <- orders %>%
  group_by(clientId) %>%
  mutate(frequency = n(),
         recency = as.numeric(today - orderdate)) %>%
  filter(orderdate == max(orderdate)) %>%
  filter(orderId == max(orderId)) %>%
  ungroup()

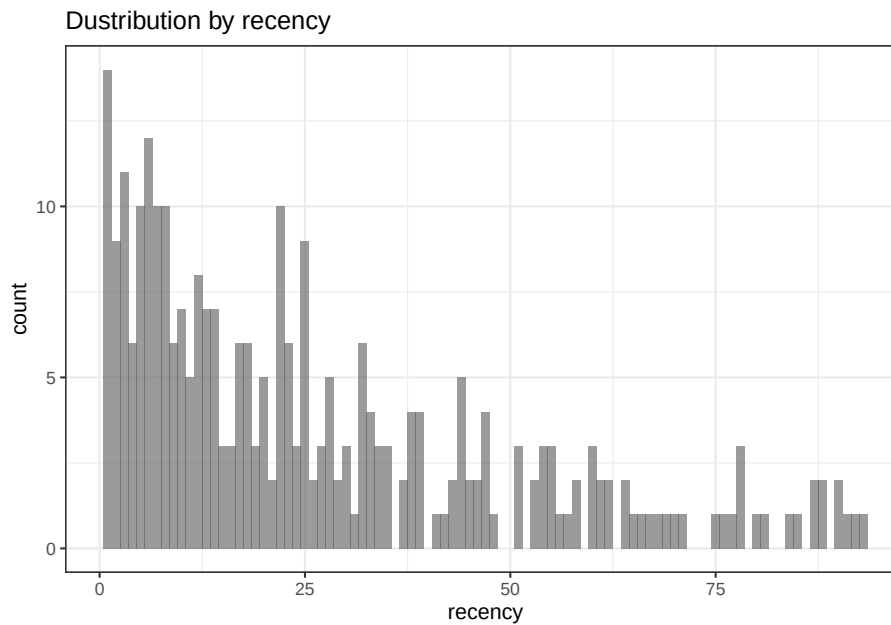
# exploratory analysis
ggplot(orders, aes(x = frequency)) +
  theme_bw() +

```

```
scale_x_continuous(breaks = c(1:10)) +
geom_bar(alpha = 0.6, width = 1) +
ggtitle("Dustribution by frequency")
```



```
ggplot(orders, aes(x = recency)) +
  theme_bw() +
  geom_bar(alpha = 0.6, width = 1) +
  ggtitle("Dustribution by recency")
```



```
orders.segm <- orders %>%
  mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
                              ifelse(
                                between(frequency, 2, 2), '2',
                                ifelse(between(frequency, 3, 3), '3',
                                      ifelse(
                                        between(frequency, 4, 4), '4',
                                        ifelse(between(frequency, 5, 5), '5', '>5')
                                      ))
                                ))) %>%
  mutate(segm.rec = ifelse(
    between(recency, 0, 6),
    '0-6 days',
    ifelse(
      between(recency, 7, 13),
      '7-13 days',
      ifelse(
        between(recency, 14, 19),
        '14-19 days',
        ifelse(
          between(recency, 20, 45),
          '20-45 days',
          ifelse(between(recency, 46, 80), '46-80 days', '>80 days')
        )
      )
    )
  )
```

```

    )
  )) %>%
  # creating last cart feature
  mutate(cart = paste(
    ifelse(a != 0, 'a', ''),
    ifelse(b != 0, 'b', ''),
    ifelse(c != 0, 'c', ''),
    sep = ''
  )) %>%
  arrange(clientId)

# defining order of boundaries
orders.segm$segm.freq <-
  factor(orders.segm$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
orders.segm$segm.rec <-
  factor(
    orders.segm$segm.rec,
    levels = c(
      '>80 days',
      '46-80 days',
      '20-45 days',
      '14-19 days',
      '7-13 days',
      '0-6 days'
    )
  )
)

```

```

lcg <- orders.segm %>%
  group_by(segm.rec, segm.freq) %>%
  summarise(quantity = n()) %>%
  mutate(client = 'client') %>%
  ungroup()

```

`summarise()` has grouped output by 'segm.rec'. You can override using the `.groups`

```

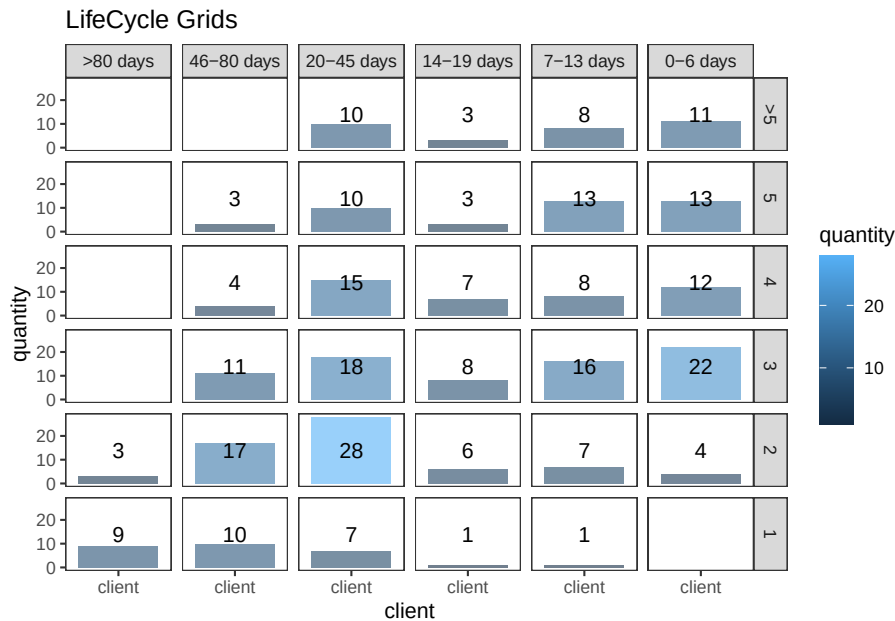
lcg.matrix <-
  dcast(lcg,
    segm.freq ~ segm.rec,
    value.var = 'quantity',
    fun.aggregate = sum)

ggplot(lcg, aes(x = client, y = quantity, fill = quantity)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(stat = 'identity', alpha = 0.6) +
  geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +

```

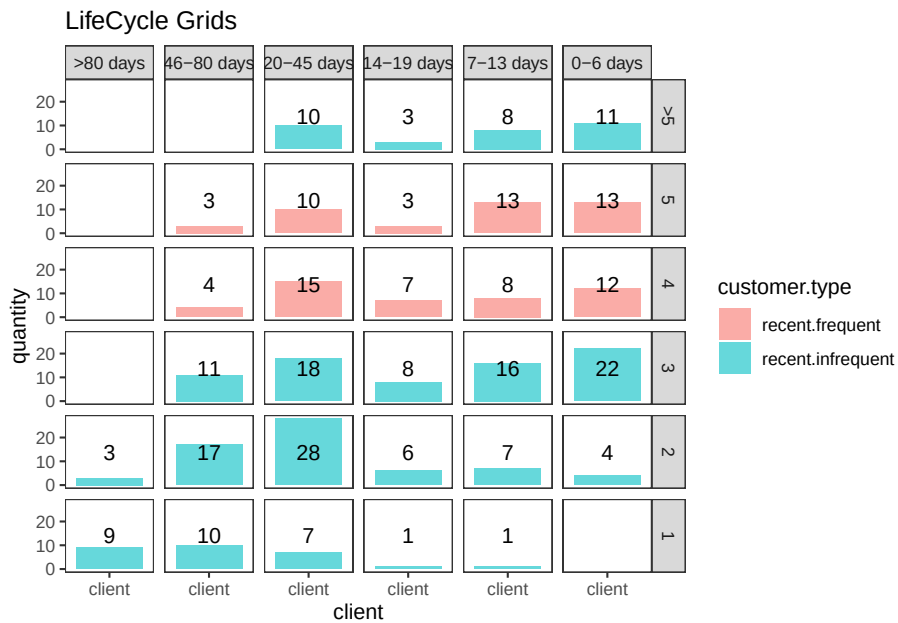


```
facet_grid(segm.freq ~ segm.rec) +
ggtitle("LifeCycle Grids")
```

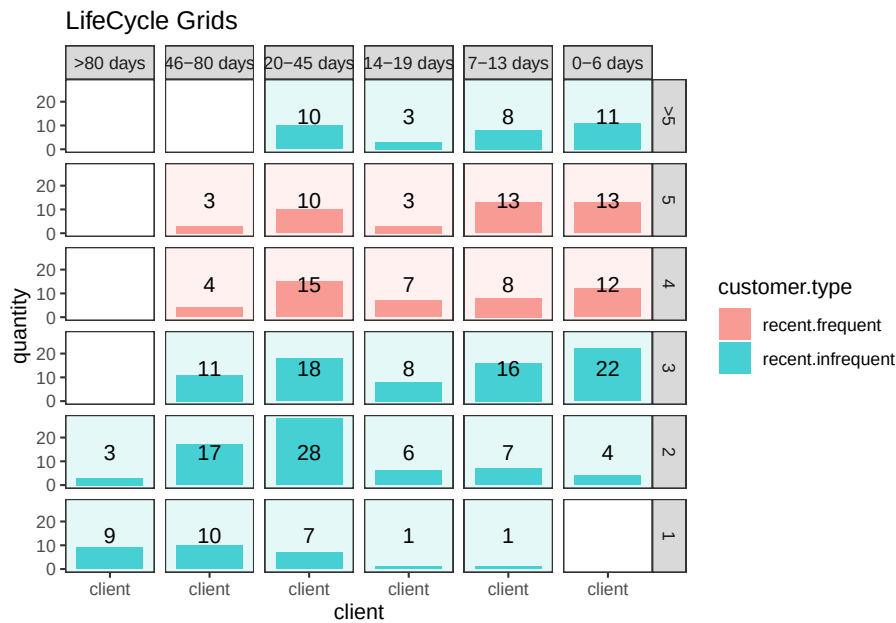


```
lcg.adv <- lcg %>%
  mutate(
    rec.type = ifelse(
      segm.rec %in% c("> 80 days", "46 - 80 days", "20 - 45 days"),
      "not recent",
      "recent"
    ),
    freq.type = ifelse(segm.freq %in% c("> 5", "5", "4"), "frequent", "infrequent"),
    customer.type = interaction(rec.type, freq.type)
  )

ggplot(lcg.adv, aes(x = client, y = quantity, fill = customer.type)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  facet_grid(segm.freq ~ segm.rec) +
  geom_bar(stat = 'identity', alpha = 0.6) +
  geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +
  ggtitle("LifeCycle Grids")
```



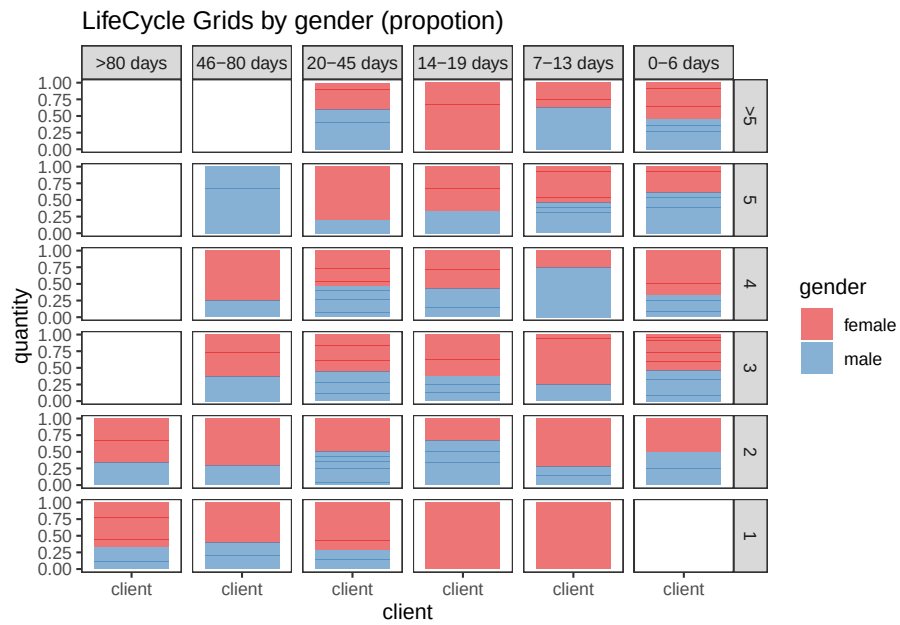
```
# with background
ggplot(lcg.adv, aes(x = client, y = quantity, fill = customer.type)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_rect(
    aes(fill = customer.type),
    xmin = -Inf,
    xmax = Inf,
    ymin = -Inf,
    ymax = Inf,
    alpha = 0.1
  ) +
  facet_grid(segm.freq ~ segm.rec) +
  geom_bar(stat = 'identity', alpha = 0.7) +
  geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +
  ggtitle("LifeCycle Grids")
```



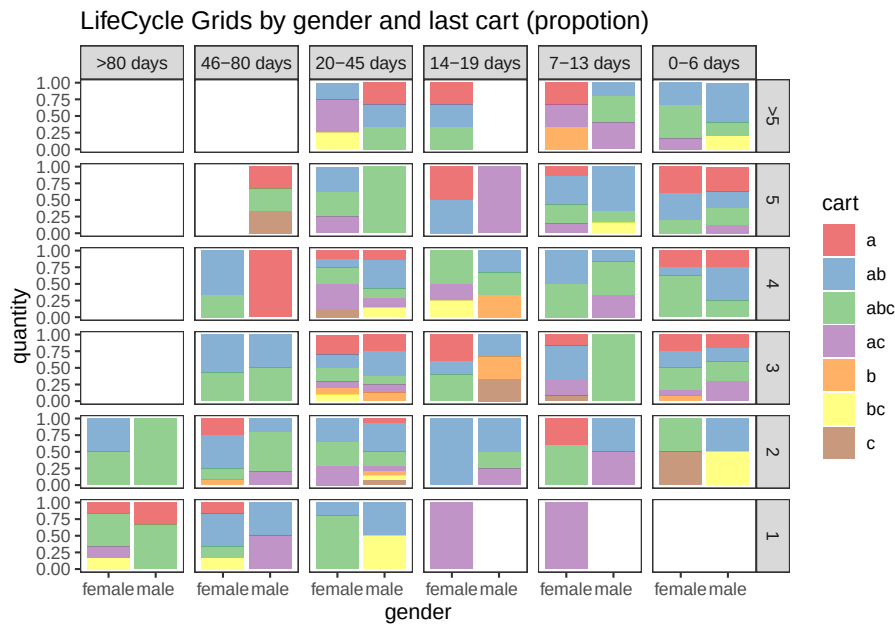
```
lcg.sub <- orders.segm %>%
  group_by(gender, cart, segm.rec, segm.freq) %>%
  summarise(quantity = n()) %>%
  mutate(client = 'client') %>%
  ungroup()
```

`summarise()` has grouped output by 'gender', 'cart', 'segm.rec'. You can override using the `

```
ggplot(lcg.sub, aes(x = client, y = quantity, fill = gender)) +
  theme_bw() +
  scale_fill_brewer(palette = 'Set1') +
  theme(panel.grid = element_blank()) +
  geom_bar(stat = 'identity',
           position = 'fill',
           alpha = 0.6) +
  facet_grid(segm.freq ~ segm.rec) +
  ggtitle("LifeCycle Grids by gender (proportion)")
```



```
ggplot(lcg.sub, aes(x = gender, y = quantity, fill = cart)) +
  theme_bw() +
  scale_fill_brewer(palette = 'Set1') +
  theme(panel.grid = element_blank()) +
  geom_bar(stat = 'identity',
           position = 'fill',
           alpha = 0.6) +
  facet_grid(segm.freq ~ segm.rec) +
  ggtitle("LifeCycle Grids by gender and last cart (propotion)")
```



15.4.2.2 CLV & CAC

calculate customer acquisition cost (CAC) and customer lifetime value (CLV)

```
# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)

# creating data sample
set.seed(10)
data <- data.frame(
  orderId = sample(c(1:1000), 5000, replace = TRUE),
  product = sample(
    c('NULL', 'a', 'b', 'c'),
    5000,
    replace = TRUE,
    prob = c(0.15, 0.65, 0.3, 0.15)
  )
)
order <- data.frame(orderId = c(1:1000),
  clientId = sample(c(1:300), 1000, replace = TRUE))
gender <- data.frame(clientId = c(1:300),
  gender = sample(
    c('male', 'female'),
```

```

        300,
        replace = TRUE,
        prob = c(0.40, 0.60)
    ))
date <- data.frame(orderId = c(1:1000),
                   orderdate = sample((1:100), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, gender, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL', ]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")

# creating data frames with CAC and Gross margin
cac <-
  data.frame(clientId = unique(orders$clientId),
             cac = sample(c(10:15), 288, replace = TRUE))
gr.margin <-
  data.frame(product = c('a', 'b', 'c'),
             grossmarg = c(1, 2, 3))

rm(data, date, order, gender)

# reporting date
today <- as.Date('2012-04-11', format = '%Y-%m-%d')

# calculating customer lifetime value
orders <- merge(orders, gr.margin, by = 'product')

clv <- orders %>%
  group_by(clientId) %>%
  summarise(clv = sum(grossmarg)) %>%
  ungroup()

# processing data
orders <-
  dcast(
    orders,
    orderId + clientId + gender + orderdate ~ product,
    value.var = 'product',
    fun.aggregate = length
  )

orders <- orders %>%
  group_by(clientId) %>%
  mutate(frequency = n(),

```

```

    recency = as.numeric(today - orderdate)) %>%
  filter(orderdate == max(orderdate)) %>%
  filter(orderId == max(orderId)) %>%
  ungroup()

orders.segm <- orders %>%
  mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
                             ifelse(
                               between(frequency, 2, 2), '2',
                               ifelse(between(frequency, 3, 3), '3',
                                      ifelse(
                                        between(frequency, 4, 4), '4',
                                        ifelse(between(frequency, 5, 5), '5', '>5')
                                      ))
                             ))) %>%
  mutate(segm.rec = ifelse(
    between(recency, 0, 6),
    '0-6 days',
    ifelse(
      between(recency, 7, 13),
      '7-13 days',
      ifelse(
        between(recency, 14, 19),
        '14-19 days',
        ifelse(
          between(recency, 20, 45),
          '20-45 days',
          ifelse(between(recency, 46, 80), '46-80 days', '>80 days')
        )
      )
    )
  )
  ) %>%
  # creating last cart feature
  mutate(cart = paste(
    ifelse(a != 0, 'a', ''),
    ifelse(b != 0, 'b', ''),
    ifelse(c != 0, 'c', ''),
    sep = ''
  )) %>%
  arrange(clientId)

# defining order of boundaries
orders.segm$segm.freq <-
  factor(orders.segm$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
orders.segm$segm.rec <-

```

```

factor(
  orders.segm$segm.rec,
  levels = c(
    '>80 days',
    '46-80 days',
    '20-45 days',
    '14-19 days',
    '7-13 days',
    '0-6 days'
  )
)

orders.segm <- merge(orders.segm, cac, by = 'clientId')
orders.segm <- merge(orders.segm, clv, by = 'clientId')

lcg.clv <- orders.segm %>%
  group_by(segm.rec, segm.freq) %>%
  summarise(quantity = n(),
            # calculating cumulative CAC and CLV
            cac = sum(cac),
            clv = sum(clv)) %>%
  ungroup() %>%
  # calculating CAC and CLV per client
  mutate(cac1 = round(cac / quantity, 2),
         clv1 = round(clv / quantity, 2))

```

`summarise()` has grouped output by 'segm.rec'. You can override using the `.groups`

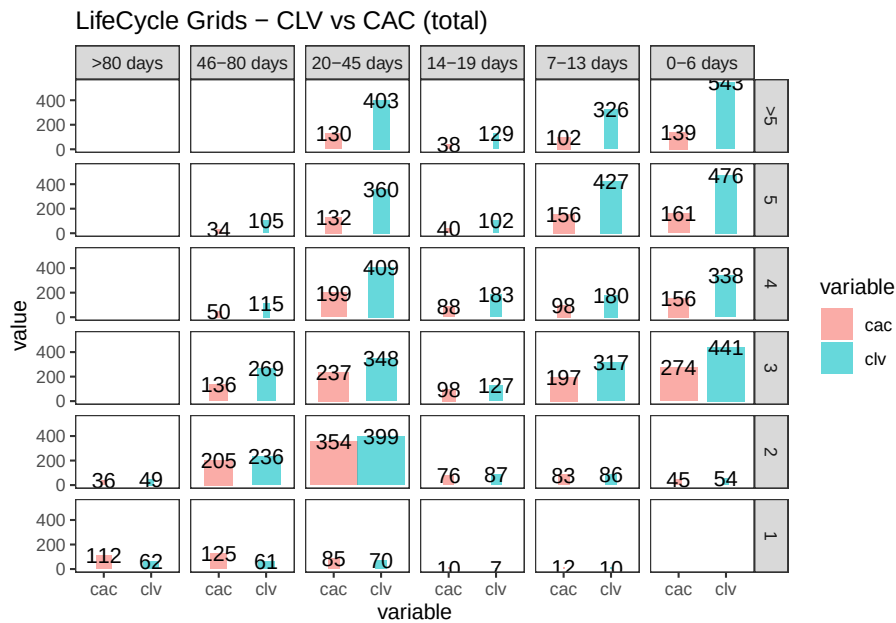
```

lcg.clv <-
  reshape2::melt(lcg.clv, id.vars = c('segm.rec', 'segm.freq', 'quantity'))

ggplot(lcg.clv[lcg.clv$variable %in% c('clv', 'cac'), ], aes(x = variable, y =
  value, fill = variable)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(stat = 'identity', alpha = 0.6, aes(width = quantity / max(quantity))) +
  geom_text(aes(y = value, label = value), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  ggtitle("LifeCycle Grids - CLV vs CAC (total)")

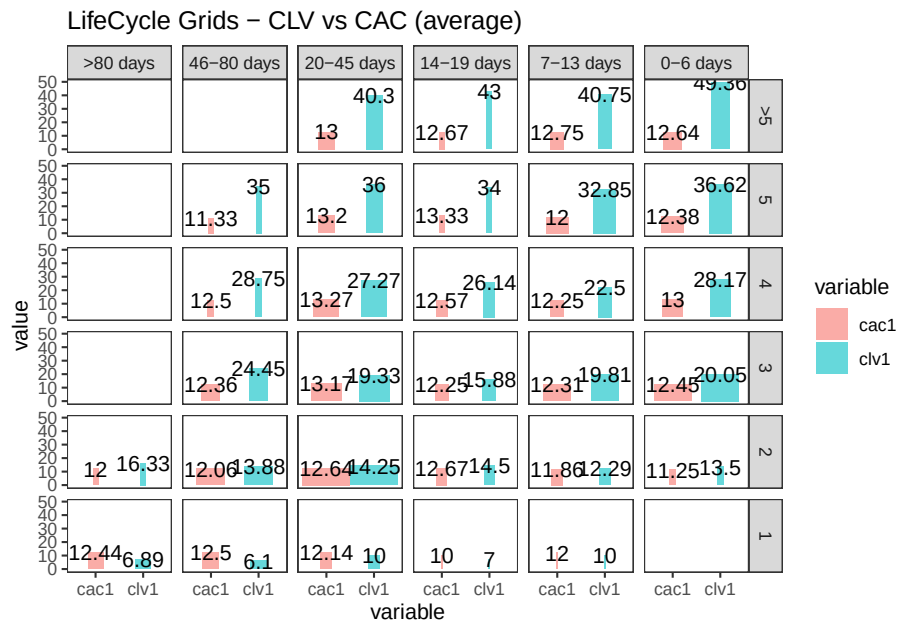
```

Warning: Ignoring unknown aesthetics: width



```
ggplot(lcg.clv[lcg.clv$variable %in% c('clv1', 'cac1'), ], aes(x = variable, y =
value, fill = variable)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(stat = 'identity', alpha = 0.6, aes(width = quantity / max(quantity))) +
  geom_text(aes(y = value, label = value), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  ggtitle("LifeCycle Grids – CLV vs CAC (average)")
```

```
## Warning: Ignoring unknown aesthetics: width
```



15.4.2.3 Cohort Analysis

link

combine customers through common characteristics to split customers into homogeneous groups

```
# loading libraries
```

```
library(dplyr)
```

```
library(reshape2)
```

```
library(ggplot2)
```

```
library(googleVis)
```

```
## Creating a generic function for 'toJSON' from package 'jsonlite' in package 'googleVis'
```

```
##
```

```
## Welcome to googleVis version 0.6.10
```

```
##
```

```
## Please read Google's Terms of Use
```

```
## before you start using the package:
```

```
## https://developers.google.com/terms/
```

```
##
```

```
## Note, the plot method of googleVis will by default use
```

```
## the standard browser to display its output.
```

```
##
```

```
## See the googleVis package vignettes for more details,
```

```
## or visit https://github.com/mages/googleVis.
```

```
##
## To suppress this message use:
## suppressPackageStartupMessages(library(googleVis))

set.seed(10)
# creating orders data sample
data <- data.frame(
  orderId = sample(c(1:5000), 25000, replace = TRUE),
  product = sample(
    c('NULL', 'a', 'b', 'c'),
    25000,
    replace = TRUE,
    prob = c(0.15, 0.65, 0.3, 0.15)
  )
)
order <- data.frame(orderId = c(1:5000),
  clientId = sample(c(1:1500), 5000, replace = TRUE))
date <- data.frame(orderId = c(1:5000),
  orderdate = sample((1:500), 5000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order)
# creating data frames with CAC, Gross margin, Campaigns and Potential CLV
gr.margin <-
  data.frame(product = c('a', 'b', 'c'),
    grossmarg = c(1, 2, 3))
campaign <- data.frame(clientId = c(1:1500),
  campaign = paste('campaign', sample(c(1:7), 1500, replace = TRUE), sep =
    ' '))
cac <-
  data.frame(campaign = unique(campaign$campaign),
    cac = sample(c(10:15), 7, replace = TRUE))
campaign <- merge(campaign, cac, by = 'campaign')
potential <- data.frame(clientId = c(1:1500),
  clv.p = sample(c(0:50), 1500, replace = TRUE))
rm(cac)

# reporting date
today <- as.Date('2013-05-16', format = '%Y-%m-%d')
```

where

- campaign, which includes campaign name and customer acquisition cost for each customer,
- margin, which includes gross margin for each product,

- potential, which includes potential values / predicted CLV for each client,
- orders, which includes orders from our customers with products and order dates.

calculating CLV, frequency, recency, average time lapses between purchases and defin

```
orders <- merge(orders, gr.margin, by = 'product')
```

```
customers <- orders %>%
```

```
  # combining products and summarising gross margin
```

```
  group_by(orderId, clientId, orderdate) %>%
```

```
  summarise(grossmarg = sum(grossmarg)) %>%
```

```
  ungroup() %>%
```

```
  # calculating frequency, recency, average time lapses between purchases and defin
```

```
  group_by(clientId) %>%
```

```
  mutate(
```

```
    frequency = n(),
```

```
    recency = as.numeric(today - max(orderdate)),
```

```
    av.gap = round(as.numeric(max(orderdate) - min(orderdate)) / frequency, 0),
```

```
    cohort = format(min(orderdate), format = '%Y-%m')
```

```
  ) %>%
```

```
  ungroup() %>%
```

```
  # calculating CLV to date
```

```
  group_by(clientId, cohort, frequency, recency, av.gap) %>%
```

```
  summarise(clv = sum(grossmarg)) %>%
```

```
  arrange(clientId) %>%
```

```
  ungroup()
```

`summarise()` has grouped output by 'orderId', 'clientId'. You can override using tl

`summarise()` has grouped output by 'clientId', 'cohort', 'frequency', 'recency'. Y

calculating potential CLV and CAC

```
customers <- merge(customers, campaign, by = 'clientId')
```

```
customers <- merge(customers, potential, by = 'clientId')
```

leading the potential value to more or less real value

```
customers$clv.p <-
```

```
  round(customers$clv.p / sqrt(customers$recency) * customers$frequency,
```

```
  2)
```

```
rm(potential, gr.margin, today)
```

adding segments

```
customers <- customers %>%
```

```
  mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
```

```
    ifelse(
```

```
      between(frequency, 2, 2), '2',
```

```
      ifelse(between(frequency, 3, 3), '3',
```

```

        ifelse(
          between(frequency, 4, 4), '4',
          ifelse(between(frequency, 5, 5), '5', '>5')
        ))
      ))) %>%
mutate(segm.rec = ifelse(
  between(recency, 0, 30),
  '0-30 days',
  ifelse(
    between(recency, 31, 60),
    '31-60 days',
    ifelse(
      between(recency, 61, 90),
      '61-90 days',
      ifelse(
        between(recency, 91, 120),
        '91-120 days',
        ifelse(between(recency, 121, 180), '121-180 days', '>180 days')
      )
    )
  )
))

# defining order of boundaries
customers$segm.freq <-
  factor(customers$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
customers$segm.rec <-
  factor(
    customers$segm.rec,
    levels = c(
      '>180 days',
      '121-180 days',
      '91-120 days',
      '61-90 days',
      '31-60 days',
      '0-30 days'
    )
  )
)

```

```

lcg.coh <- customers %>%
  group_by(cohort, segm.rec, segm.freq) %>%
  # calculating cumulative values
  summarise(
    quantity = n(),

```

```

    cac = sum(cac),
    clv = sum(clv),
    clv.p = sum(clv.p),
    av.gap = sum(av.gap)
  ) %>%
ungroup() %>%
# calculating average values
mutate(
  av.cac = round(cac / quantity, 2),
  av.clv = round(clv / quantity, 2),
  av.clv.p = round(clv.p / quantity, 2),
  av.clv.tot = av.clv + av.clv.p,
  av.gap = round(av.gap / quantity, 2),
  diff = av.clv - av.cac
)

# 1. Structure of averages and comparison cohorts

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = diff), stat = 'identity', alpha = 0.5) +
  geom_text(aes(y = diff, label = round(diff, 0)), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  theme(axis.text.x = element_text(
    angle = 90,
    hjust = .5,
    vjust = .5,
    face = "plain"
  )) +
  ggtitle("Cohorts in LifeCycle Grids - difference between av.CLV to date and av.CAC")

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = av.clv.tot), stat = 'identity', alpha = 0.2) +
  geom_text(aes(
    y = av.clv.tot + 10,
    label = round(av.clv.tot, 0),
    color = cohort
  ), size = 4) +
  geom_bar(aes(y = av.clv), stat = 'identity', alpha = 0.7) +
  geom_errorbar(aes(y = av.cac, ymax = av.cac, ymin = av.cac),
    color = 'red',
    size = 1.2) +

```

```

    geom_text(
      aes(y = av.cac, label = round(av.cac, 0)),
      size = 4,
      color = 'darkred',
      vjust = -.5
    ) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
      angle = 90,
      hjust = .5,
      vjust = .5,
      face = "plain"
    )) +
    ggtitle("Cohorts in LifeCycle Grids - total av.CLV and av.CAC")

# 2. Analyzing customer flows
# customers flows analysis (FPD cohorts)

# defining cohort and reporting dates
coh <- '2012-09'
report.dates <- c('2012-10-01', '2013-01-01', '2013-04-01')
report.dates <- as.Date(report.dates, format = '%Y-%m-%d')

# defining segments for each cohort's customer for reporting dates
df.sankey <- data.frame()

for (i in 1:length(report.dates)) {
  orders.cache <- orders %>%
    filter(orderdate < report.dates[i])

  customers.cache <- orders.cache %>%
    select(-product, -grossmarg) %>%
    unique() %>%
    group_by(clientId) %>%
    mutate(
      frequency = n(),
      recency = as.numeric(report.dates[i] - max(orderdate)),
      cohort = format(min(orderdate), format = '%Y-%m')
    ) %>%
    ungroup() %>%
    select(clientId, frequency, recency, cohort) %>%
    unique() %>%
    filter(cohort == coh) %>%
    mutate(segm.freq = ifelse(
      between(frequency, 1, 1),

```

```

      '1 purch',
    ifelse(
      between(frequency, 2, 2),
      '2 purch',
      ifelse(
        between(frequency, 3, 3),
        '3 purch',
        ifelse(
          between(frequency, 4, 4),
          '4 purch',
          ifelse(between(frequency, 5, 5), '5 purch', '>5 purch')
        )
      )
    )
  )) %>%
mutate(segm.rec = ifelse(
  between(recency, 0, 30),
  '0-30 days',
  ifelse(
    between(recency, 31, 60),
    '31-60 days',
    ifelse(
      between(recency, 61, 90),
      '61-90 days',
      ifelse(
        between(recency, 91, 120),
        '91-120 days',
        ifelse(between(recency, 121, 180), '121-180 days', '>180 days')
      )
    )
  )
)) %>%
mutate(
  cohort.segm = paste(cohort, segm.rec, segm.freq, sep = ' : '),
  report.date = report.dates[i]
) %>%
select(clientId, cohort.segm, report.date)

df.sankey <- rbind(df.sankey, customers.cache)
}

# processing data for Sankey diagram format
df.sankey <-
  dcast(df.sankey,
    clientId ~ report.date,

```



```

        value.var = 'cohort.segm',
        fun.aggregate = NULL)
write.csv(df.sankey, 'customers_path.csv', row.names = FALSE)
df.sankey <- df.sankey %>% select(-clientId)

df.sankey.plot <- data.frame()
for (i in 2:ncol(df.sankey)) {
  df.sankey.cache <- df.sankey %>%
    group_by(df.sankey[, i - 1], df.sankey[, i]) %>%
    summarise(n = n()) %>%
    ungroup()

  colnames(df.sankey.cache)[1:2] <- c('from', 'to')

  df.sankey.cache$from <-
    paste(df.sankey.cache$from, '(', report.dates[i - 1], ')', sep = '')
  df.sankey.cache$to <-
    paste(df.sankey.cache$to, '(', report.dates[i], ')', sep = '')

  df.sankey.plot <- rbind(df.sankey.plot, df.sankey.cache)
}

# plotting
plot(gvisSankey(
  df.sankey.plot,
  from = 'from',
  to = 'to',
  weight = 'n',
  options = list(
    height = 900,
    width = 1800,
    sankey = "{link:{color:{fill:'lightblue'}}}"
  )
))

# purchasing pace

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = av.gap), stat = 'identity', alpha = 0.6) +
  geom_text(aes(y = av.gap, label = round(av.gap, 0)), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  theme(axis.text.x = element_text(
    angle = 90,

```

```

      hjust = .5,
      vjust = .5,
      face = "plain"
    )) +
    ggtitle("Cohorts in LifeCycle Grids - average time lapses between purchases")

```

15.4.2.3.1 First-purchase date cohort

```

# campaign cohorts
lcg.camp <- customers %>%
  group_by(campaign, segm.rec, segm.freq) %>%
  # calculating cumulative values
  summarise(
    quantity = n(),
    cac = sum(cac),
    clv = sum(clv),
    clv.p = sum(clv.p),
    av.gap = sum(av.gap)
  ) %>%
  ungroup() %>%
  # calculating average values
  mutate(
    av.cac = round(cac / quantity, 2),
    av.clv = round(clv / quantity, 2),
    av.clv.p = round(clv.p / quantity, 2),
    av.clv.tot = av.clv + av.clv.p,
    av.gap = round(av.gap / quantity, 2),
    diff = av.clv - av.cac
  )

```

15.4.2.3.2 Campaign Cohorts

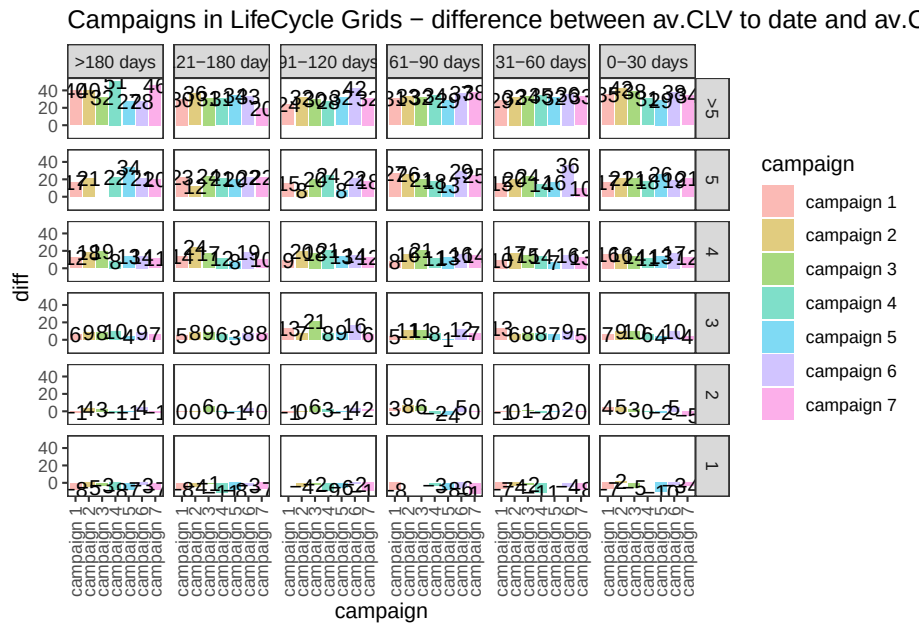
`summarise()` has grouped output by 'campaign', 'segm.rec'. You can override using `ungroup()`

```

ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = diff), stat = 'identity', alpha = 0.5) +
  geom_text(aes(y = diff, label = round(diff, 0)), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  theme(axis.text.x = element_text(
    angle = 90,
    hjust = .5,
    vjust = .5,
    face = "plain"
  )) +

```

```
ggtitle("Campaigns in LifeCycle Grids - difference between av.CLV to date and av.CAC")
```

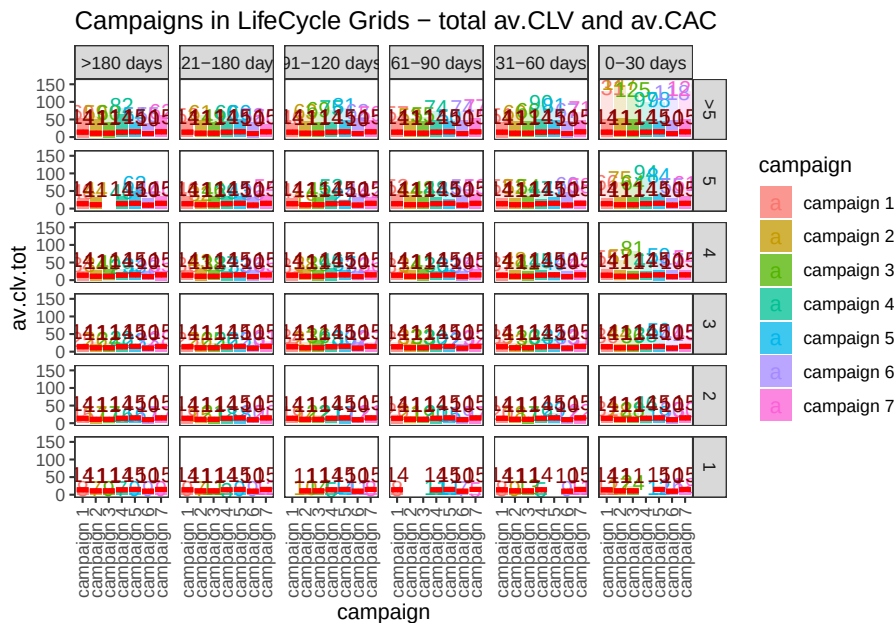


```
ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = av.clv.tot), stat = 'identity', alpha = 0.2) +
  geom_text(aes(
    y = av.clv.tot + 10,
    label = round(av.clv.tot, 0),
    color = campaign
  ), size = 4) +
  geom_bar(aes(y = av.clv), stat = 'identity', alpha = 0.7) +
  geom_errorbar(aes(y = av.cac, ymax = av.cac, ymin = av.cac),
    color = 'red',
    size = 1.2) +
  geom_text(
    aes(y = av.cac, label = round(av.cac, 0)),
    size = 4,
    color = 'darkred',
    vjust = -.5
  ) +
  facet_grid(segm.freq ~ segm.rec) +
  theme(axis.text.x = element_text(
    angle = 90,
    hjust = .5,
```

```

    vjust = .5,
    face = "plain"
  )) +
  ggtitle("Campaigns in LifeCycle Grids - total av.CLV and av.CAC")

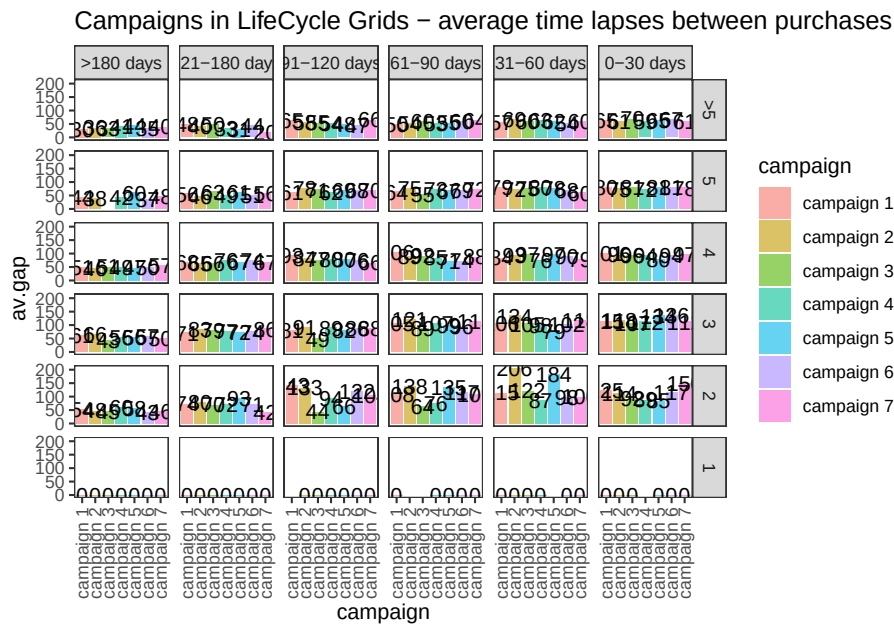
```



```

ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
  theme_bw() +
  theme(panel.grid = element_blank()) +
  geom_bar(aes(y = av.gap), stat = 'identity', alpha = 0.6) +
  geom_text(aes(y = av.gap, label = round(av.gap, 0)), size = 4) +
  facet_grid(segm.freq ~ segm.rec) +
  theme(axis.text.x = element_text(
    angle = 90,
    hjust = .5,
    vjust = .5,
    face = "plain"
  )) +
  ggtitle("Campaigns in LifeCycle Grids - average time lapses between purchases")

```



15.4.2.3.3 Retention Rate Customer Retention Rate

```
# loading libraries
```

```
library(dplyr)
```

```
library(reshape2)
```

```
library(ggplot2)
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## discard
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
## col_factor
```

```
## The following object is masked from 'package:kernlab':
```

```
##
```

```
## alpha
```

```
library(gridExtra)
```

```
##
```

```
## Attaching package: 'gridExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##      combine

# creating data sample
set.seed(10)
cohorts <-
  data.frame(
    cohort = paste('cohort', formatC(
      c(1:36),
      width = 2,
      format = 'd',
      flag = '0'
    ), sep = '_'),
    Y_00 = sample(c(1300:1500), 36, replace = TRUE),
    Y_01 = c(sample(c(800:1000), 36, replace = TRUE)),
    Y_02 = c(sample(c(600:800), 24, replace = TRUE), rep(NA, 12)),
    Y_03 = c(sample(c(400:500), 12, replace = TRUE), rep(NA, 24))
  )

# simulating seasonality (Black Friday)
cohorts[c(11, 23, 35), 2] <-
  as.integer(cohorts[c(11, 23, 35), 2] * 1.25)
cohorts[c(11, 23, 35), 3] <-
  as.integer(cohorts[c(11, 23, 35), 3] * 1.10)
cohorts[c(11, 23, 35), 4] <-
  as.integer(cohorts[c(11, 23, 35), 4] * 1.07)

# calculating retention rate and preparing data for plotting
df_plot <-
  reshape2::melt(
    cohorts,
    id.vars = 'cohort',
    value.name = 'number',
    variable.name = "year_of_LT"
  )

df_plot <- df_plot %>%
  group_by(cohort) %>%
  arrange(year_of_LT) %>%
  mutate(number_prev_year = lag(number),
         number_Y_00 = number[which(year_of_LT == 'Y_00')]) %>%
  ungroup() %>%
  mutate(
    ret_rate_prev_year = number / number_prev_year,
    ret_rate = number / number_Y_00,
    year_cohort = paste(year_of_LT, cohort, sep = '-')
  )
```

```

)

##### The first way for plotting cycle plot via scaling
# calculating the coefficient for scaling 2nd axis
k <-
  max(df_plot$number_prev_year[df_plot$year_of_LT == 'Y_01'] * 1.15) / min(df_plot$ret_rate[df_

# retention rate cycle plot
ggplot(
  na.omit(df_plot),
  aes(
    x = year_cohort,
    y = ret_rate,
    group = year_of_LT,
    color = year_of_LT
  )
) +
  theme_bw() +
  geom_point(size = 4) +
  geom_text(
    aes(label = percent(round(ret_rate, 2))),
    size = 4,
    hjust = 0.4,
    vjust = -0.6,
    fontface = "plain"
  ) +
  # smooth method can be changed (e.g. for "lm")
  geom_smooth(
    size = 2.5,
    method = 'loess',
    color = 'darkred',
    aes(fill = year_of_LT)
  ) +
  geom_bar(aes(y = number_prev_year / k, fill = year_of_LT),
    alpha = 0.2,
    stat = 'identity') +
  geom_bar(aes(y = number / k, fill = year_of_LT),
    alpha = 0.6,
    stat = 'identity') +
  geom_text(
    aes(y = 0, label = cohort),
    color = 'white',
    angle = 90,
    size = 4,
    hjust = -0.05,

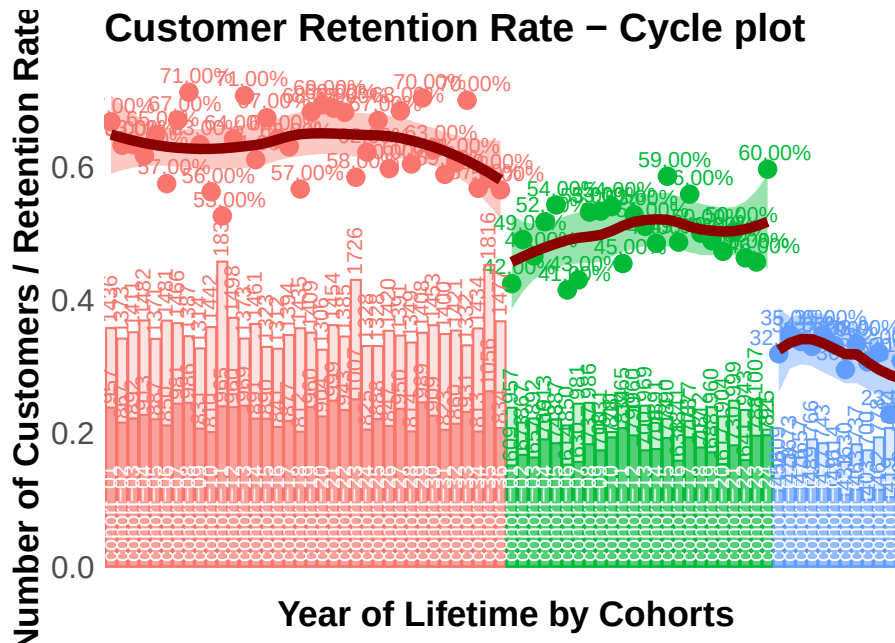
```

```

      vjust = 0.4
    ) +
    geom_text(
      aes(y = number_prev_year / k, label = number_prev_year),
      angle = 90,
      size = 4,
      hjust = -0.1,
      vjust = 0.4
    ) +
    geom_text(
      aes(y = number / k, label = number),
      angle = 90,
      size = 4,
      hjust = -0.1,
      vjust = 0.4
    ) +
    theme(
      legend.position = 'none',
      plot.title = element_text(size = 20, face = "bold", vjust = 2),
      axis.title.x = element_text(size = 18, face = "bold"),
      axis.title.y = element_text(size = 18, face = "bold"),
      axis.text = element_text(size = 16),
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank(),
      axis.ticks.y = element_blank(),
      panel.border = element_blank(),
      panel.grid.major = element_blank(),
      panel.grid.minor = element_blank()
    ) +
    labs(x = 'Year of Lifetime by Cohorts', y = 'Number of Customers / Retention Rate'),
    ggtitle("Customer Retention Rate - Cycle plot")

```

```
## `geom_smooth()` using formula 'y ~ x'
```

```
##### The second way for plotting cycle plot via multi-plotting
# plot #1 - Retention rate
p1 <-
  ggplot(
    na.omit(df_plot),
    aes(
      x = year_cohort,
      y = ret_rate,
      group = year_of_LT,
      color = year_of_LT
    )
  ) +
  theme_bw() +
  geom_point(size = 4) +
  geom_text(
    aes(label = percent(round(ret_rate, 2))),
    size = 4,
    hjust = 0.4,
    vjust = -0.6,
    fontface = "plain"
  ) +
  geom_smooth(
    size = 2.5,
    method = 'loess',
    color = 'darkred',
```

```

    aes(fill = year_of_LT)
  ) +
  theme(
    legend.position = 'none',
    plot.title = element_text(size = 20, face = "bold", vjust = 2),
    axis.title.x = element_blank(),
    axis.title.y = element_text(size = 18, face = "bold"),
    axis.text = element_blank(),
    axis.ticks.x = element_blank(),
    axis.ticks.y = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank()
  ) +
  labs(y = 'Retention Rate') +
  ggtitle("Customer Retention Rate - Cycle plot")

# plot #2 - number of customers
p2 <-
  ggplot(na.omit(df_plot),
    aes(x = year_cohort, group = year_of_LT, color = year_of_LT)) +
  theme_bw() +
  geom_bar(aes(y = number_prev_year, fill = year_of_LT),
    alpha = 0.2,
    stat = 'identity') +
  geom_bar(aes(y = number, fill = year_of_LT),
    alpha = 0.6,
    stat = 'identity') +
  geom_text(
    aes(y = number_prev_year, label = number_prev_year),
    angle = 90,
    size = 4,
    hjust = -0.1,
    vjust = 0.4
  ) +
  geom_text(
    aes(y = number, label = number),
    angle = 90,
    size = 4,
    hjust = -0.1,
    vjust = 0.4
  ) +
  geom_text(
    aes(y = 0, label = cohort),
    color = 'white',

```

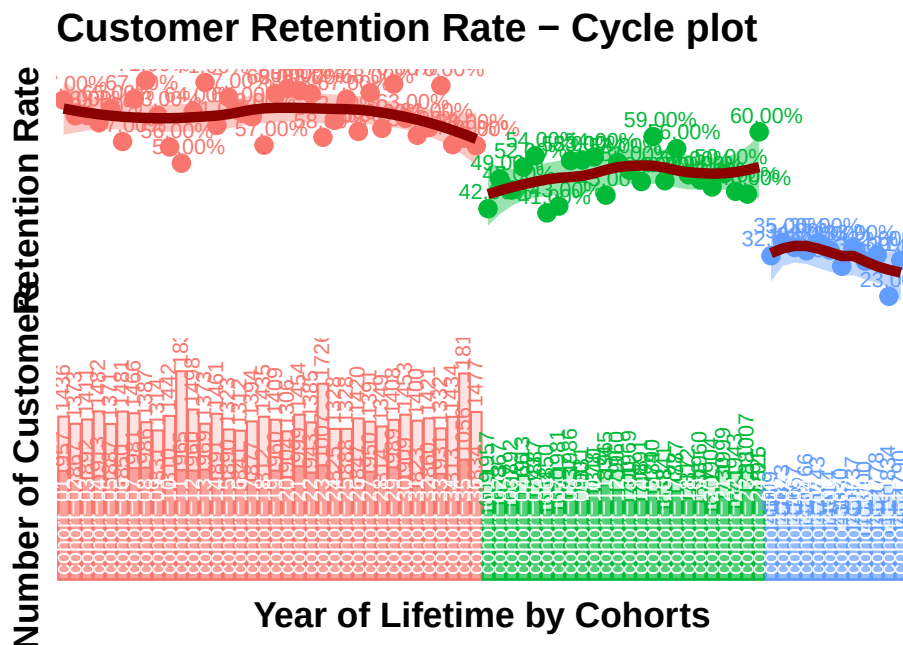
```

    angle = 90,
    size = 4,
    hjust = -0.05,
    vjust = 0.4
  ) +
  theme(
    legend.position = 'none',
    plot.title = element_text(size = 20, face = "bold", vjust = 2),
    axis.title.x = element_text(size = 18, face = "bold"),
    axis.title.y = element_text(size = 18, face = "bold"),
    axis.text = element_blank(),
    axis.ticks.x = element_blank(),
    axis.ticks.y = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank()
  ) +
  scale_y_continuous(limits = c(0, max(df_plot$number_Y_00 * 1.1))) +
  labs(x = 'Year of Lifetime by Cohorts', y = 'Number of Customers')

# multiplot
grid.arrange(p1, p2, ncol = 1)

## `geom_smooth()` using formula 'y ~ x'

```



```

# retention rate bubble chart
ggplot(na.omit(df_plot),
  aes(
    x = cohort,
    y = ret_rate,
    group = cohort,
    color = year_of_LT
  )) +
  theme_bw() +
  scale_size(range = c(15, 40)) +
  geom_line(size = 2, alpha = 0.3) +
  geom_point(aes(size = number_prev_year), alpha = 0.3) +
  geom_point(aes(size = number), alpha = 0.8) +
  geom_smooth(
    linetype = 2,
    size = 2,
    method = 'loess',
    aes(group = year_of_LT, fill = year_of_LT),
    alpha = 0.2
  ) +
  geom_text(
    aes(label = paste0(
      number, '/', number_prev_year, '\n', percent(round(ret_rate, 2))
    )),
    color = 'white',
    size = 3,
    hjust = 0.5,
    vjust = 0.5,
    fontface = "plain"
  ) +
  theme(
    legend.position = 'none',
    plot.title = element_text(size = 20, face = "bold", vjust = 2),
    axis.title.x = element_text(size = 18, face = "bold"),
    axis.title.y = element_text(size = 18, face = "bold"),
    axis.text = element_text(size = 16),
    axis.text.x = element_text(
      size = 10,
      angle = 90,
      hjust = .5,
      vjust = .5,
      face = "plain"
    ),
    axis.ticks.x = element_blank(),
    axis.ticks.y = element_blank(),
  )

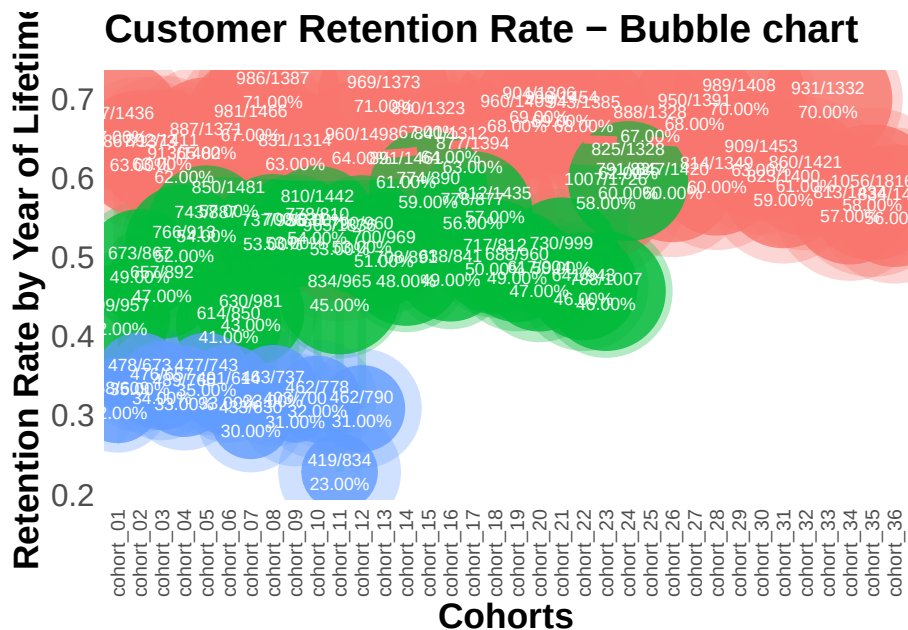
```

```

    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank()
  ) +
  labs(x = 'Cohorts', y = 'Retention Rate by Year of Lifetime') +
  ggtitle("Customer Retention Rate - Bubble chart")

```

```
## `geom_smooth()` using formula 'y ~ x'
```



```

# retention rate falling drops chart
ggplot(df_plot,
  aes(
    x = cohort,
    y = ret_rate,
    group = cohort,
    color = year_of_LT
  )) +
  theme_bw() +
  scale_size(range = c(15, 40)) +
  scale_y_continuous(limits = c(0, 1)) +
  geom_line(size = 2, alpha = 0.3) +
  geom_point(aes(size = number), alpha = 0.8) +
  geom_text(
    aes(label = paste0(number, '\n', percent(round(
      ret_rate, 2

```

```

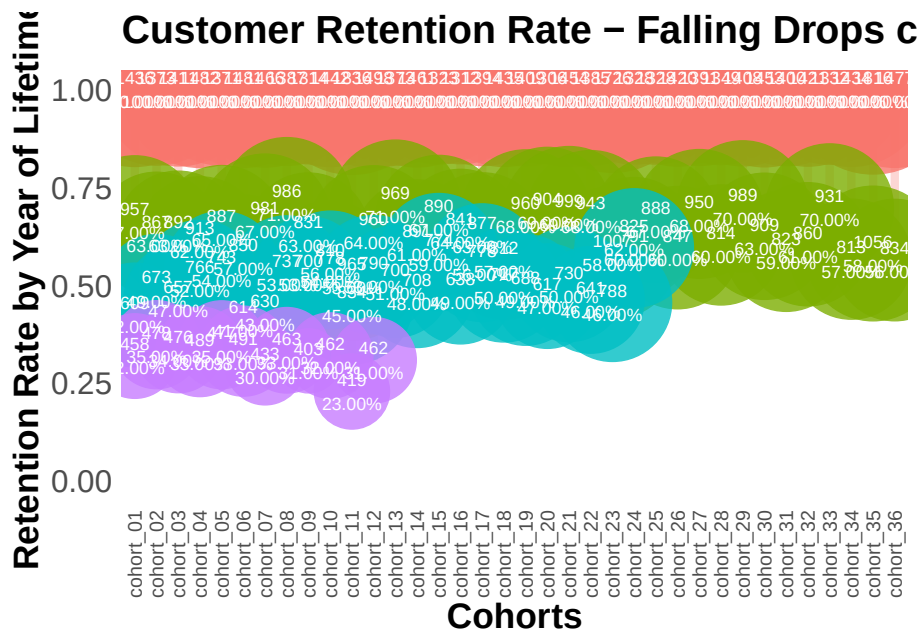
))))),
  color = 'white',
  size = 3,
  hjust = 0.5,
  vjust = 0.5,
  fontface = "plain"
) +
theme(
  legend.position = 'none',
  plot.title = element_text(size = 20, face = "bold", vjust = 2),
  axis.title.x = element_text(size = 18, face = "bold"),
  axis.title.y = element_text(size = 18, face = "bold"),
  axis.text = element_text(size = 16),
  axis.text.x = element_text(
    size = 10,
    angle = 90,
    hjust = .5,
    vjust = .5,
    face = "plain"
  ),
  axis.ticks.x = element_blank(),
  axis.ticks.y = element_blank(),
  panel.border = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank()
) +
labs(x = 'Cohorts', y = 'Retention Rate by Year of Lifetime') +
ggtitle("Customer Retention Rate - Falling Drops chart")

```

```
## Warning: Removed 36 row(s) containing missing values (geom_path).
```

```
## Warning: Removed 36 rows containing missing values (geom_point).
```

```
## Warning: Removed 36 rows containing missing values (geom_text).
```



15.4.2.3.4 Retention Charts Retention charts

```
# libraries
library(dplyr)
library(ggplot2)
library(reshape2)

cohort.clients <- data.frame(
  cohort = c(
    'Cohort01',
    'Cohort02',
    'Cohort03',
    'Cohort04',
    'Cohort05',
    'Cohort06',
    'Cohort07',
    'Cohort08',
    'Cohort09',
    'Cohort10',
    'Cohort11',
    'Cohort12'
  ),
  M01 = c(11000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  M02 = c(1900, 10000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  M03 = c(1400, 2000, 11500, 0, 0, 0, 0, 0, 0, 0, 0, 0),
```

```

M04 = c(1100, 1300, 2400, 13200, 0, 0, 0, 0, 0, 0, 0, 0),
M05 = c(1000, 1100, 1400, 2400, 11100, 0, 0, 0, 0, 0, 0, 0),
M06 = c(900, 900, 1200, 1600, 1900, 10300, 0, 0, 0, 0, 0, 0),
M07 = c(850, 900, 1100, 1300, 1300, 1900, 13000, 0, 0, 0, 0, 0),
M08 = c(850, 850, 1000, 1200, 1100, 1300, 1900, 11500, 0, 0, 0, 0),
M09 = c(800, 800, 950, 1100, 1100, 1250, 1000, 1200, 11000, 0, 0, 0),
M10 = c(800, 780, 900, 1050, 1050, 1200, 900, 1200, 1900, 13200, 0, 0),
M11 = c(750, 750, 900, 1000, 1000, 1180, 800, 1100, 1150, 2000, 11300, 0),
M12 = c(740, 700, 870, 1000, 900, 1100, 700, 1050, 1025, 1300, 1800, 20000)
)

cohort.clients.r <- cohort.clients #create new data frame
totcols <-
  ncol(cohort.clients.r) #count number of columns in data set
for (i in 1:nrow(cohort.clients.r)) {
  #for loop for shifting each row
  df <- cohort.clients.r[i,] #select row from data frame
  df <- df[, !df[] == 0] #remove columns with zeros
  partcols <-
    ncol(df) #count number of columns in row (w/o zeros)
  #fill columns after values by zeros
  if (partcols < totcols)
    df[, c((partcols + 1):totcols)] <- 0
  cohort.clients.r[i,] <- df #replace initial row by new one
}
# Retention ratio = # clients in particular month / # clients in 1st month of life-time

#calculate retention (1)
x <- cohort.clients.r[, c(2:13)]
y <- cohort.clients.r[, 2]
reten.r <- apply(x, 2, function(x)
  x / y)
reten.r <- data.frame(cohort = (cohort.clients.r$cohort), reten.r)

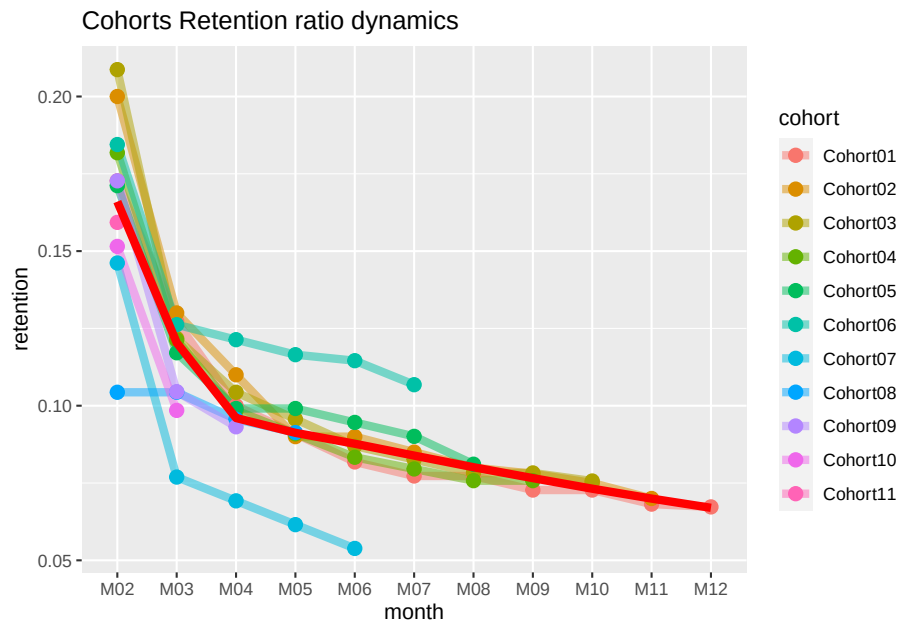
#calculate retention (2)
c <- ncol(cohort.clients.r)
reten.r <- cohort.clients.r
for (i in 2:c) {
  reten.r[, (c + i - 1)] <- reten.r[, i] / reten.r[, 2]
}
reten.r <- reten.r[, -c(2:c)]
colnames(reten.r) <- colnames(cohort.clients.r)

```



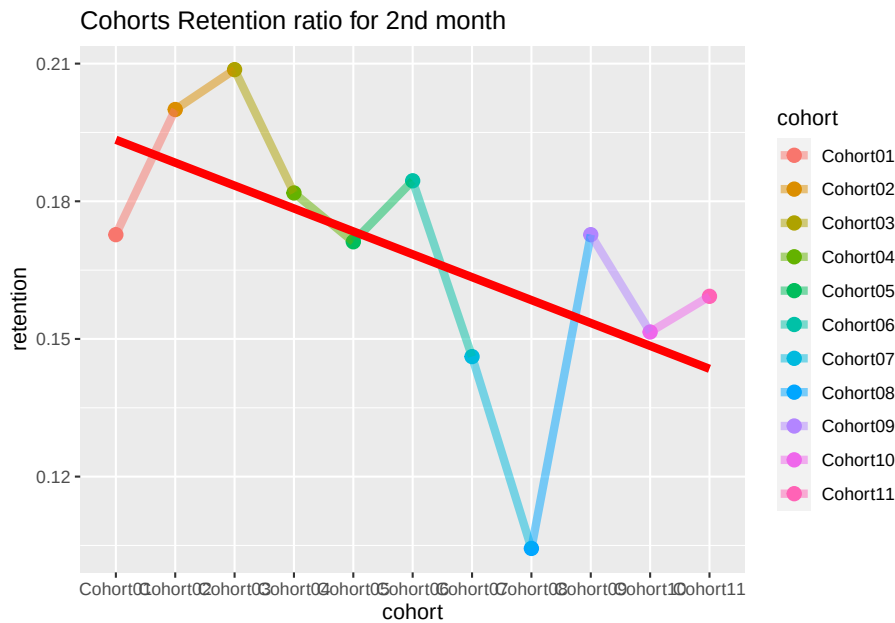
```
#charts
reten.r <- reten.r[,-2] #remove M01 data because it is always 100%
#dynamics analysis chart
cohort.chart1 <- melt(reten.r, id.vars = 'cohort')
colnames(cohort.chart1) <- c('cohort', 'month', 'retention')
cohort.chart1 <- filter(cohort.chart1, retention != 0)
p <-
  ggplot(cohort.chart1,
    aes(
      x = month,
      y = retention,
      group = cohort,
      colour = cohort
    ))
p + geom_line(size = 2, alpha = 1 / 2) +
  geom_point(size = 3, alpha = 1) +
  geom_smooth(
    aes(group = 1),
    method = 'loess',
    size = 2,
    colour = 'red',
    se = FALSE
  ) +
  labs(title = "Cohorts Retention ratio dynamics")
```

```
## `geom_smooth()` using formula 'y ~ x'
```



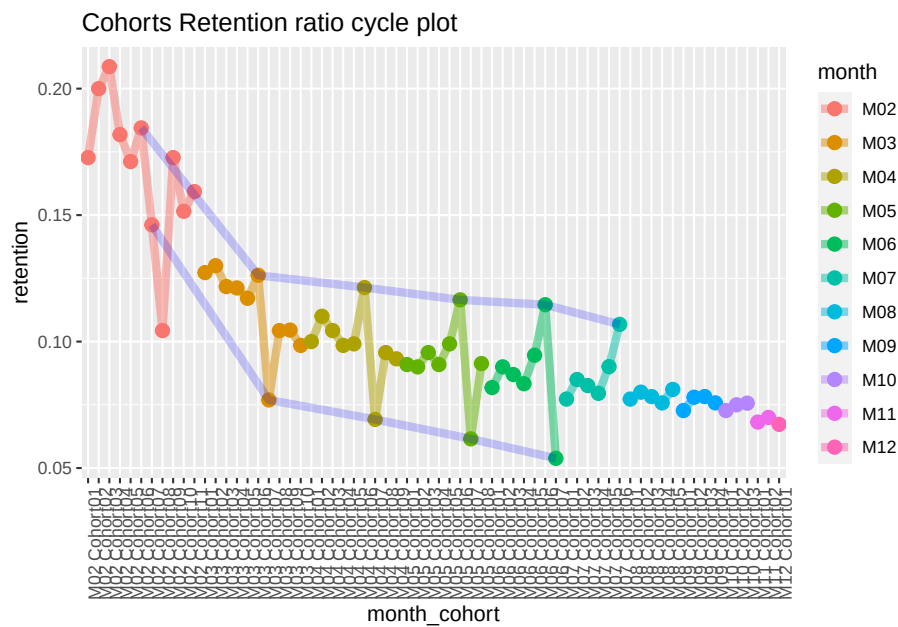
```
#second month analysis chart
cohort.chart2 <-
  filter(cohort.chart1, month == 'M02') #choose any month instead of M02
p <-
  ggplot(cohort.chart2, aes(x = cohort, y = retention, colour = cohort))
p + geom_point(size = 3) +
  geom_line(aes(group = 1), size = 2, alpha = 1 / 2) +
  geom_smooth(
    aes(group = 1),
    size = 2,
    colour = 'red',
    method = 'lm',
    se = FALSE
  ) +
  labs(title = "Cohorts Retention ratio for 2nd month")
```

```
## `geom_smooth()` using formula 'y ~ x'
```



```
#cycle plot
cohort.chart3 <- cohort.chart1
cohort.chart3 <-
  mutate(cohort.chart3, month_cohort = paste(month, cohort))
p <-
  ggplot(cohort.chart3,
    aes(
      x = month_cohort,
      y = retention,
      group = month,
      colour = month
    ))
#choose any cohorts instead of Cohort07 and Cohort06
m1 <- filter(cohort.chart3, cohort == 'Cohort07')
m2 <- filter(cohort.chart3, cohort == 'Cohort06')
p + geom_point(size = 3) +
  geom_line(aes(group = month), size = 2, alpha = 1 / 2) +
  labs(title = "Cohorts Retention ratio cycle plot") +
  geom_line(
    data = m1,
    aes(group = 1),
    colour = 'blue',
    size = 2,
    alpha = 1 / 5
  ) +
```

```
geom_line(
  data = m2,
  aes(group = 1),
  colour = 'blue',
  size = 2,
  alpha = 1 / 5
) +
theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



15.4.2.4 Lifecycle phase sequential analysis

- analyze the path patterns of each cohort
- identify cohorts that attracted customers with the path we prefer to make offers.

```
library(TraMineR)
```

```
## Warning: package 'TraMineR' was built under R version 4.0.5
```

```
##
```

```
## TraMineR stable version 2.2-2 (Built: 2021-08-14)
```

```
## Website: http://traminer.unige.ch
```

```
## Please type 'citation("TraMineR")' for citation information.
```

```

min.date <- min(orders$orderdate)
max.date <- max(orders$orderdate)

l <-
  c(seq(0, as.numeric(max.date - min.date), 10), as.numeric(max.date - min.date))

df <- data.frame()
for (i in l) {
  cur.date <- min.date + i
  print(cur.date)

  orders.cache <- orders %>%
    filter(orderdate <= cur.date)

  customers.cache <- orders.cache %>%
    select(-product, -grossmarg) %>%
    unique() %>%
    group_by(clientId) %>%
    mutate(frequency = n(),
           recency = as.numeric(cur.date - max(orderdate))) %>%
    ungroup() %>%
    select(clientId, frequency, recency) %>%
    unique() %>%

  mutate(segm =
    ifelse(
      between(frequency, 1, 2) & between(recency, 0, 60),
      'new customer',
      ifelse(
        between(frequency, 1, 2) &
          between(recency, 61, 180),
        'under risk new customer',
        ifelse(
          between(frequency, 1, 2) & recency > 180,
          '1x buyer',

          ifelse(
            between(frequency, 3, 4) &
              between(recency, 0, 60),
            'engaged customer',
            ifelse(
              between(frequency, 3, 4) &
                between(recency, 61, 180),
                'under risk engaged customer',
                ifelse(

```

```

        between(frequency, 3, 4) & recency > 180,
        'former engaged customer',

        ifelse(
          frequency > 4 & between(recency, 0, 60)
          'best customer',
          ifelse(
            frequency > 4 &
              between(recency, 61, 180),
            'under risk best customer',
            ifelse(frequency > 4 &
              recency > 180, 'former b

          )
        )
      )
    )
  )
  )) %>%

  mutate(report.date = i) %>%
  select(clientId, segm, report.date)

df <- rbind(df, customers.cache)
}

```

```

## [1] "2012-01-02"
## [1] "2012-01-12"
## [1] "2012-01-22"
## [1] "2012-02-01"
## [1] "2012-02-11"
## [1] "2012-02-21"
## [1] "2012-03-02"
## [1] "2012-03-12"
## [1] "2012-03-22"
## [1] "2012-04-01"
## [1] "2012-04-11"
## [1] "2012-04-21"
## [1] "2012-05-01"
## [1] "2012-05-11"
## [1] "2012-05-21"
## [1] "2012-05-31"
## [1] "2012-06-10"
## [1] "2012-06-20"

```

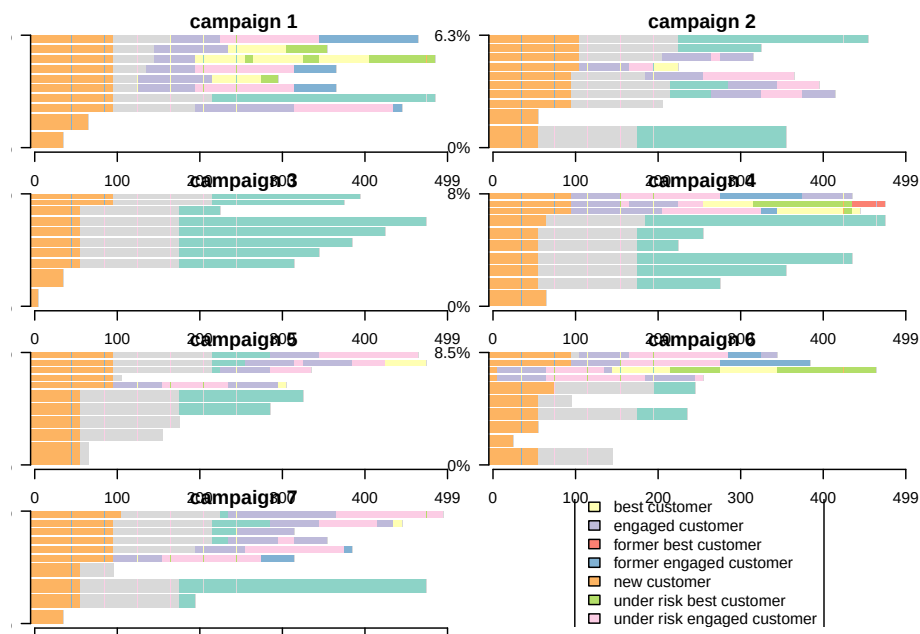
```
## [1] "2012-06-30"
## [1] "2012-07-10"
## [1] "2012-07-20"
## [1] "2012-07-30"
## [1] "2012-08-09"
## [1] "2012-08-19"
## [1] "2012-08-29"
## [1] "2012-09-08"
## [1] "2012-09-18"
## [1] "2012-09-28"
## [1] "2012-10-08"
## [1] "2012-10-18"
## [1] "2012-10-28"
## [1] "2012-11-07"
## [1] "2012-11-17"
## [1] "2012-11-27"
## [1] "2012-12-07"
## [1] "2012-12-17"
## [1] "2012-12-27"
## [1] "2013-01-06"
## [1] "2013-01-16"
## [1] "2013-01-26"
## [1] "2013-02-05"
## [1] "2013-02-15"
## [1] "2013-02-25"
## [1] "2013-03-07"
## [1] "2013-03-17"
## [1] "2013-03-27"
## [1] "2013-04-06"
## [1] "2013-04-16"
## [1] "2013-04-26"
## [1] "2013-05-06"
## [1] "2013-05-15"

# converting data to the sequence format
df <-
  dcast(df,
        clientId ~ report.date,
        value.var = 'segm',
        fun.aggregate = NULL)
df.seq <- seqdef(df,
                 2:ncol(df),
                 left = 'DEL',
                 right = 'DEL',
                 xtstep = 10)
```

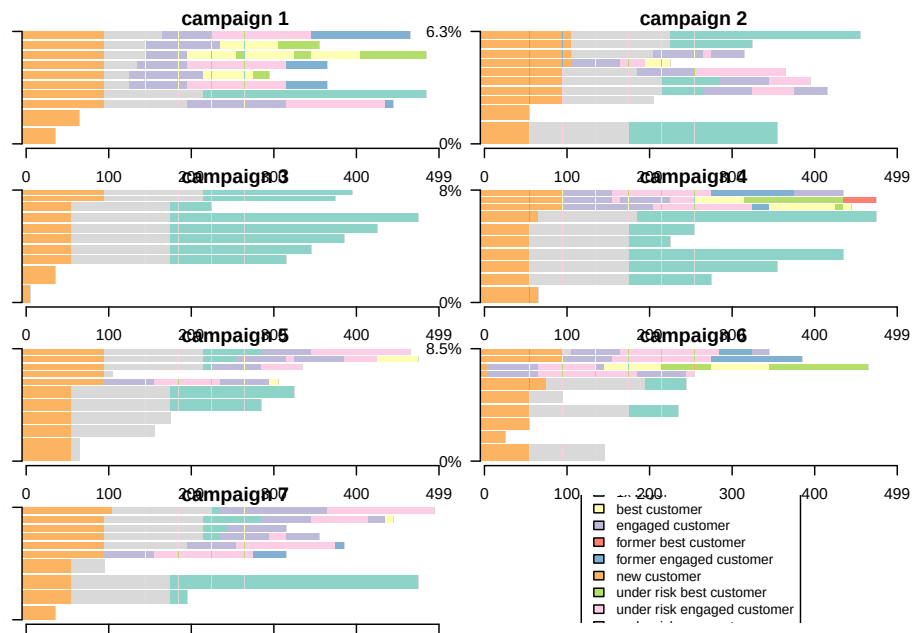
```
## [>] found missing values ('NA') in sequence data
## [>] preparing 1445 sequences
## [>] coding void elements with '%' and missing values with '*'
## [>] 9 distinct states appear in the data:
##      1 = 1x buyer
##      2 = best customer
##      3 = engaged customer
##      4 = former best customer
##      5 = former engaged customer
##      6 = new customer
##      7 = under risk best customer
##      8 = under risk engaged customer
##      9 = under risk new customer
## [>] state coding:
##      [alphabet]                [label]                [long label]
##      1  1x buyer                1x buyer                1x buyer
##      2  best customer            best customer            best customer
##      3  engaged customer          engaged customer          engaged customer
##      4  former best customer      former best customer      former best customer
##      5  former engaged customer    former engaged customer    former engaged customer
##      6  new customer              new customer              new customer
##      7  under risk best customer  under risk best customer  under risk best customer
##      8  under risk engaged customer under risk engaged customer under risk engaged customer
##      9  under risk new customer   under risk new customer   under risk new customer
## [>] 1445 sequences in the data set
## [>] min/max sequence length: 1/51
# creating df with first purch.date and campaign cohort features
feat <- df %>% select(clientId)
feat <- merge(feat, campaign[, 1:2], by = 'clientId')
feat <- merge(feat, customers[, 1:2], by = 'clientId')
par(mar = c(1, 1, 1, 1))
```



```
# plotting the 10 most frequent sequences based on campaign
seqfplot(df.seq, border = NA, group = feat$campaign)
```



```
# plotting the 10 most frequent sequences based on campaign
seqfplot(
  df.seq,
  border = NA,
  group = feat$campaign,
  cex.legend = 0.9
)
```



plotting the 10 most frequent sequences based on first purch.date cohort

```
coh.list <- sort(unique(feats$cohort))
# defining cohorts for plotting
feats.coh.list <- feats[feats$cohort %in% coh.list[1:6] ,]
df.coh <- df %>% filter(clientId %in% c(feats.coh.list$clientId))
df.seq.coh <-
  seqdef(
    df.coh,
    2:ncol(df.coh),
    left = 'DEL',
    right = 'DEL',
    xtstep = 10
  )
```

```
## [>] found missing values ('NA') in sequence data
```

```
## [>] preparing 1063 sequences
```

```
## [>] coding void elements with '%' and missing values with '*'
```

```
## [>] 9 distinct states appear in the data:
```

```
## 1 = 1x buyer
```

```
## 2 = best customer
```

```
## 3 = engaged customer
```

```
## 4 = former best customer
```

```
##      5 = former engaged customer

##      6 = new customer

##      7 = under risk best customer

##      8 = under risk engaged customer

##      9 = under risk new customer

##  [>] state coding:

##      [alphabet]                [label]                [long label]

##      1  1x buyer                1x buyer                1x buyer

##      2  best customer            best customer            best customer

##      3  engaged customer          engaged customer          engaged customer

##      4  former best customer      former best customer      former best customer

##      5  former engaged customer   former engaged customer   former engaged customer

##      6  new customer              new customer              new customer

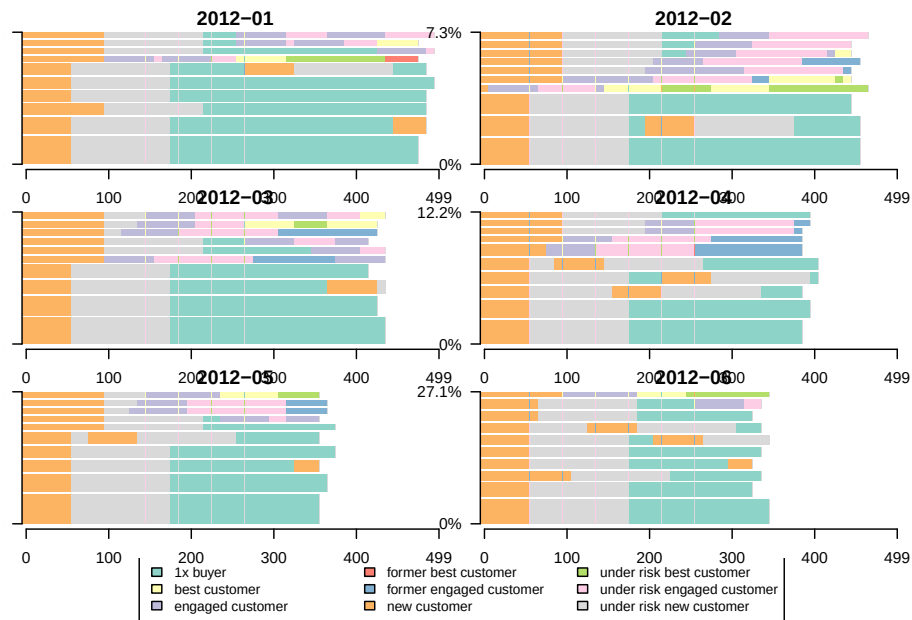
##      7  under risk best customer  under risk best customer  under risk best customer

##      8  under risk engaged customer under risk engaged customer under risk engaged customer

##      9  under risk new customer   under risk new customer   under risk new customer

##  [>] 1063 sequences in the data set

##  [>] min/max sequence length: 33/51
seqfplot(
  df.seq.coh,
  border = NA,
  group = feat.coh.list$cohort,
  cex.legend = 0.9
)
```



15.5 Shopping carts analysis

15.5.1 Multi-layer pie chart

Example by Sergey Bryl

```
# loading libraries
library(dplyr)
library(tidyverse)
library(reshape2)
library(plotrix)

##
## Attaching package: 'plotrix'

## The following object is masked from 'package:scales':
##
##   rescale

# Simulate of orders
set.seed(15)
df <- data.frame(
  orderId = sample(c(1:1000), 5000, replace = TRUE),
  product = sample(
    c('NULL', 'a', 'b', 'c', 'd'),
    5000,
```

```
      replace = TRUE,  
      prob = c(0.15, 0.65, 0.3, 0.15, 0.1)  
    )  
  )  
df <- df[df$product != 'NULL',]  
df
```

```
##      orderId product  
## 1         549      b  
## 3         874      a  
## 4         674      d  
## 5         505      a  
## 6         294      a  
## 7         177      a  
## 8         261      a  
## 9         217      b  
## 11        363      d  
## 12        596      a  
## 13        230      b  
## 14        193      b  
## 15        896      a  
## 16        652      a  
## 18         37      c  
## 19        727      a  
## 20        450      c  
## 21        793      b  
## 22        906      a  
## 23        730      c  
## 24        213      c  
## 25        181      a  
## 26        867      a  
## 27        509      a  
## 28        386      b  
## 29        500      b  
## 30        930      a  
## 31        442      c  
## 32        479      a  
## 33        531      b  
## 34        840      a  
## 36        798      a  
## 37         21      b  
## 39        677      a  
## 40        591      c  
## 41        283      a  
## 42        972      a
```

## 43	598	a
## 44	2	a
## 46	142	c
## 48	10	c
## 49	713	c
## 50	685	a
## 51	721	b
## 52	189	a
## 53	289	c
## 54	896	a
## 55	295	a
## 56	188	d
## 57	689	a
## 59	518	b
## 60	610	a
## 62	123	b
## 63	419	b
## 64	305	b
## 65	587	a
## 66	561	a
## 67	184	d
## 68	321	d
## 69	372	b
## 70	551	b
## 71	600	b
## 72	930	a
## 73	593	a
## 74	351	b
## 75	881	c
## 76	723	c
## 79	576	a
## 80	44	b
## 81	957	c
## 82	437	a
## 83	600	d
## 84	240	a
## 85	397	a
## 86	762	a
## 87	533	a
## 88	136	b
## 89	296	a
## 90	796	b
## 91	301	b
## 92	873	c
## 93	201	a
## 94	582	a

## 95	691	c
## 98	262	a
## 99	189	a
## 100	482	b
## 101	144	c
## 102	395	b
## 103	941	a
## 104	444	d
## 105	11	a
## 107	168	a
## 108	448	b
## 110	296	a
## 111	519	a
## 112	913	a
## 113	723	a
## 114	589	d
## 115	90	a
## 116	147	a
## 117	186	a
## 118	133	c
## 119	525	b
## 121	415	a
## 122	954	b
## 123	803	c
## 124	25	d
## 126	566	b
## 127	387	a
## 128	885	a
## 130	104	b
## 131	674	b
## 132	587	a
## 133	201	a
## 134	599	a
## 135	710	a
## 136	883	a
## 137	142	a
## 138	372	a
## 139	73	a
## 140	395	a
## 141	960	b
## 142	101	a
## 143	927	b
## 144	326	a
## 145	837	a
## 146	592	a
## 147	25	c

## 148	923	b
## 149	771	a
## 150	770	c
## 151	175	a
## 152	916	a
## 153	649	a
## 154	445	d
## 155	100	a
## 156	694	d
## 157	79	a
## 158	84	a
## 159	658	a
## 160	384	a
## 161	83	a
## 162	946	a
## 163	922	a
## 164	542	b
## 166	449	a
## 167	259	b
## 168	845	c
## 169	320	c
## 170	933	a
## 171	859	a
## 172	778	c
## 173	819	a
## 174	256	a
## 175	608	a
## 177	990	a
## 178	8	d
## 179	200	c
## 180	155	c
## 181	889	d
## 182	657	a
## 183	458	a
## 184	445	b
## 185	335	d
## 186	298	b
## 187	142	b
## 188	629	a
## 190	203	a
## 191	650	a
## 192	18	c
## 194	741	a
## 195	485	a
## 196	606	a
## 197	781	a

## 198	249	b
## 200	271	c
## 201	350	a
## 202	440	d
## 204	24	a
## 205	275	a
## 206	625	a
## 207	243	b
## 208	210	a
## 209	567	b
## 210	595	a
## 211	156	a
## 212	317	a
## 213	936	a
## 214	333	a
## 215	995	a
## 216	245	b
## 217	76	a
## 218	970	b
## 219	129	a
## 220	547	b
## 221	778	b
## 222	92	c
## 223	424	a
## 224	375	a
## 225	221	c
## 226	716	d
## 227	697	c
## 228	212	d
## 229	55	a
## 230	679	a
## 231	467	b
## 232	688	d
## 233	224	b
## 234	150	a
## 237	544	c
## 238	666	a
## 239	986	a
## 240	581	a
## 241	646	a
## 242	164	d
## 243	291	b
## 244	960	a
## 245	40	a
## 246	556	a
## 247	622	a

##	248	175	a
##	249	700	a
##	250	944	d
##	251	469	a
##	252	67	c
##	253	312	a
##	254	958	a
##	255	371	c
##	256	555	a
##	257	724	c
##	258	668	a
##	259	467	c
##	260	135	a
##	261	765	a
##	262	115	a
##	263	407	a
##	264	491	a
##	265	179	c
##	266	432	a
##	267	161	a
##	268	557	c
##	269	202	a
##	270	617	a
##	271	711	b
##	272	688	d
##	273	521	c
##	274	468	d
##	275	365	a
##	277	131	a
##	279	523	a
##	280	918	a
##	281	656	a
##	283	513	a
##	284	267	b
##	285	84	a
##	286	974	a
##	287	758	b
##	288	475	a
##	289	565	a
##	291	787	b
##	293	295	a
##	294	323	d
##	295	281	c
##	296	738	b
##	298	212	a
##	299	989	d

## 300	147	a
## 301	202	a
## 302	384	a
## 303	268	b
## 304	176	a
## 305	162	a
## 307	12	a
## 308	843	a
## 309	916	d
## 310	769	a
## 311	460	b
## 312	667	c
## 315	922	b
## 316	160	a
## 317	718	a
## 318	223	c
## 319	654	a
## 321	705	a
## 323	635	a
## 324	619	b
## 325	694	a
## 326	651	a
## 327	848	a
## 328	982	a
## 329	616	a
## 330	520	a
## 331	331	a
## 332	490	a
## 333	446	d
## 334	465	a
## 335	627	a
## 337	572	a
## 339	234	a
## 340	217	b
## 341	509	d
## 342	727	b
## 343	799	d
## 344	714	a
## 345	299	d
## 346	541	a
## 348	256	a
## 349	482	a
## 350	296	a
## 351	37	c
## 352	88	a
## 353	73	c

##	354	552	a
##	355	864	a
##	356	957	a
##	357	69	b
##	358	15	a
##	359	643	b
##	360	133	b
##	361	811	a
##	362	959	b
##	363	944	c
##	364	305	d
##	365	955	a
##	367	321	c
##	368	629	c
##	369	216	a
##	371	969	b
##	372	762	a
##	374	347	a
##	375	683	a
##	376	181	a
##	377	32	a
##	378	2	a
##	379	63	d
##	381	9	a
##	382	343	b
##	383	215	b
##	384	711	b
##	385	742	c
##	386	734	b
##	388	922	a
##	389	805	d
##	390	987	b
##	391	450	c
##	392	639	c
##	393	888	a
##	394	250	a
##	395	911	c
##	396	809	a
##	398	969	d
##	399	618	a
##	400	670	c
##	401	252	c
##	402	866	a
##	403	126	b
##	404	720	a
##	405	34	d

## 407	270	c
## 408	828	a
## 409	497	b
## 410	760	b
## 411	263	b
## 412	343	a
## 413	836	a
## 414	334	c
## 415	876	c
## 416	658	a
## 417	57	b
## 418	756	b
## 419	206	d
## 420	297	a
## 421	992	a
## 422	67	a
## 423	446	b
## 424	461	a
## 426	172	d
## 427	310	b
## 428	199	a
## 430	966	c
## 432	581	a
## 433	719	a
## 437	173	a
## 439	59	a
## 440	223	d
## 441	650	a
## 442	868	a
## 443	73	a
## 444	585	a
## 445	325	b
## 446	117	a
## 447	83	a
## 448	772	b
## 449	972	d
## 450	593	a
## 451	759	d
## 452	313	a
## 453	31	a
## 455	940	c
## 456	693	a
## 457	279	c
## 459	47	d
## 460	28	a
## 461	534	a

## 462	101	c
## 463	48	a
## 464	627	a
## 465	173	d
## 466	965	a
## 468	84	b
## 469	451	c
## 470	267	a
## 471	542	b
## 472	175	c
## 473	472	a
## 474	526	a
## 475	316	c
## 476	982	a
## 477	487	b
## 478	80	a
## 479	357	a
## 480	635	a
## 481	575	a
## 482	123	a
## 483	213	a
## 484	537	a
## 486	577	a
## 487	15	d
## 488	922	c
## 489	798	a
## 490	386	a
## 491	278	a
## 493	755	a
## 494	710	a
## 497	848	a
## 498	166	b
## 499	206	d
## 500	469	a
## 501	621	c
## 502	101	b
## 503	381	a
## 504	312	b
## 505	536	a
## 506	972	b
## 507	861	c
## 508	331	b
## 509	354	a
## 510	97	a
## 511	61	a
## 512	271	b

## 513	97	a
## 514	481	b
## 515	904	a
## 516	429	a
## 517	835	c
## 519	895	c
## 520	154	d
## 521	635	a
## 522	940	a
## 523	69	a
## 524	448	a
## 525	798	a
## 526	331	a
## 527	103	a
## 528	364	c
## 529	404	b
## 531	753	d
## 532	395	b
## 533	949	b
## 534	121	a
## 535	404	a
## 536	873	b
## 537	929	a
## 538	86	b
## 540	218	a
## 541	934	a
## 542	929	b
## 543	134	a
## 544	867	b
## 545	816	a
## 546	910	d
## 547	158	a
## 548	594	c
## 549	76	d
## 550	830	a
## 551	401	a
## 552	323	a
## 553	587	a
## 555	682	d
## 556	936	c
## 557	796	c
## 558	625	a
## 561	296	a
## 563	385	a
## 564	273	a
## 565	243	a

## 566	957	c
## 567	811	c
## 568	443	c
## 569	144	c
## 571	59	a
## 572	373	b
## 573	445	a
## 574	913	b
## 575	290	a
## 577	978	a
## 578	935	c
## 579	690	a
## 580	450	a
## 581	864	a
## 582	136	b
## 583	787	a
## 584	97	d
## 585	766	b
## 586	639	a
## 587	981	a
## 589	136	a
## 590	805	b
## 591	410	c
## 592	10	a
## 593	937	a
## 594	553	d
## 595	373	b
## 596	993	a
## 597	572	b
## 598	473	b
## 599	624	b
## 600	115	a
## 601	652	b
## 603	90	d
## 604	679	a
## 605	968	a
## 606	947	a
## 607	496	b
## 608	181	b
## 609	150	b
## 610	8	a
## 611	452	d
## 612	80	c
## 613	788	a
## 614	571	a
## 615	801	c

## 616	139	a
## 617	413	a
## 618	652	c
## 619	132	c
## 622	276	c
## 623	964	c
## 624	923	c
## 626	660	a
## 627	855	a
## 628	700	a
## 629	761	b
## 631	68	b
## 632	60	b
## 633	392	c
## 634	371	b
## 635	799	b
## 636	554	b
## 637	684	a
## 638	642	b
## 640	758	c
## 641	467	b
## 643	732	c
## 644	282	a
## 645	316	a
## 646	556	b
## 647	268	a
## 648	664	a
## 651	526	b
## 652	97	a
## 653	362	b
## 654	969	d
## 655	903	b
## 656	786	d
## 658	618	c
## 659	964	a
## 660	921	a
## 661	345	c
## 662	157	c
## 663	465	c
## 664	855	b
## 665	588	a
## 666	327	a
## 667	467	a
## 668	149	a
## 669	604	a
## 670	751	a

## 671	415	a
## 674	560	a
## 676	659	a
## 677	917	b
## 678	329	a
## 679	729	c
## 680	256	a
## 681	6	a
## 682	524	a
## 683	942	b
## 684	956	a
## 685	233	b
## 686	466	d
## 687	476	a
## 688	671	a
## 689	994	c
## 690	188	a
## 691	781	a
## 692	851	a
## 693	302	d
## 694	661	a
## 695	639	a
## 696	974	b
## 697	40	a
## 700	741	b
## 701	214	a
## 702	22	d
## 703	559	c
## 704	846	c
## 705	854	a
## 706	116	c
## 707	494	a
## 708	638	c
## 709	954	b
## 710	294	a
## 711	627	b
## 712	300	a
## 713	176	a
## 714	631	a
## 715	572	a
## 716	350	d
## 717	700	a
## 718	410	a
## 719	726	a
## 720	13	b
## 721	57	b

## 722	966	a
## 723	545	a
## 724	727	a
## 725	252	a
## 726	77	c
## 728	28	b
## 730	693	a
## 731	591	c
## 732	952	b
## 733	310	a
## 734	357	d
## 735	218	c
## 736	17	a
## 737	784	c
## 738	70	b
## 741	896	a
## 742	417	c
## 743	469	c
## 744	280	a
## 745	102	a
## 746	361	a
## 747	923	b
## 748	382	d
## 749	203	d
## 750	136	b
## 751	252	a
## 752	629	a
## 753	408	b
## 754	481	a
## 755	64	a
## 756	253	a
## 757	877	b
## 758	426	b
## 759	408	a
## 760	786	a
## 761	731	a
## 762	99	a
## 763	940	b
## 765	551	a
## 766	544	a
## 767	24	a
## 768	245	c
## 769	64	a
## 770	401	a
## 771	814	b
## 772	959	b

## 773	141	b
## 774	190	b
## 775	248	a
## 777	804	d
## 778	688	a
## 779	800	b
## 780	92	b
## 781	206	a
## 782	985	a
## 783	867	a
## 784	51	a
## 785	277	a
## 786	645	a
## 788	468	d
## 789	957	a
## 790	433	d
## 791	254	b
## 792	15	b
## 793	806	a
## 794	705	a
## 795	577	a
## 796	425	a
## 797	410	b
## 798	506	d
## 799	341	c
## 801	802	a
## 802	689	a
## 803	194	a
## 804	240	a
## 805	994	a
## 806	798	c
## 807	422	a
## 808	36	a
## 809	94	c
## 810	301	b
## 811	57	a
## 812	699	b
## 813	184	a
## 815	753	a
## 816	381	a
## 818	423	a
## 821	785	d
## 822	489	a
## 823	746	c
## 824	2	a
## 825	86	a

## 826	944	d
## 827	221	a
## 828	198	a
## 830	725	a
## 831	204	a
## 832	937	b
## 833	704	a
## 834	121	a
## 835	545	a
## 836	565	a
## 837	211	a
## 838	494	c
## 839	131	b
## 840	418	a
## 841	980	a
## 843	473	b
## 844	906	a
## 845	257	a
## 846	175	d
## 847	783	d
## 848	806	c
## 849	310	b
## 850	710	b
## 851	157	a
## 852	137	b
## 853	437	b
## 854	652	d
## 855	998	a
## 856	83	a
## 857	836	a
## 858	182	a
## 859	405	c
## 860	517	a
## 861	120	a
## 863	338	b
## 864	561	a
## 866	292	a
## 867	333	a
## 868	45	a
## 869	176	b
## 870	791	b
## 873	104	c
## 874	574	a
## 875	90	b
## 876	337	b
## 878	903	a

## 879	781	a
## 880	341	a
## 881	554	a
## 882	624	a
## 883	36	a
## 884	10	b
## 885	505	b
## 886	383	a
## 887	668	a
## 888	436	c
## 889	284	a
## 890	704	a
## 891	994	c
## 892	86	a
## 893	499	c
## 894	761	a
## 895	372	a
## 896	454	b
## 897	417	a
## 898	370	d
## 899	788	b
## 900	244	a
## 901	346	b
## 902	710	a
## 904	73	a
## 905	428	b
## 906	147	c
## 907	683	b
## 908	33	c
## 909	193	b
## 910	394	a
## 911	861	a
## 912	705	a
## 913	3	b
## 915	438	b
## 916	692	a
## 917	414	b
## 918	324	d
## 919	11	a
## 920	598	b
## 921	818	d
## 922	474	a
## 923	582	a
## 924	795	a
## 925	108	a
## 926	738	a

## 927	700	d
## 928	812	b
## 929	379	a
## 930	744	b
## 931	334	d
## 932	299	c
## 933	470	a
## 934	748	a
## 935	307	c
## 936	62	a
## 937	516	a
## 938	896	a
## 939	436	b
## 940	750	a
## 943	478	a
## 944	155	a
## 945	225	a
## 946	383	a
## 947	874	b
## 948	156	a
## 949	705	a
## 950	624	a
## 951	845	c
## 952	902	a
## 953	647	d
## 954	706	a
## 955	837	b
## 956	982	a
## 957	344	a
## 958	287	a
## 959	575	b
## 961	482	a
## 962	907	b
## 963	703	a
## 964	426	c
## 965	699	b
## 966	278	a
## 967	188	a
## 968	777	a
## 971	816	b
## 972	727	a
## 973	766	a
## 974	10	b
## 975	941	b
## 976	847	a
## 977	353	c

## 978	661	a
## 979	137	a
## 980	974	a
## 981	751	d
## 982	404	d
## 983	354	b
## 984	681	b
## 985	611	b
## 986	580	b
## 987	365	a
## 990	759	a
## 991	787	a
## 992	11	a
## 993	546	d
## 994	432	c
## 995	154	a
## 996	542	b
## 997	197	c
## 998	891	d
## 999	58	a
## 1000	603	b
## 1001	771	a
## 1002	731	d
## 1003	873	b
## 1004	836	b
## 1006	219	b
## 1007	871	a
## 1008	151	a
## 1009	639	a
## 1010	796	b
## 1011	96	b
## 1012	205	d
## 1013	193	a
## 1015	706	a
## 1016	224	d
## 1017	957	a
## 1019	138	c
## 1020	773	d
## 1021	206	b
## 1022	362	b
## 1024	634	a
## 1025	307	b
## 1026	596	a
## 1027	499	d
## 1028	107	a
## 1029	503	b

## 1030	298	c
## 1031	382	a
## 1032	777	b
## 1033	622	c
## 1034	584	a
## 1035	65	a
## 1036	557	c
## 1037	762	a
## 1038	260	a
## 1039	467	b
## 1040	647	a
## 1041	958	a
## 1042	508	a
## 1044	667	a
## 1045	587	a
## 1046	710	a
## 1047	256	a
## 1048	403	a
## 1050	301	b
## 1051	446	a
## 1052	881	a
## 1053	486	b
## 1054	419	a
## 1055	753	a
## 1056	256	a
## 1057	38	b
## 1058	958	b
## 1060	748	c
## 1061	664	a
## 1062	573	a
## 1063	901	a
## 1065	706	a
## 1066	580	a
## 1067	501	a
## 1068	336	c
## 1069	387	d
## 1070	306	d
## 1071	462	c
## 1072	913	b
## 1073	83	b
## 1074	153	a
## 1075	505	a
## 1076	397	a
## 1077	958	c
## 1078	771	d
## 1079	960	a

## 1080	642	b
## 1081	94	a
## 1082	511	b
## 1083	238	a
## 1085	411	c
## 1086	616	c
## 1087	235	b
## 1088	890	d
## 1089	859	a
## 1090	860	a
## 1091	478	b
## 1092	228	b
## 1093	267	c
## 1094	201	a
## 1096	20	a
## 1097	348	a
## 1098	5	b
## 1099	306	a
## 1100	257	b
## 1101	456	b
## 1102	195	a
## 1103	805	a
## 1104	931	a
## 1105	971	b
## 1106	398	b
## 1107	100	a
## 1108	230	a
## 1109	398	a
## 1111	715	b
## 1112	681	a
## 1113	829	a
## 1114	369	a
## 1115	913	a
## 1116	552	a
## 1117	439	b
## 1118	409	b
## 1119	5	a
## 1120	672	b
## 1121	991	b
## 1122	487	c
## 1123	365	a
## 1124	564	a
## 1125	959	a
## 1126	898	a
## 1127	655	a
## 1128	146	c

## 1129	556	a
## 1130	852	b
## 1131	830	a
## 1132	888	b
## 1133	664	a
## 1134	284	b
## 1135	75	d
## 1136	657	a
## 1137	569	a
## 1138	20	a
## 1139	868	a
## 1141	721	b
## 1142	230	a
## 1143	100	a
## 1144	403	a
## 1145	721	d
## 1146	760	a
## 1147	837	a
## 1148	957	a
## 1149	159	c
## 1150	232	a
## 1151	723	a
## 1152	73	a
## 1154	689	a
## 1155	416	b
## 1156	680	a
## 1157	768	d
## 1158	926	a
## 1159	216	d
## 1160	708	b
## 1161	812	a
## 1162	592	a
## 1163	511	d
## 1164	800	d
## 1165	663	a
## 1166	447	a
## 1167	556	b
## 1168	11	b
## 1169	967	a
## 1170	984	a
## 1171	880	a
## 1172	600	a
## 1173	782	b
## 1174	414	a
## 1175	549	b
## 1176	76	a

## 1177	524	b
## 1178	560	d
## 1179	488	b
## 1180	958	c
## 1181	734	a
## 1182	672	a
## 1183	68	a
## 1184	628	a
## 1186	136	a
## 1187	124	b
## 1188	632	a
## 1189	693	c
## 1190	937	a
## 1191	187	b
## 1192	176	a
## 1193	884	a
## 1194	426	a
## 1195	917	b
## 1196	35	b
## 1197	923	c
## 1198	59	b
## 1199	142	a
## 1200	535	b
## 1201	587	a
## 1202	276	c
## 1203	278	d
## 1204	995	a
## 1205	253	a
## 1206	534	a
## 1207	409	d
## 1208	1	c
## 1209	297	c
## 1210	922	a
## 1211	124	b
## 1212	441	c
## 1216	322	a
## 1217	686	d
## 1219	841	b
## 1220	356	b
## 1221	977	b
## 1222	421	a
## 1223	892	c
## 1224	535	a
## 1225	443	a
## 1226	478	c
## 1227	954	c

##	1228	539	b
##	1229	33	a
##	1230	456	b
##	1231	745	a
##	1232	66	a
##	1233	323	d
##	1234	756	b
##	1235	969	b
##	1236	442	a
##	1237	493	a
##	1238	441	a
##	1239	913	a
##	1240	657	d
##	1241	13	d
##	1242	331	a
##	1243	182	a
##	1244	45	a
##	1246	631	a
##	1247	138	a
##	1248	977	a
##	1249	774	b
##	1250	563	b
##	1251	388	b
##	1252	644	a
##	1253	195	a
##	1254	741	b
##	1255	193	a
##	1257	615	b
##	1258	106	a
##	1259	347	a
##	1260	808	a
##	1261	992	b
##	1263	137	a
##	1264	926	c
##	1265	413	a
##	1266	193	b
##	1268	78	a
##	1269	78	a
##	1270	963	b
##	1271	875	c
##	1272	119	b
##	1273	500	a
##	1274	321	a
##	1276	707	a
##	1277	509	a
##	1279	828	a

## 1280	11	a
## 1281	507	a
## 1282	197	b
## 1283	593	a
## 1284	739	a
## 1285	568	a
## 1286	494	c
## 1287	528	a
## 1289	16	a
## 1290	906	a
## 1291	695	b
## 1292	970	a
## 1293	342	a
## 1294	213	a
## 1296	840	c
## 1297	899	b
## 1298	937	b
## 1299	490	a
## 1300	660	b
## 1301	636	c
## 1302	767	c
## 1304	364	b
## 1306	367	b
## 1308	957	b
## 1309	170	a
## 1310	113	a
## 1311	440	b
## 1312	405	d
## 1313	702	b
## 1314	710	a
## 1315	427	a
## 1316	606	b
## 1317	599	a
## 1318	532	b
## 1319	403	b
## 1320	290	b
## 1321	107	c
## 1322	540	a
## 1324	903	a
## 1325	825	a
## 1326	151	a
## 1327	787	a
## 1328	201	d
## 1329	621	a
## 1330	230	b
## 1331	6	a

##	1332	554	a
##	1333	791	b
##	1334	235	b
##	1335	699	b
##	1337	995	b
##	1338	900	d
##	1339	226	a
##	1340	983	d
##	1341	528	a
##	1342	480	a
##	1343	814	b
##	1344	64	b
##	1345	346	d
##	1346	2	c
##	1347	707	b
##	1348	155	b
##	1349	61	a
##	1350	420	a
##	1352	789	b
##	1353	8	c
##	1354	202	a
##	1355	327	d
##	1356	393	b
##	1357	306	a
##	1358	463	b
##	1359	48	b
##	1362	688	b
##	1363	150	b
##	1364	131	a
##	1365	292	a
##	1366	291	a
##	1367	937	b
##	1368	851	a
##	1369	64	a
##	1370	247	b
##	1371	334	a
##	1372	343	b
##	1373	684	a
##	1374	67	a
##	1375	146	a
##	1376	625	c
##	1377	37	c
##	1378	55	b
##	1379	931	c
##	1380	231	d
##	1381	220	a

## 1382	212	c
## 1383	187	c
## 1384	368	a
## 1385	522	c
## 1386	220	a
## 1387	962	a
## 1388	17	d
## 1389	724	a
## 1391	845	c
## 1392	737	a
## 1394	27	a
## 1395	881	a
## 1396	70	b
## 1397	923	a
## 1398	92	b
## 1400	407	a
## 1401	872	a
## 1402	806	a
## 1403	430	a
## 1404	950	d
## 1405	745	a
## 1406	512	b
## 1407	741	b
## 1408	252	a
## 1410	926	c
## 1411	151	c
## 1412	8	b
## 1413	416	a
## 1414	690	a
## 1415	923	c
## 1417	482	c
## 1418	240	b
## 1419	497	a
## 1420	252	a
## 1421	922	a
## 1422	853	a
## 1423	287	b
## 1424	808	a
## 1425	207	a
## 1426	339	a
## 1427	28	b
## 1428	366	a
## 1429	672	c
## 1430	82	b
## 1431	500	b
## 1432	278	a

## 1433	19	a
## 1434	394	a
## 1435	166	d
## 1436	772	a
## 1437	455	a
## 1439	885	b
## 1440	685	a
## 1441	368	b
## 1442	652	b
## 1444	959	a
## 1445	881	a
## 1446	615	a
## 1448	817	a
## 1449	269	c
## 1450	13	a
## 1451	654	c
## 1453	272	c
## 1454	517	a
## 1456	765	d
## 1457	916	b
## 1458	483	a
## 1459	889	a
## 1460	142	a
## 1461	629	a
## 1462	81	a
## 1463	442	a
## 1464	732	a
## 1466	637	a
## 1467	115	b
## 1469	290	c
## 1470	946	a
## 1471	358	a
## 1472	331	c
## 1473	235	c
## 1474	950	a
## 1476	83	a
## 1477	958	a
## 1478	813	c
## 1479	381	b
## 1480	102	a
## 1481	639	a
## 1482	492	a
## 1483	407	a
## 1484	612	a
## 1485	375	a
## 1486	193	c

## 1487	700	b
## 1488	987	c
## 1489	516	a
## 1490	937	d
## 1492	425	a
## 1494	118	b
## 1495	353	c
## 1496	983	a
## 1497	57	a
## 1498	934	c
## 1499	775	c
## 1500	48	c
## 1501	412	c
## 1502	604	b
## 1503	914	c
## 1504	387	b
## 1505	275	a
## 1506	948	a
## 1507	230	a
## 1508	874	a
## 1509	43	a
## 1510	249	a
## 1511	275	a
## 1513	66	a
## 1514	907	a
## 1516	411	a
## 1517	223	b
## 1518	115	a
## 1519	636	a
## 1520	460	b
## 1521	921	b
## 1522	201	a
## 1523	130	a
## 1524	829	a
## 1525	183	a
## 1526	863	b
## 1527	490	a
## 1528	452	b
## 1529	215	a
## 1530	184	a
## 1531	389	c
## 1532	607	a
## 1533	693	a
## 1534	531	b
## 1536	505	a
## 1537	746	b

##	1538	870	b
##	1539	684	d
##	1540	730	a
##	1541	844	c
##	1542	636	a
##	1543	904	b
##	1544	714	b
##	1546	264	c
##	1547	699	a
##	1548	14	d
##	1549	940	b
##	1550	186	a
##	1551	28	a
##	1552	305	a
##	1553	286	c
##	1554	315	d
##	1555	782	b
##	1556	914	a
##	1558	885	b
##	1559	564	a
##	1560	641	c
##	1561	270	a
##	1562	782	a
##	1563	656	c
##	1564	916	a
##	1568	678	d
##	1569	112	a
##	1570	567	a
##	1571	205	a
##	1572	510	d
##	1573	745	a
##	1574	627	b
##	1575	610	b
##	1577	779	b
##	1578	130	a
##	1579	551	d
##	1580	637	b
##	1581	844	b
##	1582	466	b
##	1583	756	a
##	1584	444	d
##	1585	695	a
##	1586	645	b
##	1588	200	a
##	1589	703	b
##	1590	646	a

##	1591	561	a
##	1592	322	b
##	1593	682	a
##	1595	352	b
##	1596	724	a
##	1597	332	c
##	1598	236	b
##	1599	81	b
##	1600	978	c
##	1601	132	a
##	1603	721	c
##	1604	670	b
##	1605	612	c
##	1606	353	b
##	1608	666	a
##	1609	508	b
##	1611	45	b
##	1613	749	c
##	1614	339	b
##	1615	519	d
##	1616	918	a
##	1620	732	b
##	1621	491	b
##	1622	386	c
##	1623	677	c
##	1624	677	a
##	1625	141	a
##	1626	561	a
##	1627	280	b
##	1628	979	b
##	1629	286	a
##	1630	548	b
##	1631	322	d
##	1632	176	a
##	1633	140	a
##	1634	694	b
##	1635	528	c
##	1636	326	a
##	1638	433	b
##	1639	328	b
##	1640	106	b
##	1641	384	a
##	1643	945	c
##	1644	645	a
##	1645	687	d
##	1646	607	b

## 1647	443	a
## 1648	518	a
## 1649	585	a
## 1650	854	a
## 1651	123	b
## 1652	815	a
## 1653	312	a
## 1654	611	b
## 1655	137	a
## 1656	33	a
## 1657	712	b
## 1658	80	a
## 1660	399	a
## 1662	916	b
## 1663	729	a
## 1664	612	b
## 1665	692	a
## 1666	809	a
## 1668	950	a
## 1669	58	b
## 1670	77	a
## 1671	309	a
## 1672	857	a
## 1673	444	a
## 1674	540	b
## 1675	655	a
## 1677	889	b
## 1678	851	b
## 1679	60	b
## 1680	925	a
## 1681	708	d
## 1682	23	a
## 1683	624	b
## 1684	907	b
## 1685	188	a
## 1686	616	b
## 1688	885	a
## 1689	771	a
## 1691	741	a
## 1692	901	b
## 1695	875	a
## 1696	453	a
## 1697	486	a
## 1698	706	a
## 1699	419	a
## 1700	262	a

## 1701	117	b
## 1702	477	a
## 1703	954	d
## 1704	53	a
## 1705	732	b
## 1706	122	a
## 1707	605	a
## 1708	129	a
## 1709	678	a
## 1710	256	a
## 1711	37	a
## 1712	133	a
## 1713	691	b
## 1714	418	d
## 1715	956	a
## 1717	49	a
## 1718	459	a
## 1719	785	a
## 1720	793	a
## 1721	239	c
## 1722	7	b
## 1723	339	a
## 1724	506	d
## 1725	966	a
## 1726	691	a
## 1727	684	a
## 1728	356	c
## 1730	955	b
## 1731	824	c
## 1732	967	a
## 1733	223	b
## 1735	891	a
## 1736	884	a
## 1737	732	a
## 1739	154	b
## 1740	294	b
## 1742	514	b
## 1743	247	a
## 1744	125	b
## 1745	67	a
## 1746	646	b
## 1747	947	a
## 1748	110	d
## 1749	206	a
## 1750	79	a
## 1751	226	a

## 1752	554	a
## 1753	197	a
## 1754	700	a
## 1755	958	b
## 1756	778	a
## 1757	28	b
## 1758	621	a
## 1759	457	a
## 1760	642	a
## 1761	907	b
## 1762	15	b
## 1763	23	a
## 1764	508	d
## 1766	229	d
## 1767	69	a
## 1768	407	a
## 1770	315	a
## 1771	561	a
## 1772	333	a
## 1774	607	b
## 1775	654	a
## 1776	883	a
## 1777	728	b
## 1778	174	a
## 1779	473	a
## 1780	21	b
## 1782	493	a
## 1784	6	d
## 1786	696	c
## 1789	127	a
## 1790	338	a
## 1791	710	a
## 1792	840	c
## 1793	155	b
## 1794	312	c
## 1795	294	b
## 1796	609	b
## 1797	892	d
## 1798	702	a
## 1799	247	c
## 1800	383	a
## 1801	779	a
## 1802	452	a
## 1803	224	a
## 1804	944	a
## 1807	364	a

## 1808	978	a
## 1809	973	d
## 1810	3	a
## 1811	212	b
## 1812	658	c
## 1813	88	a
## 1816	881	a
## 1817	885	d
## 1819	230	c
## 1820	677	a
## 1821	570	b
## 1823	731	c
## 1824	194	a
## 1825	362	b
## 1827	767	a
## 1828	76	b
## 1829	203	b
## 1830	567	a
## 1832	142	a
## 1834	102	b
## 1835	398	a
## 1836	917	b
## 1837	217	a
## 1838	153	a
## 1839	257	c
## 1840	27	b
## 1842	679	d
## 1843	54	a
## 1844	794	a
## 1845	349	b
## 1846	456	b
## 1847	741	b
## 1848	456	b
## 1850	966	a
## 1851	42	a
## 1852	843	a
## 1853	367	a
## 1854	530	a
## 1855	236	a
## 1857	526	a
## 1858	326	d
## 1859	136	c
## 1860	978	c
## 1861	665	a
## 1862	592	a
## 1863	191	c

##	1864	822	a
##	1865	726	a
##	1867	697	b
##	1868	969	a
##	1869	887	b
##	1870	95	a
##	1871	732	a
##	1872	807	b
##	1873	582	a
##	1874	820	a
##	1875	878	a
##	1876	614	a
##	1877	394	a
##	1878	123	a
##	1879	656	a
##	1880	336	d
##	1881	533	a
##	1882	676	a
##	1883	312	a
##	1884	440	a
##	1885	10	a
##	1886	300	b
##	1887	431	b
##	1888	310	c
##	1889	166	a
##	1890	922	a
##	1891	275	b
##	1892	590	a
##	1893	425	d
##	1894	443	a
##	1895	479	b
##	1896	386	a
##	1897	901	a
##	1898	539	a
##	1899	160	a
##	1900	596	a
##	1901	997	c
##	1902	111	b
##	1903	3	a
##	1904	617	c
##	1905	442	a
##	1906	535	a
##	1907	563	a
##	1908	478	c
##	1909	529	d
##	1910	621	a

##	1911	157	b
##	1912	817	c
##	1913	164	c
##	1914	281	c
##	1915	776	a
##	1916	357	c
##	1917	959	a
##	1919	530	a
##	1921	937	b
##	1923	57	c
##	1924	781	b
##	1925	74	b
##	1926	544	a
##	1927	212	a
##	1928	890	c
##	1929	593	a
##	1930	891	a
##	1931	309	a
##	1932	56	a
##	1933	486	a
##	1934	444	c
##	1935	845	a
##	1936	256	a
##	1937	921	c
##	1938	687	a
##	1939	23	a
##	1940	104	a
##	1942	568	a
##	1943	456	a
##	1944	619	a
##	1945	54	a
##	1946	313	c
##	1947	764	a
##	1948	326	a
##	1949	383	a
##	1950	971	b
##	1951	492	a
##	1952	170	b
##	1953	883	b
##	1954	750	a
##	1955	853	a
##	1956	699	a
##	1957	170	a
##	1958	617	a
##	1959	369	a
##	1960	354	a

## 1961	114	b
## 1963	881	b
## 1965	776	c
## 1966	940	c
## 1967	38	a
## 1968	614	b
## 1969	472	a
## 1970	168	a
## 1971	443	d
## 1973	816	b
## 1974	756	a
## 1975	256	a
## 1976	71	b
## 1977	541	b
## 1978	204	d
## 1979	495	a
## 1980	179	a
## 1981	105	a
## 1982	479	a
## 1983	815	a
## 1984	955	a
## 1985	936	a
## 1986	154	b
## 1988	265	d
## 1989	149	a
## 1991	575	a
## 1992	322	b
## 1993	253	c
## 1994	167	d
## 1995	735	a
## 1996	832	a
## 1998	462	a
## 2000	539	a
## 2001	140	b
## 2002	4	b
## 2003	371	c
## 2006	953	b
## 2007	887	a
## 2008	777	c
## 2009	87	a
## 2010	904	a
## 2012	959	d
## 2013	831	a
## 2014	368	b
## 2016	832	a
## 2017	172	a

##	2018	729	a
##	2021	863	a
##	2022	817	b
##	2023	190	a
##	2024	657	d
##	2025	699	a
##	2026	392	b
##	2027	647	b
##	2029	280	b
##	2030	357	b
##	2031	607	a
##	2032	9	a
##	2033	124	c
##	2034	920	d
##	2035	24	d
##	2036	308	b
##	2037	1	a
##	2038	792	a
##	2039	196	c
##	2040	7	a
##	2042	48	a
##	2043	401	b
##	2045	563	b
##	2046	629	d
##	2047	825	a
##	2048	990	a
##	2049	576	a
##	2050	888	b
##	2051	724	c
##	2052	131	b
##	2053	769	b
##	2054	295	a
##	2055	955	a
##	2056	715	a
##	2057	567	b
##	2058	680	a
##	2059	237	d
##	2061	253	a
##	2063	149	b
##	2064	655	a
##	2065	592	a
##	2066	975	a
##	2068	524	a
##	2069	781	c
##	2070	274	d
##	2071	229	a

##	2073	178	a
##	2074	395	a
##	2075	574	c
##	2076	861	d
##	2078	906	a
##	2079	173	a
##	2080	939	a
##	2081	499	c
##	2082	202	b
##	2083	667	b
##	2084	6	a
##	2085	572	b
##	2086	70	d
##	2087	459	a
##	2088	231	a
##	2089	238	b
##	2091	640	b
##	2092	527	a
##	2093	869	a
##	2094	251	a
##	2095	489	b
##	2096	491	a
##	2097	411	d
##	2098	604	a
##	2099	62	b
##	2100	406	a
##	2101	255	a
##	2102	341	a
##	2103	144	c
##	2105	233	a
##	2106	817	b
##	2107	654	a
##	2108	456	c
##	2109	971	a
##	2110	232	c
##	2111	997	b
##	2112	921	c
##	2113	168	a
##	2114	651	a
##	2115	502	b
##	2116	628	d
##	2117	702	b
##	2119	888	a
##	2120	674	a
##	2121	651	a
##	2122	429	a

##	2123	932	c
##	2124	708	a
##	2126	292	a
##	2127	290	a
##	2128	742	a
##	2130	346	a
##	2131	247	a
##	2132	610	a
##	2133	653	a
##	2134	878	b
##	2135	666	c
##	2137	129	a
##	2138	404	b
##	2139	340	a
##	2140	949	a
##	2141	492	b
##	2142	849	b
##	2143	507	a
##	2144	480	a
##	2145	480	a
##	2146	949	a
##	2147	682	c
##	2148	167	a
##	2151	869	a
##	2152	153	a
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##	2154	12	a
##	2156	949	b
##	2157	213	a
##	2158	368	b
##	2159	356	b
##	2160	692	a
##	2161	696	a
##	2162	29	a
##	2163	194	b
##	2164	66	a
##	2165	804	a
##	2166	830	c
##	2167	687	a
##	2168	58	a
##	2169	889	b
##	2170	787	c
##	2171	719	c
##	2172	850	c
##	2173	520	a
##	2174	288	b

##	2175	35	b
##	2176	617	a
##	2177	140	a
##	2178	162	a
##	2179	267	b
##	2181	546	b
##	2182	778	a
##	2184	463	b
##	2185	901	b
##	2186	848	b
##	2187	764	b
##	2188	307	a
##	2189	12	d
##	2190	65	a
##	2191	990	a
##	2192	948	a
##	2193	55	a
##	2194	836	a
##	2195	53	a
##	2196	693	d
##	2197	574	a
##	2198	595	a
##	2199	962	a
##	2200	835	a
##	2201	973	a
##	2203	200	b
##	2205	325	a
##	2206	532	a
##	2207	106	b
##	2209	272	a
##	2210	346	d
##	2211	347	d
##	2212	701	a
##	2213	662	a
##	2214	34	a
##	2215	605	a
##	2216	874	b
##	2217	857	b
##	2218	894	a
##	2219	584	a
##	2220	425	b
##	2222	618	a
##	2223	56	b
##	2224	392	a
##	2225	984	a
##	2228	805	c

##	2229	486	b
##	2230	420	b
##	2231	562	a
##	2232	690	a
##	2233	966	b
##	2234	434	b
##	2235	488	a
##	2236	6	b
##	2238	751	a
##	2240	679	b
##	2241	877	b
##	2242	492	a
##	2243	731	a
##	2244	15	b
##	2245	327	a
##	2246	83	c
##	2247	421	a
##	2248	699	a
##	2249	297	a
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##	2252	709	b
##	2253	168	b
##	2254	865	a
##	2255	75	a
##	2257	923	a
##	2258	971	a
##	2259	629	b
##	2260	78	a
##	2261	280	a
##	2262	927	a
##	2263	620	c
##	2265	731	d
##	2266	511	c
##	2267	75	a
##	2268	466	a
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##	2270	563	b
##	2271	22	c
##	2272	430	d
##	2273	702	a
##	2274	245	b
##	2275	85	b
##	2276	696	a
##	2277	914	a
##	2278	79	a
##	2279	566	c

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##	2281	561	a
##	2282	473	a
##	2283	212	b
##	2284	569	a
##	2285	60	a
##	2286	41	a
##	2287	936	a
##	2288	328	b
##	2289	707	d
##	2291	68	a
##	2292	146	b
##	2293	264	b
##	2294	377	c
##	2296	932	a
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##	2298	223	b
##	2299	425	a
##	2300	536	c
##	2302	406	d
##	2303	870	a
##	2304	872	c
##	2306	176	a
##	2307	721	d
##	2308	793	a
##	2309	225	a
##	2310	219	a
##	2311	607	a
##	2312	787	d
##	2313	436	a
##	2315	919	b
##	2316	123	a
##	2317	306	d
##	2318	686	a
##	2319	168	a
##	2320	298	a
##	2321	584	a
##	2324	475	c
##	2325	194	b
##	2326	685	b
##	2327	768	a
##	2329	408	a
##	2331	555	c
##	2332	821	a
##	2333	886	c
##	2335	758	a

##	2336	595	a
##	2337	242	a
##	2338	917	a
##	2339	991	d
##	2340	831	a
##	2341	458	b
##	2342	72	c
##	2343	323	b
##	2344	871	b
##	2345	20	a
##	2346	658	c
##	2347	346	a
##	2348	725	b
##	2349	621	c
##	2351	253	a
##	2352	44	b
##	2353	611	b
##	2354	17	a
##	2355	921	a
##	2356	564	c
##	2357	634	a
##	2358	970	a
##	2359	219	c
##	2360	597	a
##	2361	127	a
##	2362	769	a
##	2363	846	b
##	2364	152	c
##	2365	129	a
##	2366	274	a
##	2367	597	b
##	2368	406	a
##	2370	226	a
##	2371	325	a
##	2373	733	b
##	2374	843	b
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##	2376	153	a
##	2377	913	a
##	2378	795	a
##	2379	227	a
##	2380	362	a
##	2381	211	b
##	2382	918	c
##	2383	176	d
##	2384	726	d

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##	2386	25	d
##	2388	288	d
##	2389	830	a
##	2390	614	c
##	2391	878	b
##	2392	560	b
##	2393	18	b
##	2394	347	b
##	2395	703	a
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##	2397	976	a
##	2398	737	a
##	2399	178	d
##	2400	203	a
##	2401	334	c
##	2402	895	b
##	2403	610	d
##	2405	591	d
##	2406	909	d
##	2407	785	d
##	2408	720	b
##	2409	509	b
##	2410	408	b
##	2411	229	b
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##	2413	773	a
##	2414	144	b
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##	2417	580	a
##	2418	433	b
##	2419	916	a
##	2420	537	b
##	2421	676	a
##	2422	76	b
##	2423	438	b
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##	2425	809	a
##	2427	604	d
##	2428	286	a
##	2429	599	a
##	2430	934	a
##	2432	798	b
##	2433	812	d
##	2434	122	a

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##	2438	486	c
##	2439	800	b
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##	2441	463	a
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##	2447	874	a
##	2448	545	b
##	2449	868	a
##	2450	957	a
##	2451	831	b
##	2452	86	a
##	2453	912	b
##	2454	891	a
##	2455	107	d
##	2456	60	c
##	2457	254	b
##	2458	64	a
##	2459	687	a
##	2460	16	a
##	2461	402	c
##	2462	542	b
##	2463	406	a
##	2464	842	a
##	2465	497	a
##	2466	309	a
##	2467	58	a
##	2468	520	d
##	2470	24	a
##	2471	73	b
##	2472	897	c
##	2473	339	a
##	2474	781	c
##	2476	667	b
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##	2478	628	c
##	2479	72	a
##	2480	461	a
##	2481	377	b
##	2482	758	a
##	2483	872	b
##	2487	462	b

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##	2489	23	c
##	2491	163	a
##	2492	52	b
##	2493	75	a
##	2494	167	b
##	2495	712	b
##	2498	993	a
##	2499	537	a
##	2500	85	a
##	2501	848	a
##	2502	667	b
##	2503	985	d
##	2504	216	a
##	2505	343	a
##	2506	557	b
##	2507	219	b
##	2508	17	a
##	2509	43	c
##	2510	347	a
##	2512	685	a
##	2513	39	d
##	2514	848	a
##	2515	111	b
##	2516	873	d
##	2517	360	a
##	2518	505	a
##	2519	818	b
##	2520	106	a
##	2521	118	b
##	2522	560	b
##	2523	181	b
##	2524	261	b
##	2525	10	b
##	2526	791	b
##	2527	962	a
##	2528	206	b
##	2529	124	c
##	2530	105	a
##	2531	880	c
##	2532	382	d
##	2533	793	c
##	2534	26	c
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##	2536	617	b
##	2538	428	a

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##	2540	541	a
##	2541	696	a
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##	2543	735	a
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##	2545	576	a
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##	2554	262	a
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##	2556	125	b
##	2557	598	b
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##	2559	116	a
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##	2574	433	d
##	2578	765	d
##	2579	458	b
##	2580	508	a
##	2581	894	b
##	2582	597	a
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##	2585	58	d
##	2586	919	a
##	2587	732	b
##	2588	968	d
##	2589	113	a
##	2590	20	a
##	2591	979	a

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##	2594	915	d
##	2595	374	a
##	2596	903	a
##	2597	585	a
##	2598	23	a
##	2599	124	a
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##	2603	861	a
##	2604	817	a
##	2605	250	a
##	2606	473	a
##	2607	216	a
##	2608	305	c
##	2609	592	d
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##	2625	577	b
##	2626	331	c
##	2627	855	b
##	2628	900	a
##	2629	695	b
##	2630	392	b
##	2631	459	a
##	2632	390	c
##	2634	951	a
##	2636	627	a
##	2637	997	d
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##	2639	784	a
##	2640	563	a
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##	2642	659	b
##	2643	915	a

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##	2646	947	d
##	2647	171	b
##	2648	694	d
##	2649	26	c
##	2650	502	a
##	2651	838	a
##	2652	509	a
##	2653	866	c
##	2655	955	a
##	2656	662	a
##	2657	231	a
##	2659	159	c
##	2660	870	c
##	2662	707	b
##	2663	414	a
##	2664	651	a
##	2665	56	d
##	2666	190	a
##	2667	983	b
##	2668	484	b
##	2669	543	b
##	2670	408	a
##	2671	384	a
##	2674	964	b
##	2675	198	c
##	2676	98	c
##	2677	243	a
##	2679	788	b
##	2680	254	a
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##	2682	786	b
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##	2685	325	a
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##	2688	314	b
##	2689	178	a
##	2690	157	c
##	2691	922	d
##	2692	819	c
##	2693	877	a
##	2694	270	d
##	2695	214	d
##	2696	427	a
##	2697	71	c
##	2698	235	b

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##	2701	520	b
##	2702	86	a
##	2703	97	b
##	2704	569	d
##	2706	385	b
##	2707	198	d
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##	2709	681	a
##	2710	907	c
##	2711	253	a
##	2713	84	c
##	2715	270	b
##	2716	744	b
##	2717	62	a
##	2718	474	d
##	2719	191	a
##	2720	398	a
##	2721	203	b
##	2722	589	c
##	2723	369	b
##	2724	64	a
##	2725	613	a
##	2726	838	a
##	2727	422	a
##	2728	228	a
##	2729	228	d
##	2730	174	a
##	2731	414	b
##	2732	807	b
##	2733	441	a
##	2734	412	a
##	2735	282	a
##	2737	305	a
##	2738	453	a
##	2739	952	a
##	2740	763	a
##	2741	819	a
##	2742	190	b
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##	2744	810	c
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##	2748	667	a

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##	2752	111	a
##	2753	413	a
##	2754	239	a
##	2755	437	b
##	2757	457	d
##	2758	358	b
##	2760	458	a
##	2761	271	c
##	2762	474	b
##	2763	51	c
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##	2765	880	a
##	2766	31	b
##	2767	962	d
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##	2771	311	c
##	2772	902	a
##	2773	671	b
##	2774	72	a
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##	2778	415	a
##	2779	799	a
##	2780	292	a
##	2781	956	d
##	2782	660	a
##	2783	877	a
##	2784	614	a
##	2785	963	a
##	2786	980	a
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##	2792	548	d
##	2794	76	a
##	2795	706	a
##	2796	534	b
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## 2801	151	c
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## 2803	991	a
## 2804	123	a
## 2806	702	b
## 2807	897	a
## 2808	18	c
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## 2816	330	b
## 2817	847	c
## 2818	155	c
## 2819	716	a
## 2820	930	a
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## 2822	937	c
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## 2825	702	b
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## 2839	154	b
## 2840	698	a
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## 2846	106	b
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## 2849	373	a
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##	2885	930	a
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##	2902	718	a
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##	2941	358	b
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##	2961	21	b

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## 3167	132	b
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## 3188	871	b
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##	3844	504	a
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##	3846	9	b
##	3847	263	a
##	3848	805	a
##	3850	822	c
##	3851	381	d
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##	3856	249	b
##	3857	646	a
##	3858	881	b
##	3859	175	c
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##	3862	983	a
##	3864	936	c
##	3865	136	a
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##	3886	831	a
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##	3890	134	a
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##	3894	222	b
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##	3918	128	c
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##	3920	101	a
##	3921	78	a
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##	3926	9	a
##	3927	46	a
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##	3931	373	a

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##	3945	85	c
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##	4007	162	c
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##	4029	365	a
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##	4037	116	a

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##	4069	220	b
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##	4077	940	a
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##	4097	186	a
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##	4099	424	d
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##	4104	266	a
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## 4150	804	b
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## 4157	121	a
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## 4160	976	b
## 4161	539	d
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## 4163	258	a
## 4164	961	b
## 4165	271	b
## 4166	138	a
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## 4168	879	a
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## 4170	559	a
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## 4172	878	b
## 4175	994	a
## 4176	891	a
## 4178	370	a
## 4179	503	d
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## 4188	42	d
## 4190	309	a
## 4192	160	a
## 4193	138	a
## 4194	557	b
## 4195	99	a
## 4197	541	b
## 4198	137	d
## 4199	893	d

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##	4223	627	d
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##	4229	776	b
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##	4253	278	d

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## 4263	392	a
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## 4268	557	b
## 4269	562	a
## 4270	182	c
## 4271	359	c
## 4274	928	a
## 4275	789	a
## 4276	525	d
## 4277	801	d
## 4278	385	a
## 4279	84	c
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## 4282	301	d
## 4283	436	c
## 4284	56	a
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## 4287	670	a
## 4288	275	a
## 4289	180	a
## 4290	460	d
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## 4295	836	c
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## 4298	593	a
## 4299	954	a
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## 4304	291	b
## 4305	799	d
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## 4370	539	c
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## 4373	992	c
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## 4377	262	a
## 4378	97	a
## 4379	761	a
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## 4388	627	b
## 4389	807	a
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## 4392	383	a
## 4393	661	a
## 4394	694	c
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## 4396	148	a
## 4397	707	a
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## 4400	665	a
## 4401	626	a
## 4402	290	a
## 4403	377	a
## 4404	85	a
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## 4407	603	b
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##	4503	936	d
##	4505	220	c
##	4506	608	a
##	4507	421	b
##	4508	267	a
##	4509	833	b
##	4510	998	a
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##	4512	847	a
##	4513	328	a
##	4514	142	a

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##	4521	858	b
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##	4524	671	d
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##	4529	258	a
##	4530	565	a
##	4531	748	a
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## 4577	70	c
## 4578	969	a
## 4579	755	b
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## 4581	885	a
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## 4583	330	a
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## 4585	992	b
## 4586	825	d
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## 4590	759	a
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## 4597	741	a
## 4598	668	a
## 4599	566	d
## 4600	840	d
## 4601	900	c
## 4602	825	d
## 4603	509	b
## 4604	672	d
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## 4606	563	a
## 4607	611	d
## 4609	962	c
## 4610	885	d
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## 4612	113	a
## 4613	272	b
## 4614	86	b
## 4615	818	a
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## 4618	564	b

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## 4674	781	a
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## 4692	127	c
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## 4697	961	a
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## 4714	163	a
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## 4717	213	b
## 4718	618	a
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## 4727	185	a
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## 4758	269	a
## 4759	650	b
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## 4762	471	a
## 4763	667	a
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## 4773	297	a
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## 4775	12	d
## 4777	149	a
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## 4781	113	d
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## 4796	889	b
## 4797	910	d
## 4798	805	a
## 4799	652	b
## 4800	819	a
## 4801	687	a
## 4803	855	d
## 4804	734	a
## 4805	616	a
## 4806	462	a
## 4807	742	a
## 4808	67	c
## 4809	650	a
## 4811	524	b
## 4812	480	a
## 4813	23	c
## 4814	578	a
## 4815	149	a
## 4816	605	b
## 4817	26	b
## 4819	107	a
## 4820	308	c
## 4821	714	b
## 4822	433	a
## 4823	880	a
## 4824	585	a
## 4825	991	a
## 4826	641	d
## 4827	943	a
## 4829	608	b
## 4830	174	a

##	4831	699	a
##	4832	761	d
##	4833	276	a
##	4834	307	b
##	4835	596	a
##	4836	690	b
##	4837	641	a
##	4838	416	a
##	4839	473	b
##	4840	267	a
##	4841	735	a
##	4842	634	c
##	4843	596	b
##	4844	711	b
##	4845	242	b
##	4846	799	a
##	4847	422	b
##	4848	802	a
##	4849	254	d
##	4850	415	b
##	4851	297	a
##	4852	861	b
##	4853	150	b
##	4854	798	c
##	4855	31	c
##	4856	708	b
##	4857	805	a
##	4858	357	a
##	4859	384	b
##	4861	413	a
##	4862	140	a
##	4864	519	a
##	4865	593	a
##	4866	582	b
##	4867	827	c
##	4868	519	a
##	4869	706	b
##	4870	71	b
##	4871	794	a
##	4872	340	d
##	4873	724	b
##	4874	906	b
##	4875	244	b
##	4876	143	a
##	4877	503	b
##	4878	61	d

## 4880	23	a
## 4881	150	a
## 4882	580	a
## 4883	895	a
## 4884	644	a
## 4885	822	a
## 4886	939	a
## 4887	596	c
## 4888	811	c
## 4889	772	c
## 4890	342	b
## 4891	791	d
## 4892	563	a
## 4893	606	a
## 4895	842	a
## 4896	810	d
## 4897	165	b
## 4898	45	c
## 4899	9	b
## 4900	707	a
## 4901	777	b
## 4902	784	a
## 4903	875	a
## 4904	330	c
## 4905	87	c
## 4906	331	a
## 4907	146	a
## 4908	573	a
## 4909	684	a
## 4910	734	a
## 4911	54	c
## 4912	528	a
## 4914	91	b
## 4916	712	a
## 4917	949	a
## 4918	499	b
## 4920	813	a
## 4921	708	a
## 4922	269	c
## 4923	580	a
## 4924	150	c
## 4925	135	b
## 4926	500	a
## 4927	10	a
## 4928	353	a
## 4929	123	b

##	4930	625	b
##	4931	111	d
##	4932	370	c
##	4933	107	a
##	4934	962	a
##	4935	711	a
##	4936	773	a
##	4937	141	c
##	4938	521	a
##	4939	889	a
##	4940	798	b
##	4942	967	a
##	4943	793	a
##	4944	31	a
##	4945	898	a
##	4946	188	a
##	4947	32	a
##	4948	902	d
##	4950	323	a
##	4951	779	c
##	4952	830	c
##	4953	940	a
##	4954	802	a
##	4955	258	b
##	4956	503	a
##	4957	166	a
##	4958	467	a
##	4959	584	a
##	4960	192	a
##	4961	916	a
##	4962	490	b
##	4964	486	a
##	4965	668	b
##	4966	647	a
##	4967	256	a
##	4969	931	a
##	4970	548	a
##	4971	938	b
##	4972	825	a
##	4973	283	a
##	4976	651	a
##	4977	296	a
##	4978	986	a
##	4979	896	b
##	4980	629	a
##	4981	766	a

```
## 4982      738      a
## 4983      314      a
## 4984      828      b
## 4985       43      a
## 4987      571      a
## 4990      775      a
## 4993      223      a
## 4995      509      b
## 4997      711      a
## 4998      409      a
## 4999      931      b
## 5000      855      a
```

```
# processing initial data
# we need to be sure that product's names are unique
df$product <- paste0("#", df$product, "#")

prod.matrix <- df %>%
  # removing duplicated products from each order (exclude the effect of quantity)
  group_by(orderId, product) %>%
  arrange(product) %>%
  unique() %>%
  # combining products to cart and calculating number of products
  group_by(orderId) %>%
  summarise(cart = paste(product, collapse = ";"),
            prod.num = n()) %>%
  # calculating number of carts
  group_by(cart, prod.num) %>%
  summarise(num = n()) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'cart'. You can override using the `.groups` argument.
head(prod.matrix)
```

```
## # A tibble: 6 x 3
##   cart          prod.num  num
##   <chr>          <int> <int>
## 1 #a#             1    115
## 2 #a#;#b#         2    248
## 3 #a#;#b#;#c#     3    172
## 4 #a#;#b#;#c#;#d# 4     78
## 5 #a#;#b#;#d#     3    119
## 6 #a#;#c#         2     98
```

```
# calculating total number of orders/carts
tot <- sum(prod.matrix$num)
```

```

# splitting orders for sets with 1 product and more than 1 product
one.prod <- prod.matrix %>% filter(prod.num == 1)

sev.prod <- prod.matrix %>%
  filter(prod.num > 1) %>%
  arrange(desc(prod.num))

# defining parameters for pie chart
iniR <- 0.2 # initial radius
cols <- c(
  "ffffff",
  "fec44f",
  "fc9272",
  "a1d99b",
  "fee0d2",
  "2ca25f",
  "8856a7",
  "43a2ca",
  "fdbb84",
  "e34a33",
  "a6bddb",
  "dd1c77",
  "ffeda0",
  "756bb1"
)
prod <- df %>%
  select(product) %>%
  arrange(product) %>%
  unique()
prod <- c('NO', c(prod$product))
colors <- as.list(setNames(cols[c(1:(length(prod)))], prod))

# 0 circle: blank
pie(
  1,
  radius = iniR,
  init.angle = 90,
  col = c('white'),
  border = NA,
  labels = ''
)

# drawing circles from last to 2nd
for (i in length(prod):2) {
  p <- grep(prod[i], sev.prod$cart)

```

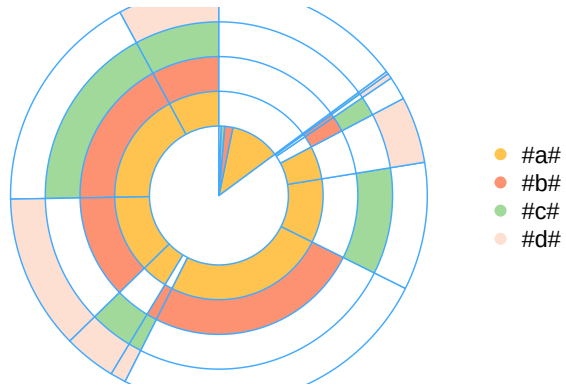
```

col <- rep('NO', times = nrow(sev.prod))
col[p] <- prod[i]
floating.pie(
  0,
  0,
  c(sev.prod$num, tot - sum(sev.prod$num)),
  radius = (1 + i) * iniR,
  startpos = pi / 2,
  col = as.character(colors [c(col, 'NO')]),
  border = "#44aaff"
)
}

# 1 circle: orders with 1 product
floating.pie(
  0,
  0,
  c(tot - sum(one.prod$num), one.prod$num),
  radius = 2 * iniR,
  startpos = pi / 2,
  col = as.character(colors [c('NO', one.prod$cart)]),
  border = "#44aaff"
)

# legend
legend(
  1.5,
  2 * iniR,
  gsub("_", " ", names(colors)[-1]),
  col = as.character(colors [-1]),
  pch = 19,
  bty = 'n',
  ncol = 1
)

```



```
# creating a table with the stats
stat.tab <- prod.matrix %>%
  select(-prod.num) %>%
  mutate(share = num / tot) %>%
  arrange(desc(num))

library(scales)
stat.tab$share <-
  percent(stat.tab$share) # converting values to percents

# adding a table with the stats
# addtable2plot(
#   -2.5,
#   -1.5,
#   stat.tab,
#   bty = "n",
#   display.rownames = FALSE,
#   hlines = FALSE,
#   vlines = FALSE,
#   title = "The stats"
# )
```


15.5.2 Sankey Diagram

```
# loading libraries
library(googleVis)
library(dplyr)
library(reshape2)

# creating an example of orders
set.seed(15)
df <- data.frame(
  orderId = c(1:1000),
  clientId = sample(c(1:300), 1000, replace = TRUE),
  prod1 = sample(
    c('NULL', 'a'),
    1000,
    replace = TRUE,
    prob = c(0.15, 0.5)
  ),
  prod2 = sample(
    c('NULL', 'b'),
    1000,
    replace = TRUE,
    prob = c(0.15, 0.3)
  ),
  prod3 = sample(
    c('NULL', 'c'),
    1000,
    replace = TRUE,
    prob = c(0.15, 0.2)
  )
)

# combining products
df$cart <- paste(df$prod1, df$prod2, df$prod3, sep = ';')
df$cart <- gsub('NULL;|;NULL', '', df$cart)
df <- df[df$cart != 'NULL',]

df <- df %>%
  select(orderId, clientId, cart) %>%
  arrange(clientId, orderId, cart)

head(df)
```

```
##   orderId clientId  cart
## 1    181         1     b
## 2    282         1     a
```

```
## 3      748      1 a;b;c
## 4       27      2  a;b
## 5      209      2  b;c
## 6      244      2 a;b;c
```

```
orders <- df %>%
  group_by(clientId) %>%
  mutate(n.ord = paste('ord', c(1:n()), sep = '')) %>%
  ungroup()
```

```
head(orders)
```

```
## # A tibble: 6 x 4
##   orderId clientId cart  n.ord
##   <int>    <int> <chr> <chr>
## 1     181        1 b     ord1
## 2     282        1 a     ord2
## 3     748        1 a;b;c ord3
## 4      27        2 a;b   ord1
## 5     209        2 b;c   ord2
## 6     244        2 a;b;c ord3
```

```
orders <-
  dcast(orders,
        clientId ~ n.ord,
        value.var = 'cart',
        fun.aggregate = NULL)
```

```
# choose a number of carts/orders in the sequence we want to analyze
```

```
orders <- orders %>%
  select(ord1, ord2, ord3, ord4, ord5)
```

```
orders.plot <- data.frame()
```

```
for (i in 2:ncol(orders)) {
  ord.cache <- orders %>%
    group_by(orders[, i - 1], orders[, i]) %>%
    summarise(n = n()) %>%
    ungroup()
```

```
colnames(ord.cache)[1:2] <- c('from', 'to')
```

```
# adding tags to carts
```

```
ord.cache$from <- paste(ord.cache$from, '(', i - 1, ')', sep = '')
ord.cache$to <- paste(ord.cache$to, '(', i, ')', sep = '')
```

```

    orders.plot <- rbind(orders.plot, ord.cache)
  }

## `summarise()` has grouped output by 'orders[, i - 1]'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'orders[, i - 1]'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'orders[, i - 1]'. You can override using the `.groups` argument
## `summarise()` has grouped output by 'orders[, i - 1]'. You can override using the `.groups` argument

plot(gvisSankey(
  orders.plot,
  from = 'from',
  to = 'to',
  weight = 'n',
  options = list(
    height = 900,
    width = 1800,
    sankey = "{link:{color:{fill:'lightblue'}}}"
  )
))

## starting httpd help server ...

## done

# The bandwidths correspond to the weight of sequence

```

15.5.3 Sequence in-depth analysis

```

library(dplyr)
library(TraMineR)
library(reshape2)
library(googleVis)

# creating an example of shopping carts
set.seed(10)
data <- data.frame(
  orderId = sample(c(1:1000), 5000, replace = TRUE),
  product = sample(
    c('NULL', 'a', 'b', 'c'),
    5000,
    replace = TRUE,
    prob = c(0.15, 0.65, 0.3, 0.15)
  )
)

order <- data.frame(orderId = c(1:1000),
  clientId = sample(c(1:300), 1000, replace = TRUE))

```

```
sex <- data.frame(clientId = c(1:300),
                  sex = sample(
                    c('male', 'female'),
                    300,
                    replace = TRUE,
                    prob = c(0.40, 0.60)
                  ))
date <- data.frame(orderId = c(1:1000),
                  orderdate = sample((1:90), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, sex, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order, sex)

# combining products to the cart
df <- orders %>%
  arrange(product) %>%
  select(-orderId) %>%
  unique() %>%
  group_by(clientId, sex, orderdate) %>%
  summarise(cart = paste(product, collapse = ";")) %>%
  ungroup()
```

`summarise()` has grouped output by 'clientId', 'sex'. You can override using the `

```
max.date <- max(df$orderdate) + 1
ids <- unique(df$clientId)
df.new <- data.frame()

for (i in 1:length(ids)) {
  df.cache <- df %>%
    filter(clientId == ids[i])

  ifelse(nrow(df.cache) == 1,
    av.dur <- 30,
    av.dur <-
      round(((
        max(df.cache$orderdate) - min(df.cache$orderdate)
      ) / (
        nrow(df.cache) - 1
      )) * 1.5, 0))

  df.cache <-
    rbind(
```

```

        df.cache,
        data.frame(
          clientId = df.cache$clientId[nrow(df.cache)],
          sex = df.cache$sex[nrow(df.cache)],
          orderdate = max(df.cache$orderdate) + av.dur,
          cart = 'nopurch'
        )
      )
    }
    ifelse(max(df.cache$orderdate) > max.date,
          df.cache$orderdate[which.max(df.cache$orderdate)] <- max.date,
          NA)

    df.cache$to <- c(df.cache$orderdate[2:nrow(df.cache)] - 1, max.date)

    # order# for Sankey diagram
    df.cache <- df.cache %>%
      mutate(ord = paste('ord', c(1:nrow(df.cache)), sep = ''))

    df.new <- rbind(df.new, df.cache)
  }
  # filtering dummies
  df.new <- df.new %>%
    filter(cart != 'nopurch' | to != orderdate)
  rm(orders, df, df.cache, i, ids, max.date, av.dur)

  ##### Sankey diagram #####

  df.sankey <- df.new %>%
    select(clientId, cart, ord)

  df.sankey <-
    dcast(df.sankey,
          clientId ~ ord,
          value.var = 'cart',
          fun.aggregate = NULL)

  df.sankey[is.na(df.sankey)] <- 'unknown'

  # chosing a length of sequence
  df.sankey <- df.sankey %>%
    select(ord1, ord2, ord3, ord4)

  # replacing NAs after 'nopurch' for 'nopurch'
  df.sankey[df.sankey[, 2] == 'nopurch', 3] <- 'nopurch'
  df.sankey[df.sankey[, 3] == 'nopurch', 4] <- 'nopurch'

```

```

df.sankey.plot <- data.frame()
for (i in 2:ncol(df.sankey)) {
  df.sankey.cache <- df.sankey %>%
    group_by(df.sankey[, i - 1], df.sankey[, i]) %>%
    summarise(n = n()) %>%
    ungroup()

  colnames(df.sankey.cache)[1:2] <- c('from', 'to')

  # adding tags to carts
  df.sankey.cache$from <-
    paste(df.sankey.cache$from, '(', i - 1, ')', sep = '')
  df.sankey.cache$to <- paste(df.sankey.cache$to, '(', i, ')', sep = '')

  df.sankey.plot <- rbind(df.sankey.plot, df.sankey.cache)
}

```

`summarise()` has grouped output by 'df.sankey[, i - 1]'. You can override using the

`summarise()` has grouped output by 'df.sankey[, i - 1]'. You can override using the

`summarise()` has grouped output by 'df.sankey[, i - 1]'. You can override using the

```

plot(gvisSankey(
  df.sankey.plot,
  from = 'from',
  to = 'to',
  weight = 'n',
  options = list(
    height = 900,
    width = 1800,
    sankey = "{link:{color:{fill:'lightblue'}}}"
  )
))

```

```
rm(df.sankey, df.sankey.cache, df.sankey.plot, i)
```

```

df.new <- df.new %>%
  # choosing a length of sequence
  filter(ord %in% c('ord1', 'ord2', 'ord3', 'ord4')) %>%
  select(-ord)

# converting dates to numbers
min.date <- as.Date(min(df.new$orderdate), format = "%Y-%m-%d")
df.new$orderdate <- as.numeric(df.new$orderdate - min.date + 1)
df.new$to <- as.numeric(df.new$to - min.date + 1)

```

```

df.form <-
  seqformat(
    as.data.frame(df.new),
    id = 'clientId',
    begin = 'orderdate',
    end = 'to',
    status = 'cart',
    from = 'SPELL',
    to = 'STS',
    process = FALSE
  )

## [>] time axis: 1 -> 91
## [>] converting SPELL data into 288 STS sequences (internal format)
df.seq <-
  seqdef(df.form,
    left = 'DEL',
    right = 'unknown',
    xtstep = 10) # xtstep - step between ticks (days)

## [>] found missing values ('NA') in sequence data
## [>] preparing 288 sequences
## [>] coding void elements with '%' and missing values with '*'
## [>] 8 distinct states appear in the data:
##      1 = a
##      2 = a;b
##      3 = a;b;c
##      4 = a;c
##      5 = b
##      6 = b;c
##      7 = c
##      8 = nopurch
## [>] adding special state(s) to the alphabet: unknown
## [>] state coding:
##      [alphabet] [label] [long label]
##      1  a      a      a

```

```

##      2  a;b          a;b      a;b
##      3  a;b;c        a;b;c    a;b;c
##      4  a;c          a;c      a;c
##      5  b            b        b
##      6  b;c          b;c      b;c
##      7  c            c        c
##      8  nopurch      nopurch  nopurch
##      9  unknown      unknown  unknown

##  [>] 288 sequences in the data set

##  [>] min/max sequence length: 4/91
summary(df.seq)

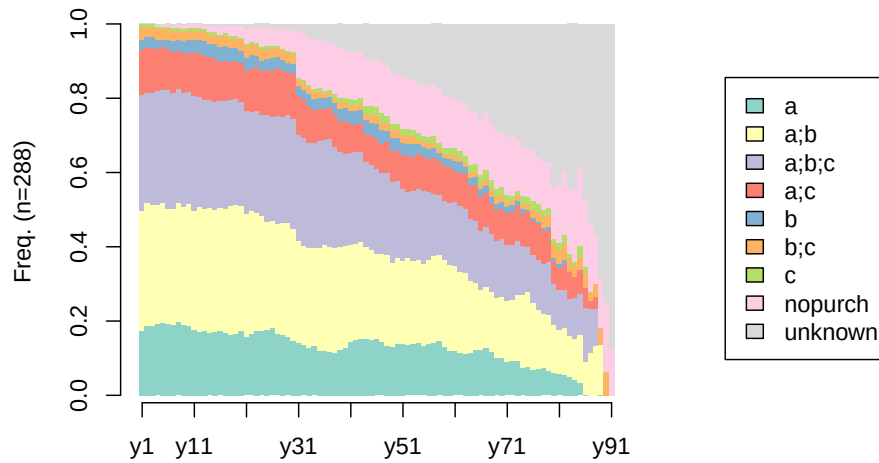
##  [>] sequence object created with TraMineR version 2.2-2
##  [>] 288 sequences in the data set, 286 unique
##  [>] min/max sequence length: 4/91
##  [>] alphabet (state labels):
##      1=a (a)
##      2=a;b (a;b)
##      3=a;b;c (a;b;c)
##      4=a;c (a;c)
##      5=b (b)
##      6=b;c (b;c)
##      7=c (c)
##      8=nopurch (nopurch)
##      9=unknown (unknown)
##  [>] dimensionality of the sequence space: 728
##  [>] colors: 1=#8DD3C7 2=#FFFFB3 3=#BEBADA 4=#FB8072 5=#80B1D3 6=#FDB462 7=#B3DE69 8=#FDB462
##  [>] symbol for void element: %

df.feats <- unique(df.new[, c('clientId', 'sex')])

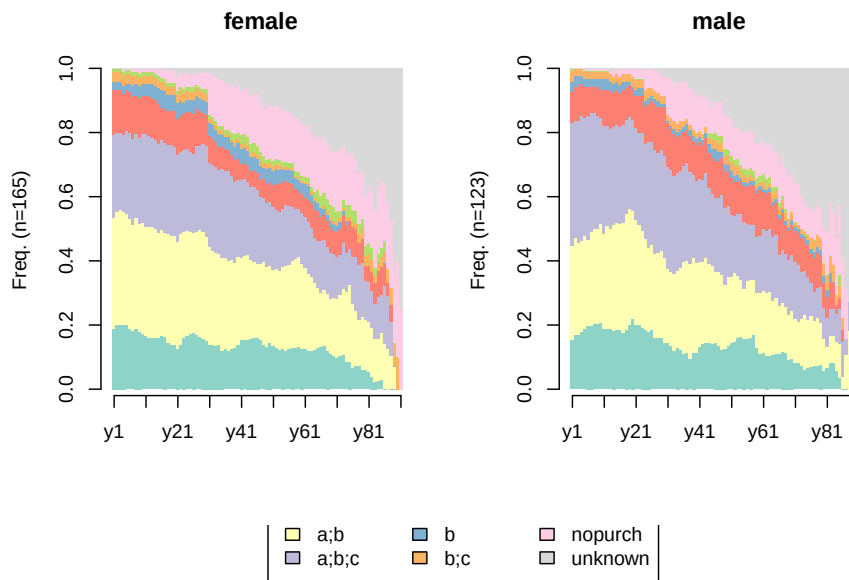
# distribution analysis
seqdplot(df.seq, border = NA, withlegend = 'right')

##  [!!] In local() : withlegend is deprecated, use with.legend instead.

```

```
seqdplot(df.seq, border = NA, group = df.feats$sex) # distribution based on gender
```



```
seqstatd(df.seq)
```

```
## [State frequencies]
```

##	y1	y2	y3	y4	y5	y6	y7	y8	y9	y10	y11	
## a	0.174	0.1875	0.1875	0.1910	0.1951	0.192	0.1888	0.196	0.1895	0.187	0.176	
## a;b	0.323	0.3299	0.3264	0.3229	0.3240	0.311	0.3147	0.323	0.3193	0.331	0.320	
## a;b;c	0.312	0.2986	0.3021	0.3056	0.3031	0.318	0.3112	0.305	0.3053	0.299	0.310	
## a;c	0.122	0.1215	0.1215	0.1146	0.1150	0.112	0.1119	0.102	0.1088	0.106	0.116	
## b	0.028	0.0278	0.0278	0.0243	0.0209	0.024	0.0280	0.032	0.0316	0.035	0.035	
## b;c	0.031	0.0278	0.0278	0.0278	0.0279	0.028	0.0280	0.025	0.0246	0.025	0.025	
## c	0.010	0.0069	0.0069	0.0069	0.0070	0.007	0.0070	0.007	0.0070	0.007	0.007	
## nopurch	0.000	0.0000	0.0000	0.0035	0.0035	0.007	0.0070	0.011	0.0105	0.011	0.011	
## unknown	0.000	0.0000	0.0000	0.0035	0.0035	0.000	0.0035	0.000	0.0035	0.000	0.000	
##	y12	y13	y14	y15	y16	y17	y18	y19	y20	y21		
## a	0.173	0.1725	0.1696	0.1702	0.1744	0.1685	0.1619	0.1655	0.1727	0.1583		
## a;b	0.335	0.3310	0.3357	0.3298	0.3310	0.3333	0.3489	0.3489	0.3381	0.3309		
## a;b;c	0.296	0.2923	0.2898	0.2908	0.2883	0.2903	0.2842	0.2842	0.2806	0.2770		
## a;c	0.120	0.1197	0.1166	0.1170	0.1032	0.1075	0.1043	0.1043	0.1043	0.1079		
## b	0.035	0.0387	0.0353	0.0390	0.0427	0.0430	0.0360	0.0324	0.0324	0.0360		
## b;c	0.025	0.0246	0.0247	0.0248	0.0249	0.0251	0.0252	0.0252	0.0288	0.0288		
## c	0.007	0.0070	0.0071	0.0071	0.0071	0.0072	0.0072	0.0072	0.0072	0.0072		
## nopurch	0.011	0.0106	0.0177	0.0177	0.0214	0.0215	0.0288	0.0288	0.0324	0.0396		
## unknown	0.000	0.0035	0.0035	0.0035	0.0071	0.0036	0.0036	0.0036	0.0036	0.0144		
##	y22	y23	y24	y25	y26	y27	y28	y29	y30	y31		
## a	0.1667	0.1739	0.1745	0.1758	0.1801	0.1661	0.1624	0.1587	0.1481	0.1407		
## a;b	0.3297	0.3188	0.3091	0.2930	0.2904	0.2952	0.3026	0.3063	0.3000	0.2741		
## a;b;c	0.2681	0.2754	0.2727	0.2857	0.2794	0.2915	0.2841	0.2915	0.2963	0.2852		
## a;c	0.1159	0.1159	0.1164	0.1209	0.1250	0.1292	0.1255	0.1144	0.1222	0.1074		
## b	0.0362	0.0290	0.0291	0.0293	0.0331	0.0295	0.0295	0.0258	0.0259	0.0259		
## b;c	0.0290	0.0290	0.0291	0.0293	0.0257	0.0258	0.0258	0.0258	0.0259	0.0148		
## c	0.0072	0.0072	0.0073	0.0073	0.0074	0.0037	0.0037	0.0037	0.0074	0.0074		
## nopurch	0.0399	0.0399	0.0509	0.0476	0.0478	0.0480	0.0554	0.0590	0.0556	0.1259		
## unknown	0.0072	0.0109	0.0109	0.0110	0.0110	0.0111	0.0111	0.0148	0.0185	0.0185		
##	y32	y33	y34	y35	y36	y37	y38	y39	y40	y41	y42	
## a	0.1370	0.1296	0.1338	0.1199	0.1199	0.1170	0.114	0.122	0.127	0.144	0.149	
## a;b	0.2667	0.2667	0.2639	0.2772	0.2846	0.2830	0.284	0.279	0.277	0.261	0.259	
## a;b;c	0.2889	0.2852	0.2825	0.2884	0.2846	0.2906	0.273	0.256	0.246	0.249	0.247	
## a;c	0.1037	0.1000	0.0892	0.0861	0.0824	0.0830	0.087	0.084	0.088	0.078	0.075	
## b	0.0259	0.0296	0.0297	0.0300	0.0300	0.0302	0.030	0.031	0.035	0.035	0.035	
## b;c	0.0185	0.0185	0.0186	0.0187	0.0112	0.0113	0.011	0.019	0.015	0.016	0.016	
## c	0.0074	0.0074	0.0074	0.0075	0.0075	0.0075	0.011	0.011	0.012	0.016	0.016	
## nopurch	0.1259	0.1296	0.1338	0.1311	0.1348	0.1283	0.136	0.134	0.131	0.128	0.129	
## unknown	0.0259	0.0333	0.0409	0.0412	0.0449	0.0491	0.053	0.065	0.069	0.074	0.075	
##	y43	y44	y45	y46	y47	y48	y49	y50	y51	y52	y53	y54
## a	0.151	0.151	0.151	0.153	0.145	0.133	0.130	0.140	0.140	0.137	0.140	0.142
## a;b	0.263	0.247	0.241	0.227	0.236	0.246	0.227	0.230	0.221	0.232	0.223	0.222
## a;b;c	0.243	0.231	0.229	0.231	0.219	0.221	0.218	0.209	0.196	0.180	0.188	0.191
## a;c	0.076	0.076	0.078	0.079	0.079	0.083	0.084	0.081	0.089	0.094	0.092	0.093
## b	0.036	0.036	0.037	0.041	0.037	0.029	0.034	0.034	0.034	0.034	0.035	0.027

```

## b;c      0.016 0.016 0.020 0.021 0.021 0.021 0.021 0.021 0.021 0.021 0.021 0.022 0.013
## c        0.016 0.024 0.024 0.025 0.025 0.021 0.017 0.017 0.017 0.017 0.017 0.017 0.018
## nopurch  0.127 0.127 0.122 0.132 0.136 0.133 0.134 0.136 0.140 0.137 0.135 0.133
## unknown  0.072 0.092 0.098 0.091 0.103 0.112 0.134 0.132 0.140 0.146 0.148 0.160
##          y55  y56  y57  y58  y59  y60  y61  y62  y63  y64  y65
## a        0.135 0.140 0.142 0.1475 0.1308 0.1190 0.121 0.1127 0.1133 0.116 0.122
## a;b      0.220 0.226 0.233 0.2304 0.2336 0.2333 0.227 0.2206 0.2167 0.192 0.180
## a;b;c    0.197 0.181 0.169 0.1613 0.1589 0.1667 0.174 0.1765 0.1724 0.167 0.169
## a;c      0.090 0.090 0.096 0.0922 0.0935 0.0952 0.082 0.0882 0.0936 0.091 0.095
## b        0.027 0.027 0.027 0.0230 0.0234 0.0238 0.029 0.0294 0.0197 0.020 0.021
## b;c      0.013 0.014 0.014 0.0092 0.0093 0.0095 0.014 0.0098 0.0099 0.015 0.011
## c        0.018 0.018 0.018 0.0184 0.0187 0.0190 0.019 0.0196 0.0197 0.020 0.026
## nopurch  0.139 0.140 0.137 0.1429 0.1402 0.1381 0.135 0.1373 0.1379 0.146 0.143
## unknown  0.161 0.163 0.164 0.1751 0.1916 0.1952 0.198 0.2059 0.2167 0.232 0.233
##          y66  y67  y68  y69  y70  y71  y72  y73  y74  y75  y76
## a        0.123 0.127 0.117 0.102 0.101 0.091 0.094 0.093 0.075 0.0786 0.0682
## a;b      0.160 0.166 0.162 0.159 0.166 0.164 0.163 0.179 0.197 0.2000 0.1818
## a;b;c    0.160 0.160 0.156 0.153 0.154 0.152 0.150 0.146 0.129 0.1214 0.1288
## a;c      0.091 0.094 0.084 0.085 0.083 0.085 0.087 0.093 0.088 0.0929 0.0909
## b        0.021 0.017 0.017 0.017 0.018 0.018 0.019 0.013 0.014 0.0071 0.0076
## b;c      0.011 0.017 0.017 0.017 0.018 0.012 0.012 0.013 0.014 0.0143 0.0227
## c        0.027 0.028 0.028 0.023 0.024 0.018 0.012 0.013 0.020 0.0214 0.0227
## nopurch  0.150 0.155 0.151 0.153 0.148 0.158 0.163 0.139 0.129 0.1286 0.1364
## unknown  0.257 0.238 0.268 0.290 0.290 0.303 0.300 0.311 0.333 0.3357 0.3409
##          y77  y78  y79  y80  y81  y82  y83  y84  y85  y86  y87
## a        0.0714 0.0726 0.0672 0.0603 0.0588 0.057 0.049 0.042 0.032 0.000 0.000
## a;b      0.1587 0.1452 0.1513 0.1293 0.1176 0.125 0.111 0.111 0.129 0.091 0.116
## a;b;c    0.1349 0.1371 0.1429 0.1121 0.1078 0.102 0.099 0.111 0.113 0.145 0.116
## a;c      0.0952 0.0887 0.0840 0.0603 0.0588 0.068 0.074 0.056 0.065 0.055 0.023
## b        0.0079 0.0081 0.0084 0.0086 0.0098 0.011 0.000 0.000 0.000 0.000 0.000
## b;c      0.0238 0.0323 0.0336 0.0345 0.0392 0.045 0.037 0.028 0.032 0.036 0.023
## c        0.0238 0.0161 0.0168 0.0172 0.0196 0.023 0.012 0.014 0.032 0.018 0.000
## nopurch  0.1270 0.1290 0.1261 0.1379 0.1569 0.170 0.173 0.194 0.210 0.182 0.186
## unknown  0.3571 0.3710 0.3697 0.4397 0.4314 0.398 0.444 0.444 0.387 0.473 0.535
##          y88  y89  y90  y91
## a        0.000 0.000 0.000 0.00
## a;b      0.133 0.136 0.000 0.00
## a;b;c    0.100 0.000 0.000 0.00
## a;c      0.033 0.000 0.000 0.00
## b        0.000 0.000 0.000 0.00
## b;c      0.033 0.045 0.062 0.00
## c        0.000 0.000 0.000 0.00
## nopurch  0.133 0.136 0.188 0.12
## unknown  0.567 0.682 0.750 0.88
##
##          [Valid states]

```

```

##          y1 y2 y3 y4 y5 y6 y7 y8 y9 y10 y11 y12 y13 y14 y15 y16 y17 y18
## N      288 288 288 288 287 286 286 285 285 284 284 284 283 282 281 279 278
##          y19 y20 y21 y22 y23 y24 y25 y26 y27 y28 y29 y30 y31 y32 y33 y34 y35 y36
## N      278 278 278 276 276 275 273 272 271 271 271 270 270 270 270 269 267 267
##          y37 y38 y39 y40 y41 y42 y43 y44 y45 y46 y47 y48 y49 y50 y51 y52 y53 y54
## N      265 264 262 260 257 255 251 251 245 242 242 240 238 235 235 233 229 225
##          y55 y56 y57 y58 y59 y60 y61 y62 y63 y64 y65 y66 y67 y68 y69 y70 y71 y72
## N      223 221 219 217 214 210 207 204 203 198 189 187 181 179 176 169 165 160
##          y73 y74 y75 y76 y77 y78 y79 y80 y81 y82 y83 y84 y85 y86 y87 y88 y89 y90
## N      151 147 140 132 126 124 119 116 102 88 81 72 62 55 43 30 22 16
##          y91
## N      8
##
##          [Entropy index]
##          y1 y2 y3 y4 y5 y6 y7 y8 y9 y10 y11 y12 y13 y14 y15
## H      0.7 0.7 0.7 0.71 0.71 0.71 0.72 0.71 0.72 0.71 0.72 0.72 0.73 0.73 0.74
##          y16 y17 y18 y19 y20 y21 y22 y23 y24 y25 y26 y27 y28 y29
## H      0.75 0.74 0.74 0.74 0.75 0.77 0.77 0.77 0.78 0.78 0.78 0.77 0.78 0.77
##          y30 y31 y32 y33 y34 y35 y36 y37 y38 y39 y40 y41 y42 y43
## H      0.78 0.8 0.81 0.82 0.82 0.82 0.81 0.81 0.82 0.84 0.85 0.85 0.86 0.86
##          y44 y45 y46 y47 y48 y49 y50 y51 y52 y53 y54 y55 y56 y57
## H      0.87 0.88 0.89 0.89 0.88 0.89 0.89 0.89 0.89 0.9 0.88 0.88 0.88 0.88
##          y58 y59 y60 y61 y62 y63 y64 y65 y66 y67 y68 y69 y70 y71
## H      0.87 0.87 0.87 0.88 0.88 0.87 0.88 0.88 0.88 0.88 0.87 0.86 0.86 0.85
##          y72 y73 y74 y75 y76 y77 y78 y79 y80 y81 y82 y83 y84 y85
## H      0.84 0.84 0.83 0.82 0.83 0.83 0.82 0.82 0.78 0.79 0.81 0.75 0.74 0.78
##          y86 y87 y88 y89 y90 y91
## H      0.69 0.6 0.6 0.43 0.32 0.17

df.seq <- seqdef(df.form,
  left = 'DEL',
  right = 'DEL',
  xtstep = 10)

## [>] found missing values ('NA') in sequence data

## [>] preparing 288 sequences

## [>] coding void elements with '%' and missing values with '*'

## [>] 8 distinct states appear in the data:

##      1 = a
##      2 = a;b
##      3 = a;b;c
##      4 = a;c

```

```

##      5 = b

##      6 = b;c

##      7 = c

##      8 = nopurch

##  [>] state coding:

##      [alphabet]  [label]  [long label]

##      1  a          a      a

##      2  a;b        a;b    a;b

##      3  a;b;c      a;b;c  a;b;c

##      4  a;c        a;c    a;c

##      5  b          b      b

##      6  b;c        b;c    b;c

##      7  c          c      c

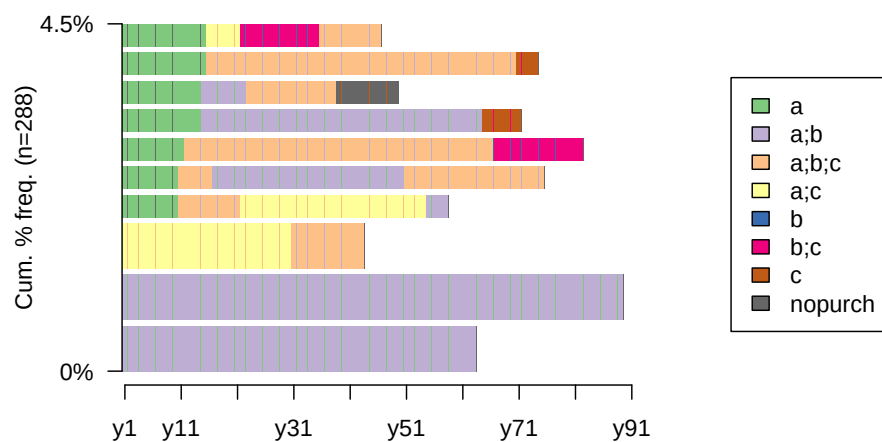
##      8  nopurch    nopurch nopurch

##  [>] 288 sequences in the data set

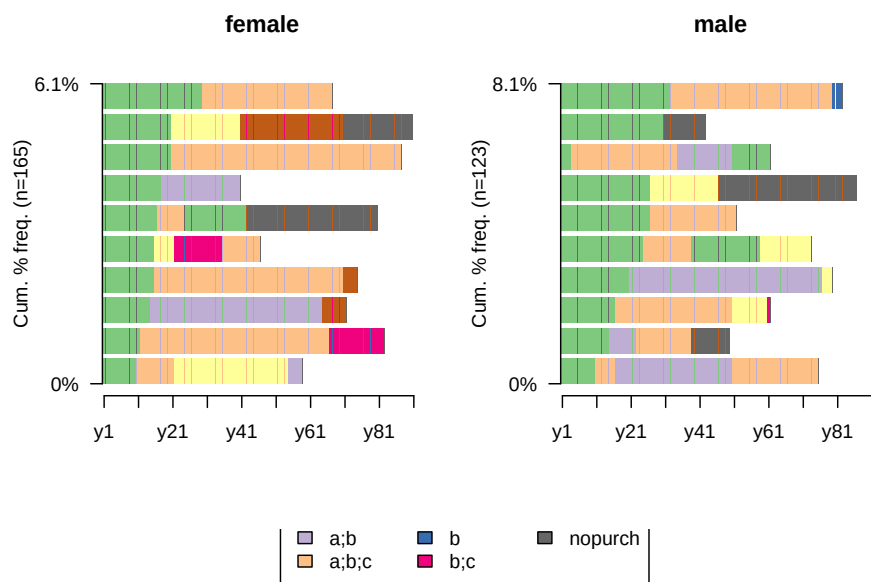
##  [>] min/max sequence length: 3/91
# the 10 most frequent sequences
seqfplot(df.seq, border = NA, withlegend = 'right')

##  [!!] In local() : withlegend is deprecated, use with.legend instead.

```



```
# the 10 most frequent sequences based on gender
seqfplot(df.seq, group = df.feats$sex, border = NA)
```



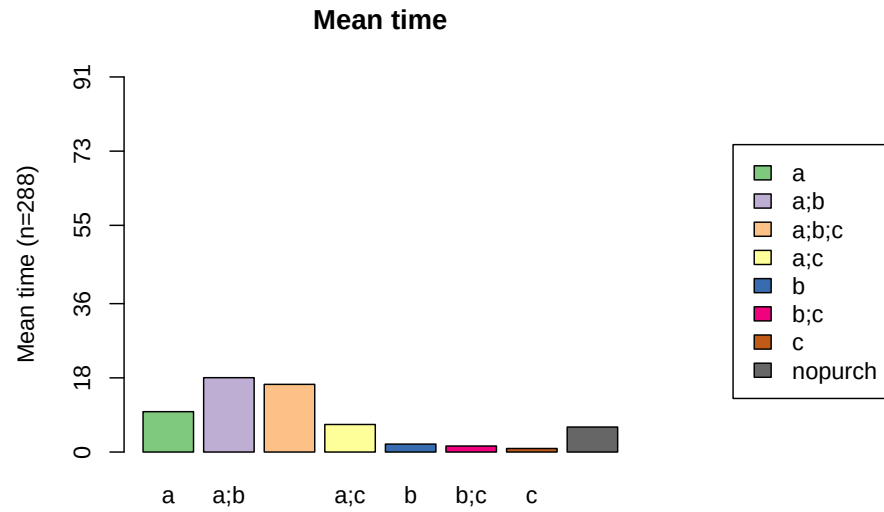
```
# returning the frequency stats
seqtab(df.seq) # frequency table
```

```
##                               Freq Percent
## a;b/63                        2      0.69
## a;b/89                        2      0.69
## a;c/30-a;b;c/13              2      0.69
## a/10-a;b;c/11-a;c/33-a;b/4   1      0.35
## a/10-a;b;c/6-a;b/34-a;b;c/25 1      0.35
## a/11-a;b;c/55-b;c/16         1      0.35
## a/14-a;b/50-c/7              1      0.35
## a/14-a;b/8-a;b;c/16-nopurch/11 1      0.35
## a/15-a;b;c/55-c/4            1      0.35
## a/15-a;c/6-b;c/14-a;b;c/11   1      0.35
seqtab(df.seq[, 1:30]) # frequency table for 1st month
```

```
##                               Freq Percent
## a;b;c/30                      41    14.24
## a;b/30                        40    13.89
## a/30                         24     8.33
## a;c/30                       15     5.21
## b;c/30                        4     1.39
## a;b/1-a;b;c/29                2     0.69
## a;b;c/28-a;b/2                2     0.69
## a;c/15-a;b/15                 2     0.69
## a;c/7-a;b/23                  2     0.69
## b/30                          2     0.69
# mean time spent in each state
seqmplot(df.seq, title = 'Mean time', withlegend = 'right')
```

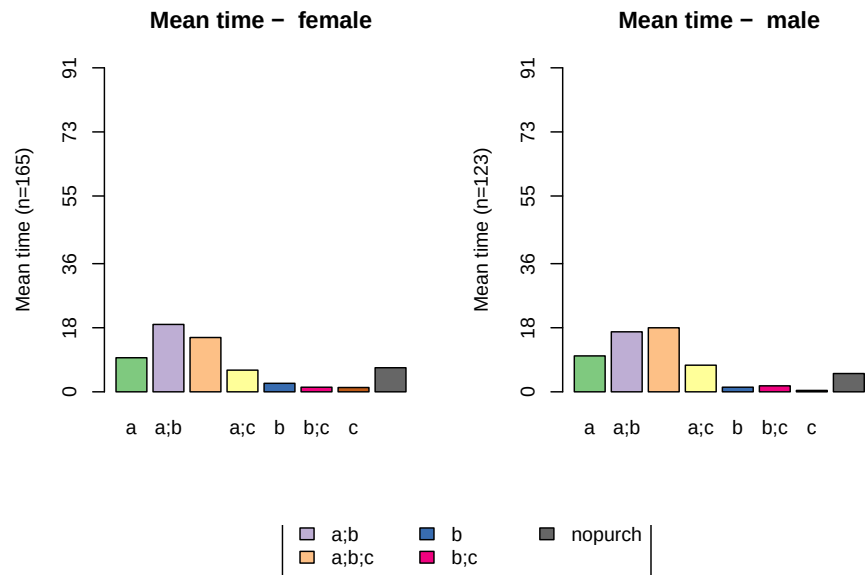
```
## [!!] In local() : title is deprecated, use main instead.
```

```
## [!!] In local() : withlegend is deprecated, use with.legend instead.
```



```
seqmtplot(df.seq, group = df.feats$sex, title = 'Mean time')
```

```
## [!!] In local() : title is deprecated, use main instead.
```



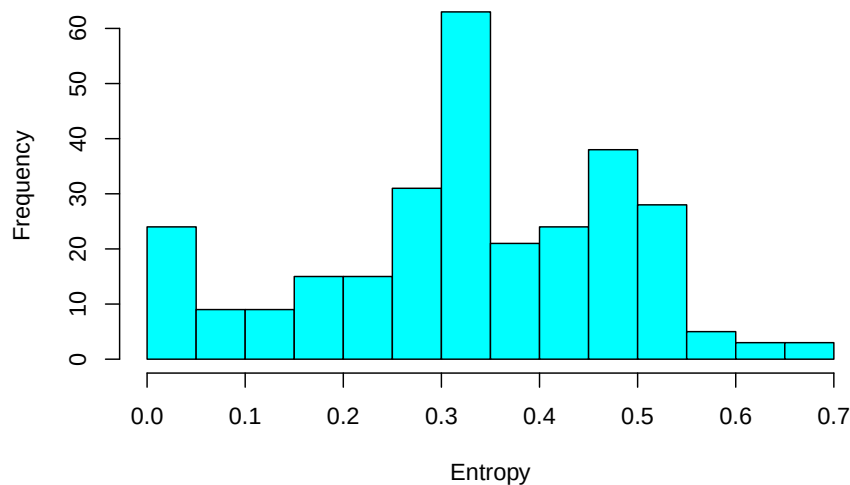

```

statd <-
  seqistatd(df.seq) #function returns for each sequence the time spent in the different states

## [>] computing state distribution for 288 sequences ...
apply(statd, 2, mean) #We may be interested in the mean time spent in each state

##          a          a;b          a;b;c          a;c          b          b;c          c
## 9.7916667 18.0416667 16.4201389  6.6840278  1.9236111  1.4583333  0.8611111
##  nopurch
## 6.0694444
# calculating entropy
df.ient <- seqient(df.seq)
hist(df.ient,
     col = 'cyan',
     main = NULL,
     xlab = 'Entropy') # plot an histogram of the within entropy of the sequences

```

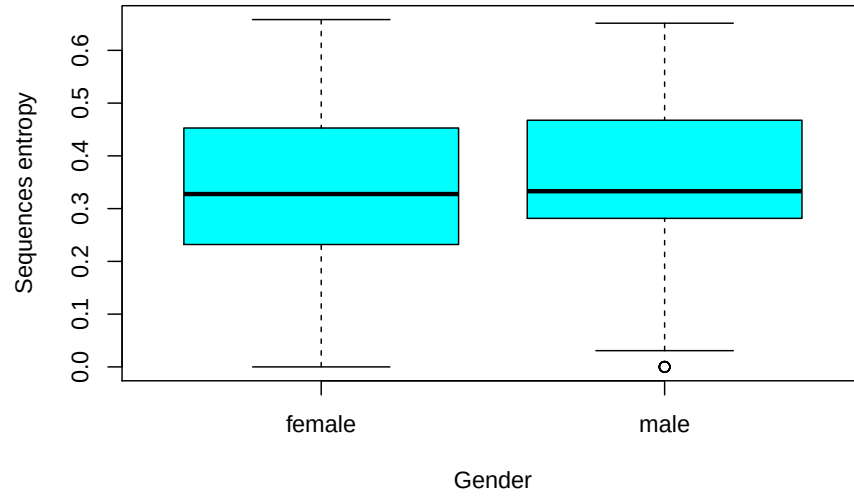


```

# entropy distribution based on gender
df.ent <- cbind(df.seq, df.ient)
boxplot(
  Entropy ~ df.feats$sex,
  data = df.ent,
  xlab = 'Gender',
  ylab = 'Sequences entropy',

```

```
col = 'cyan'  
)
```



Chapter 16

Qualitative Research

16.1 Inter-rate reliability methods

Calculation to judge the degree of agreement between the choices made by two or more independent judges

Other packages are

- vcd for visualization
- DescTools

16.1.1 Percent Agreement

$$\frac{\text{number of agreement}}{\text{number of total}} \times 100$$

```
library("irr")
```

```
## Loading required package: lpSolve
```

```
data("diagnoses", package = "irr")
```

```
data("anxiety", package = "irr")
```

```
head(diagnoses,10)
```

##	rater1	rater2	rater3
## 1	4. Neurosis	4. Neurosis	4. Neurosis
## 2	2. Personality Disorder	2. Personality Disorder	2. Personality Disorder
## 3	2. Personality Disorder	3. Schizophrenia	3. Schizophrenia
## 4	5. Other	5. Other	5. Other
## 5	2. Personality Disorder	2. Personality Disorder	2. Personality Disorder
## 6	1. Depression	1. Depression	3. Schizophrenia
## 7	3. Schizophrenia	3. Schizophrenia	3. Schizophrenia

```
## 8          1. Depression          1. Depression          3. Schizophrenia
## 9          1. Depression          1. Depression          4. Neurosis
## 10         5. Other              5. Other              5. Other
##          rater4              rater5              rater6
## 1          4. Neurosis          4. Neurosis          4. Neurosis
## 2          5. Other              5. Other              5. Other
## 3 3. Schizophrenia 3. Schizophrenia          5. Other
## 4          5. Other              5. Other              5. Other
## 5          4. Neurosis          4. Neurosis          4. Neurosis
## 6 3. Schizophrenia 3. Schizophrenia 3. Schizophrenia
## 7 3. Schizophrenia          5. Other              5. Other
## 8 3. Schizophrenia 3. Schizophrenia          4. Neurosis
## 9          4. Neurosis          4. Neurosis          4. Neurosis
## 10         5. Other              5. Other              5. Other
```

```
agree(diagnoses)
```

```
## Percentage agreement (Tolerance=0)
```

```
##
```

```
## Subjects = 30
```

```
## Raters = 6
```

```
## %-agree = 16.7
```

```
# library(vcd)
```

```
# # Create the plot
```

```
# p <- agreementplot(anxiety)
```

16.1.2 Cohen's Kappa

(Cohen, 1960)

$$k = \frac{p_o - p_e}{1 - p_e} = 1 - \frac{1 - p_o}{1 - p_e}$$

where

- p_o = relative observed agreement among raters
- p_e = hypothetical probability of chance agreement

strict agreements between raters

appropriate in case of 2 ordinal or nominal variables

Based on (Landis and Koch, 1977)'s guide, we have

Degree	Decision
0.01 – 0.20	slight agreement

Degree	Decision
0.21 – 0.40	fair agreement
0.41 – 0.60	moderate agreement
0.61 – 0.80	substantial agreement
0.81 – 1.00	almost perfect or perfect agreement

```

# Unweighted kappa for 2 nominal or 2 ordinal categorical
kappa2(diagnoses[, c("rater1", "rater2")], weight = "unweighted") # two ordinal variables only, c

## Cohen's Kappa for 2 Raters (Weights: unweighted)
##
## Subjects = 30
## Raters = 2
## Kappa = 0.651
##
## z = 7
## p-value = 2.63e-12

# Weighted kappa ordinal scales
kappa2(diagnoses[, c("rater1", "rater2")], weight = "equal") # linear weightes of the differences

## Cohen's Kappa for 2 Raters (Weights: equal)
##
## Subjects = 30
## Raters = 2
## Kappa = 0.633
##
## z = 5.43
## p-value = 5.52e-08

kappa2(diagnoses[, c("rater1", "rater2")], weight = "squared") # squared weightes of the differences

## Cohen's Kappa for 2 Raters (Weights: squared)
##
## Subjects = 30
## Raters = 2
## Kappa = 0.655
##
## z = 3.91
## p-value = 9.37e-05

```

p-value less than 0.05, mean that raters agree more than what you would expect by chance.

16.1.3 Fleiss'kappa

for two or more categorical variables (nominal or ordinal)

for three or more raters

- 0 = no agreement
- 1 = perfect agreement

```
# no assumption of same raters for all subjects
kappam.fleiss(diagnoses[, 1:3])
```

```
## Fleiss' Kappa for m Raters
##
## Subjects = 30
## Raters = 3
## Kappa = 0.534
##
## z = 9.89
## p-value = 0
```

16.2 Krippendorff's Alpha

calculates disagreement among raters.

$$\frac{\text{observed disagreement}}{\text{expected disagreement}}$$

Rules of thumb by (Shelley and Krippendorff, 1984)

- $\alpha \geq 0.08$ = good
- $\alpha \geq 0.667$ = acceptable lower limit

```
nmm <-
  matrix(
    c(
      1,
      1,
      NA,
      1,
      2,
      2,
      3,
      2,
      3,
      3,
      3,
      3,
      3,
      3,
      3,
```

```
      3,
      3,
      2,
      2,
      2,
      2,
      1,
      2,
      3,
      4,
      4,
      4,
      4,
      4,
      1,
      1,
      2,
      1,
      2,
      2,
      2,
      2,
      NA,
      5,
      5,
      5,
      NA,
      NA,
      1,
      1,
      NA,
      NA,
      3,
      NA
    ),
    nrow = 4
  )
  # first assume the default nominal classification
  kripp.alpha(nmm)

## Krippendorff's alpha
##
## Subjects = 12
## Raters = 4
## alpha = 0.743
```

```
# now use the same data with the other three methods
kripp.alpha(nmm, "ordinal")
```

```
## Krippendorff's alpha
##
## Subjects = 12
## Raters = 4
## alpha = 0.815
kripp.alpha(nmm, "interval")
```

```
## Krippendorff's alpha
##
## Subjects = 12
## Raters = 4
## alpha = 0.849
kripp.alpha(nmm, "ratio")
```

```
## Krippendorff's alpha
##
## Subjects = 12
## Raters = 4
## alpha = 0.797
```

16.2.1 Kendall's W

Continuous ordinal ratings

```
rtr1 <- c(1, 6, 3, 2, 5, 4)
rtr2 <- c(1, 5, 6, 2, 4, 3)
rtr3 <- c(2, 3, 6, 5, 4, 1)
ratings <- cbind(rtr1, rtr2, rtr3)
library(DescTools)
```

```
## Warning: package 'DescTools' was built under R version 4.0.5
KendallW(ratings, test=TRUE)
```

```
##
## Kendall's coefficient of concordance W
##
## data: ratings
## Kendall chi-squared = 8.5238, df = 5, subjects = 6, raters = 3, p-value
## = 0.1296
## alternative hypothesis: W is greater 0
## sample estimates:
## W
```



```
## 0.568254
```

16.2.2 Intraclass correlation coefficients

(Shrout and Fleiss, 1979)

16.2.2.1 Continuous scales

Decision:

- model:
 - "oneway" when subjects are random effects
 - "twoway" when subjects and raters are random effects
- type:
 - "agreement" differences in mean ratings among raters are of interest
 - "consistency" is default
- unit
 - "single" single value
 - "average" mean of several ratings

```
icc(anxiety,
    model = "twoway", # can be "oneway"
    type = "agreement", # can be "consistency"
    unit = "single" # can be "average"
)

## Single Score Intraclass Correlation
##
## Model: twoway
## Type : agreement
##
## Subjects = 20
## Raters = 3
## ICC(A,1) = 0.198
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(19,39.7) = 1.83 , p = 0.0543
##
## 95%-Confidence Interval for ICC Population Values:
## -0.039 < ICC < 0.494

DescTools package
```

```

rtr1 <- c(9, 6, 8, 7, 10, 6)
rtr2 <- c(2, 1, 4, 1, 5, 2)
rtr3 <- c(5, 3, 6, 2, 6, 4)
rtr4 <- c(8, 2, 8, 6, 9, 7)
ratings <- cbind(rtr1, rtr2, rtr3, rtr4)
DescTools::ICC(ratings)

##
## Intraclass correlation coefficients
##
##          type    est F-val df1 df2    p-val lwr.ci upr.ci
## Single_raters_absolute ICC1 0.166  1.79   5  18 0.164769    NA    NA
## Single_random_raters   ICC2 0.290 11.03   5  15 0.000135    NA    NA
## Single_fixed_raters    ICC3 0.715 11.03   5  15 0.000135    NA    NA
## Average_raters_absolute ICC1k 0.443  1.79   5  18 0.164769    NA    NA
## Average_random_raters  ICC2k 0.620 11.03   5  15 0.000135    NA    NA
## Average_fixed_raters   ICC3k 0.909 11.03   5  15 0.000135    NA    NA
##
## Number of subjects = 6      Number of raters = 4

```

16.2.3 Light's kappa

- average of Cohen's Kappa if using more than two categorical variables

```

# Light's kappa: multiple raters
kappam.light(diagnoses[, 1:3])

## Warning in sqrt(varkappa): NaNs produced
## Light's Kappa for m Raters
##
## Subjects = 30
## Raters = 3
## Kappa = 0.555
##
## z = NaN
## p-value = NaN

```

Chapter 17

Nudges

(Garbinsky et al., 2020) uncovers that people under save because they have positive illusion of financial responsibility (i.e., they thought they are financially more responsible than the average).

(Mrkva et al., 2021) found that choice architecture influences inequities across people. More specifically, people with the following dimensions enjoy the most benefit from nudges:

- Low socioeconomic status (SES)
- Low numerical ability
- Low knowledge

Hence, we should use nudges to reduce SES disparities.

Chapter 18

Visualization

Chapter 19

Surveys

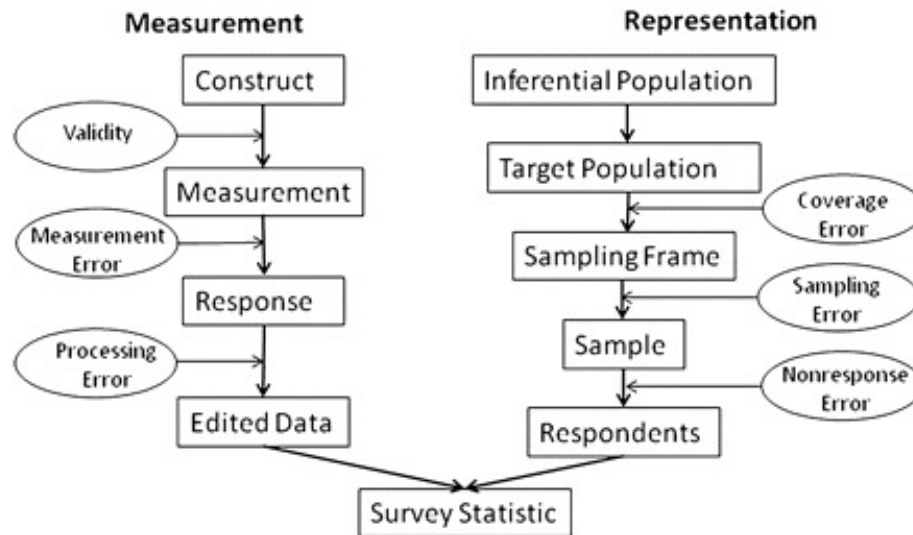
Table 19.1: SICSS Summer 2020 -Survey Research in the Digital Age by professor **Matthew Salganik**

	Sampling	Interviews	Data environment
1st era	area probability	face-to-face	stand-alone
2nd ear	random digital dial probability	telephone	stand-alone
3rd era	non-probability	computer-administered	linked

Total survey error framework (Groves and Lyberg, 2010)

Insight:

- 1. Errors can come from bias or variance
- 2. Total survey error = Measurement error + representation error



(Groves and Lyberg, 2010, Fig. 3)

Probability and Non-probability Sampling

- Probability sample: every unit from a frame population has a known and non-zero probability of inclusion
- With weighting, we can recover bias in your sampling.
- Non-response problem

Horvitz-Thompson estimator (or bias estimator):

$$\hat{y} = \frac{\sum_{i \in s} y_i / \pi_i}{N}$$

where π_i = person i 's probability of inclusion (we have to estimate)

Wiki Survey

- Create a survey that leverages the power of people

Mass Collaboration

- Human Computation: Train People -> Train Lots of People -> Train Machine
 - Cleaning
 - De-biasing
 - Combining
- Open Call:
 - solutions are easier to check than generate

- required specialized skills
- Distributed Data Collection:
 - People go out and collect data
 - quality check

Fragile family challenges

Chapter 20

Construct vs. Variable

Table 20.1: measured or perceived vs. reality

		Reality	
Measured	True	True Correct	False Type 1 error (False positive)
	False	Type 2 error (False Negative)	Correct

picture from (Irvine, 2019)

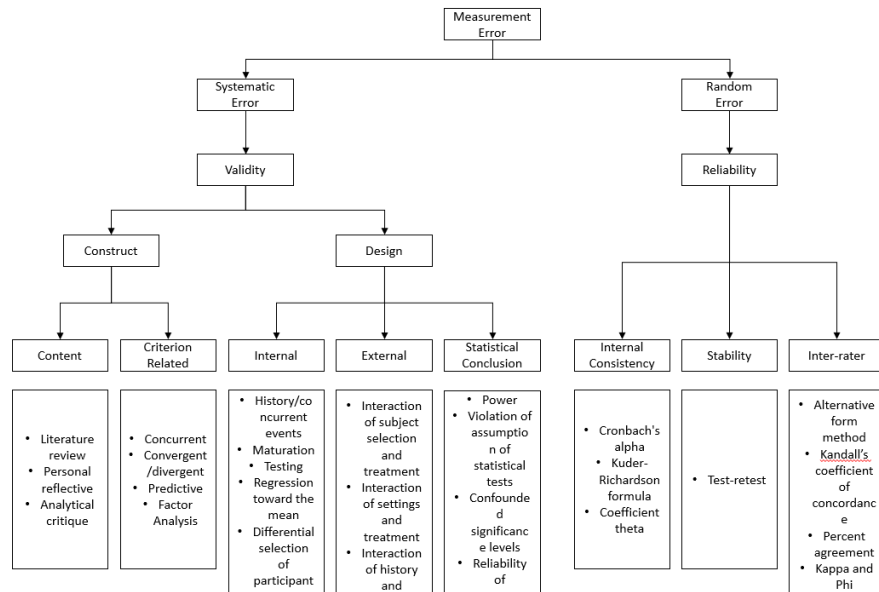


Figure 20.1: Measurement Error

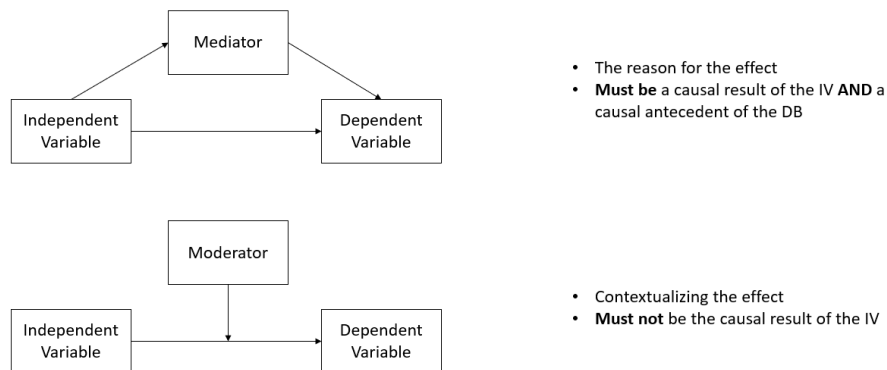


Figure 20.2: Moderator vs. Mediator

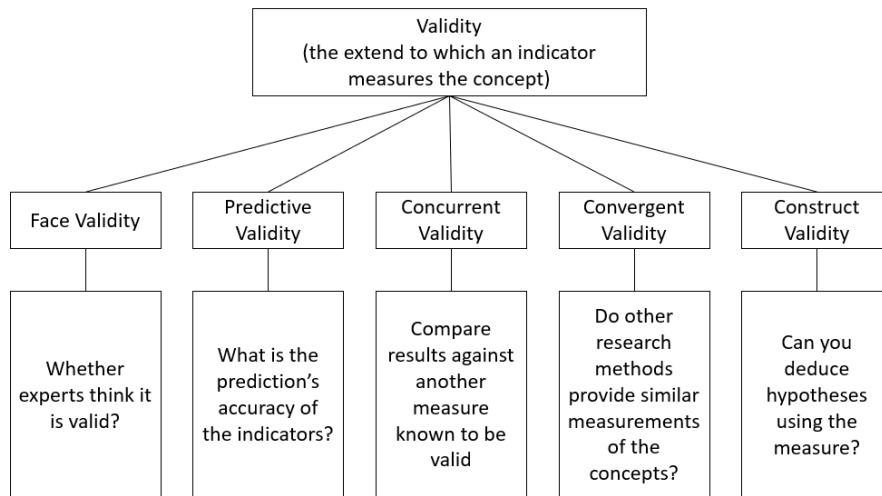


Figure 20.3: Validity

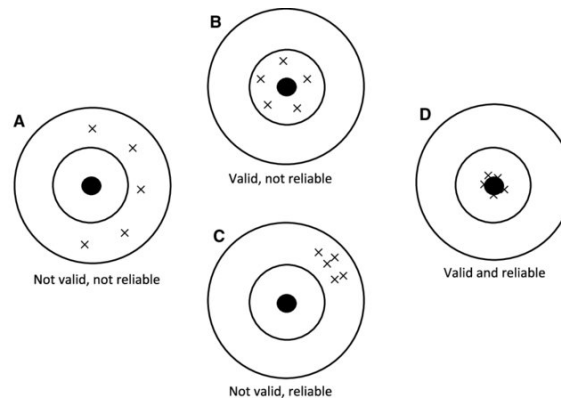


Figure 20.4: Reliability vs. Validity

Chapter 21

Experiment

Things to consider:

- Validity
 - statistical conclusion validity
 - internal validity
 - construct validity: whether you operationalize your construct correctly
 - external validity: generalization
- Heterogeneity of treatment effects
 - average (aggregate) effect can be artificially constructed.
- Mechanisms

More things to consider

- cost
- control
- realism
- ethics

The Principles of Humane Experimental Technique by Russell and Burch (1959):
3Rs

- Replace: with natural (less invasive)
- Refine: less harmful
- Reduce: min number of participants

Before running experiments, to make sure that you did not tamper with the data, or hypotheses (i.e., change them after the fact), you should always pre-register your research. A website that allows you to do this and automatically provides you with time stamp is aspredicted

This way then you can show to reviewers that you did hypothesize before running analysis.

To combat replication crisis in psychology, we can advocate for open research in which researchers public their data and analyses. Popular websites include:

- ResearchBox
- <https://osf.io/>

To see whether authors temper with their data, we can use p-curve to inspect.
1 2 3 4 5

Interaction Effects Need Interaction Controls (basically, you can't just control for main effects).

Chapter 22

Report

1. Introduction

1. Note on managerial relevance/ context
2. Note on the gap in current managerial understanding and academic literature
3. State your research questions
4. Summarize your intended contributions to both theory and practice (at least 3 contributions)

Chapter 23

Surveys

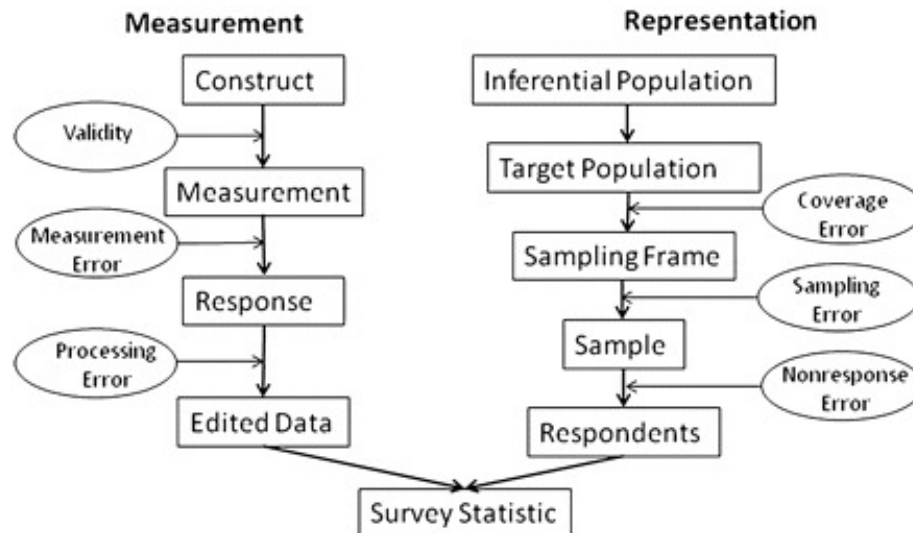
Table 23.1: SICSS Summer 2020 -Survey Research in the Digital Age by professor **Matthew Salganik**

	Sampling	Interviews	Data environment
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3rd era	non-probability	computer-administered	linked

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Insight:

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(Groves and Lyberg, 2010, Fig. 3)

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Horvitz-Thompson estimator (or bias estimator):

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- Human Computation: Train People -> Train Lots of People -> Train Machine
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 - Combining
- Open Call:
 - solutions are easier to check than generate

- required specialized skills
- Distributed Data Collection:
 - People go out and collect data
 - quality check

Fragile family challenges

Chapter 24

Construct vs. Variable

Table 24.1: measured or perceived vs. reality

		Reality	
Measured	True	True Correct	False Type 1 error (False positive)
	False	Type 2 error (False Negative)	Correct

picture from (Irvine, 2019)

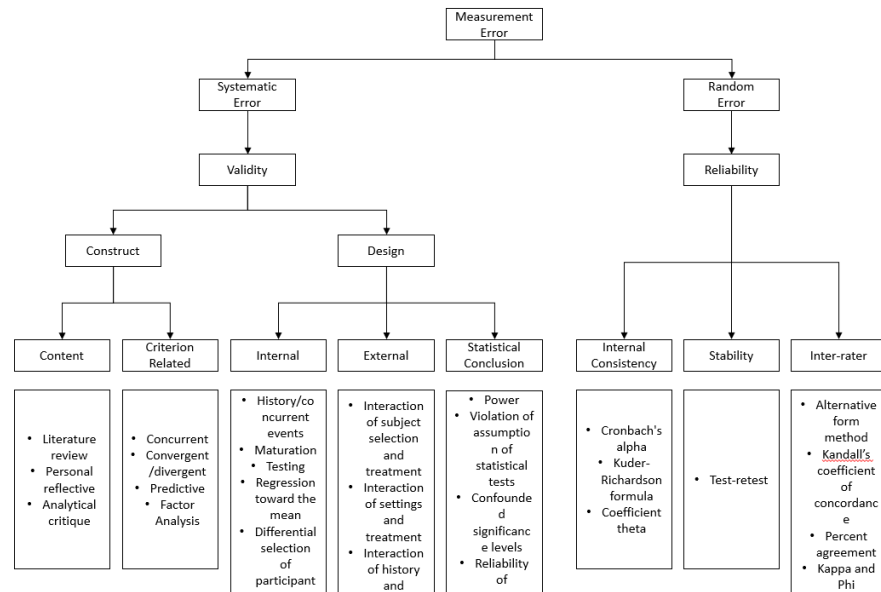


Figure 24.1: Measurement Error

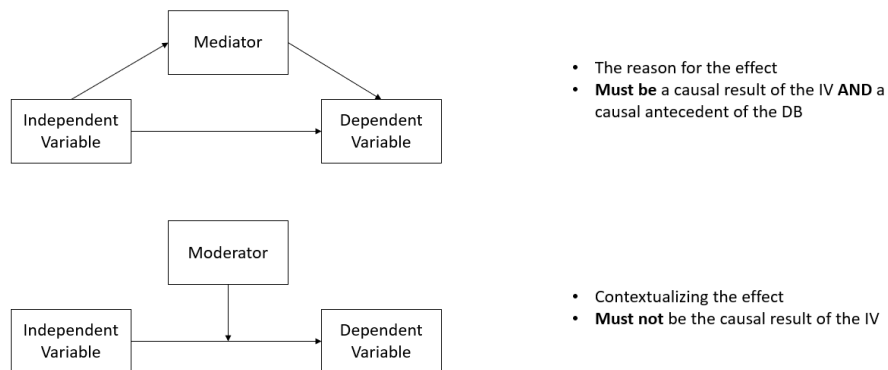


Figure 24.2: Moderator vs. Mediator

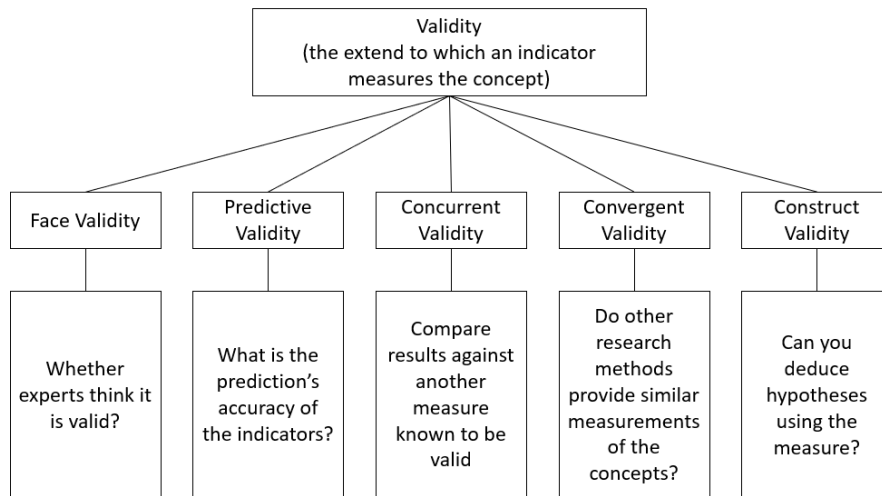


Figure 24.3: Validity

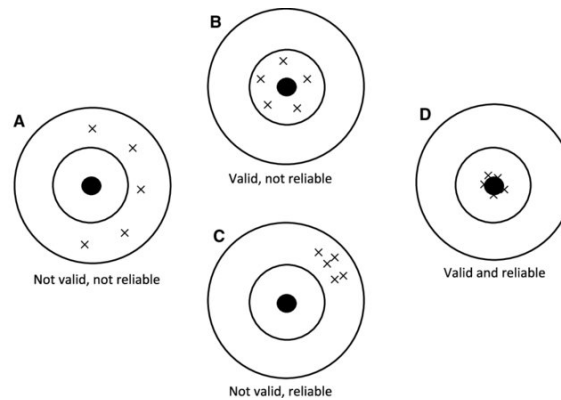


Figure 24.4: Reliability vs. Validity

Chapter 25

Experiment

Things to consider:

- Validity
 - statistical conclusion validity
 - internal validity
 - construct validity: whether you operationalize your construct correctly
 - external validity: generalization
- Heterogeneity of treatment effects
 - average (aggregate) effect can be artificially constructed.
- Mechanisms

More things to consider

- cost
- control
- realism
- ethics

The Principles of Humane Experimental Technique by Russell and Burch (1959):
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- Replace: with natural (less invasive)
- Refine: less harmful
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Before running experiments, to make sure that you did not tamper with the data, or hypotheses (i.e., change them after the fact), you should always pre-register your research. A website that allows you to do this and automatically provides you with time stamp is aspredicted

This way then you can show to reviewers that you did hypothesize before running analysis.

To combat replication crisis in psychology, we can advocate for open research in which researchers public their data and analyses. Popular websites include:

- ResearchBox
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To see whether authors temper with their data, we can use p-curve to inspect.
1 2 3 4 5

Interaction Effects Need Interaction Controls (basically, you can't just control for main effects).

Chapter 26

Image

Transdermal Optical Imaging:

- <https://www.nuralogix.ai/technology>
- extract facial blood flow information
- <https://github.com/nuralogix>
- <https://www.deepaffex.ai/>
- <https://www.deepaffex.ai/media>

Chapter 27

Sarcasm

- AI-based Learning Techniques for Sarcasm Detection of Social Media Tweets
- Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media
-

Chapter 28

Formative vs. Reflective Measurements

	Reflective	Formative
Root	Common factor model	Principal Component Analysis (Weighted Linear Composites)
	Construct as causes of measures	Measures as causes of construct
Dimensionality	Unidimensionality	Multidimensionality of formative measures (this is a liability because it is analogous to multibarralled items)
	Measures = reflection of construct in its entirety (Unidimensionality of reflective measures)	because the variance of the latent variable is determined by the paths (i.e., strength) from the measures, and variances and covariances of the measures.
	Useful redundancy	
Internal consistency	Reflective measures should correlate positively	No needs for the formative measures to correlate
		Correlation among measures can be problem (multicollinearity). Higher the correlation (among measures), more unstable the loadings on construct, larger standard errors. The lack of internal consistency does not provide evidence for a set of measures to be formative.

Reflective	Formative
Identification is identified only when either • at least 3 measures, and fix scale variance to 1 or fix a loading to 1. • 2 measures, and loadings, are set equal to each other, or latent variable covary with another latent variables that has at least 2 reflective measures.	are not identified, unless 2 additional reflective measures are added to be caused by the construct- multiple indicator multiple cause (MIMIC) model. Hence, the choice of reflective measures for identification concern can change the loadings relating the measures to its latent variable, which in turn, change the construct itself. Hence, specifying the reflective measures in MIMIC model is monumental
Measurement independent measurement errors Error	No measurement error (i.e., error-free indicators)
Construct validity is manifested by the the magnitudes of the loadings relating the measures to the construct, which determined by the covariances among measures (e.g., the higher covariances the stronger correspondence)	1. strength between construct and its measures 2. proportion of variance in the construct attributed to the residual ζ (smaller better) 3. In relation to other variables (nomological validity and criterion-oriented validity) 4. Indirect effect of formative measure on reflective measure via the construct.
	Drawbacks: • estimates are sensitive to the chose of reflective measures as mentioned in the identification section. • Nomological network could not be empirically validated because of the conceptual heterogeneity in the formative measures.

Reflective	Formative
Construct to measures construct = latent variable From a realist perspective, construct = real entity (i.e., could influence) (e.g., psychological state), measures = scores	measures to construct construct = composite variable From an operationalist or instrumentalist perspective, formative measures define the construct (i.e., composite score) I'd suggest to think of formative constructs as processes Under constructivist perspective, constructs can be ascribed theoretical meaning, but would only be a conceptual entity But the causal thinking is inappropriate (because measures and construct are not 2 distinct entities), they are part-whole correspondence

According to (Edwards, 2010),

$$x_i = \lambda_i \xi + \delta_i$$

where

- x_i = reflective measure
- ξ = construct
- λ_i = effect of ξ on x_i
- δ_i = uniqueness of the measure (error)

$$\eta = \gamma_i x_i + \zeta$$

where

- η = construct
- γ_i = effect of x_i on η
- ζ = residual (unexplained part of η)

In case that η is a perfect weighted linear combination of x_i , then

$$\eta = \gamma_i x_i$$

28.1 Alternative to formative model

- Model with first-order constructs instead of measures, and each with single reflective measures
 - can't estimate the loadings and unique variances of the reflective measures
- Model with first-order constructs instead of measure, but each with three reflective measures, and second-order construct also has two reflective measures.
 - sometimes, first-order construct can't be measured multiple times (e.g., income, age)

Chapter 29

Scientific Writing

Section	Length (words)	Purpose	Verb Tense	Elements
Title	8 - 15			
Abstract	200-250	Paper's summary	Simple past (work done) Present (general fact, paper itself or analysis)	- Objectives - Methods - Results - Conclusions
Introduction	500-1000	Rationale for the paper	Present (current state-of-art, general background) Present perfect (previous research)	The problem
Literature Review	1000-2000	Understanding of the current knowledge landscape	Present (current state-of-art)	- Review of the literature - Hypothesis

Section	Length (words)	Purpose	Verb Tense	Elements
Methods	500-1000	What was done	Simple past (work done) Present (explain diagrams/figures)	• Approach/methods used • Data description • Procedure description
Results	500-1500	presents data, facts	Simple past (what was found) present (explain diagrams/figures)	• Results of analysis
Discussion	1000-1500	discuss relationships among constructs, and insights	Present (established knowledge) past (summarize findings)	• Relationships • Outliers • Comparison to previous studies
Conclusion	500	summarize findings	Present (implications and future search) Past (completed research)	

Rules of thumb:

- Past: events that have been completed
 - We found
 - Method proposed by X was followed
- Present Perfect: events that started in the past but are still ongoing
 - Many studied have researched on
- Present: events that are general fact, and current

Journal paper types:

1. Research Paper: hypotheses and findings

2. Methods Paper: new method
3. Review Paper: consolidate a domain
4. Discussion Paper: opinions regarding other papers

Chapter 30

Review Process

Communicate your review by O. C. Ferrell:

1. Paragraph
 1. contextualize the research
 2. convey the key message of the research
 3. be positive about what was done well
2. Paragraph:
 1. Is the paper well written, interesting and important?
 2. Are the methods, data, etc. appropriate?
 3. Does the data support the findings?
3. Paragraphs: Numbers, topics-examples of an issue:
 1. Factual errors
 2. Invalid arguments
 3. length/journal guidance
 4. Paths to improve the research

Documenting your reviews

- Use publons.com to document manuscripts you reviewed for promotion and CV
- to verify your review, send the “thank you for reviewing” email to reviews@publons.com

The review process:

- Journals always look for reviewers
 - Get involved and email your CV to editor, or area editor.
- Do not wait to answer whether you review or not.

Responding to reviews

- Turn around in 3 months
- R & R: reject and resubmit (new). The turn around will be sent to a new set of reviewers
- R & R: review and resubmit.

Chapter 31

Behavioral Seminar

“Cognitive perspective-taking refers to the ability to make inferences about others’ thoughts and beliefs.”

“Affective perspective-taking is the ability to make inferences about others’ emotions and feelings” (Healey and Grossman, 2018)

Chapter 32

CB Seminar

32.1 Overview

(Simonson et al., 2001):

- Behavioral Decision Theory (i.e., study judgment and decisions) (Kahneman & Tversky) vs Social Cognition
 - Both use mediation and path analysis
- Positivist (focuses on causation) Versus Postmodern (focuses on interpretation)

	Behavioral Decision Theory	Social Cognition
Underlying model	decision-making model, (e.g., determinants of choice)	response hierarchy model (i.e., how judgments and attitudes are formed)
Phenomena	Stimulus-based	Memory-based
Task		
Measures	Information acquisition, verbal protocols, and response time	Cognitive response

Types fo Consumer Research:

- Theory Development
- Theory Application
 - limited impact on managerial practice

Theory-Testing vs. Substantive Phenomena-Driven Consumer Research

(Janiszewski et al., 2016)

- Deductive-Conceptual Knowledge
 - A theory is “a statement of concepts and their interrelationships that shows how and/or why a phenomenon occurs.” (Corley and Gioia, 2011)
 - Knowledge Creation Strategies:
 - * Bridging disciplines
 - * Challenging assumptions
 - * Introducing mediators and moderators
 - * Contrastive explanation
 - * Borrowing and blending
 - Knowledge Appreciation
 - * Quality of the contribution
 - depends on rigor, and execution of research method
 - * Benefit of the contribution
 - Originality / Novelty
 - Interestingness
 - Changes in core understanding/beliefs
- Integration (i.e., tree of knowledge)

32.2 Social Influence

(McFerran et al., 2010)

- body type of others influences one’s food consumption
 - moderated by body type (undesirable reference group) of the other consumers
 - low appearance self-esteem or cognitive load also moderate this relationship
- Anchoring and adjustment processes
- Social Influence and Food Choice
 - people conform to group average (anchor)

- Study 1: Social influence effects is present regardless of food perception (healthy vs. unhealthy). People eat more when there is another person, but the magnitude is moderated by the other's body type (consume less when seeing obese others - undesirable group). The effect persists after social influence.
- Study 2: Presence of other increases food quantity.
 - High anchor (obese person takes high quantity), participants take less.
 - Low anchor (obese person takes small quantity), participants take more
- Study 3: People with low appearance self-esteem and high processing resources will attenuate more

(Moreau and Herd, 2010)

- Social comparison: people want to gain self-knowledge.
 - Self-designed products are based on:
 - * comparison with the characteristics of other comparable products
 - * comparison with skills, talents, and expertise of the other designers.
- Since professional designers are expert, consumer designers face upward comparison
- Study 1:
 - a. Self-deigned product evaluation is lower when compared to professional default product, than consumer designed default design
 - b. Firm guidance moderates this effect
 - There is a premium for self-designed product.
- Study 2: Defensive vs. Nondefensive processing
 - need to protect self-esteem/ self-image.
 - a. At low defensive processing, evaluations of self-designed products are lower when the default product is professionally designed than consumer designed
 - defensive processing reduces negative comparison
- Study 3: The behavioral consequences of social comparison in self-design.
 - Public prize (visibility) enhance self-evaluations.
 - a. Repair opportunity affects the evaluation of self-designed products

(Hamilton, 2003)

- Influence based on context effects (suggesting other unlikely alternative)
 - moderated by the belief whether others are trying to manipulate the choice context.
- People understand context effects (e.g., choice set construction).
- To influence others, people construct a set of unattractive alternatives (attraction effect), or a compromise between two other alternatives (compromise effect).
- Author hypothesized: People correct the contextual effects when they know constructors' intentions (to influence them).
- Results: "Subjects' beliefs that menus had been created (**by their friends**) to influence their choices seemed to enhance rather than limit the effectiveness of the menus." (p. 498) likely due to homophily bias.
 - Evaluated the target more favorably.
- Source credibility, expertise, trustworthiness, attractiveness, and similarity can influence the effectiveness of persuasion attempt (Wilson and Sherrell, 1993)
- Hypothesis: ulterior motive of persuader moderates increase the resistant effect of chooser than group-oriented motive.
 - Subjects are likely to agree (choose) when they know the menu is created by their friends
 - knowing that the menu has been created by a stranger, under compromise strategy, subjects still choose the middle alternative.
- Attraction strategy is more likely to be used than compromise strategy
- Even if subjects understand contextual effects (manipulation), they do not resist because
 - it's hard to recognize characteristics of a local set from the characteristics of a global set. (Simonson et al., 1993)
 - If subjects assume that the most attractive option has been eliminated, they have to choose the best among alternatives.

(Argo et al., 2005)

- Non interactive social presence in the consumption context.
- social forces has greatest influence (Latan? and Wolf, 1981):
 - number (social size) (large > small):
 - * emotions and behaviors: (e.g., stage fright, crowding)

- immediacy (proximity) (closer > far)
- social source strength (importance) (high > low)
- Social Impact Theory
- Study 1:
 - 0 -> 1 person increased positive emotions
 - 1 -> 3 person decreased positive emotions
- Study 2:
 - proximity moderates the impact of social size on emotions and brand selection.

32.3 Interpersonal perception and consumer lay beliefs

(Folkes and Patrick, 2003)

- Info valence: **Negativity bias for products** (negative info about a product has greater influence on brand perceptions than positive info). (Herr et al., 1991)
- **Positivity bias for services** (positive experience from a service provider can lead customers to infer positivity about the brand as a whole than negative experience).
 - Could be because service is more heterogeneous.
 - this effect is greater for novice consumer.
- Study 1: First time car insurance buyers
 - Alternative explanation: Subtyping (Weber and Crocker, 1983), or occupational stereotype consistency, question order are all ruled out.
- Study 2: replicate study 1: compare the effect of info about individual service provider to that of others who is not from the firm.
 - Subjects perceive focal firm employee more positively and consider negative experience from the other firms' agents as outliers.
- Because consumers are predisposed to think that a service encounter in general should be more positive than negative (for, 2005), this assumption could explain why consumers consider positive experience as typical and generalize it to the brand, and negative experience as outliers.
- 3 sources that affect consumers' beliefs:
 - general perceptions of services

- beliefs specific to a firm
- beliefs specific to an occupation
- Study 3 used a critical incident method to examine the following hypothesis: the positive effect is moderated by consumer experience (e.g., stronger for less experience consumers).
 - Valence affects perceived typicality, but experience with a firm did not influence typicality (can't find support for H3)
- Study 4: Replicate study 1 and 2 in a natural setting

(Brough et al., 2016)

- Men are less likely to buy environmentally friendly products than women because of the close association between green behavior and femininity (perceived by both men and women, and both users and observers).
 - Prior research attributes this gap to personality traits (women are more prosocial/altruistic (Lee and Holden, 1999), and stronger ethic of care (Zelezny et al., 2000))
 - The link between greenness and femininity could be explained by
 - * target of green marketing usually involved women (e.g., cleaning, laundry).
 - * caring and nurturing from greenness are feminine traits
- Since men are more concerned with gender-identity maintenance (which is kinda surprising from my view), they avoid purchasing green products which might jeopardize their macho image.
 - From social-identity theory: self-concept is also derived from perceived group membership (Turner and Oakes, 1986)

(Wang et al., 2016)

- Smile affects 2 social judgments: warmth and competence
 - Broader smile leads to more warmth, but less competent
 - * Consumers focus moderates this effect
 - Promotion-focused consumers (and low-risk consumption) like bigger smile on warmth
 - Prevention-focused consumers (and high-risk consumption) like slight smile (signal competent)
- Based on stereotype content model (SCM) of social judgments (Fiske et al., 2002)
- Facial configuration has evolutionary purposes.

(Haws et al., 2017)

- Consumers believe healthier foods are more expensive than less healthy foods.
- Due to the dual process model, intuitions based on biased information (processed heuristically) will lead consumers evaluating healthy claims with higher standard (e.g., higher standard for intuition-inconsistent than consistent claim).
- Lay theories or lay beliefs about food = expensive intuition affects consumer decision making process, because this process is low involvement one.

32.4 Emotions, mood and affect

32.5 Persuasion and attitude change

32.6 Judgment and decision making and behavioral pricing

32.7 Goals and Motivation

32.8 Brand personality and consumer -brand relationships

32.9 Culture and consumer behavior

32.10 Prosocial behavior and morality

32.11 Consumer well-being

32.12 Digital marketing and WOM

32.13 Experiential consumption and time

Chapter 33

Model Building

$$U_A = V - P_A - t|x - a| U_B = V - P_B - t|b - x|$$

where

- $\bar{x}$ = marginal consumer = indifferent consumer
- $-t|x - a|$ = Disutility of consuming a product away from ideal
 - transportation cost parameters, weights on the disutility incur when they purchase a product different from the ideal point.
- V = reservation price = max max that the customer is willing to take
- x = location of marginal customer - customer who is indifference from buying from firm A and firm B.

To find the point that a consumer is indifference from buying from firm A and buying from firm B, we equate $U_A = U_B$

$$V - P_A - t(\bar{x} - a) = V - P_B - t(b - \bar{x}) P_B - P_A = 2t\bar{x} + at - tb\bar{x} = \frac{b - a}{2} + \frac{P_B - P_A}{2t}$$

Market Share for firm A = $\bar{x}$

Market Share for firm B = $1 - \bar{x}$

$$\pi_A = P_A \left(\frac{b - a}{2} + \frac{P_B - P_A}{2t} \right) \pi_B = P_B \left(1 - \frac{b - a}{2} - \frac{P_B - P_A}{2t} \right)$$

$$\frac{d\pi_A}{dP_A} = \frac{d\pi_B}{dP_B} = 0$$

$$\frac{P_B}{2t} - \frac{2P_A}{2t} + \frac{b-a}{2} = 0$$

and

$$\frac{d^2\pi_A}{dP_A^2} = \frac{1}{t} < 0$$

$$1 - \frac{b-a}{2} - \frac{P_B}{t} + \frac{P_A}{2t} = 0 \quad \frac{b-a}{4} - \frac{3P_A}{4t} = 0 \quad \frac{t(2+b-a)}{3} = P_A \quad \frac{1}{2} - \frac{b-a}{4} + \frac{2+b-a}{12} = \frac{P_B}{2t} \quad \frac{4/3}{-} \frac{b-a}{3} = -\frac{P_A}{t}$$

Chapter 34

Sales

(Habel et al., 2021) found that consistent with traditional idea, having variable factor in salesperson's compensation plan will increase performance, but also introduce health hazard (more stress which results in emotional exhaustion and more sick days). In cases where salespeople have higher personal ability and more social resources, this harmful effect is mitigated.

Chapter 35

Satisfaction

Customer satisfaction measured using sentiment analysis (Kumar and Zymbler, 2019; Wei et al., 2020).

Another implementation can be found [here](#)

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